

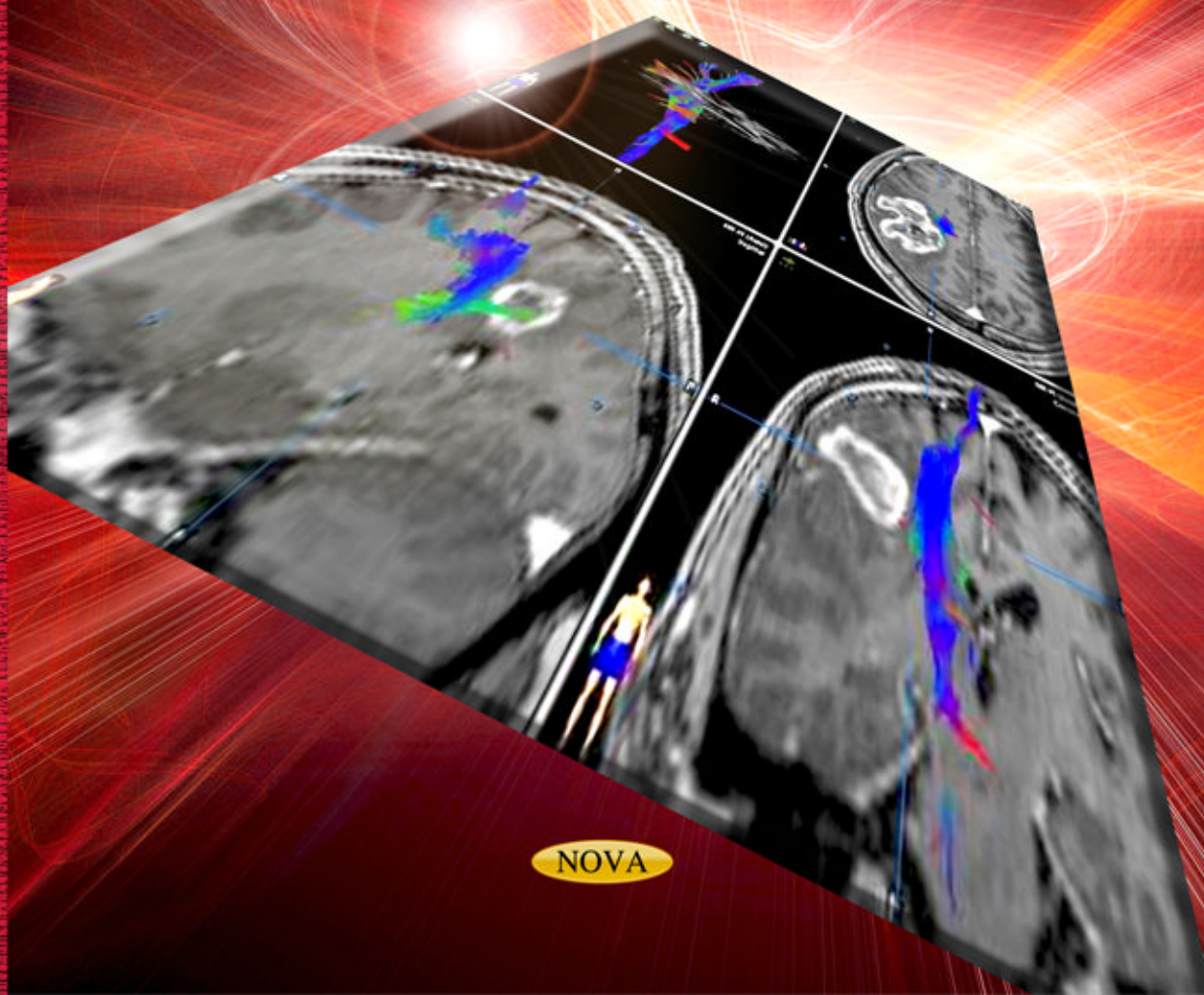
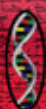
*Cancer Etiology, Diagnosis and Treatments*

# GLIOMAS

Symptoms, Diagnosis and Treatment Options

*Marzenna Wiranowska*  
*Frank D. Vrionis*  
*Editors*

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**CANCER ETIOLOGY, DIAGNOSIS AND TREATMENTS**

# **GLIOMAS**

## **SYMPTOMS, DIAGNOSIS AND TREATMENT OPTIONS**

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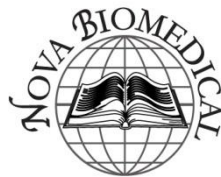
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**CANCER ETIOLOGY, DIAGNOSIS AND TREATMENTS**

**GLIOMAS**

**SYMPTOMS, DIAGNOSIS  
AND TREATMENT OPTIONS**

**MARZENNA WIRANOWSKA**  
**AND**  
**FRANK D. VRIONIS**  
**EDITORS**



*New York*

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# INTRODUCTION

Despite years of research, malignant glioma remains a disease with high mortality and limited treatment options. Once symptoms occur the disease is already in late stage of progression and glioma cells have spread throughout the brain. Therefore, standard therapies, which include surgery, radiation and chemotherapy, are only partially effective resulting in limited patient survival. There is an urgent necessity for further research of glioma biology to identify new biomarkers and targets of this disease for a better diagnosis and new more effective treatments. A summary of the most current knowledge relating to glioma symptoms, diagnosis and treatment options is provided in this book. The objective is to assist basic and clinical researchers in evaluating and developing of new antiglioma approaches. The book provides a comprehensive review of the biology, current treatment modalities and new emerging therapeutic approaches and targets for this highly aggressive and invasive brain tumor.

This book would not be possible without participation of numerous authors, basic and/or clinical scientists who contributed book chapters presenting the most up-to-date state of the art in this field. It contains 21 chapters organized into three main sections.

In the first section entitled **“Symptoms and Diagnosis”** the pathologic features of the most frequently WHO classified gliomas encountered in the clinical practice, are described. The molecular genetics and specifically the mutations that underlie brain tumors are outlined. The inherited genetic susceptibility and lifestyle risk factors reflecting on glioma epidemiology are discussed. With regard to “Diagnosis”, the clinical features of glioma as well as diagnostic methods that include conventional and advanced magnetic resonance imaging (MRI) and brain mapping are described. These brain mapping techniques are used pre and intraoperatively in glioma surgery and allow the identification of the cortical and subcortical functional pathways to maximize the extent of glioma resection while decreasing the risk of postoperative neurological deficits.

In the second section entitled **“Current Treatment Options”** the available therapies are discussed. They include surgery, chemotherapy, and radiation with specific focus on newly developed, dose rate modulated re-treatment irradiation for recurrent glioma patients. In this approach the radiation is delivered below standard dose-rates resulting in reduced normal tissue toxicity but significant tumor regression. Targeted therapies are discussed especially the anti-angiogenic therapies for which expectations were high, but so far have not matched clinical outcome. In addition, therapies for cerebrospinal fluid disseminated disease or leptomeningeal metastasis (LMD), a common problem in neuro-oncology patients are

discussed. Finally, treatments for the neurological complications of glioma management that are not directly related to tumor eradication are outlined.

In the third and final section of this book entitled **“Future Treatment Options: New Targets and Experimental Therapies”** some of the new emerging therapeutic approaches and targets for glioma are reviewed. These include targeting cancer stem cells, the vascular niche and the extracellular matrix components that provide support for these cells. Other innovative therapeutic approaches include targeting of glioma-associated antigens and the development of immunotherapeutics and oncolytic viruses. In addition, targeting of glioma pathways such as PTEN/AKT involved in cell proliferation, adhesion and apoptosis and the thiol-based redox system, the basis of chemotherapeutic resistance characteristic of glioma cell biology are discussed as well. Also, reviewed are new approaches (with some being in clinical trials) for targeting glioma with anti-cancer cytotoxins, such as interleukin 13 (IL-13)-based cytotoxin and glioma targeting peptides such as scorpion venom derived peptide, chlorotoxin, evaluated in anti-glioma therapeutics delivery. Finally, nanoparticles based systems are reviewed representing a new modality for the delivery of therapeutics to brain tumors with some systems such as magnetic nanoparticles being currently in clinical trials. This new rapidly developing field of nanotechnology-enabled delivery often employs targeting molecules such as chlorotoxin and antibodies against glioma. In summary, this book provides a compilation of the most recent information in the field of glioma biology, diagnosis and an emerging treatment options for malignant gliomas.

The editors would like to extend their gratitude and appreciation to NovaScience Publishers for the invitation to write this book. As editors, with a combined background in the basic science and clinical areas, we had the distinct honor to work with other authors who contributed to this book. Overall, this was an exciting opportunity not only to edit but also to contribute our own three chapters for this book. The editors sincerely believe that this book provides an up-to-date information on “Glioma: Symptoms, Diagnosis and Treatment Options” and will assist other researchers and clinicians in their pursuit of new anti-glioma treatments.

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## **SECTION I. SYMPTOMS AND DIAGNOSIS**



## **Chapter 1**

# **EPIDEMIOLOGY OF GLIOMAS**

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## **ABSTRACT**

Gliomas are the most common malignant adult brain tumor and one of the most lethal human cancers. The etiology of these highly aggressive tumors is still largely unknown. Associations with rare hereditary disorders, family history and common variants identified in genome wide association studies indicate that inherited genetic susceptibility explains a share of the interindividual risk for developing glioma. The contribution of environmental risk factors and lifestyle is not clear though available evidence suggests a less prominent role for exogenous exposures in the onset of glioma when compared to other tumors. This chapter provides an overview of the descriptive epidemiology of glioma and summarizes current knowledge on genetic and lifestyle risk factors and directions for future research.

## **INTRODUCTION**

An estimated 22,900 persons develop a primary brain tumor every year in the United States [1]. Gliomas encompass all tumors that originate from glial cells in the brain and account for the majority of primary malignant brain tumors [2]. Astrocytomas and oligodendrogliomas are the two primary types of gliomas. Glioblastoma multiforme (GBM), which is the most common type of astrocytoma, accounts for about 50% of gliomas diagnosed in adults [3]. The majority of GBM occur de novo (~95%), and a small proportion of GBM progress from low-grade or anaplastic astrocytoma (~5%) [4]. Glioma is a

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heterogeneous disease with different genetic alterations common for each histologic subtype. LOH 10q (~70%) and TP53 mutations (>65%) are the most common genetic alterations in astrocytomas, along with EGFR amplifications (40-50%) and PTEN mutations [4]. For oligodendrogliomas the most common alteration is LOH 1p/19q, which is observed in the majority of tumors of this type [4]. The etiology of these tumors remains poorly understood. Known genetic syndromes account for rare cases and common genetic variants that confer a modestly elevated risk have emerged in recent genome-wide scans. More recent studies are beginning to shed light on whether rare variants may account for the remaining heritability of these tumors. The role of lifestyle and environmental exposures in gliomagenesis remains unclear. High dose ionizing radiation is known to cause brain tumors of all types; however, whether lower radiation doses in diagnostic radiography increase occurrences of glioma is unknown. Similarly, diet and lifestyle, body weight, reproduction and hormonal risk factors have not been well studied owing in part to the low occurrence and rapid progression of these tumors hampering efforts to study large numbers of patients in epidemiologic investigations. In this chapter we review recent advances in the genetic epidemiology of glioma and state of knowledge on the potential role of lifestyle and modifiable risk factors as well as directions for future research.

## **INCIDENCE AND MORTALITY**

The annual incidence of adult glioma in the United States (US) is approximately 6 per 100,000 people [2]. Glioma incidence is higher in males (7 per 100,000 people) than females (5 per 100,000 people) and is twice as common in Caucasians than African-Americans [2]. Incidence for all gliomas is highest among individuals ages 75-84 years old; however, the distribution of tumor types differs across age groups with the median age at onset highest for primary GBM (64 years) and younger for anaplastic astrocytomas (51 years), mixed gliomas (42 years), and oligodendrogliomas (41 years) [2]. Malignant brain tumors are the second most common cancer in children after leukemia. Pilocytic astrocytoma (0.80 per 100,000 people), malignant glioma, not otherwise specified (0.56 per 100,000 people), and ependymoma (0.25 per 100,000 people) are the glioma subtypes most common among children aged 0-19 [2]. Glioma incidence rates increased slightly between 1980 and 1987, likely due to the use of better diagnostic procedures, but there has been no major consistent time trend over the last 30 years [5]. Survival rates vary by tumor type and age at diagnosis [6]. High grade glioma are aggressive and have poor prognosis: grade III anaplastic astrocytomas have a median survival time of 2-3 years, while grade IV GBM have a median survival time of only about 12 months [3]. Survival is lowest in GBM, with a 5-year survival rate of 3% among cases diagnosed at  $\geq 65$  years and 30% for cases diagnosed at  $\leq 45$  years. In contrast, oligodendrogliomas have five-year survival rates that exceed 70% [2].

## **GENETIC FACTORS**

Familial clustering and strong associations with rare genetic syndromes suggest that heredity plays a role in the onset of glioma. Rare autosomal dominant genetic conditions

associated with familial glioma include tuberous sclerosis (involving *TSC1* and *TSC2*), neurofibromatosis types 1 and 2 (involving *NF1* and *NF2*), Turcot syndrome (involving DNA repair genes *APC*, *hMLH1*, *hMSH2*, *PMS2*, and *PTEN*) and Li-Fraumeni syndrome (involving *TP53*) [7, 8].

Multiple studies have observed familial aggregation of glioma [9-15]. A study in Sweden and Norway calculated the standardized incidence ratio (SIR) for glioma by comparing incidence among cases with a first degree relative with a nervous system tumor to cases without a family history of such tumors [15]. The SIR for glioma was 1.8 for cases with a parent or sibling with a nervous system tumor and 11.2 in multiplex families where a case had both a parent and sibling with a nervous system tumor. In a population-based study in Utah that examined familial clustering for different glioma tumor types, significantly higher than expected rates of GBM and astrocytomas were observed in first degree relatives of probands with a glioma of the same histology type [14]. Familial clustering is observed in about 5% of glioma cases and rare genetic syndromes are present in about 1% of cases [11]. Given that there is a low prevalence of rare inherited mutations among cases, inherited risk for glioma is likely the result of co-inheritance of multiple low-risk variants.

Common polymorphisms in genes involved in DNA stability and repair, cell cycle, and carcinogen metabolism have been studied in relation to glioma (reviewed in [16-18]). Variants significantly associated with glioma risk have been identified in one or more studies among the DNA repair genes *XRCC7*, *XRCC1*, *PARP1*, *MGMT*, *ERCC1*, *ERCC2*, and the cell cycle gene *EGF*. No clear associations have been observed for genes involved in carcinogen metabolism (*NAT2* and *GST*); two meta-analyses found no associations between *GST* variants and glioma risk [19, 20]. Several studies have also investigated genetic variation in immune response genes (described in the Allergies section below). In genome-wide association studies (GWAS), common genetic variants in *CCDC26*, *PHLB1*, telomere maintenance genes *TERT* and *RTEL1*, and cell cycle genes *CDKN2A/B* and *EGFR*, were associated with glioma risk [21-24].

The associations with genetic variants differ by glioma histology, suggesting that there are different etiologic pathways for the recognized glioma subtypes. *TERT*, *RTEL1* and *CDKN2A/B* are most strongly associated with astrocytomas and GBM, whereas variants in *CCDC26* and *PHLB1* are primarily associated with astrocytomas and oligodendrogliomas [25, 26]. An uncommon variant in *TP53* found in approximately 1% of Caucasians is associated with a more than 2-fold excess risk in all subtypes of glioma [27]; additional rare variants conferring susceptibility will likely be discovered with the completion of next generation sequencing-based studies.

## ADULT STATURE AND BODY WEIGHT

Two large cohort studies observed a significant linear trend between self-reported adult height and increasing glioma risk [28, 29]. A prospective study that measured height directly however did not find an association [30]. The relationship between adult obesity and glioma risk is unclear. Several cohort studies did not find adult BMI, or other measures of body fatness including waist circumference and waist-to-hip ratio, to be associated with glioma risk

[28-30]. However, in one study, obesity at age 18 was significantly associated with an increased risk of developing glioma later in life [28].

## **HORMONAL AND REPRODUCTIVE FACTORS**

It is unclear whether reproductive factors and exogenous hormone use play a role in the development of glioma. Incidence of glioma is lower in women with *in vitro* evidence that estrogen inhibits proliferation of glioblastoma cells [31, 32] suggesting that female hormones may play a protective role. A positive association between later age at menarche and glioma has been observed in case-control [33-35] and cohort [36, 37] studies. Associations with other reproductive factors such as parity, age at first live birth, menopause status, and exogenous hormone use from oral contraceptives and hormone replacement therapy have been inconsistently reported [33-35, 37-40], although most studies lacked sufficient statistical power to detect an association.

## **ALLERGIES AND OTHER MEDICAL CONDITIONS**

Immunologic factors may be involved in glioma pathogenesis. Studies have consistently observed a significant inverse association between history of allergies and glioma risk [41-47] and several studies noted a significant dose-response relationship with a greater risk reduction observed with increasing number of allergic conditions [43, 46]. In the European INTERPHONE Study that pooled data from a series of independent case-control studies, individuals who reported ever having asthma, hay fever, eczema, or other type of allergy were less likely to develop glioma [45]. A history of autoimmune disease was also associated with a reduced risk of glioma in a large hospital based case-control study [42]. A meta-analysis of eight observational studies that included 3,450 glioma cases found a significant 30% to 40% reduction in the risk of glioma associated with a history of any allergy, asthma, or eczema [48]. Consistent with case-control studies, a prospective study with a median 19 year period of observation between self-reported allergy history and glioma diagnosis observed an inverse but statistically nonsignificant 50% reduction in glioma risk associated with self-report of any allergy [47]. Given that most case-control studies included proxy interviews in a proportion of cases, the inverse association could potentially reflect under-reporting of these conditions by proxy respondents: one study found a stronger association in proxy-reported cases than self-reported cases [43]; however, in a meta-analysis, the association remained statistically significant even after excluding studies that included proxy interviews [48].

The biological mechanisms through which allergies influence glioma risk are unclear. Individuals with allergies may have more efficient tumor immunosurveillance that limits proliferation of abnormal cells [49]. Consistent with this mechanism, two studies found glioma cases to have significantly lower levels of IgE [50, 51], the antibody responsible for atopic disease. A reduced risk for glioma has also been associated with higher levels of varicella-zoster virus IgG [52, 53]. Significant associations with glioma have been observed for single nucleotide polymorphism (SNPs) in interleukin (*IL*)-3 and *IL*-4, cytokine genes that are known to be associated with allergic conditions [54-56]. SNPs in the *SELP* gene which is



involved in innate immunity pathways was also associated with risk of GBM in one report [57].

History of seizures and epilepsy has consistently been associated with glioma [58, 59]. In a large case-control study that included 6,088 glioma cases, individuals admitted to a hospital with epilepsy two to three years before the reference date had a 5-fold higher odds of high-grade glioma [59]. It is unclear whether epilepsy and seizures are etiologically relevant to the onset of glioma, or instead represent early symptoms of the evolving tumor.

A number of studies have considered head injury as a potential risk factor for glioma [60-64]. In a study of 228,055 Danish individuals hospitalized for a concussion, skull fracture, or other head injury, investigators reported a slightly higher than expected incidence of intracranial tumors developing in this population [61]. However, the excess risk was confined to the period immediately following hospitalization for the head injury; this suggests that the detection of prevalent brain tumors during the diagnostic workup may have given rise to the association.

## SMOKING AND ALCOHOL

All available evidence suggests that cigarette smoking has no etiologic role in glioma. In a prospective study that pooled data from the Health Professionals Follow-up Study and Nurses' Health Study cohorts, neither smoking status, number of cigarettes smoked, nor duration of smoking contributed to glioma risk [65]. Null findings for smoking were demonstrated in a meta-analysis based on 6 cohort and 11 case-control studies [66].

Fewer studies have examined a possible role of alcohol consumption in the onset of glioma. One prospective study found a significant positive relationship with the incidence of GBM [67]. However, this finding was not confirmed in one case-control [68] and 3 other cohort [29, 69, 70] studies.

## N-NITROSO COMPOUNDS AND OTHER DIETARY FACTORS

N-nitroso compounds (NOCs) are comprised of nitrosamines and nitrosamides which are produced endogenously in the stomach when nitrite interacts with secondary amines from exogenous sources [71]. NOCs are established neuro-carcinogens in animal models [72, 73]. Processed and cured meat is the primary dietary source of NOCs [74]. Grains and some vegetables are an additional source produced by the conversion of nitrates in these foods to nitrite in the oral cavity [74]. High maternal consumption of cured meat during pregnancy has been associated with the development of childhood brain tumors [75]. However, no consistent association has been demonstrated between dietary NOC exposure and glioma in adults. Several case-control studies have suggested a significant positive association with consumption of cured or processed meat [76-79], whereas other case-control studies found no association [80-84]. A meta-analysis incorporating nine observational studies suggested that individuals with a high intake of cured meat have an increased risk for glioma [85]. However, as noted by the authors, small numbers of cases, a high rate of proxy respondents, and failure to control for total energy intake were limitations of the analysis. Prospective studies which

are less subject to methodologic bias have demonstrated no association between processed meat consumption and glioma risk [86, 87]. Drinking water is an additional potential source of exogenous NOCs; 2 studies found no association of nitrates in drinking water with glioma risk [88, 89].

Vitamins C and E can inhibit endogenous formation of NOCs [90]. In a large case-control study with 802 glioma cases, a significant inverse association was observed between dietary vitamin C intake and glioma [91]. Prospective studies, however, did not observe an association between dietary or supplemental vitamin C and E intake and glioma risk [86, 87, 92]. The findings for fruit and vegetable consumption have also been inconsistent. An analysis based on the INTERPHONE Study that included 1,185 glioma cases found that intake of leafy green and yellow-orange vegetables was significantly inversely associated with glioma risk, with the strongest associations observed for astrocytomas and GBM [84]. Prospective studies, however, have not supported these observations [86, 92]. Among other dietary factors studied in relation to glioma, carotenoids were found to be significantly inversely associated with glioma risk in 2 studies [83, 91] and a population based case-control study found that a high calcium intake was significantly associated with a reduced risk of astrocytomas in women only [93]. Further studies are needed to determine whether dietary modifications may lower glioma risk.

## IONIZING RADIATION

Exposure to ionizing radiation is the only well-established environmental risk factor for glioma [49, 94, 95]. Several studies have shown a significant increased risk of brain tumors in adults that had undergone therapeutic radiation for the treatment of childhood cancer [96-99]. In a retrospective cohort of 2,169 with childhood acute lymphoblastic leukemia (ALL) and followed a median of 18 years after therapeutic cranial radiation, an SIR of 46 was observed for subsequent brain tumors (most commonly GBM and astrocytoma) when compared to the incidence in the general US population [98]. The median time from ALL diagnosis to brain tumor development was 8.8 years (range: 1.7 – 14.1 years). Similarly, in a cohort of 14,361 children who were 5-year cancer survivors, the children who received therapeutic radiation had almost a seven-fold increased odds of primary glioma compared to children who did not receive radiation; the median latency period was 9 years following the original cancer diagnosis [99]. An excess risk for brain tumors was also observed in studies of children that received low-dose ionizing radiation to the head and neck region for treatment of tinea capitis [100-102] and hemangioma [103]. In a cohort of 10,834 Israeli children followed a median of 40 years after receiving scalp radiation for tinea capitis, a significant linear trend was observed between radiation dose and excess relative risk of malignant brain tumors including glioma [100]. A similar dose-response relationship was seen in a Swedish cohort of 28,008 infants followed exposure to ionizing radiation for treatment of skin hemangioma [103]. In several studies the excess risk for brain tumors was greatest among children exposed to radiation at younger ages [99, 100, 102].

Few studies have explored whether ionizing radiation from medical diagnostic procedures increases glioma risk. Evidence to date suggests either no association [60, 104], or a significant inverse association [105, 106] associated with exposure to computerized

tomography scans or medical X-rays. However, use of proxy respondents for the cases in these studies could have produced the inverse associations observed. No link to glioma risk has been demonstrated with occupational exposure to ionizing radiation [105, 107-109].

## CELL PHONE USE

The dramatic increase in cell phone use over the last several decades has prompted concern that exposure to radiofrequency (RF) electromagnetic fields emitted from cell phones may contribute to the onset of brain tumors. However, with few exceptions [110-112], the majority of studies to date (reviewed in [113, 114]) including a recent meta-analysis of 10 case-control studies [115] found no overall association between cell phone use and glioma risk. The possibility that long-term or intensive cell phone use might contribute to risk has been less exhaustively studied: in an analysis from the INTERPHONE Study that included 2,708 glioma cases, a non-significant excess risk was observed only among long term users ( $\geq 10$  years) with the highest cumulative call times [116]. However, the possibility of recall bias cannot be ruled out given well publicized concerns regarding a potential etiologic role for cell phone use in glioma. In an analysis confined to glioma patients in which the investigators sought to quantify RF dose to different regions of the brain after accounting for cumulative call time and cell phone type [117], an excess of brain tumors was observed in brain regions projected to be most heavily exposed to RF; risk was confined to persons that started using cell phones at least 10 years before the diagnosis consistent with a latency effect between exposure and onset of the tumor. However, the results were based on a small number of long-term cell phone users and need to be confirmed in larger studies. It has been noted that incidence of glioma has not increased over the last 20 years, a period that saw a dramatic rise in exposure to RF from cell phones, arguing against an important causal role associated with their use [118].

## CONCLUSION

Glioma is highly aggressive and rapidly fatal tumor with a poorly understood etiology. A number of rare hereditary syndromes account for only a small fraction of cases. Furthermore, high-dose ionizing irradiation, the only documented environmental risk factor, accounts for few modern cases of the disease, leaving most occurrences of glioma unexplained. There is sufficient evidence to rule out smoking as a risk factor and alcohol is unlikely to play a major contributing role. However, diet and other lifestyle factors including body weight and exercise remain largely unexplored. A strong inverse association between allergies and glioma risk as well as significant associations with IgE antibodies suggest a role for immunity in gliomagenesis although the underlying mechanism remains to be elucidated. Future studies that consider glioma heterogeneity at the histologic or molecular level are needed to shed further light on the etiology of these tumors and identify population subgroups at greatest risk.

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## *Chapter 2*

# **PATHOLOGY OF GLIOMAS**

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## **ABSTRACT**

This chapter seeks to provide the reader with a concise, yet comprehensive and current review of the pathologic features of the most frequently encountered gliomas in clinical practice.

Gliomas are tumors of neuroepithelial tissue origin and the term glioma refers to a broad class of neoplasms which includes all tumors thought to arise from glial cells, i.e. astrocytes, oligodendrocytes and ependyma.

The World Health Organization (WHO) classification and grading of Central Nervous System (CNS) tumors will be used all through the chapter, since it is the preferred scheme used by most clinical and research disciplines to define, diagnose, make therapeutic decisions and dictate prognosis.

## **INTRODUCTION**

With an incidence rate of 6.5 per 100,000 population per year, an estimated 22,340 cases of CNS tumors will be diagnosed in the U.S. in 2011 [1]. Gliomas represent about 30% of all CNS tumors, and although grouped into this broad category, they differ greatly in morphologic features, molecular abnormalities and behavior.

The WHO Classification of Tumors of the CNS [2], includes all gliomas within the neuroepithelial tumor group and categorizes them by cell of origin and histologic grade (Table 1). This chapter will detail the morphologic features of the most commonly diagnosed gliomas.

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**Table 1. Classification and grading of glial tumors adapted from the WHO Classification of Tumors of the CNS, fourth edition [2]**

|                                                         | WHO Grade |    |     |    |
|---------------------------------------------------------|-----------|----|-----|----|
|                                                         | I         | II | III | IV |
| <i>Astrocytic Tumors</i>                                |           |    |     |    |
| Subependymal giant cell astrocytoma                     | ▲         |    |     |    |
| Pilocytic astrocytoma                                   | ▲         |    |     |    |
| Pilomyxoid astrocytoma                                  |           | ▲  |     |    |
| Pleomorphic xanthoastrocytoma                           |           | ▲  |     |    |
| Diffuse astrocytoma                                     |           | ▲  |     |    |
| Anaplastic astrocytoma                                  |           |    | ▲   |    |
| Gliomatosis cerebri                                     |           |    | ▲   |    |
| Glioblastoma (including Gliosarcoma and Giant Cell GBM) |           |    |     | ▲  |
| <i>Oligodendroglial Tumors</i>                          |           |    |     |    |
| Oligodendroglioma                                       |           | ▲  |     |    |
| Anaplastic oligodendroglioma                            |           |    | ▲   |    |
| <i>Oligoastrocytic Tumors</i>                           |           |    |     |    |
| Oligoastrocytoma                                        |           | ▲  |     |    |
| Anaplastic oligoastrocytoma                             |           |    | ▲   |    |
| <i>Ependymal Tumors</i>                                 |           |    |     |    |
| Subependymoma                                           | ▲         |    |     |    |
| Myxopapillary ependymoma                                | ▲         |    |     |    |
| Ependymoma                                              |           | ▲  |     |    |
| Anaplastic ependymoma                                   |           |    | ▲   |    |

## ASTROCYTIC TUMORS

### Glioblastoma (GBM), WHO Grade IV

#### Introduction

Previously termed “Glioblastoma multiforme”, glioblastoma (GBM), including all variants and gliosarcoma, is a highly invasive tumor of astrocytic differentiation that can arise de novo, also called “primary” GBM or progress from a lower grade, previously known astrocytoma, in which case it is referred to as “secondary” GBM. There are no distinct morphological differences between primary and secondary GBM. It is the most common and most aggressive brain tumor with a median survival of 1 year, despite the use of a great variety of treatment modalities. It is slightly more common in men than women with a ratio of 3:2. GBM occurs at any age, but its incidence is highest between the 5<sup>th</sup> and 7<sup>th</sup> decades. Spinal cord and extra CNS metastases are rare but several case reports have been published [3, 4]. A known risk factor associated with GBM in the past was diagnostic X-rays, doses of which are now much lower. The association of GBM and brain tumors in general, with

cellular telephony has been a subject of intense debate and investigation since the mid 90's, and several contradicting studies have been published without conclusive evidence, raising more questions than answers [5-11]. Although not completely ruled out, more studies are needed to definitely ascertain a true association between cellular telephone usage and GBM. In addition, perhaps more advanced diagnostic imaging techniques and more consistent means of reporting have given a false impression of an increased incidence of brain tumors in general.

Clinical presentation of course, depends on the location of the tumor. It may vary from visual disturbances if located in the occipital lobe, motor or sensory deficits if involving motor or sensory cortices respectively, to stroke-like symptoms, seizures and headache among others. Traditionally GBMs continued to be grouped under the astrocytic tumors; however, newer but still unproven concepts are being considered in regards to a more undifferentiated cell of origin, either an immature precursor cell or neural stem cells.

### ***Macroscopic Features***

Typically, the epicenter of most GBMs is located in the subcortical white matter of the cerebral hemispheres. Grossly it has a variegated appearance as its previous term "multiforme" implies.

There is poor demarcation between the tumor and the surrounding parenchyma with an outer firm and grayish area and a center, which may contain yellowish discoloration due to areas of tumor necrosis, or brownish foci consistent with recent and/or previous hemorrhage. (Figure 1).

Usually, tumors are already very large at presentation, even in cases with very short clinical history. It most often involves a single lobe, but many cases will track antero-posteriorly through the fornices, to the adjacent hemisphere through the corpus callosum. Some may involve the leptomeninges and attach to the dura, occasionally posing a challenge to radiologically differentiate them from meningiomas.

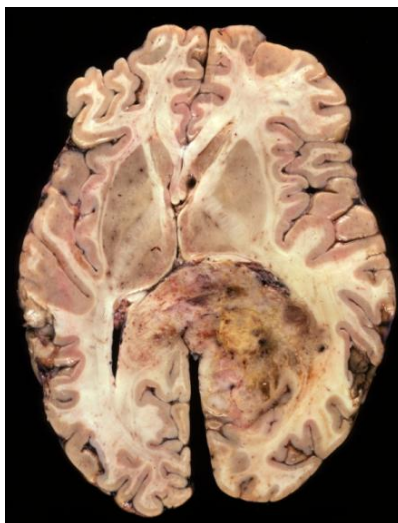


Figure 1. Axial section of a brain with a GBM extending from the right to the left hemisphere through the corpus callosum. Necrosis and punctate hemorrhage are evident Image courtesy of Dr. Leroy R. Sharer.

### ***Microscopic Features***

GBMs are one of the most heterogeneous tumors in humans, which has led to the prior description of several variants as separate entities, i.e. small and giant cell GBM, as well as gliosarcoma. These are all now included under the umbrella of Glioblastoma and will be detailed further in the chapter. The necessary criteria for the diagnosis of GBMs are: high cellularity with anaplasia, abundant mitotic figures, robust microvascular proliferation and/or necrosis, pseudopalisading or otherwise.

The variation in both cellular density and pathologic features of GBMs is well known, hence proper sampling of the tumor is always recommended. The amount and viability of the neoplastic cells vary, from the outer zones of the tumor to a usually necrotic center. A small biopsy from either of these areas has the potential to misinform the diagnosis. The periphery of the tumor exhibits low cellularity, with tumor cells infiltrating recognizable normal structures. In such areas an erroneous diagnosis of a lower grade tumor, i.e. fibrillary or anaplastic astrocytoma is a dangerous pitfall. A biopsy of the necrotic center can make it impossible to differentiate a primary CNS tumor and a metastasis from a morphological or even immunohistochemical standpoint. Intraoperative consultation with proper radiological correlation plays a crucial role in assuring that representative material is present at the time of the surgery, and may guide the neurosurgeon in avoiding increased patient morbidity and additional surgery.

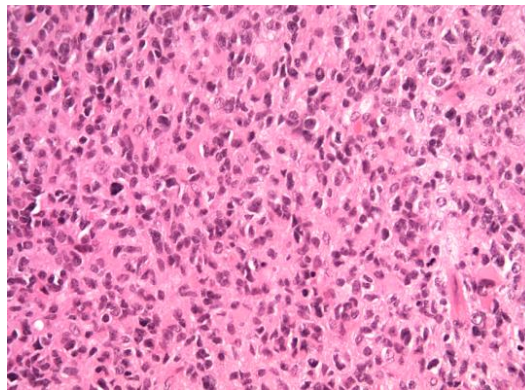


Figure 2. Glioblastoma with high cellularity and mitotic activity.

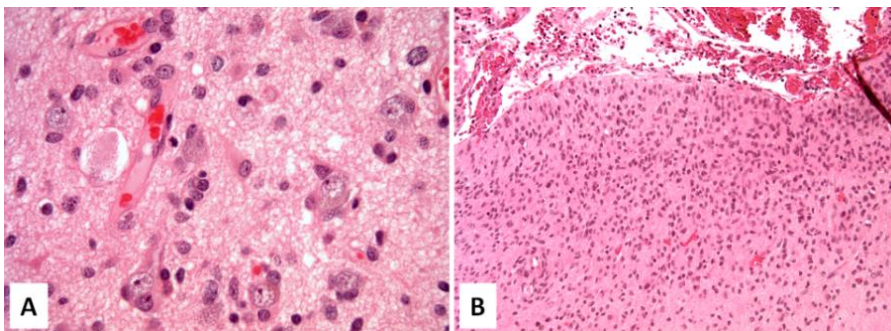


Figure 3. Secondary structures in Glioblastoma A) Neoplastic cells around blood vessels and neurons 'neuronal satellitosis'. B) Neoplastic cells forming subpial nodules.

GBMs are composed of a mixture of cells varying in morphology from uniform and monotonous, medium-sized with obvious astrocytic phenotype to highly variable, ranging from small with scanty cytoplasm and hyperchromatic nuclei to large with vesicular nuclei, occasional prominent nucleoli and abundant eosinophilic cytoplasm (Figure 2).

Binucleate and gemistocytic forms are not uncommon. These neoplastic cells infiltrate the CNS parenchyma in an insidious manner; they surround neurons, a phenomenon called neuronal satellitosis, they also surround vessels, migrate through white matter tracts and when a boundary is reached, e.g. the pial surface, they accumulate in small nodules until such edge is also breached, extending into the subarachnoid space with potential CSF spread (Figure 3). This pattern of cell migration was described in 1940 as secondary structures [12].

The WHO defines microvascular proliferation, previously termed endothelial cell hyperplasia, another hallmark of GBM as “multilayered, mitotically active endothelial cells together with smooth muscle cells/pericytes” [2]. These vessels, often arranged in a spheroid fashion, referred to as glomeruloid tufts because of their resemblance to renal glomeruli, are often thrombosed (Figure 4).

Tumor necrosis, probably the most distinguishing characteristic of GBMs, can be seen as sheets of necrotic tumor cells and as bands of necrotic tumor surrounded by fusiform and hyperchromatic tumor cells arranged in a pseudo-palisading pattern (Figure 5).

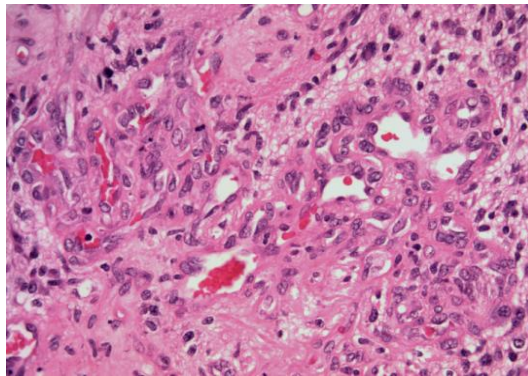


Figure 4. Classical appearance of microvascular proliferation as multilayer ‘glomeruloid tufts’ in GBM.

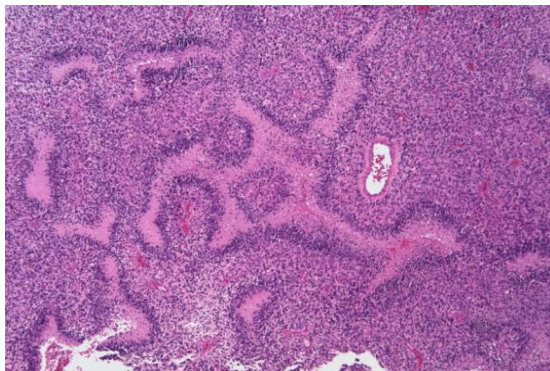


Figure 5. Pseudopalisading necrosis which is the most recognizable hallmark of GBM.



Immunohistochemically, most GBMs show variable patterns of expression of GFAP and a high proliferation index (Ki-67 nuclear antigen labeling index identified with MIB-1 antibody), average of 12.4% [13], but may vary significantly from case to case. Expression of the nuclear antigen p53 is often seen in GBMs, although more commonly noted in secondary GBMs, while EGFR amplification is frequently present in primary or de novo GBMs. Despite fairly similar clinical and prognostic characteristics, some variants of GBM have neoplastic cells that display a predominant morphology and hence warrant a separate description. Included in this group are small and giant cell GBM and gliosarcoma.

### ***Small Cell GBM***

Small cell GBMs are highly cellular with a predominance of small cells with scanty cytoplasm and hyperchromatic, relatively monomorphic nuclei. Immunohistochemistry for GFAP is scarce; the proliferation index is usually high and expression of p53 abundant (Figure 6).

Metastatic small cell carcinoma and other small blue cell tumors must be considered within the differential diagnoses and must be carefully excluded, particularly when sample size is limited, a common challenge with many neurosurgical specimens.

### ***Giant Cell GBM***

Giant cell GBM is relatively well circumscribed and superficial, occasionally posing a diagnostic challenge radiographically with metastatic carcinoma. It is composed predominantly of very large cells, although medium and small cells are also part of the tumor. These large cells have single nuclei or are multinucleated with abundant eosinophilic cytoplasm. Very bizarre cells, atypical mitoses, cytoplasmic lipidization, nuclear inclusion can all be observed in these enormous cells. The proliferation rate is similar to that of most GBMs. Necrosis is usually encountered, however, pseudopalisading pattern of necrosis is not a feature of giant cell GBMs. Microvascular proliferation and glomeruloid tufts are seldom seen. (Figure 7). Expression of p53 antigen is characteristic.

### ***Gliosarcoma***

As with giant cell GBM, gliosarcomas tend to appear well circumscribed and may attach to the dura, prompting consideration of meningioma as well as metastatic carcinoma within the differential diagnosis. Gliosarcomas in addition to the glial component have a sarcomatous component that is thought to derive from the stroma of the prominently proliferating vessels of GBMs (Figure 8).

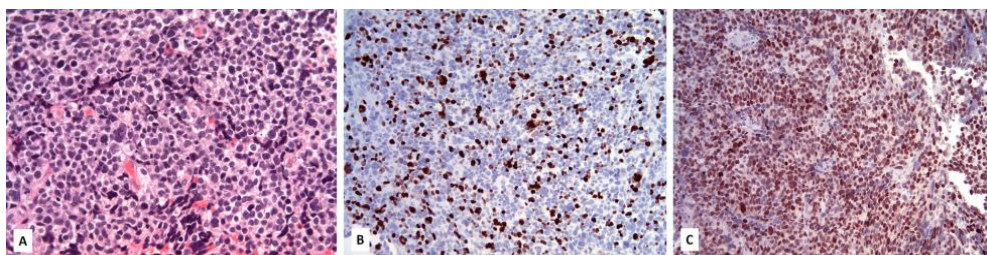


Figure 6. Small cell GBM A) Primitive small cells with abundant mitoses B) Elevated MIB-1 C) High p53 immunoexpression.



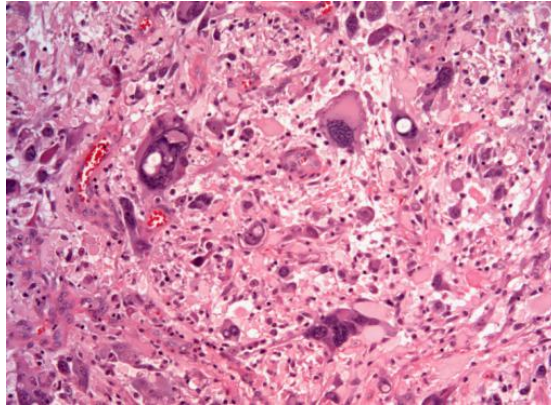


Figure 7. Giant cell GBM showing high degree of atypia and nuclear pleomorphism including multinucleated and bizarre cells.

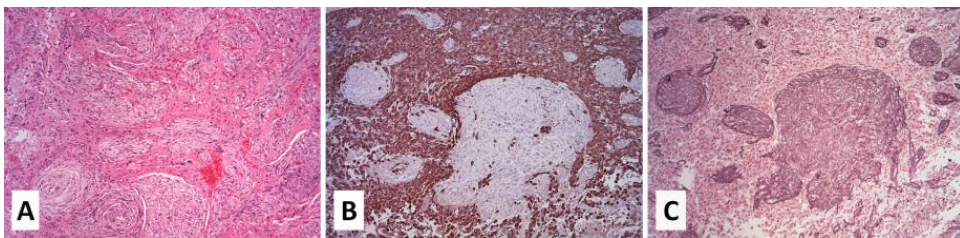


Figure 8. Gliosarcoma A) Gliomatous component mixed with well delineated nodules of sarcomatous component B) Strong and diffuse GFAP expression of the gliomatous component C) Reticulin-rich sarcomatous component (same microscopic field as B).

These sarcomatous elements, which give gliosarcomas their characteristic biphasic pattern of growth, as implied by the term “sarcomatous” are also malignant. Mitoses, necrosis and marked nuclear atypia are all found in these areas, which are commonly arranged in bundles of spindle cells similar to that of soft tissue sarcomas. Cartilaginous [14], smooth muscle [15], and osteosarcomatous [16] elements have also been described in the sarcomatous component. Several studies have shown that both components have similar genetic profile such as TP53 [17] and PTEN [18] mutations, as well as other studies, which have shown that both components shared 57% of the chromosomal imbalances detected by CGH [19]. EGFR amplification is rarely seen in this entity.

## Anaplastic Astrocytoma, WHO Grade III

### *Introduction*

Anaplastic astrocytoma is an infiltrating neoplasm that like GBMs can arise from a previous lower grade diffuse astrocytoma or de novo, and invariably progress to a higher, grade IV or GBM. It is therefore considered to be an intermediary stage in the continuum that is glioma with astrocytic differentiation. It arises, like other gliomas, from the subcortical white matter of the cerebral hemispheres and is characterized by high cellularity, with increased nuclear pleomorphism and mitotic activity, however necrosis and microvascular proliferation are not present. Proper sampling of the tumor is essential to avoid a misdiagnosis

of a lower grade astrocytoma. The tumor has a peak incidence in the fifth decade and a slight male predilection.

### ***Macroscopic Features***

Typically a firm homogenous expansion of the hemisphere without circumscription is seen, and distinction from a lower grade diffuse astrocytoma is often difficult on macroscopic grounds alone. However, the presence of granular, friable and discolored hypervascular areas, seen in more cellular regions, may provide some clue to the diagnosis.

### ***Microscopic Features***

Higher cellularity, nuclear atypia and mitoses are the main differences between anaplastic astrocytomas and the lower grade fibrillary astrocytomas. Neoplastic cells have irregular pleomorphic nuclei and mitotic figures, including atypical forms, are essential for the diagnosis even in the presence of low cellularity (Figure 9A). Although some degree of endothelial cell hyperplasia is often seen, microvascular proliferation, defined as multilayered vessels with multiple lumina or the presence of necrosis are absent. Expression of GFAP by immunohistochemistry is common. The proliferation index (MIB-1 immunohistochemistry) ranges between 5-10% (Figure 9B) although less cellular areas may have lower values. Mutations of p53 are identified in 50-60% of anaplastic astrocytoma. [19]

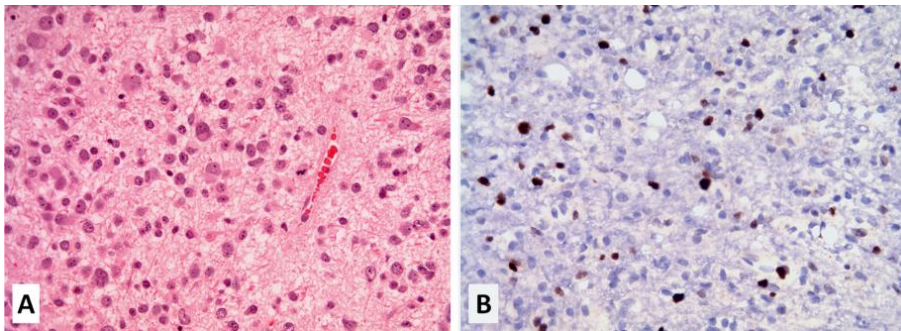


Figure 9. Anaplastic astrocytoma A) Increase cellularity, nuclear pleomorphism and mitotic figures. B) Increased proliferation index (MIB-1).

## **Diffuse Astrocytoma, WHO Grade II**

### ***Introduction***

Diffuse astrocytoma includes morphologic variants of low-grade astrocytic neoplasms such as fibrillary, protoplasmic and gemistocytic astrocytoma. Gemistocytic astrocytoma is considered by some authors [20, 21] to be anaplastic or WHO Grade III, because of its more aggressive behavior, progression to anaplastic astrocytoma and GBM and increased mortality. However the WHO does not recommend its inclusion in the anaplastic grade III category, as these tumors lack the increased mitotic activity that is typical for anaplastic astrocytoma. They arise more commonly in the white matter of cerebellar hemispheres, with the brainstem and spinal cord following as the next most common locations. Their peak incidence is in the

fourth decade with approximately 10% manifesting in individuals younger than 20 [2] and, like other astrocytic tumors, have a slight male predominance.

### ***Macroscopic Features***

These infiltrating neoplasms typically show a distended zone of grayish or slightly yellowish color involving brain parenchyma without clear borders and obscuring the gray-white junction (Figure 10). Microcystic and gelatinous areas can also be seen.



Figure 10. Axial section of a brain with a WHO grade II astrocytoma. Expansion of the white matter by a homogeneous mass that also involves the cortex, focally. Image courtesy of Dr. Leroy R. Sharer.

### ***Microscopic Features***

Diffuse astrocytomas have low to moderate cellularity, at times difficult to differentiate from reactive conditions and a true challenge during intraoperative consultations. The neoplastic cells resemble their non-neoplastic fibrillary, protoplasmic or gemistocytic astrocyte counterpart and are distributed uniformly in a background of fine processes and microcysts (Figure 11). The nuclei are round and uniform and mitotic figures are exceptionally rare. Except for the gemistocytic variant which has abundant eosinophilic cytoplasm, the neoplastic cells have scanty cytoplasm and fine cell processes. Perivascular accumulation of lymphocytes is a common feature and is seen in many other low grade tumors. Diffuse expression of GFAP is the rule and the proliferation index (MIB-1 immunohistochemistry) is usually below 4%. Like anaplastic astrocytomas p53 mutations are seen in over 60% of cases. In gemistocytic astrocytomas, the gemistocytes should represent more than 20% of the neoplastic cells.

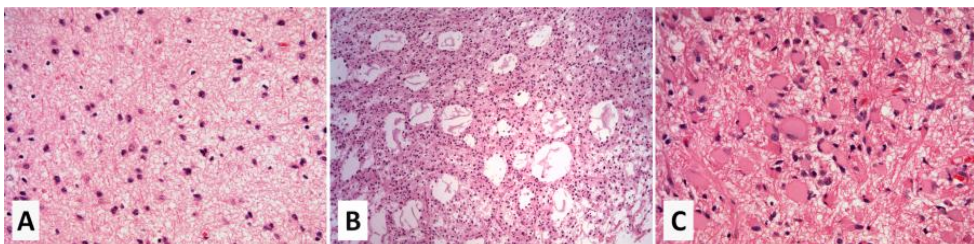


Figure 11. Diffuse astrocytoma WHO Grade II. A) Fibrillary astrocytoma with low cellularity and B) microcyst formation C) Gemistocytic astrocytoma with the classical astrocytes with abundant eosinophilic cytoplasm and peripherally placed nuclei.

Distinction from oligodendrogliomas is, at times, difficult, but the presence of p53 immunoreexpression and the absence of loss of heterozygosity (LOH) in chromosomes 1p and 19q should support morphologic diagnoses.

## OLIGODENDROGLIAL TUMORS

### Oligodendroglioma (WHO Grade II)

#### *Introduction*

Oligodendrogliomas are infiltrating gliomas in which neoplastic cells recapitulate the morphology of normal oligodendrocytes. First described by Bailey and Cushing in 1926 [22], oligodendrogliomas are thought to arise from a cell committed to oligodendrocyte lineage and typically present in middle-aged adults with a history of seizures [23]. Loss of heterozygosity (LOH) in chromosomes 1p and 19q is the most relevant molecular abnormality described in oligodendrogliomas. No clear etiology is recognized for oligodendrogliomas. A few case reports have identified links between oligodendrogliomas and irradiation [24, 25], nonetheless, this association only accounts for an insignificant portion of all oligodendrogliomas. Experimental animal models have implicated carcinogens, but the influence, if any that these substances have in human gliomagenesis remains undetermined. Studies have also identified viral etiologies [26], but subsequent investigators have not been able to reproduce these findings [27].

Approximately 2.0% of brain tumors and 6.5% of gliomas are oligodendrogliomas [28]. Typically, these cancers arise in adults, typically between the ages of 40 and 45 [29, 30]. Younger patients are rarely afflicted, with an estimated incidence of 1.4% of all brain tumors in children under the age of 14. According to data from 2004-2007 reported by the Central Brain Tumor Registry of the United States, oligodendrogliomas may arise more frequently in males than females with a ratio of 1.28 to 1. Overall, the age-adjusted (2000 U.S. standard population) annual incidence of oligodendrogliomas was 0.28, with a 95% confidence interval of 0.27 to 0.29, per 100,000 people [28].

#### *Macroscopic Features*

Oligodendrogliomas are relatively well demarcated, grayish soft tumors which commonly occur in the cortex and the white matter of the cerebral hemispheres, with the frontal lobe being involved most often [31]. Gelatinous change can be seen in cases with mucoid degeneration and intratumoral calcification may confer a coarse or gritty texture.

#### *Microscopic Features*

Oligodendrogliomas are low to moderately cellular, diffusely infiltrating gliomas. Focal nodules with higher cellularity are often seen in large resection specimens. The neoplastic cells, although larger, resemble the non-neoplastic oligodendrocyte and have round, relatively uniform and slightly hyperchromatic nuclei. The classical clearing of the cytoplasm also referred to as “fried egg appearance” is very useful in the diagnosis of oligodendrogliomas, however this is an artifactual change related to fixation and therefore is not present in frozen or smear preparations.



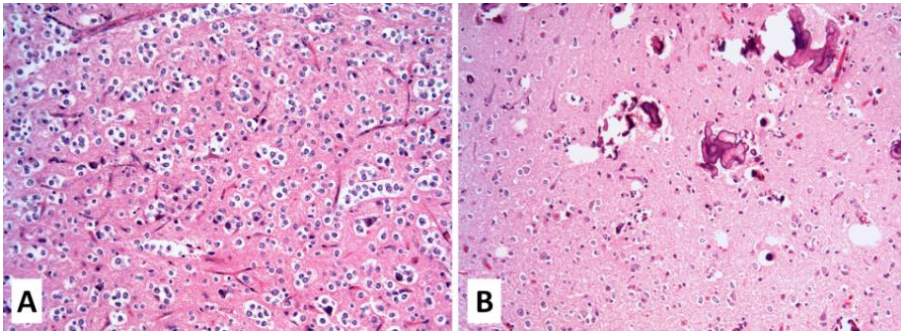


Figure 12. Oligodendroglioma WHO Grade II A) network of branching capillaries, relatively uniform nuclei and classical clearing of the cytoplasm also referred to as “fried egg appearance”. B) Microcalcifications at the tumor edge abundant eosinophilic cytoplasm and peripherally placed nuclei.

Oligodendrogliomas also contain tumor cells which morphologically resemble small gemistocytes known as minigemistocytes. The vessels are characteristically arranged as a dense network of branching capillaries, commonly referred to as “chicken wire vasculature” (Figure 12A).

Microcalcifications, although not specific, are often present and may be associated with blood vessels. These microcalcifications may be found within the bulk of the tumor or the peripherally invaded brain (Figure 12B).

A few mitotic figures and nuclear atypia can be seen, however the presence of abundant mitoses, microvascular proliferation and necrosis are indicative of a WHO Grade III or Anaplastic Oligodendroglioma.

The diagnosis of oligodendroglioma is made mostly on a morphologic basis, since histological features are, in most cases, classical and no specific oligodendroglioma marker is available. GFAP expression is focal and variable, although it is more diffusely expressed in the minigemistocytes. MIB-1 proliferation index is usually under 5%. The presence of 1p and 19q LOH and the absence of p53 immunoexpression can be helpful in difficult cases.

## Anaplastic Oligodendroglioma (WHO Grade III)

### *Introduction*

Anaplastic oligodendrogliomas may arise de novo or secondary to WHO grade II oligodendrogliomas, with features of malignancy and consequently a worse prognosis than WHO Grade II oligodendrogliomas. According to 2004-2007 data reported by the Central Brain Tumor Registry of the United States, approximately 0.6% of primary brain tumors are anaplastic oligodendrogliomas [28].

Typically, these tumors arise in adults, typically between the ages of 45 and 50 [29, 30]. Overall, the age-adjusted (2000 U.S. standard population) annual incidence of oligodendrogliomas was 0.12, with a 95% confidence interval of 0.11 to 0.13, per 100,000 people [30].

### ***Macroscopic Features***

Although anaplastic oligodendrogliomas may exhibit areas of tumor necrosis, overall, the macroscopic features are quite similar to those of WHO grade II oligodendrogliomas.

### ***Microscopic Features***

Anaplastic oligodendrogliomas are also diffusely infiltrating gliomas. The neoplastic cells still bear resemblance to oligodendrocytes with perinuclear halos and few cellular processes, however, larger cells with nuclear pleomorphism, increased mitotic activity, vascular proliferation (Figure 13 a and B) and necrosis are characteristic. Microcalcifications, minigemistocytes and branching “chicken-wire” capillary network can still be identified in anaplastic oligodendrogliomas.

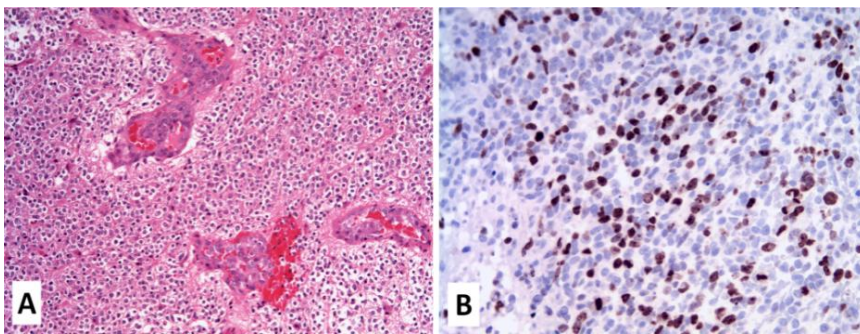


Figure 13. Anaplastic Oligodendroglioma WHO Grade III A) Increased cellularity, mitotic activity and vascular proliferation. B) High proliferation index (MIB-1).

## **CONCLUSION**

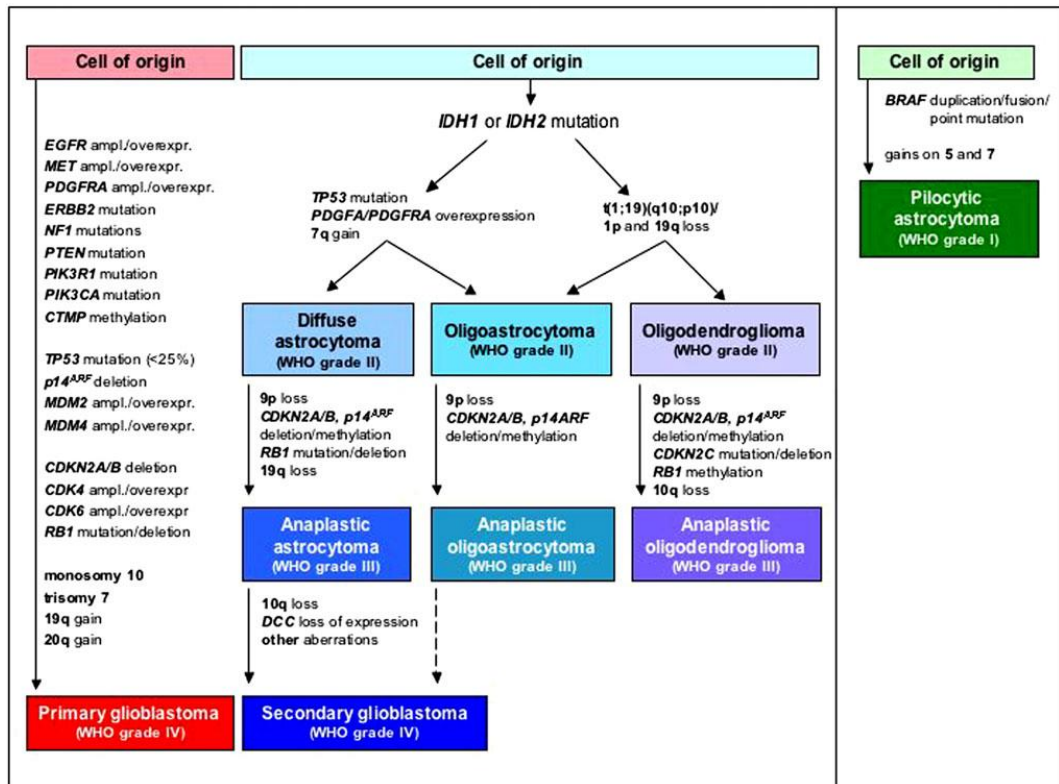
The prognosis of brain tumors, most prominently GBM remains poor, despite extensive investigative efforts to identify etiology and effective treatment modalities.

Different molecular abnormalities have been discovered and have led to attempts to classify gliomas based on their molecular profile. Studies of gene expression patterns of WHO Grade II and IV astrocytomas revealed three major patterns (mesenchymal, proliferative and proneural) that relate to survival and resemble key stages of neurogenesis [32].

Table 2 shows the molecular evolution and differences among the more common gliomas [33] including more recently described IDH mutations in lower grade gliomas and the secondary GBM that results from their malignant evolution as well as different pathways for primary and secondary GBM.

More recent advances using targeted therapies directed toward tumors with specific molecular alterations which have yielded some success in the management breast and lung cancer among others, have not yet proven efficacious in gliomas. The proper classification, molecular progression, prognostic determinants and therapeutic options in the management of gliomas thus continue to remain both challenging and elusive targets.

Table 2. Reproduced with permission [33]



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### *Chapter 3*

## **MOLECULAR GENETICS OF BRAIN TUMORS**

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### **ABSTRACT**

Alterations in a brain cell genome, within genes that control cell growth, cell cycle, and cell death, are the basis for the formation of brain tumors. These mutations usually arise spontaneously during cell division or as a result of failure to properly correct DNA damage. External causes may significantly increase the risk for development of a brain tumor; yet this evidence regarding this malignancy remains limited. In this chapter, the identified gene alterations that are implicated in the development of a brain cancer are reviewed. However, further research is of significant importance to acquire a complete picture of the various gene mutation patterns for each cancer.

### **INTRODUCTION**

The genetic understanding of brain tumors has substantially increased over the last several years. The basic principle of tumorigenesis is that cells accumulate DNA alterations and damage that as a result mutations of critical genes that then allows for uncontrolled cell division. A normal cell will tightly regulate cell replication but a cancer cell has uncontrolled growth driven by pathologic instructions from the mutated genome. Normally, cells are driven to maintain homeostasis with the surrounding environment, whereas cancer cell will invade and eventually destroy the surrounding brain.

Although for a small percentage of brain tumors the first mutation is inherited as part of a hereditary syndrome such as Von Hippel Lindaeu, in most tumors mutations arise spontaneously during cell division and/or as a result of improper DNA repair. In both

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scenarios subsequent mutations are needed to initiate tumor formation. According to Knudson theory, the first hit dramatically increases the risk of tumor formation, but by itself it is not sufficient. The paradigm for hereditary cancer syndromes involves the inheritance of a mutation in a tumor suppressor gene in the germline and a second spontaneous mutation inactivating the remaining allele.

The way to distinguish a somatic from a hereditary mutation is to sequence the suspected gene mutation in both the tumor and normal tissue, such as lymphocyte DNA. If the mutation is detected only in the tumor sample and not in normal tissue, it is evidence for a somatic mutation. Neurofibromatosis type 1, Li-Fraumeni Syndrome, familial adenomatous polyposis are examples of syndromes that are associated with inherited tumor suppressor gene mutation.

## **TUMOR SUPPRESSORS, ONCOGENES AND MUTATOR GENES**

Gene mutations that are associated with tumorigenesis can be classified by how the mutation contributes to tumor formation. Two main categories exist: Gain or loss of function. Gain of function mutations are activating and convert a proto-oncogene to an oncogene. Activation of only one allele of an oncogene can contribute functionally to tumor progression in concert with other gene mutations. Several mechanisms of proto-oncogene activation have been proposed, such as amplification, splicing-related mutations that add or delete gene exons and point mutations that increase activity of the protein.

Loss of function mutations are typically associated with tumor suppressor genes. These genes function as genome guardians and they normally prevent malignant transformation. Missense or non-sense mutations, deletions or insertions can render the protein non-functional.

Lastly, mutator genes (i.e. DNA repair enzymes) can increase the mutation rate in the cell's DNA. When DNA repair enzymes are altered due to inactivating mutations, the cell subsequently loses its ability to adequately repair errors that occur during DNA replication. The increased rate of DNA mutations then can lead to critical changes in oncogenes and tumor suppressor genes. For instance, in GBMs about 5% of tumors will have dysfunction of the mismatch repair enzymes machinery.

## **MUTATIONS THAT UNDERLIE BRAIN TUMORS**

With the completion of the human genome project and the advent of next generation sequencing technology, genome wide genetic studies are now feasible. One of the key principles from these landmark studies is that the number and combination of mutations that can be used by the cancer cell to escape the normal control mechanism are far larger and more complex than originally envisioned by scientists.

One such global sequencing study is from Parson et al. [1] who studied 22 glioblastoma samples. They sequenced 20,661 protein coding genes in order to determine the presence of alterations within the coding regions of the genome. In addition, amplifications and deletions were detected using high-density oligonucleotide arrays and gene expression analyses were performed to correlate genetic changes to mRNA levels. This study revealed that some of the

pathways known to be altered in GBMs affect a larger fraction of genes and patients than previously anticipated. A majority of the tumors analyzed had alterations in genes encoding components of each of the TP53, RB1, and PI3K pathways. The comprehensive nature of their study led to the identification of IDH1 as an unexpected target of genetic alteration in 5 patients with GBM. All mutations in this gene resulted in amino acid substitutions at position 132. All 5 had the same heterozygous point mutation, a change of a guanine to an adenine at position 395 of the IDH1 transcript (G395A), leading to the replacement of an arginine with a histidine at amino acid residue 132 of the protein (R132H).

## GLIOBLASTOMAS

Glioblastoma multiforme (GBM) is the most common and lethal type of brain cancer [1] and therefore the genetic basis for this disease has been better characterized than for other central nervous system neoplasms. They are classified as grade I to grade IV on the basis of histopathological and clinical criteria established by the World Health Organization (WHO) [2]. Historically, GBMs have been categorized into two groups “primary” and “secondary” [3] on the basis of clinical presentation. Primary GBMs first occur as a grade IV tumor, whereas secondary GBMs are defined as cancers that have clinical, radiologic, or histopathologic evidence of malignant progression from a preexisting lower-grade tumor [3].

Substantial research effort has focused on the identification of genetic alterations in GBMs that might help define subclasses of GBM patients with differing prognoses and responses to specific therapies [1]. For example, distinctions between the genetic lesions found in primary and secondary GBMs have been made. TP53 mutation appear to be a relatively early event during the development of an astrocytoma, whereas the loss or mutation of PTEN and amplification of EGFR are characteristic of higher-grade tumors. Specifically, secondary GBMs have a higher frequency of TP53 alterations rather than primary; 65% versus 28%, lower frequency of PTEN mutation; 4% versus 25% and lower rate of EGFR amplification 8% versus 36% [4-6]. It has been reported that the types of TP53 mutations differ between primary and secondary GBMs. In the first population-based study on glioblastomas that includes incidence, survival rates, and key genetic alterations by Oghaki et al. [7], in 57% of secondary glioblastomas the mutations were located in the two-hotspot codons, 248 and 273. In primary glioblastomas, mutations were more equally distributed through exons, only 17% occurring in codons 248 and 273.

Glioblastomas cell lines have been the historical standard both for exploring the biology of human tumors and as preclinical models for screening potential therapeutic agents [8]. It has become increasingly clear, however, that phenotypic characteristics and the multitude of genetic aberrations found within repeatedly *in vitro* passaged cancer cell lines often bear little resemblance to those found within the corresponding primary human tumor [8]. The concept of tumor stem cells (TSCs) provides a new paradigm for understanding tumor biology, although it remains unclear whether TSCs will prove to be a more robust model than traditional cancer cell lines. TSCs may be a more reliable model than many commonly utilized cancer cell lines for understanding the biology of primary human tumors [8]. By using a model system derived from primary GBMs, it has been demonstrated that NBE-

cultured (“NBE” conditions: serum free Neurobasal media supplemented with basic FGF and EGF) cells derived from primary GBMs bear remarkable similarity to normal NSCs.

In a screening of 20,661 genes, 5 out of 22 GBMs tumors had the same heterozygous point mutation, a change of a guanine to an adenine at position 395 of the IDH1 transcript (G395A), leading to the replacement of an arginine with a histidine at amino acid residue 132 of the protein (R132H) [1]. The mutation in IDH1 preferentially occurs in younger GBM patients, with a mean age of 33 years for IDH1-mutated patients, as opposed to 53 years for patients with wild-type IDH1 ( $P < 0.001$ , *t* test). Interestingly, mutations in IDH1 are found in nearly all of the patients with secondary GBMs (mutations in 5 of 6 secondary GBM patients, as compared to 7 of 99 patients with primary GBMs) ( $P < 0.001$ , binomial test). Lastly, patients with IDH1 mutations have a significantly improved prognosis, with a median overall survival of 3.8 years as compared to 1.1 years for patients with wild-type IDH1.

To assess the enzymatic activity of wild-type and mutant IDH1 and IDH2 proteins, Yan et al. utilized a human oligodendroglioma line without IDH1 or IDH2 mutations that was transfected with a vector containing the coding sequences of the wild-type IDH1, wild-type IDH2, or mutant IDH genes (corresponding to the most common IDH1 mutation, R132H, or the IDH2 mutations R172G, R172K, and R172M) [2]. They measured the enzymatic activity (reduction of NADP<sup>+</sup> to NADPH) of IDH1 and IDH2 proteins in an oligodendroglioma line that had been transfected with wild-type or mutant IDH1 or IDH2 genes. These mutants represented 88% of the IDH1 mutations and 100% of the IDH2 mutations found in patients. Data showed that exogenous expression of wild-type IDH1 or IDH2 significantly increased the production of NADPH, whereas only endogenous IDH activity was observed in cells that had been transfected with mutant IDH1 or IDH2 genes. Wild-type (WT) IDH1 converts isocitrate and NADP<sup>+</sup> to  $\alpha$ -ketoglutarate ( $\alpha$ -KG) and NADPH. Mutated amino acids in IDH1 and IDH2 reside in the catalytic pocket and result in a neo-enzymatic activity:  $\alpha$ -KG + NADPH  $\rightarrow$  D-2-hydroxyglutarate (2-HG) + NADP<sup>+</sup>. Although the role of IDH1 mutation in tumorigenesis has not been determined, decreased NADPH production from loss of IDH1 WT function coupled with increased 2-HG levels could lead to oxidative stress. Furthermore, 2-HG interferes with the electron transport chain and could alter mitochondrial physiology and drive cells toward aerobic glycolysis [9-11].

Mutations that arise during tumorigenesis may provide a selective advantage to the tumor cell (driver mutations) or have no net effect on tumor growth (passenger mutations). Identification of GBM candidate cancer genes (CAN-genes) are worth further investigation since they are most likely drivers. The CAN-genes reported in the literature include TP53, PTEN, CDKN2A, RB1, EGFR, NF1, PIK3CA, and PIK3R1. Of these genes, the most frequently altered are CDKN2A (altered in 50% of GBMs); TP53, EGFR, and PTEN (altered in 30 to 40%); NF1, CDK4, and RB1 (altered in 12 to 15%); and PIK3CA and PIK3R1 (altered in 8 to 10%).

The phosphatidylinositol 3-kinases (PI3K) are a family of enzymes that relay important cellular growth control signals. In half of GBMs there is either a PI3K-activating mutation or a deletion/mutation of its negative regulator PTEN. A mutated insulin-mediated activator of PI3K kinase, insulin substrate receptor-1 has also been reported [12]. This observation is confirmed in a study by Gallia et al. [12]. The authors found that this mutation occurs in a significant number of human glioblastomas, further indicating that therapeutic targeting of this pathway in glioblastomas is of value. Moreover, PIK3CA mutations are identified in 21% of pediatric primary glioblastomas and in 17% of adult patients.

The PTEN gene importantly encodes a protein that regulates cell proliferation, apoptosis, and tumor invasion. PTEN mutations have been reported in 15% to 40% of glioblastomas [4, 13]. Most missense mutations are located in exons 1 to 6, the region homologous to tensin, auxilin, and dual-specificity phosphatases, whereas nonsense mutations and deletions or insertions leading to stop codons and protein truncation are located more evenly throughout the gene. In several studies, PTEN mutations were not associated with prognosis of glioblastoma patients, and this was confirmed at the population level [7].

One important gene that deserves special attention is the EGFR, as is genomically amplified in 40% of tumors [14]. After genomic amplification, the EGFR gene may undergo further rearrangement. The most frequently identified deletion, is loss of the internal exons that correspond to the coding sequence for amino acids 6 to 273 [15]. This is referred to as the EGFR type 3 rearrangement, or EGVRvIII [16]. In another study, the authors show that this common activating epidermal growth factor receptor (EGFR) mutation (EGFRvIII) stimulates mTORC2 kinase activity, which is partially suppressed by PTEN. The mTORC2 signaling promotes GBM growth and survival, and activates NF- $\kappa$ B. Importantly, this mTORC2-NF- $\kappa$ B pathway renders GBM cells resistant to chemotherapy in a manner independent of Akt [17]. The predictive value of EGFR amplification has been unclear. Studies have reported that EGFR amplification is a significant predictor of poorer overall survival in glioblastoma patients and that the EGFR gene status is a more significant prognostic factor in younger patients [18]. Other studies found EGFR amplification to be a predictor of longer survival in older glioblastoma patients [19].

Data on the predictive value of TP53 mutations in glioblastomas have been contradictory. While some hospital-based studies showed no association between TP53 status and outcome of glioblastoma patients [20], one study showed that the presence of TP53 mutations was a favorable prognostic factor [21]. In another study however, age-adjusted multivariate analysis revealed no difference in survival between patients with and without TP53 mutations [7].

In a recent comprehensive study, cancer-specific DNA methylation of CpG dinucleotides located in CpG islands within the promoters of 2,305 genes were measured relative to normal brain [22]. The promoter methylation status of MGMT, a DNA repair enzyme that removes alkyl groups from guanine residues, is associated with GBM sensitivity to alkylating agents. Twenty-one percent of the cases were found to contain MGMT promoter methylation [22]. Among the treated samples lacking MGMT methylation (29%) of the validated somatic mutations occurred as G:C to A:T transitions in CpG dinucleotides (characteristic of spontaneous deamination of methylated cytosines), and a comparable 23% of all mutations occurred as G:C to A:T transitions in non-CpG dinucleotides. In contrast, in the treated samples *with* MGMT methylation, 81% of all mutations turned out to be of the G:C to A:T transition type in *non*-CpG dinucleotides whereas only 4% of all mutations were G:C to A:T transition mutations within CpGs. That pattern is consistent with a failure to repair alkylated guanine residues caused by treatment. In other words, MGMT methylation shifted the mutation spectrum of treated samples to a preponderance of G:C to A:T transition at *non*-CpG sites [22].

In a previously described study [22], all seven mutations in Mismatch repair (MMR) genes found in six MGMT methylated hypermutated (treated) tumors occurred as G:C to A:T mutations at non-CpG sites. No MMR mutations in nonmethylated hypermutated tumors had this characteristic. Therefore, MMR deficiency and MGMT methylation together, in the context of treatment, exert a powerful influence on the overall frequency and pattern of

somatic point mutations in GBM tumors [22]. Microsatellite instability (MSI) has been identified in various human cancers, particularly those associated with the hereditary nonpolyposis colorectal cancer syndrome (HNPCC). Although gliomas have been reported in a few hereditary nonpolyposis colorectal cancer syndromes, data on the incidence of MSI in gliomas are conflicting, and the nature of the mismatch repair defect is not entirely known [23]. Five percent of GBMs have microsatellite instability, a sign of functional MMR mutation. This mutation is more commonly identified in relapsing patients and is associated with rapid acquisition of drug resistance [24, 25].

The link between age and genetic alterations in glioblastomas has been investigated in the literature. Glioblastomas with a TP53 mutation were observed in younger patients (mean: 53 years), in particular in patients younger than 35 years [4]. The mean age of patients with glioblastomas carrying a PTEN mutation and LOH 10q were older than those without these alterations. Interestingly, EGFR amplification was never observed in any glioblastoma that developed in patients below 35 years of age [4]. Hence, the poor prognosis of older patients cannot be explained by the frequency of specific genetic alterations or the combination.

## LOW-GRADE ASTROCYTOMAS AND OLIGODENDROGLIOMAS

Low-grade gliomas, defined by the World Health Organization (WHO) as grade I or II oligodendroglioma, astrocytoma, or mixed oligoastrocytoma, accounts for about 10% of all primary central nervous system tumors and 25% of gliomas [26]. Because WHO grades I–II gliomas are significantly less common than high-grade gliomas, availability of tissue for histologic and molecular analysis has made such investigations more difficult than similar research for high-grade gliomas. Although most low-grade gliomas are relatively slow-growing, they can behave heterogeneously, and therefore outcomes vary widely. Surgical resection is the primary modality of treatment, and the role of adjuvant radiation therapy remains controversial. Although at a higher rate, the TP53 mutations are found in grade II astrocytomas similar to GBMs. TP53 mutations were most frequent in gemistocytic astrocytomas (88%), followed by fibrillary astrocytomas and oligoastrocytomas, but infrequent (13%) in oligodendrogliomas [4].

Pilocytic astrocytomas are the most frequent brain tumor affecting children. They are classified as grade I astrocytomas by the World Health Organization (WHO). They occur predominantly in children and have a better prognosis than higher-grade astrocytomas, with a 10-year survival rate as high as 96% [4]. This tumor is usually not infiltrating and progression to higher grades occurs in less than 20% of the patients. Although gross total resection may result in cure, recurrence is seen in 19% of cases [27]. Recent studies highlight the importance of BRAF alterations resulting in mitogen activated protein kinase (MAK/ERK) pathway activation in low-grade CNS tumors [28]. For this purpose, Lin et al. studied 106 low-grade CNS neoplasms in a cohort of primarily pediatric patients to identify the prevalence and clinicopathologic significance of these alterations [28]. They found that BRAF alterations included KIAA1549:BRAF fusions in 51 (48%) and BRAF exon 15 (BRAF<sup>V600E</sup>) point mutations in 8 (8%). These alterations were more common in tumors arising from the cerebellum and optic pathways.



Oligodendroglioma, is a glioma subtype marked by unique clinical, pathological, and genetic characteristics. Unlike other gliomas such as astrocytomas and ependymomas, oligodendrogliomas are chemosensitive and often progress in a slow and predictable manner [29]. They display a classical appearance of cells, with round, regular nuclei associated with clearing of the cytoplasm and in close proximity to fine branching vasculature.

Oligodendrogliomas are the second most common malignant brain tumor in adults and often exhibit characteristic losses of chromosomes 1p and 19q [30]. They account for 20% of brain tumors in adults and, as their name implies, they consist primarily of cells resembling oligodendroglia [31]. To date, the best biomarker for oligodendrogliomas is loss of heterozygosity (LOH) of chromosomes 1p and 19q. Assessment for LOH events is now commonly performed in patients with oligodendrogliomas because of their important implications for therapeutic responses [30, 32, 33]. The chromosome losses occur in 50% to 70% of tumors and are often associated with a pericentromeric translocation of chromosomes 1 and 19, producing marker chromosome der (1;19) (q10;p10). This translocation is unbalanced, leaving the cells with one copy of the short arm of chromosome 1 and one copy of the long arm of chromosome 19. This suggests that the basis for the t (1;19) translocation is the unmasking of a tumor suppressor gene(s) on either chromosome 1p or 19q.

Recently, Bettegowda and colleagues performed exomic sequencing of seven tumors with 1p/19q co-deletion in order to better understand the biologic basis of the der (1;19) (q10;p10) translocation. The authors found that the *CIC* gene on chromosome 19q was somatically mutated in six cases and that the *FUBP1* gene [encoding far upstream element (FUSE) binding protein] on chromosome 1p was somatically mutated in two tumors. Further examination of 27 additional oligodendrogliomas revealed 12 and 3 more tumors with mutations of *CIC* and *FUBP1*, respectively. The vast majority of mutations in these two genes were predicted to inactivate function of the proteins, suggesting that both act as tumor suppressor genes. One compelling possibility is that 1p/19q co-deletion is a mechanism of inactivating one copy of these tumor suppressor genes. The second copy could then be inactivated via other epigenetic or genetic mechanisms. Furthermore, the loss of 1p/19q in oligodendrogliomas accurately predicts sensitivity to chemotherapy [34, 35]. It remains to what role *CIC* and *FUBP1* play in conferring this chemosensitivity.

In addition, oligodendrogliomas are associated with a constellation of positive prognostic markers including methylation of the *MGMT* promoter, *IDH1* mutations, and the recently described CpG island methylator phenotype (G-CIMP) [36]. These markers are also present in glioblastomas that arise from low-grade astrocytomas. An important divergence in the molecular pathogenesis of low-grade oligodendrogliomas and astrocytomas is 1p/19q co-deletion in the former and TP53 mutations in the latter. The mutual exclusivity of these events underscores the distinct molecular characteristics of oligodendrogliomas [29]. These findings indicate that LOH 1p/19q and TP53 mutations are genetic alterations that clearly distinguish two pathways leading to oligodendrogliomas and to low-grade astrocytomas, respectively [4].

TP53 mutations are significantly more frequent in secondary than primary glioblastomas. In the pathway leading to secondary glioblastomas, TP53 mutations are early genetic events, since they are already present in low-grade and anaplastic gliomas at similar frequencies, while LOH 10q is a late genetic event [4]. It is intriguing that recurrent mutation in *CIC*, located on chromosome 19q, is found almost exclusively in 1p/19q co-deleted oligodendrogliomas with *IDH1* mutation, yet loss of chromosome 1p is more strongly

associated with the oligodendrogliomatous phenotype and clinical behavior than 19q loss [35].

## MEDULLOBLASTOMAS

Medulloblastomas are the most common malignant brain tumor of children [37]. Medulloblastomas arise in the cerebellum, have a propensity to disseminate throughout the central nervous system, and are diagnosed in approximately 1 in 200,000 children less than 15 years old each year [37]. Over the last 30 years, medulloblastoma has been the subject of a number of studies undertaken to improve the disease's prognosis by combining chemotherapy with postoperative radiotherapy [38]. Although aggressive multimodal therapy has improved the prognosis for children with medulloblastomas, a significant proportion of patients are currently incurable [39].

Moreover, survivors often suffer significant treatment-related morbidities, including neurocognitive deficits related to radiation therapy. The benefit in terms of limited neuropsychological sequelae, that derive from craniospinal irradiation dose reduction is not entirely clear [38].

So far, gene-based research has identified two subgroups of medulloblastomas, one associated with mutated genes within the Hedgehog pathway (SHH) and the other associated with altered WNT pathway genes [40, 41]. In 2009, Northcott et al. studied a large cohort of medulloblastomas to determine how many subgroups of the disease exist, how they differ, and the extent of overlap between subgroups [42]. They identified four distinct, nonoverlapping molecular variants: WNT, SHH, group C, and group D. The difference between group C and group D is that although MYC is highly expressed in group C and WNT tumors and MYCN is highly expressed in SHH tumors, neither MYC nor MYCN is highly expressed in group D tumors. Their analysis of overall survival demonstrated a marked reduction in survival for children with Group C medulloblastoma regardless of metastatic stage [42].

Previously, the molecular basis of medulloblastomas emerged from the study of hereditary tumor syndromes. For instance, individuals with Gorlin or Turcot syndrome possess germline mutations in the PTCH1 and APC tumor suppressor genes, respectively, and are predisposed to medulloblastoma, among other cancers [43-45]. Studies of the PTCH1 gene in Gorlin syndrome and sporadic medulloblastomas, as well as knockout studies of its mouse homolog, Ptc, have helped to establish a role for aberrant SHH signaling in 25-35% of medulloblastomas [45].

Similarly, the identification of APC mutations in Turcot syndrome and more frequent mutations of CTNNB1 in sporadic cases have implicated the Wnt signaling cascade in 10–15% of patients with medulloblastomas. Furthermore, patients with Li-Fraumeni syndrome have germline TP53 mutations and can have a broad spectrum of cancer types, including medulloblastoma [44, 46].

New insights into the pathogenesis of these tumors have driven Parsons et al. [37] to sequence 225,752 protein coding exons, adjacent intronic splice donor and acceptor sites, and miRNA genes in 22 pediatric medulloblastomas samples (17 samples extracted directly from primary tumors, 4 samples passaged in nude mice as xenografts, and 1 cell line). A total of

225 somatic mutations were identified in this manner including point mutations, small insertions, duplications or deletions, ranging from 1 to 48b pin length. Two interesting observations were made from their research. First, the average number of non-silent somatic mutations (non-synonymous mis-sense, nonsense, indels, or splice site alterations) per medulloblastoma patient was only 8.3, which is 5 to 10-fold less than the average number of alterations detected in the previously studied solid tumors. Second, the proportion of nonsense mutations was over two fold higher than expected given the mutation spectra observed in this tumor type, and the relative fraction of nonsense, insertion, and duplication alterations was higher in MBs than in any of the adult solid tumors analyzed.

Several genes have been identified as candidate cancer genes (CAN-genes) such as MBPTCH1, MLL2, CTNNB1, TP53, MYC, PTEN, OTX2, SMARCA4, MLL3 in medulloblastomas [37]. However, the pathways most highly enriched for genetic alterations had not previously been implicated in medulloblastomas, involved genes responsible for chromatin remodeling and transcriptional regulation, particularly the histone-lysine N-methyltransferase MLL2. Up to twenty percent of tumors harbored a mutation in a gene within these pathways or in a related gene member: the histone-lysine-N methyltransferases MLL2 and MLL3; the SWI/SNF-related matrix-associated actin-dependent regulator of chromatin members SMARCA4 and ARID1A; and the histone lysine demethylase KDM6B [37].

## SUMMARY: PERSPECTIVES IN BRAIN TUMORS GENOMICS

Tremendous advances have been made into the genetic and biological understanding of brain tumors. There has been a significant increase in the rate of discovery of mutated genes in brain cancers due to advances in automated sequencing technology and completion of the human genome sequence [47]. Soon it will be possible and affordable to perform global mutational profiling on every patient. One imminent challenge will be how to analyze and store these data in ways that allows for a meaningful impact in the way we treat patients. Deciphering the roles that the myriad of changes harbored by an individual tumor cell play will be central in developing more rational and targeted diagnostic and therapeutic strategies.

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## *Chapter 4*

# **CLINICAL FEATURES OF CENTRAL NERVOUS SYSTEM TUMORS**

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## **ABSTRACT**

Central nervous system neoplasms are rare and account for less than 3% of tumors in the adult population. The low incidence and broad spectrum of overlapping clinical features present a diagnostic challenge. Clinical manifestations of CNS tumors depend in part on their location, size and growth rate. Common generalized signs include headache, memory loss, altered mental status, nausea, vomiting, and seizures, and reflect an increase in intracranial pressure (ICP). On the other hand, focal signs and symptoms reflect compression of specific neural structures, resulting in cranial nerve palsy, weakness or sensory deficits. Tumors often express clinical features of both generalized and focal localizing signs.

A thorough history and complete neurologic exam can aid in diagnosing tumors of the central nervous system by narrowing the cohort that would benefit from further radiographic workup.

## **INTRODUCTION**

Despite advances in the field of brain tumor research and treatment, making the initial clinical diagnosis remains a formidable task. This is in part due to the broad spectrum of signs and symptoms in which central nervous (CNS) tumors manifest. For example, a glioblastoma may present as acute onset seizure with associated hemiparesis in one individual, yet present as progressive headache without neurologic deficits in another. The degree and severity of

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symptoms are dependent in part to the tumor location, size, and growth rate. While advances in radiographic imaging (i.e. MRI, CT) allow for rapid assessment and diagnosis, obtaining imaging on all symptomatic individuals would not be feasible and would place an undue financial burden on the health care system. The goal of this chapter is to highlight the most common clinical features of brain tumors in order to aid the clinician in selecting the cohort of patients for additional workup.

The annual incidence of CNS tumors ranges from 10 to 17 cases per 100,000 persons. In the US, approximately 17,000 patients are diagnosed with primary brain tumors annually. They comprise less than 3% of tumors in the adult population. A study on the frequency of gliomas by anatomic location revealed that 86% were located in the cerebral hemispheres, 40% within the frontal lobe, 29% within the temporal lobe, 14% within the parietal lobe, and only 3% within the occipital lobe. Approximately 6% were identified within deep brain structures, 2% within the ventricular system, 4% within the brainstem, and only 1.5% within the cerebellum. They also occur more commonly in the right hemisphere (51%) as compared to the left (40%).<sup>1</sup> In contrast to the adult population, gliomas represent over 90% of central nervous system neoplasms in the pediatric population, 70% of which occur in the infratentorial region [2].

Presenting signs and symptoms of gliomas are in part dependent on their anatomic location and size within the CNS. They cause neurologic dysfunction by infiltration along nerve fiber tracts with disruption of normal brain or spinal cord architecture.

This mechanism often results in seizures or focal neurologic deficits. Their size and rate of growth can cause obstruction of cerebrospinal fluid flow, or mass effect on adjacent normal brain structures. This mechanism of action can produce both focal deficits due to mass effect on the surrounding neuropil and generalized symptoms due to increased intracranial pressure. The common symptoms include headache, nausea, gait disturbance, seizure, visual disturbances, and altered mental status.

Table 1 outlines the initial presenting symptoms of grade III and IV gliomas [3]. Table 2 shows the most common presentations of intracranial masses in children. The significant overlap in presenting symptoms and frequency reflect the fact that gliomas represent the majority of pediatric intracranial masses [4]. The generalized signs and symptoms of gliomas will be discussed below, followed by focal signs depending on location within the CNS.

**Table 1. Common signs and symptoms of malignant gliomas. Adopted from Smith M. et al., 2005<sup>3</sup>**

| Symptom             | Grade III (Percent) | Grade IV (Percent) | Total (Percent) |
|---------------------|---------------------|--------------------|-----------------|
| Headache            | 53                  | 57                 | 56              |
| Memory Loss         | 26                  | 39                 | 36              |
| Cognitive Changes   | 22                  | 39                 | 34              |
| Motor Weakness      | 25                  | 36                 | 33              |
| Language Deficit    | 22                  | 36                 | 33              |
| Personality Changes | 11                  | 27                 | 23              |
| Seizure             | 56                  | 23                 | 32              |

|                          |    |    |    |
|--------------------------|----|----|----|
| Visual Symptoms          | 23 | 21 | 22 |
| Changes in Consciousness | 11 | 18 | 16 |
| Nausea and Vomiting      | 8  | 15 | 13 |
| Sensory Deficit          | 5  | 12 | 13 |
| Papilledema              | 5  | 5  | 5  |
| Other                    | 14 | 19 | 17 |

**Table 2. Common presenting signs and symptoms of intracranial mass in pediatrics.**  
Adopted from Wilne S. et al., 2007<sup>4</sup>

| Symptom                       | Percentage |
|-------------------------------|------------|
| Headache                      | 33         |
| Nausea and Vomiting           | 32         |
| Abnormal Gait or Coordination | 27         |
| Papilledema                   | 13         |
| Seizures                      | 13         |
| Signs of Increased ICP        | 10         |
| Squint                        | 7          |
| Behavioral Change             | 7          |
| Macrocephaly                  | 7          |
| Cranial Nerve Palsy           | 7          |
| Lethargy                      | 6          |
| Abnormal Eye Movements        | 6          |
| Hemiplegia                    | 6          |
| Weight Loss                   | 5          |
| Visual Abnormalities          | 5          |
| Altered Consciousness         | 5          |

## CLINICAL PRESENTATION

Up to 90% of Americans suffer headache during any given year [5]. Headache represents one of the most common reasons for visit to a neurologist. Nevertheless, new onset headache can be a cardinal sign of a serious underlying problem. Headache is the most common manifestation of brain tumors and is present in over half of patients at some point during the course of their illness [6, 7]. Headache was the initial symptom in 37% of patients with supratentorial tumors and 56% of infratentorial tumors [8] Patients with elevated intracranial pressure, or large tumors with mass effect ,are more likely to present with tumor-related headache.<sup>6</sup> Headache occurs as a result of traction on intracranial arteries, large draining

veins, and dural structures. Additionally, direct pressure on cranial nerves with pain afferent fibers from the head can produce symptoms [9, 10].

Tumor pathology can influence headache prevalence. Larger tumors, such as glioblastoma, are often associated with increased severity of headache. However, slower growing tumors, such as meningiomas and low grade astrocytomas are less likely to cause headache. Posterior fossa tumors, such as medulloblastoma and ependymoma can cause headache in 60-83% of patients, presumably due to obstructive hydrocephalus and associated increased intracranial pressures [8, 11]. Furthermore, primary and metastatic brain tumors are equally likely to cause headache [5, 8]. Leptomeningeal carcinomatosis can cause headache due to communicating hydrocephalus along, in addition to neck stiffness, cranial nerve palsy, and other neurologic deficits [12].

The headaches are typically described as intermittent and dull in nature. Severe headaches usually indicate increased intracranial pressure and may be associated with nausea and vomiting. Headaches are often aggravated by maneuvers that increase intracranial pressures, including Valsalva, straining and posture. Any change in headache pattern, increased symptoms with Valsalva, and an abnormal neurological exam are highly suggestive of an intracranial mass [5].

Although headaches can localize to the side of the tumor, they may also be generalized or even tension-type in nature with pain in the back of the neck and a band-like distribution. Additionally, pain location was not useful in distinguishing tumor-related headache [6]. The 'classic tumor headache' with increased severity in the morning and awakens the patient from sleep [6] occur less commonly in only 17% of patients [13]. Its mechanism of action is thought to be due to increased intracranial pressures from the recumbent position.

Features of headaches that warrant further investigation include, new onset headache in any adult, headache that has changed in quality from previous headache, headache associated with exertion, straining or neurologic deficit, progressive headache, or any headache in the pediatric population.

The nausea and vomiting associated with gliomas is usually secondary to increased intracranial pressure from tumor mass, tumor-associated edema, or obstructive hydrocephalus. Elevated intracranial pressures cause irritation of the area postrema, a medullary center in the floor of the 4<sup>th</sup> ventricle that controls vomiting. Indicators of increased ICP as an underlying cause of vomiting include association with changes in body position or maneuvers that increase ICP (e.g. sneezing or coughing). The nausea and vomiting may also be present in the context of other signs of intracranial mass like headaches, focal neurological deficit, or abnormality on neurological examination, although this is not always the case [14].

Mental status changes are common features of intracranial tumors. Psychomotor retardation, or the slowing of thought processes and physical motor activity, is among the most common presenting symptom of gliomas. These symptoms are initially subtle and usually only identifiable to the patient, family members or close friends. Cognitive impairment typically involves the frontal lobe. Patients with bilateral frontal lobe involvement will demonstrate impairment earlier in the disease course. Additionally, bilateral frontal lobe dysfunction can manifest as a loss of executive function. Other subtle mental status changes include forgetfulness, loss of insight, indifference to social interactions, decreased initiative, or blunted affect. Occasionally these symptoms are erroneously attributed to dementia or old age. Further hindrance to a proper diagnosis comes from the patient as he or she will conceal or deny these changes due to embarrassment or health concerns. Patients will commonly

admit to decreased energy, changes in sleep patterns, and a loss of interest in previously enjoyed activities.

Personality changes such as apathy, disinterest, and inattentiveness are common; however, more extreme changes ranging from depression and emotional lability to impulsive and loss of inhibition can also be observed [15, 16]. Confusion and dementia typically occur later in the disease course and is associated with other neurologic deficits.

Seizures may be either the initial presenting symptom of glioma, or may subsequently develop later as the disease progresses. New onset seizures in the adult population should raise suspicion for focal underlying disease, especially neoplasm. Incidence of seizures has been reported in 25-50% of brain tumors [17].

A number of studies suggests that seizures are more likely to manifest as the initial presenting symptom in low grade tumors[17]. Nevertheless, other studies have demonstrated a higher prevalence of seizures among high-grade gliomas (III or IV) as opposed to low-grade gliomas, and are present in up to 30 percent of patients with glioblastoma multiforme at the time of diagnosis [18-21]. Much like the overall symptomatology of gliomas, the seizures vary by focus of origin.

Seizures may be either focal or generalized. Generalized seizures can occur with tumors anywhere within the CNS. Seizures in the frontal lobes may cause tonic-clonic movements. Temporal lobe seizures often involve sensory auras with or without changes in behavior. Parietal lobe foci may lead to sensory deficits or paresthesias. Occipital lobe seizures may cause visual disturbances. Partial complex seizures are more frequently associated with temporal lobe tumors. Focal seizures are more commonly associated with tumors in the motor or sensory cortex. New onset seizure in an adult warrants a thorough workup, including brain imaging studies. Patients who have had a first time seizure as a result of an intracranial mass are at significantly increased risk of having subsequent seizures, and prophylactic anticonvulsant therapy is warranted in most cases [22, 23].

While it is difficult to localize tumor location based on generalizing signs, it should be emphasized that these presenting symptoms are still grounded on anatomic location. Focal signs and symptoms reflect direct or indirect effects of tumor growth on specific functional centers or pathways within the CNS. Discussion of the pathophysiologic etiology requires that the clinician have an in depth understanding of the structural and functional anatomy of the CNS. The focal signs and symptoms of gliomas will be discussed separately below with detailed focus given to optic pathway gliomas and aphasia as examples of the nuances in presentation depending on location of the lesion.

The optic pathway begins with the retina where retinal ganglion cells converge to form the optic nerve. The optic nerve exits the eye at the optic disc and cross at the optic chiasm, located just anterior to the pituitary stalk. The axons from the nasal retina cross in the optic chiasm, while temporal fibers remain ipsilateral so that the right optic tract carries information from the left visual field and vice versa. After the optic chiasm, the collective retinal ganglion cells become the optic tract and travel posteriorly to the lateral geniculate nucleus of the thalamus. Some axons project to the superior colliculi of the brainstem and the suprachiasmatic nucleus of the hypothalamus, which aid in controlling eye movements and controlling circadian rhythms, respectively. From the lateral geniculate nucleus, axons carrying information from the superior visual field (inferior retina) course along the lateral ventricle in the temporal lobe, a bundle collectively known as Meyer's loop, before terminating in the lingual gyrus of the occipital lobe. Fibers carrying information from the

inferior visual fields (superior retina) travel through the parietal lobe before reaching the cuneus of the occipital lobe. As can be imagined, lesions in different regions along this extensive pathway may cause various symptoms ranging from proptosis to visual disturbances.

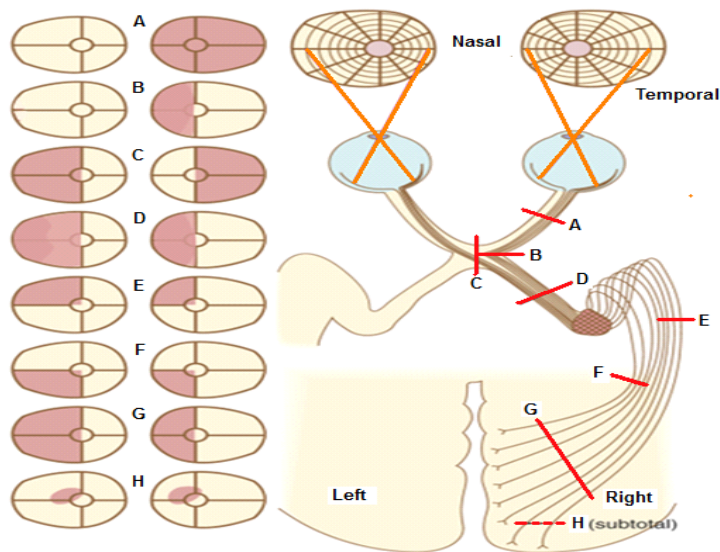


Figure 1. Visual pathway with points of potential interruption.

**Table 3. Corresponding symptom associated with lesions in figure 1**

| Location                                                          | Symptom                                                                |
|-------------------------------------------------------------------|------------------------------------------------------------------------|
| 1. Optic Nerve                                                    | Unilateral Blindness                                                   |
| 2. Lateral Optic Chiasm                                           | Incomplete contralateral Hemianopia                                    |
| 3. Central Optic Chiasm                                           | Bitemporal Hemianopia                                                  |
| 4. Optic Tract                                                    | Homonymous Hemianopia                                                  |
| 5. Temporal (Meyer’s Loop) Optic Radiation                        | Homonymous Superior Quadrantanopia                                     |
| Location                                                          | Symptom                                                                |
| 6. Parietal (superior) Optic Radiation                            | Congruous Partial or Complete Homonymous Inferior Quadrantanopia       |
| 7. Complete Parieto-Occipital Interruption of the Optic Radiation | Homonymous Hemianopia with Sparing of Central Vision (Macular Sparing) |
| Location                                                          | Symptom                                                                |
| 8. Incomplete Damage to the Visual Cortex                         | Homonymous Scotomas, Usually Encroaching at on Central Vision          |

The most common signs are strabismus, visual loss, and decreased acuity [24]. There is significant variation in symptoms based upon whether or not the glioma is associated with neurofibromatosis. In NF patients, proptosis is more common, whereas in sporadic cases, nystagmus and hydrocephalus are more common [25]. Anterior optic pathway gliomas occur before retinal fibers reach the chiasm, whereas posterior lesions occur either in or posterior to the chiasm. Most anterior gliomas are classified as low grade astrocytomas, and are most commonly found in children [26]. About half occur in children with neurofibromatosis type 1,

and the other half are sporadic [24]. Figure 1 and Table 3 below demonstrate the presentation of visual symptoms expected with interruption of the visual pathway in various points along its course.

Aphasia is defined as an impaired ability to understand or generate language, whether written, verbal, or even sign language. The language center is a very complex network of connections that is usually located in the left cerebral hemisphere [27, 28]. While tumors such as gliomas rarely cause a ‘classic’ aphasia syndrome, it is important for the clinician to be cognizant of the nuances of the different aphasias, as well as be aware that any disturbance in language function warrants investigation into a possible mass occupying lesion. In assessing aphasia, fluency, comprehension, and repetition should be evaluated. Fluency can typically be assessed by listening to the patient’s speech. The spectrum of abnormalities range from a readily apparent inability to initiate speech, to a paucity of speech, or to a more subtle compensated use of shortened sentences. Comprehension may be evaluated by instructing the patient to follow increasingly difficult multi-step commands. Similarly, repetition can be assessed by initially asking the patient to repeat short, commonly used phrases, and then progressively difficult phrases. The assessment of these three categories of language will allow the clinician to distinguish between the eight main types of aphasia, while an evaluation of reading, writing, and naming will enable further discrimination between pure word deafness, aphemia, pure alexia, and pure agraphia. Table 4 provides a breakdown of the different types of aphasia based on either a positive or negative finding in each of these categories [29].

Motor impairment is a common complaint in patients with nearly all intracranial lesions. The spectrum of deficits encompasses weakness, balance difficulties, and incoordination. The weakness may be due to direct tumor extension to the motor cortex in the posterior frontal lobe. Contralateral focal weakness will localize to specific areas of the homunculus affected within the motor cortex. Disruption of the white matter tracts of the descending upper motor neurons prior to the pyramidal decussation will result in a contralateral hemiparesis. Disruption of the subcortical input from the basal ganglia can produce a more complex picture of contralateral hemiplegia, athetosis, chorea, or hemiballism. Whereas, disruption of the cerebellar function can produce ipsilateral weakness, ataxia, or dysmetria. As with cognitive changes, the motor dysfunction may be subtle early on with progression as either the lesion itself or the surrounding edema enlarges. In the upper extremities, weakness is typically more pronounced in the flexors of the forearm and wrist, whereas in the lower extremities, the flexors are preferentially affected. Indirect disruption may be due to tumor-associated edema, which tends to improve with steroids or osmotic agents (i.e., mannitol, hypertonic saline) [30]. Alternatively, tumors involving the frontal or parietal lobe near the motor cortex may provoke seizures with a resultant transient postictal hemiparesis or hemiplegia.

Hearing impairment includes tinnitus and loss of auditory acuity. While these symptoms can present with cranial nerve VIII tumor of the cerebellopontine angle tumors, tinnitus is a common symptom and can occur in as many as 20% of the population between the ages of 55-65 [31]. Less frequently, hearing impairment can be due to leptomeningeal carcinomatosis; however, this is associated with other generalized and focal signs.

**Table 4. Clinical features of aphasia**

| Aphasia Syndrome                   | Fluency | Comprehension | Repetition | Reading | Writing | Brain Location                                                                 |
|------------------------------------|---------|---------------|------------|---------|---------|--------------------------------------------------------------------------------|
| Anomia (inability to name objects) | +       | +             | +          | +/-     | +/-     | Temporal, parietal, occipital, without affecting classical language areas      |
| Conduction                         | +       | +             | -          | +       | +/-     | Arcuate fasciculus in the superior temporal gyrus                              |
| Transcortical Sensory              | +       | -             | +          | -       | -       | Dominant posterior watershed, PCA/MCA territory                                |
| Wernicke's                         | +       | -             | -          | -       | -       | Wernicke's area. Dominant superior temporal/inferior parietal region           |
| Transcortical Motor                | -       | +             | +          | +       | -       | Dominant mesial frontal. Supplementary motor area.                             |
| Broca's                            | -       | +             | -          | +       | -       | Broca's area. Dominant inferior frontal                                        |
| Transcortical Mixed                | -       | -             | +          | -       | -       | Anterior and posterior watershed. Disconnect of perisylvian cortex.            |
| Global                             | -       | -             | -          | -       | -       | Large region of dominant hemisphere, rarely seen with neoplasm-induced aphasia |



| Aphasia Syndrome                | Fluency | Comprehension | Repetition | Reading | Writing | Brain Location                                                |
|---------------------------------|---------|---------------|------------|---------|---------|---------------------------------------------------------------|
| Aphemia (inability to speak)    | -       | +             | +          | +       | +       | Motor cortex to muscles of articulation                       |
| Pure Alexia (inability to read) | +       | +             | +          | -       | +       | Dominant occipital lobe involving splenium of corpus callosum |
| Pure word deafness              | +       | -             | +          | +       | +       | Dominant/Bilateral superior temporal gyrus                    |
| Pure agraphia                   | +       | +             | +          | +       | -       | Dominant inferior frontal                                     |

## CONCLUSION

The clinical presentation of CNS neoplasm is very broad and may initially be either overt or quite subtle. A common feature is the progressive nature of the symptoms. Generalized symptoms, such as headache, nausea, vomiting, altered mental status, and seizure, are manifestations of tumor size and tumor-associated edema causing elevated intracranial pressures. Focal symptoms are often anatomically dependent. They arise from direct compression or herniation of specific neural structures. Any change in mental status or neurological functioning warrants further investigation into a possible space occupying lesion of the central nervous system. A thorough neurological examination should be performed to evaluate for possibility of a focal lesion. If clinical examination correlates with presenting symptomatology, imaging may be justified. Although these tumors may present in unusual manners, a thorough workup up including biopsy may reveal an unsuspected diagnosis of glioma.

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## *Chapter 5*

# NEURO IMAGING OF MALIGNANT GLIOMAS

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## ABSTRACT

Magnetic resonance imaging is an essential tool in the diagnosis and management of brain tumors. However, conventional MRI is not perfect and the clinician is often faced with confounding findings both at initial diagnosis and during the post-treatment period. Recent advances in MR imaging, when done in conjunction with conventional MRI, can prove useful in the workup and management of brain tumors.

## INTRODUCTION

Imaging plays an essential role in the diagnosis, treatment, and follow-up of primary brain tumors. Computed tomography (CT) is useful in the initial diagnosis and is critical in the peri-operative period. The bulk of imaging, however, is done with magnetic resonance imaging (MRI). Conventional MRI which can show tumor location, edema, cystic changes, and patterns of tumor spread can serve as valuable non-invasive tool in diagnosis. The excellent spatial resolution of MRI is critical in determining if a tumor is resectable and what the best trajectory and maximal safe volume of tumor resection may be. Periodic MRI in the post-treatment and post-operative time frame provides a safe and effective means by which to detect tumor recurrence or residual tumor.

Recent advances in MRI provide clinicians with an improved means by which to diagnose and manage brain tumors relative to conventional MRI. These tools, often collectively referred to as “advanced neuroimaging”, include perfusion imaging, spectroscopy, and diffusion weighted/diffusion tensor imaging. These tools are an important adjunct to MRI and are often complementary when performed in conjunction with one or

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several other functional studies. Brain mapping to locate eloquent tissue in relation to pathology with functional MRI (fMRI) has become an integral part of pre-operative tumor surgery planning. This chapter will describe these newer MRI applications and their role in the diagnosis and management of gliomas.

## **PERFUSION IMAGING**

Perfusion imaging is a broad term that encompasses several MRI techniques capable of imaging tumor at a microscopic level. Imaging techniques can be done both with and without the use of gadolinium based contrast agents. Techniques that require contrast administration include Dynamic Susceptibility Contrast (DSC) and Dynamic Contrast Enhancement (DCE), whereas Arterial Spin Labeling (ASL) is a non-contrast technique.

DSC is perhaps the most commonly used perfusion based imaging technique. Gadolinium-based imaging agents have both a T1-shortening (makes T1 images brighter) and T2-shortening (makes T2 images darker) effect. DSC technique uses gradient echo or spin echo-echo planar techniques. These T2 weighted sequences provide somewhat limited spatial resolution but can acquire very rapidly large volumes of data and are very susceptible to the T2-shortening effects of gadolinium-based contrast agents. Gradient or echo planar imaging of the area of interest is performed in rapid succession during the injection of a relatively large bolus of gadolinium based contrast agent. The areas of greatest gadolinium concentration are depicted as the areas with the most signal loss. Studies have shown a proportional relationship between the degree of signal drop and the concentration of blood vessels in the area of interest and, from this, we are able to subjectively and objectively determine the degree of vascularity of a certain lesion. There are several ways in which to quantify the degree of vascularity, but perhaps the most important parameter in oncologic imaging is the computation of cerebral blood volume (CBV). This value, measured in milliliters of blood/100g of brain tissue, is more often given as a relative value, or rCBV, in which the volume of blood in the lesion is compared to an area of “normal” brain, usually in the contralateral cerebral hemisphere. rCBV values have been shown to be useful in tumor grading, differentiating primary brain tumors from metastatic disease, guiding biopsy and surgical resection, differentiation of treatment related effects from tumor as well as in prognosis.

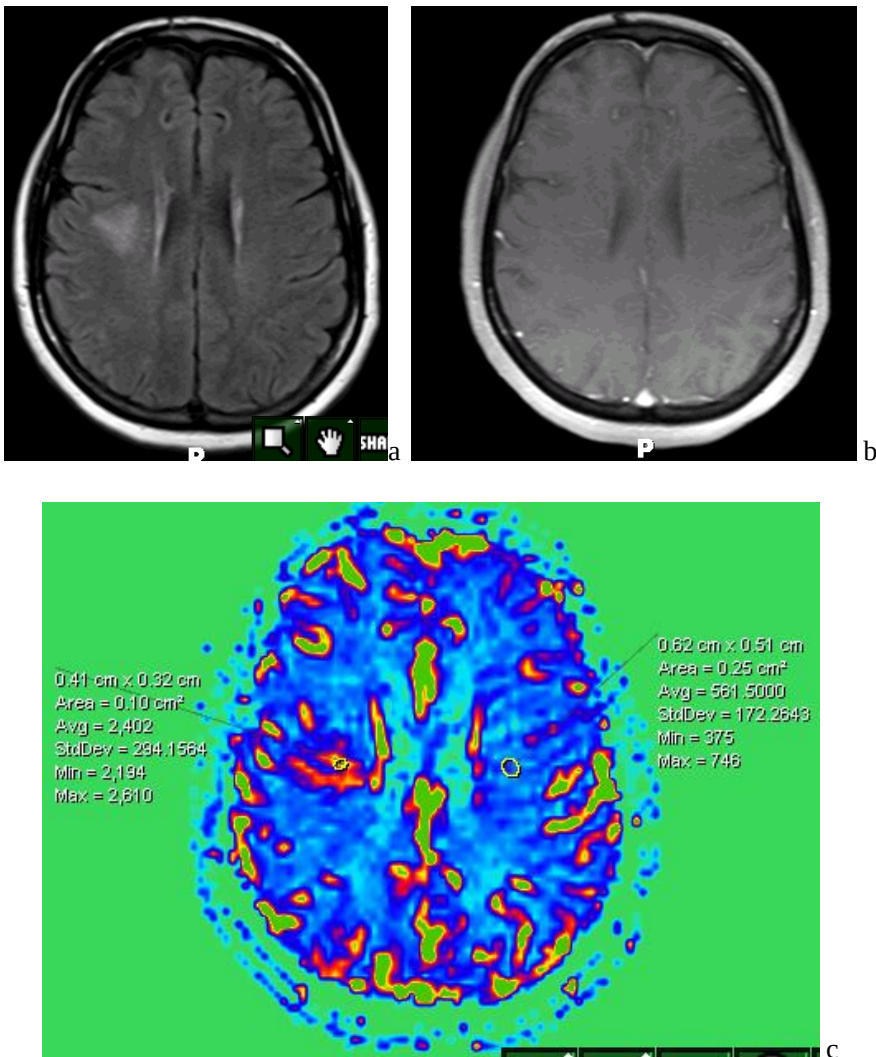
## **Tumor Grading**

Tumor grading has important implications in the treatment of primary brain tumors. Low grade brain tumors are generally managed less aggressively and can often be followed. Higher grade tumors are typically treated with a more aggressive approach that may include surgery, radiation, chemotherapy, or a combination of these approaches. Conventional MR imaging is limited in the grading of brain tumors.

There are certain characteristics that favor one type of tumor histology over another, but these are not always reliable. For example, low grade tumors typically manifest as areas of increase T2 signal intensity with relatively little mass effect. These also do not typically

enhance. Higher grade tumors, on the other hand, often present with mass effect, a large amount of surrounding edema, and often with areas of abnormal enhancement. There are frequent exceptions, however, in which low grade tumors may enhance and high grade tumors present with relatively little edema and enhancement. Studies have shown limited sensitivity in the differentiation of low versus high grade glioma using conventional imaging alone, ranging from 55.1 to 83.3% in some studies [1].

Several studies have shown improved sensitivity and specificity in the pre-operative, non-invasive differentiation of low and high grade brain tumors using perfusion imaging alone as well as perfusion imaging in conjunction with spectroscopy. A study by Meng, et al., in 2003 looked at 160 patients with newly diagnosed glioma using conventional MR imaging, perfusion, and spectroscopy.



Figures 1a-c. MRI of the brain shows an area of increased signal in the right frontal lobe on FLAIR (a) with no abnormal enhancement (b). Perfusion imaging shows increased rCBV of 2.4. The lesion was biopsied and found to be pathologically proven grade III astrocytoma.

Using the conventional imaging alone they found a sensitivity and specificity of 72.5% and 65% respectively. When using a threshold rCBV value of 1.75 they were able to differentiate low-grade from high-grade glioma with a significantly increased sensitivity of 95% and from this they concluded that perfusion imaging with threshold rCBV values can be an effective tool in the grading of brain tumors [1].

## **Guiding Treatment**

Perfusion imaging plays a significant role in guiding therapy, both before and after cytoreductive treatment. Using conventional MRI, areas of enhancement are used as a surrogate marker for the identification of aggressive, actively growing tumor and are subsequently targeted for treatment. Enhancement is attributed to “leaky” or abnormal blood brain barrier often found with tumor angiogenesis. High grade tumors frequently show a heterogeneous picture with regions of enhancing and non-enhancing tumor. The areas of enhancement often represent only the “tip of the iceberg” and the isolation of these areas for biopsy or treatment purposes may leave a significant population of tumor cells inadequately staged or without adequate, aggressive treatment. In fact, the areas of greatest tumor cellularity may not be within the area of enhancement, but instead may be within the area of peritumoral edema/T2 signal abnormality.

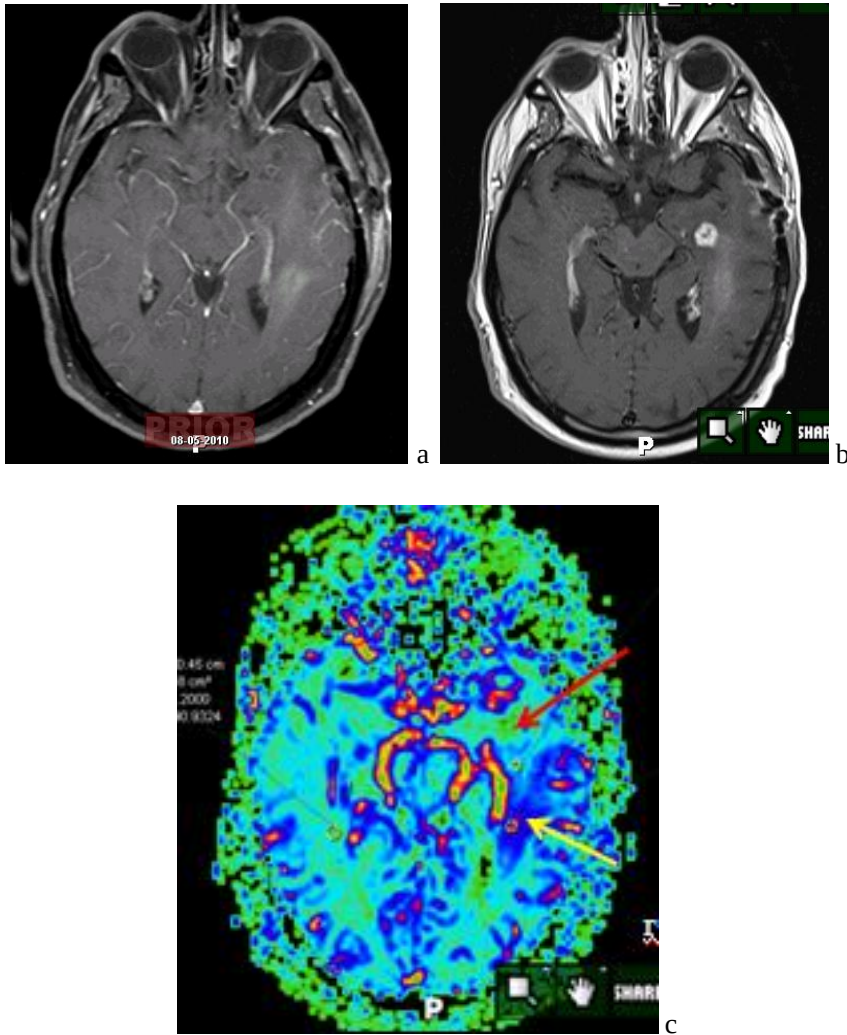
Conventional MRI alone may lead to inadequate biopsy leading to understaging and potentially less aggressive therapy. Perfusion imaging, when included in the pre-treatment workup, can help in the identification of the most critical areas to biopsy, resect or treat. A 2008 study by Sadeghi, et al., compared biopsy results to rCBV values in 18 patients with low and high grade brain tumors. They showed a positive correlation between rCBV values and both cell and microvascular density, further supporting the role of perfusion in guiding therapy [2].

## **Radiation Injury vs Recurrent Tumor**

Radiation therapy is an important and frequently used tool in the treatment of high grade glioma. The tumor angiogenesis necessary for rapid growth of a high grade tumor lead to appearance of blood vessels with an abnormal, porous endothelium resulting in leakage of gadolinium based contrast agents and in enhancement on T1 weighted images. Radiation treatment can cause vascular injury with leaky blood brain barrier and may similarly result in enhancement.

The result is often a confusing picture in the post-treatment period in patients with high grade glioma that have been treated with radiation. Perfusion has been shown to be an effective tool in the differentiation of post operative enhancement from radiation injury. While radiation induced injury and recurrent tumor may share similar imaging features of enhancement and edema on conventional MRI, perfusion imaging shows a relative decrease in vascularity (manifest as a decrease in rCBV) with radiation injury and increased vascularity (increased rCBV) in recurrent tumor.





Figures 2a-c. MRI of the brain in patient with glioblastoma treated with temozolomide and radiation. First image (a) is post treatment image showing ill defined enhancement in the posterior left temporal lobe. The second image (b) is from 4 months later, showing a new enhancing nodule anteriorly. Perfusion imaging (c) was performed showing low rCBV value(0.4) corresponding to the new nodule consistent with an area of radiation injury. Posteriorly there is an area of increased perfusion consistent with an area of residual/recurrent tumor that subsequently grew on follow up imaging.

A 2009 study by Barajas, et al., retrospectively reviewed 57 patients with a diagnosis of glioblastoma who underwent radiation therapy and were followed with perfusion imaging. The authors compared rCBV and two other perfusion parameters (peak height and percentage of signal intensity recovery) between patients with histologically proven radiation necrosis and recurrent glioblastoma and found a significant difference in values between the two cohorts. Specifically, they found a significantly increased rCBV and peak height (peak intensity) in patients with recurrent glioblastoma versus those with radiation induced injury. The percentage of signal intensity recovery was also significantly lower in recurrent glioblastoma. [3] Another study by Hue, et al. in 2010 reviewed 40 sites in 13 patients with high grade glioma treated with radiation. They found that a threshold rCBV value of 0.71

resulted in differentiation of recurrent tumor from radiation necrosis with a sensitivity of 91% and a specificity of 100% [4].

## Primary Disease versus Metastasis

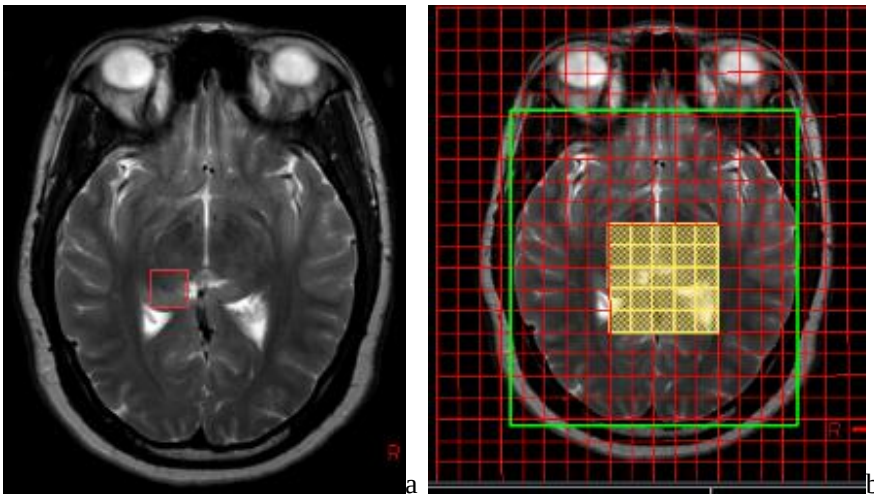
Perfusion imaging has been shown to be useful in differentiating primary brain neoplasm from metastatic disease. In the case of solitary lesions it is often difficult to differentiate a primary brain lesion from a metastasis based on conventional MRI alone. Management of these two lesions is different and therefore differentiation of the two entities is imperative. Law, et al., use a combination of perfusion imaging and spectroscopy to investigate the area of T2 signal abnormality along the margins of the enhancing tumor, the so-called “peri-tumoral” edema. In metastasis this area of T2 signal abnormality is the result of increased interstitial water resulting from increased capillary permeability. In primary brain tumors this area of peri-tumoral edema consists of a combination of increased interstitial water and infiltrating tumor cells along blood vessels. As a result, Law et al. found a significant difference in peritumoral relative cerebral blood volume values in glioma ( $rCBV = 1.31$ ) versus metastatic lesions ( $rCBV = 0.39$ ). They also noted elevated choline levels in the peri-tumoral edema of gliomas but not in metastatic lesions. [5] A study by Cha, et al., in 2006 looked at peak height values (a DSC parameter that has been shown to correlate strongly with  $rCBV$ ) in the peri-tumoral edema and showed a significant increase in peak height in the areas surrounding GBM when compared to metastasis. They also showed a significant increase in the permeability of microvasculature in metastatic lesions using average percentage of signal intensity recovery, another useful DSC parameter [6].

## Prognosis and Tumor Progression

The prognosis for patients with primary brain tumors varies greatly, even in patients with similar histologic diagnoses. A 2007 study by Hirai, et al., looked at maximal  $rCBV$  values in patients with high grade gliomas (WHO III and IV) and found that the  $rCBV$  values were a useful biomarker in predicting survivability in this patient population. Using a threshold  $rCBV$  value of 2.3, they found that patients with  $rCBV$  of less than 2.3 had an overall survival rate of 67% at 2 years, but only a 9% survival rate if the  $rCBV$  was greater than 2.3. [7] A 2008 study by Law, et al., showed that perfusion imaging could be used to predict median time to progression of gliomas, independent of histologic diagnosis. In 189 patients they found that patients with a brain tumor with an  $rCBV$  value of less than 1.75 had a median time to progression of 3585 days, whereas those with  $rCBV$  values of greater than 1.75 had median time to progression of 265 days. [8] In 2008, Danchaivijitr, et al., followed a cohort of 13 patients with low grade glioma for 3 years using conventional MRI and perfusion imaging. They found that 7 patients progressed to high grade glioma and those 7 patients had a slightly higher  $rCBV$  at entry into the study. They also found that the ones who progressed “the transformers” presented with a continuous increase in  $rCBV$  values at consecutive follow up visits whereas “the non-transformers” remained relatively stable. They concluded that perfusion weighted imaging can demonstrate evidence of transformation up to 12 months earlier than conventional MRI alone [9].

## SPECTROSCOPY

Spectroscopy is a non-invasive means by which to identify the major metabolites in an area of interest. Imaging is based on the characteristics of hydrogen protons found in all organic material. Imaging is done without contrast, can be done in a single-voxel (region of interest) or multiple voxels at the same time, and can be obtained using long and short echo time (TE) technique.



Figures 3a and b. Examples of single-voxel (a) and multi-voxel technique in spectroscopy.

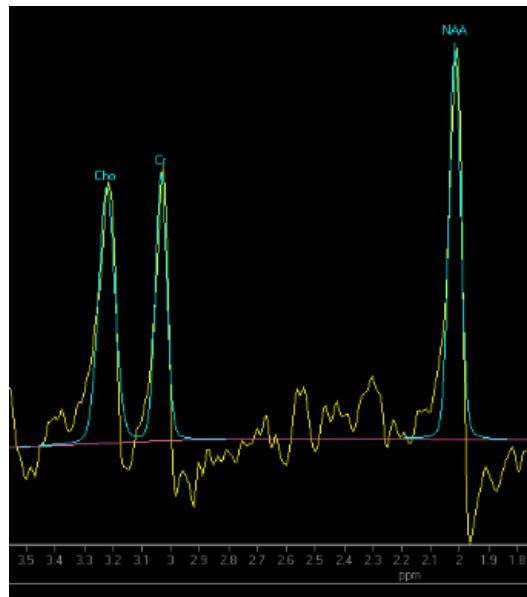


Figure 4. An example of normal spectroscopy acquired on a 3.0 tesla MRI. Parts per million (ppm) are given on the x-axis. While there are regional changes in what is considered normal in the brain, this is a fairly typical pattern in which there is an increase in height from choline to creatine and from creatine to NAA.

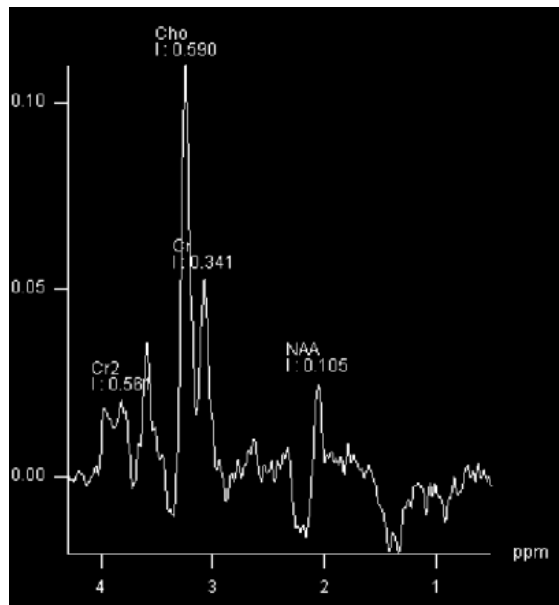


Figure 5. Spectroscopy of high grade (WHO IV) glioma use short TE technique (35 ms). Note markedly elevated choline and decreased NAA. There is an inverted lactate doublet at around 1.3 ppm.

The most useful metabolites used in diagnostic neuroimaging are N-acetyl aspartate (NAA), Choline (Cho) and Creatine (Cr). Lactate, lipids, myoinositol, glutamine, glutamate, leucine and alanine can also be seen but do not play as significant a role.

- NAA is a marker found in normal neurons. It resonates at around 2.0 ppm. Tumor, infarction, atrophy or other processes that destroy or replace normal brain tissue will present with a decrease in NAA.
- Choline plays a role in the synthesis and degradation of cell membranes. It typically resonates around 3.2 ppm. Processes that increase cell turnover, like tumor, will present with an increase in Cho.
- Creatine typically resonates at around 3.0 ppm but can have a second peak at 3.9 ppm. Creatine is an energy source and generally remains stable. It is useful as a baseline or benchmark for comparison to the other two metabolites (i.e. elevation of choline relative to creatine). While generally stable, it can change with hypermetabolic or hypometabolic conditions.
- Lactate is seen as a doublet at around 1.3 ppm (to the right of NAA). The direction of the peak changes depending on the spectroscopic technique. Specifically, with short TE (35ms) the peak is below baseline and with long technique (135ms) the peak is above baseline. Lactate is seen in the presence of anaerobic metabolism such as necrotic tumor or infarcted brain parenchyma.

Unfortunately, there is no single tumor specific metabolite that can be used to consistently identify or stage brain tumors. There are, however, certain patterns of metabolite ratios and absolute metabolite concentrations that have been shown to be helpful in non-invasive, pre-operative diagnosis and planning.

Specifically, spectroscopy has been shown to be useful in differentiating neoplasm from non-neoplasms, as a predictor for the degree of malignancy, and as a useful tool in the differentiation between tumor and treatment related changes seen on conventional MRI.

Benign brain lesions, such as hamartomas, are difficult to differentiate from low grade gliomas on imaging alone. The clinical picture, such as a history of Neurofibromatosis I, can be helpful, but it is often difficult, if not impossible, to differentiate a benign lesion from a low grade malignancy. These lesions often require serial follow up or, in many cases, biopsy. A study by Castillo, et al., showed that the spectral pattern on NF-I related hamartomas did not differ significantly from normal brain tissue. [10] Gliomas, on the other hand, had lower NAA/Cho, NAA/Cr, and Cr/Cho ratios compared to normal brain. Vuorl, et al., compared spectroscopic findings in 10 patients with low grade glioma to 8 patients with focal cortical developmental malformations (FCDM's) and found that changes in the NAA/Cr and NAA/Cho ratios could help differentiate glioma from FCDM. They also postulate that relative changes in Cho and Cr could help differentiate astrocytoma from oligodendroglioma and oligoastrocytoma. [11] Specifically they noted that oligodendrogliomas and oligoastrocytomas presented with an increase in both Cho and Cr, whereas astrocytomas presented typically with modest increase in Cho but decrease in Cr. A study by Hourani, et al., (2007) performed spectroscopy and perfusion in 69 adult patients with untreated brain tumors. They found that these imaging modalities could differentiate tumor from non-neoplastic lesions with a sensitivity of 72.2% and a specificity of 91.7% using cutoff points of NAA/Cho <0.61 and rCBV <1.50 [12].

Law, et al., performed spectroscopy on 160 gliomas in an effort to differentiate spectral patterns between low and high grade glioma. Using a threshold Cho/NAA ratio of 0.75, the authors were able to differentiate between low and high grade glioma with a sensitivity of 96.7% and specificity of only 10%. From this authors concluded that spectroscopy could serve as a valuable adjunct to conventional MRI (sensitivity 72.5%, specificity 65%) in preoperative tumor grading [13].

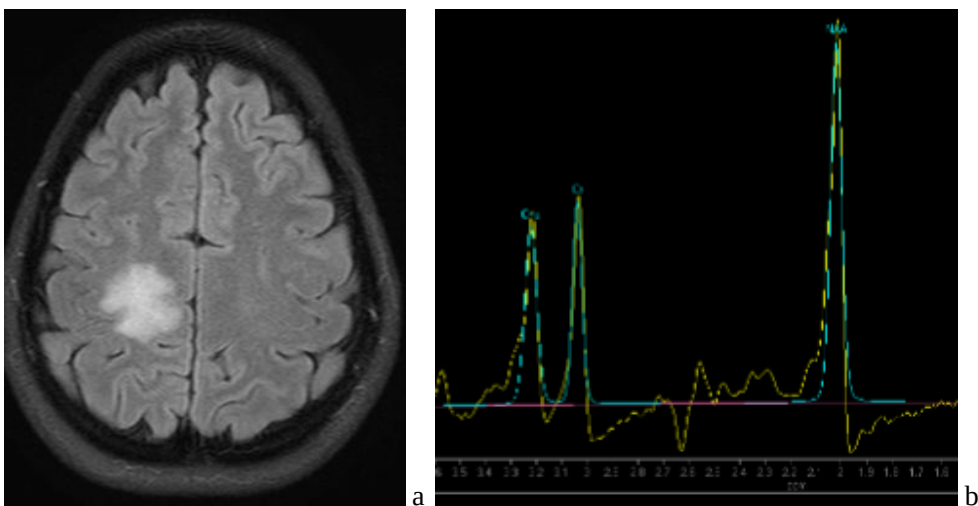


Figure 6a and b. FLAIR image (a) shows a lesion that did not enhance and was stable in size for several years. Spectroscopy (b) shows normal spectral pattern, supporting a diagnosis of hamartoma.

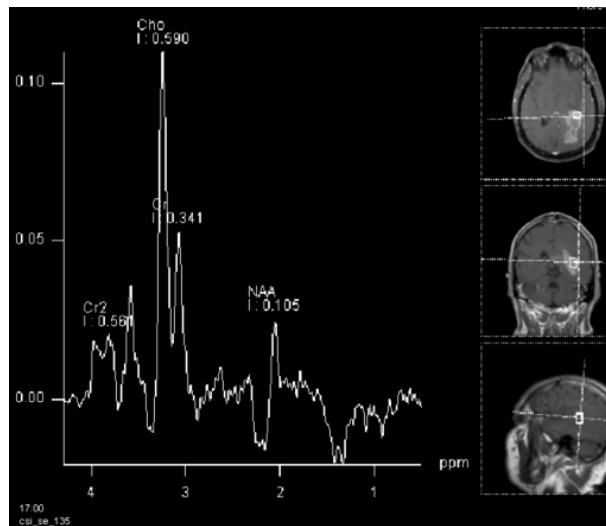


Figure 7. 45 year old patient with anaplastic astrocytoma treated with surgery followed by radiation and chemotherapy. Large rim enhancing lesion appeared 8 months after radiation treatment. Spectroscopy was performed to differentiate recurrent tumor from radiation necrosis. Note markedly elevated Cho/NAA ratio ( $>5.6$ ) consistent with recurrent tumor. Radiation necrosis typically presents with a uniform decrease in concentration of all metabolites.

MR spectroscopy has also shown to be useful in the differentiation of radiation induced injury versus recurrent tumor in the previously treated patient. Both radiation induced changes and recurrent tumor can present as areas of abnormal enhancement in the brain. In these cases it is often difficult to determine if the patient merits additional, more aggressive treatment or simply careful observation. A study by Rhabinov, et al., used spectroscopy in 17 patients previously treated with high dose ( $>54$  Gy) radiation. The authors were able to non-invasively predict recurrent tumor versus radiation injury in 16 of these 17 patients [14].

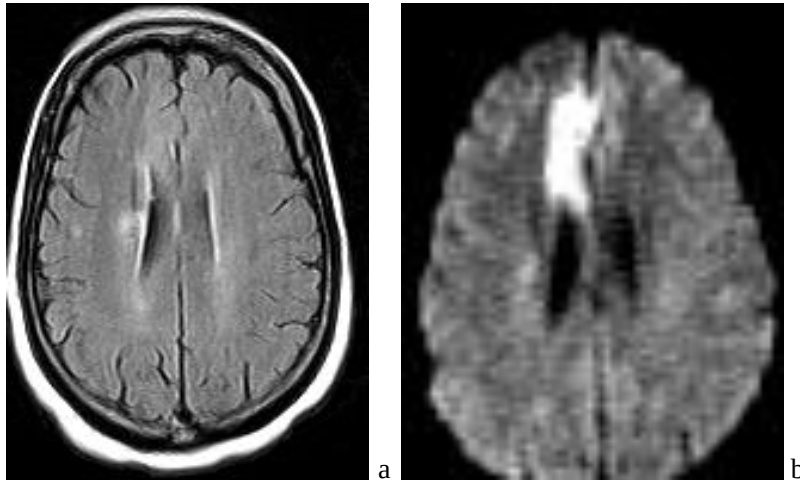
## DIFFUSION WEIGHTED IMAGING

Diffusion weighted imaging, or DWI, is an MRI technique (like T1, T2, FLAIR) that, in essence, allows us to track the movement of water. Normal or Brownian motion of water is equally likely in all directions (aka “isotropic” movement). Applying this theory we know that a drop of water in the middle of a container of other water molecules is equally likely to move in any direction unless acted on by an outside force. Conversely, water molecules that are restricted or contained, such as water molecules moving in the highly structured white matter tracts of the brain, will not have freedom of motion, a finding known as “anisotropic” movement.

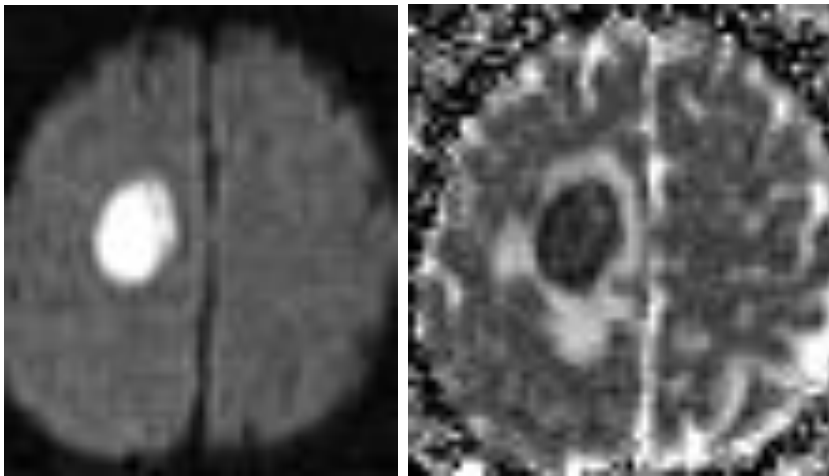
Researchers noted that hydrogen protons moving in a heterogeneous magnetic field lost their signal as they moved across the field. This forms the basis of diffusion weighted imaging. DWI artificially creates a heterogeneous magnetic field in which the signal changes of water molecules are observed. DWI is the simplest form of diffusion imaging and can be done with as little as a single sequence in a single plane. For clinical purposes, however, at least three planes are typically used. Diffusion weighted imaging is useful clinically in the

workup of conditions that limit the motion of water, such as stroke, necrotic tumor, or very dense tumor.

Diffusion imaging produces a value, known as the apparent diffusion coefficient (ADC). An ADC “map”, or series of axial images depicting ADC values, is created from these values and is useful in the interpretation of findings on the DWI sequence. In essence, areas that are dark, or “low”, on the ADC maps correspond to areas in which there is restricted diffusion on the DWI series.



Figures 8a-b. Figure a, is a FLAIR image in a patient with mental status changes. Note mild chronic microvascular changes with very subtle signal abnormality in the right frontal lobe near the cingulate gyrus. In the diffusion image (Figure b) we see marked hyperintensity corresponding to an acute infarction in the anterior cerebral artery distribution.



Figures 9 a-b. Figure A shows increased DWI signal corresponding to a ring enhancing lesion seen on conventional MRI. Figure b shows low (or dark) ADC values corresponding to the lesion consistent with an area of restricted diffusion. This lesion was an abscess.

ADC values play an important role in the non-invasive management of brain tumors. Perhaps the most useful application of ADC values in the management of brain tumors is in the non-invasive diagnosis and staging of brain tumors. Brain tumors are classified using the World Health Organization staging system, of which one of the main criteria is tumor cellularity. Specifically, higher cellularity implies higher tumor grade. Increased cell density will limit the free motion of water molecules and therefore, ADC values can serve as a non-invasive surrogate for cellularity and, subsequently, tumor grade. The relationship of ADC values to tumor grade has been shown in several studies. A study by Higano, et al., retrospectively reviewed ADC values in 37 patients with high grade glioma (WHO grade III or IV). These patients all underwent biopsy or resection. Subsequently, ADC values were compared to Ki-67 labeling index (LI), an indicator of cellular proliferation. The authors found a significant negative relationship between ADC values and the Ki-67 LI. They found that the mean ADC value in glioblastoma was significantly lower than in AA. In addition they found that the ADC values in the progressive group were significantly lower than in the stable group. Finally, they proposed an ADC cutoff value of  $0.9 \times 10^{-3} \text{ mm}^2/\text{sec}$  for differentiation of patients with a favorable diagnosis versus those more likely to progress.<sup>15</sup> Yamasaki, et al., retrospectively reviewed MRIs in 275 patients with brain tumors. They noted a significant negative correlation between ADC values and grade of glioma. Using ADC values, the authors were able to differentiate WHO grade II tumors from grades III and IV with an accuracy of 91.3% and grade III from grade IV with an accuracy of 82.4%. They also found that ADC values could differentiate lymphoma from glioblastoma with an accuracy of 83.6% [16].

Diffusion weighted imaging also plays a role in the evaluation of the post-operative patient. A recent study by Smith, et al., showed that areas of restricted diffusion are often seen along the corticectomy margins. These areas will eventually evolve into areas of encephalomalacia and gliosis but can often enhance in the early subacute period. It is imperative that the area of diffusion abnormality is recognized and that this is identified as a benign process, so as not to be confused with residual or recurrent tumor in the post operative patient [17].

Diffusion imaging also plays a significant role in differentiating radiation induced injury from residual or recurrent tumor in patients who have undergone radiation therapy. Radiation can result in increased permeability of the blood brain barrier and, as a result, normal radiation induced changes are often indistinguishable from enhancing tumor on follow up imaging. Al Sayyari, et al., studied 17 patients with high grade glial neoplasm with both perfusion and diffusion imaging in an effort to differentiate enhancement resulting from radiation induced injury from recurrent tumor. They found significantly lower ADC values in patients with recurrent tumor and increased ADC values in patients with proven radiation induced injury and proposed that diffusion weighted imaging could be a useful adjunct tool in the follow up of treated brain tumors [18].

Diffusion *tensor* imaging is a more complex form of DWI in which water motion is observed in multiple planes, generally between 6 to 256 planes. Using this technique, the observer is able to create a *tensor*, or vector, of the water motion, giving both direction and strength/uniformity of flow. White matter tracts in the brain are water containing structures with very strong predilection for anisotropic movement. Using DTI we are able to extrapolate the size, integrity, and direction of the white matter fiber tracts in the brain.



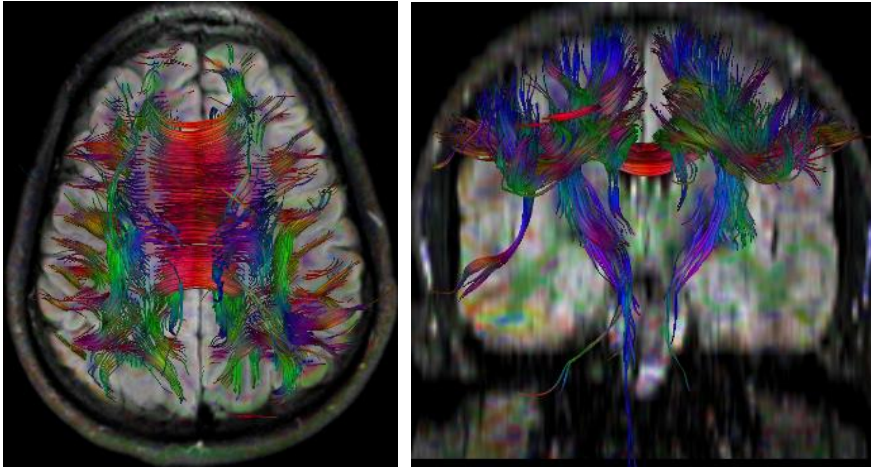


Figure 10a-b. Diffusion tensor imaging of the brain superimposed on FLAIR sequence. Colors indicated directionality: red fibers travel in transverse fashion, the blue fibers travel craniocaudally, and the green fibers travel anteroposteriorly.

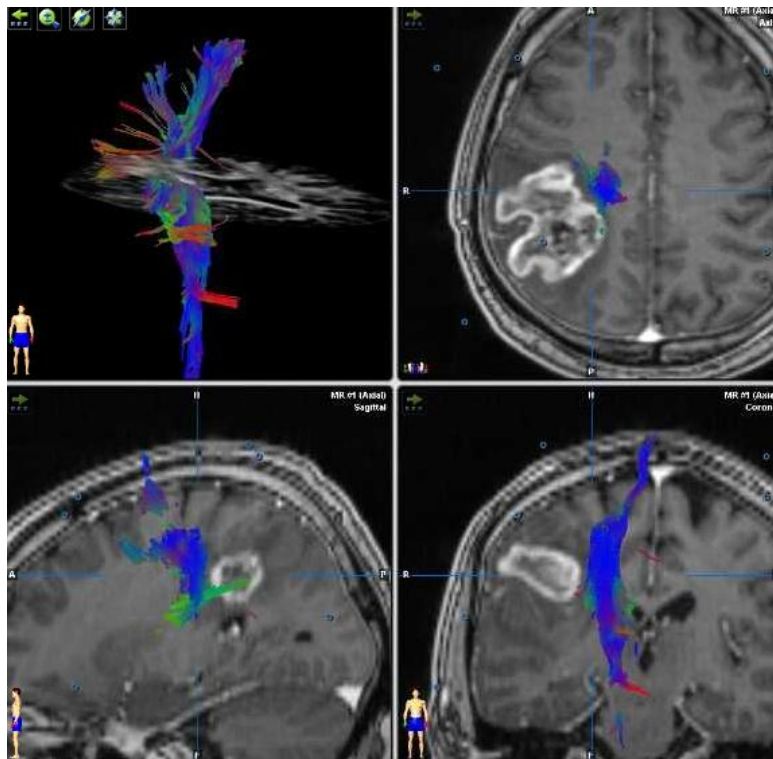


Figure 11. Diffusion tensor imaging of the right corticospinal (CS) tracts. The CS tracts are superimposed on conventional MRI T1-post gadolinium images showing a large, rim-enhancing glioblastoma. DTI shows that the CS tracts are located along the anteromedial border of the tumor. The enhancing portions of the tumor were safely resected with preservation of CS tracts.

Perhaps the most useful role of DTI in the management of brain tumors is in determining the approach and volume of intended tumor resection. Metastatic lesions often displace, rather

than invade, the white matter tracts in the brain. Glial tumors, on the other hand, are more invasive and can disrupt the white matter tracts. DTI can be superimposed on conventional MRI to show the relationship of the relevant white matter tracts to the tumor prior to determining the approach and amount of tumor to be resected.

## BRAIN MAPPING WITH FUNCTIONAL MRI (fMRI)

The term “Functional MRI” or “Functional Neuroimaging” technically only specifically refers to the neural activation techniques used to locate eloquent brain but has unfortunately become a commonly used term to order any or all of the techniques described above and in this section as well. Clinical requests for “Functional MRI” must be questioned to determine exactly which of these newer imaging techniques are in fact, ordered. The shorthand “fMRI” is typically relative to the brain mapping endeavor. It is extremely useful in locating eloquent brain, such as motor cortex and speech centers as part of modern surgical planning to improve outcomes.

fMRI scans require a cooperative patient and tolerant MRI technologists as well as high field strength MRI scanners for ideal results. Patients in the scanner are requested to perform a simple paradigm (finger movement for motor cortex, counting out loud for efferent speech) during echo planar image acquisitions that gather hundreds of images per second. Alternating the paradigm performance for a full 30 seconds with an equal 30 second period of total rest, with repetition up to three minutes will produce sufficient differences in blood flow in eloquent cortex for the sensitive MRI to detect and then project onto a more detailed anatomic series of axial images.

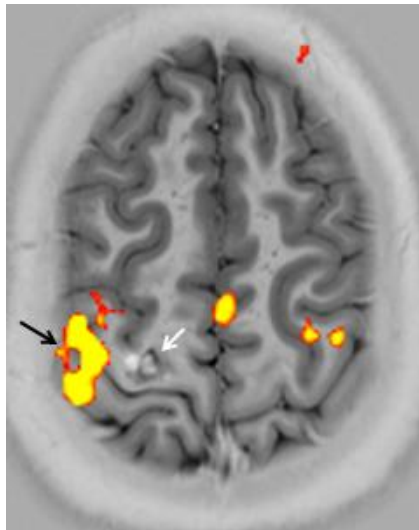


Figure 12. BOLD imaging superimposed on axial T1-weighted MR image. T1 image shows a cavernous malformation (white arrow) with subacute blood products. Functional MRI shows activation in the right motor cortex. The motor cortex is lateral to the lesion and, knowing this, an appropriate surgical path and resection volume can be planned. Note small amount of activation on the patient's left hemisphere as the subject almost invariably moves his/her contralateral hand to a small degree while performing the task.

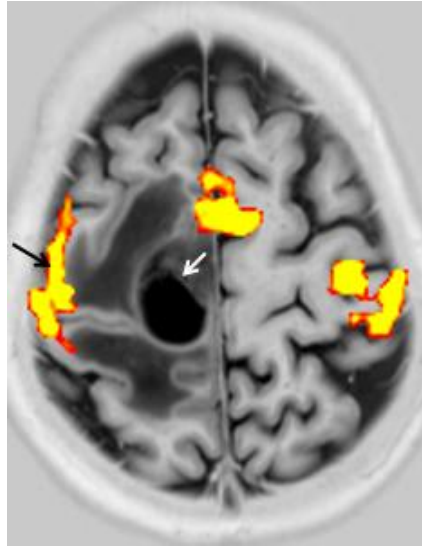


Figure 13. BOLD imaging superimposed on T1-weighted MR image. There is a cystic right frontal mass with large amount of surrounding edema (white arrow). Functional MRI shows lateral and some anterior displacement of the motor cortex. The mass was safely resected from a superior approach with preservation of left sided motor function. Note: activity on the left from contralateral hand motion on functional imaging.

The phenomenon of neurovascular coupling is key to this technique, with the minute increased capillary blood supply following the invocation of Betz cells in the targeted locations. Image detail depends upon the differences of blood flow in conjunction with the increased oxygen extraction that takes place during the task performance. Although this is known as the BOLD technique (Blood Oxygen Level Detection), it actually is a combination of that and the FLOW technique (blood FLOW Detection); both ultimately depending upon the fact that arterial blood is diamagnetic and de-oxygenated blood is paramagnetic and eminently detectable on fMRI.

It is important for the neurosurgeon to know the exact relationship of either the primary motor cortex or Broca's area to the bulk of a tumor which is to be removed as it decreases the chances for post-operative morbidity or deficit. Coupled with DTI, connectivity can be assessed with what can be termed "Eloquent Diffusion Tractography". However, there are other uses for this technique which are currently being explored by the Psychiatric community, including connectivity studies, drug response, and determination of veracity.

## CONCLUSION

Perfusion, spectroscopy, and diffusion weighted imaging are advanced MRI techniques that can play a useful role in the diagnosis and management of brain tumors. While these techniques can be useful as stand alone techniques, it should be noted that they are more frequently helpful when used in conjunction with other advanced MRI techniques, conventional MRI, and the clinical presentation.

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## **Chapter 6**

# **INTRA-OPERATIVE BRAIN MAPPING IN GLIOMA SURGERY**

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## **ABSTRACT**

Mapping of cortical and subcortical functional pathways is a mainstay of modern glioma surgery that allows the neurosurgeon to maximize the extent of resection while decreasing the risk of permanent neurological morbidity. Intraoperative direct cortical stimulation remains the gold standard for localization of motor and language circuits. However, the recent development of preoperative adjuncts such as magnetoencephalography, diffusion tensor imaging, and transcranial magnetic stimulation are becoming increasingly important for improving the accuracy and efficiency of localizing critical functional circuits during glioma surgery.

## **INTRODUCTION**

Given the evidence that extent of resection improves outcome in glioma patients, the primary goal of glioma surgery is to maximize the extent of resection while minimizing morbidity. To this end, it is critical for neurosurgeons to have a precise understanding of the basic cortical and subcortical circuits responsible for functional integrity. Despite a number of important advances in preoperative functional assessment in brain tumor patients, including functional MRI, diffusion tensor imaging, magnetoencephalography, and transcranial magnetic stimulation, direct cortical stimulation in the operating room remains the gold standard for identifying indispensable functional pathways. In this chapter, we begin by

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reviewing the relevant cortical and subcortical functional anatomy in the human brain, with a particular focus on recent data from intraoperative direct cortical stimulation. We then discuss the basic principles of preoperative evaluation, management, and surgical planning for patients with intrinsic brain tumors, with an emphasis on indications for intraoperative mapping. Finally, we provide a detailed description of the surgical technique and established cortical/subcortical mapping protocols that allow for optimal surgical resection.

## LOCALIZATION OF HUMAN CORTICAL AND SUBCORTICAL FUNCTION

### 1. Motor

The majority of motor function in the human cortex is located within the frontal lobe. The primary motor cortex, which corresponds to Brodmann's Area 4 (A4) and is also known as the precentral gyrus, is bounded by the central sulcus posteriorly and the precentral sulcus anteriorly (Figure 1). Electrical stimulation of A4 in either hemisphere, a mainstay of intraoperative human cortical mapping, produces contralateral movement of the contralateral body musculature in a topographically organized fashion. This phenomenon was first demonstrated in humans by Sir Victor Horsley in 1891 [1] and followed by the elegant demonstration of the motor homunculus by Penfield in 1937 [2].

Clinically, stimulation of A4 produces movement of a contralateral single muscle or limited group of muscles via the corticospinal tract (CST), which the patient is unable to suppress. The exception to this rule is the palate/pharynx/masseter/tongue which show bilateral contractions due to crossed and uncrossed fibers originating from A4. In addition to the contribution of A4 to the CST, premotor and supplementary motor cortices in the frontal lobe also contribute to the CST. The exact function of the "higher-order" motor areas are less well understood but recent studies have given some insight into their functional contributions. The premotor cortex (PMC), which corresponds to Brodmann Area 6, is defined as the portion of the superior frontal gyrus immediately anterior to the primary motor cortex and posterior to the frontal eye field (Brodmann Area 8). PMC lesions produce proximal muscle weakness and limb-kinetic apraxia (difficulty making precise movements with the contralateral limb) [3].

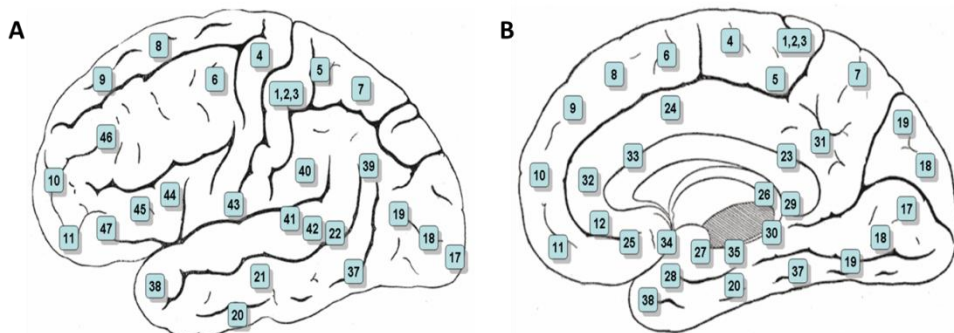


Figure 1. Diagram of Brodmann's areas on the (a) lateral and (b) medial cortical surfaces [43].



Intraoperative stimulation of the SMC produces contralateral limb movements that differ from A4 stimulation by (a) higher stimulation threshold, (b) larger domain of muscles recruited and (c) slower movements, consistent with the proposed role of PMC in selecting movements. The supplementary motor area (SMA) is located anterior to the foot motor cortex along the medial frontal lobe. The SMA is thought to be involved in motor planning. SMA stimulation causes contralateral movements that resemble postures, and resection of the SMA produces a reproducible syndrome with impairment of volitional movements, hemineglect, and dyspraxia/apraxia of the contralateral limbs with preservation of muscle tone. If involving the dominant language hemisphere, difficulty initiating speech can also occur. SMA syndrome inevitably recovers by 6 months postoperatively [4, 5]. The frontal eye field (FEF) of the frontal lobe, is located in the posterior portion of the middle frontal gyrus anterior to the face and arm motor cortex of A4. The FEF is involved in saccadic eye movement, and intraoperative stimulation causes conjugate movements toward the contralateral side. Lesions of the FEF cause an inability to voluntarily gaze to the opposite side [6, 7]. In addition to motor regions located in the frontal lobe, the parietal lobe is involved in movement preparation, control, adjustment, and intention. Brodmann areas 1, 2, and 3, also termed primary or principal somatosensory area (S1), are bounded anteriorly by the central sulcus and posteriorly by the postcentral sulcus. These areas when stimulated can produce complex motor movements. Other proposed motor function of the parietal lobe include visuomotor transformation and coding of motor acts such as grasping [8]. Subcortically, the major component of the motor system descends through the centrum semiovale, corona radiata, posterior limb of the internal capsule, cerebral peduncle, pyramidal decussation in the medulla, and then descends as the lateral corticospinal spinal tract in the spinal cord. The other major descending motor system is the corticobulbar tracts, which originate in the cortical motor areas described above and descend with the corticospinal tracts, eventually projecting to brainstem cranial nerve nuclei bilaterally. The major exception to this rule is innervation to the lower face, which is unilateral.

## **2. Sensory**

Sensory information from the periphery reaches the thalamus via relays at the level of the spinal cord and brainstem. Major ascending sensory systems include the spinothalamic, trigeminothalamic, and medial lemniscus pathways. From the thalamus, information is transmitted via the posterior limb of the internal capsule to S1, which comprises the postcentral gyrus of the parietal lobe. Electrical stimulation of S1 causes sensory perception, typically described as tingling or numbness, in a localized region of the contralateral body [9].

## **3. Language**

### **3.1. Cerebral Dominance**

The concept of left hemispheric dominance for language was first proposed by Paul Broca in 1865 [10]. Current data show that in 85% of the population the left hemisphere is dominant, 9% has bilateral representation, and 6% is right-sided dominant. A more clinically

relevant statistic is that 98% of right-handed patients have left-sided dominance. Thus, clinical investigation of dominance, the gold-standard being the Wada test, is typically reserved for left-handed or ambidextrous patients. In addition to language, the left hemisphere is typically involved in logical problem solving and calculation. Conversely, the right hemisphere is specifically devoted to facial recognition, tasks involving visuospatial manipulation, and musicality [11].

### ***3.2. Language Comprehension***

The comprehension of language was described by Wernicke as residing in the posterior superior temporal gyrus corresponding to Brodmann Area 44 [12]. Direct cortical stimulation studies have further expanded the region of language comprehension to include the posterior portion of both the superior and middle temporal gyri, as well as the inferior parietal cortex just superior to the sylvian fissure. Damage to any of these areas can result in a receptive aphasia, in which the patient can still produce written or oral language with normal grammar/syntax/prosody, but the word content is defective, often with neologisms or “word salad.” In addition, the ability to repeat words and name pictures is compromised, although naming and language comprehension may be mediated by distinct regions of the posterior-superior temporal lobe [13]. Also, the ability to sing and to recite memorized passages is maintained. If similar areas are damaged in the nondominant hemisphere, dysprosody may occur, which is the inability to detect the pitch, rhythm, or emotional content of speech. More recent studies have aimed to further dissect various aspects of language within the parietal-temporal-occipital junction. From these studies, we have learned that the posterior temporal lobe (superior and inferior) is involved in reading and word retrieval but not particularly involved in word repetition. The supramarginal gyrus, inferior parietal lobe, and arcuate fasciculus are responsible for appropriate transmission of received language to the frontal lobe language areas, with injury to these areas producing a conductive aphasia (fluent speech with relatively intact grammar/syntax, paraphasic errors for which the patient is aware but cannot correct, and poor repetition).

### ***3.3. Language Expression***

Classically, the final common pathway of language output is known as Broca’s area, which encompasses the pars triangularis and pars opercularis within the posterior third of the inferior frontal gyrus. This area corresponds to Brodmann’s Areas 44 and 45. Stimulation of Broca’s area intraoperatively causes speech arrest and lack of movement. In contrast, stimulation of mouth motor cortex can also produce speech arrest but with concomitant vocalization. Lesions of Broca’s area produce non-fluent aphasia, where comprehension is maintained but the patient is unable to express verbal or written language. Direct cortical stimulation studies in humans have demonstrated that while the classical anatomic landmarks are generally accurate with respect to localizing cortex necessary for language output, there is considerable variation among human subjects [14]. For example, Duffau et al., reported a case of intact language function following resection of Broca’s area [15], demonstrating the crucial importance of direct surface stimulation to localize functionally required tissue. Interestingly, the inferior frontal gyrus of the nondominant hemisphere appears to be involved in the speech prosody (rhythm/stress/intonation of speech), with lesions to this area causing flat, unemotional speech [16]. In addition, as mentioned earlier in the chapter, the SMA of the dominant hemisphere is involved in speech initiation, and stimulation can cause temporary

speech arrest or vocalization. Finally, the dominant insular cortex may play an important role in speech planning [17]. Recent insight from cortical stimulation studies has refined our understanding of the relationship of writing. While circuits involved in writing do correspond to the same hemisphere as oral language, at least some of the writing sites are partially distinct, as evidenced by the presence of writing deficits despite negative mapping at traditional language sites. Areas shown to be important for writing function include the dominant hemispheres superior parietal lobe, supramarginal gyrus, insula, 2<sup>nd</sup>/3<sup>rd</sup> frontal convolutions, and SMA [18].

### ***3.4. Subcortical Language Representation***

The major subcortical pathways subserving oral language function as identified by direct intraoperative stimulation are [1] superior longitudinal fasciculus (SLF), [2] inferior occipito-frontal fasciculus, and [3] subcallosal fasciculus. The SLF connects the parietal/temporal region with the frontal lobe and is composed of two functionally distinct white matter pathways – the arcuate fasciculus (AF), which connects the classical Wernicke’s and Broca’s areas, and an indirect pathway parallel and lateral to the AF termed the lateral SLF. The AF (also termed the dorsal phonologic stream) connects the posterior/superior temporal region to the posterior/inferior frontal lobe and distributes the phonological aspects of language, with interruption at any site along the pathway resulting in phonological disturbances. The lateral SLF, which connects the posterior temporal, inferior parietal, and Broca’s regions, is involved in speech perception and articulation. The inferior occipitofrontal fasciculus, also termed the ventral semantic stream, is involved with semantic aspects of speech and stimulation of the pathway intraoperatively produces semantic paraphasias. Finally, the subcallosal fasciculus, connecting the mesial frontal lobe structures (SMA, cingulate) to the caudate nucleus may mediate the control of language, with lesions resulting in transcortical motor aphasias characterized by nonfluent aphasia with intact repetition.

### ***3.5 Bilingual Representation***

A number of studies have investigated the representation of languages in multilingual patients. A consistent finding is that both shared and language-specific sites are present in bilingual patients. While primary language and shared language sites are found throughout the temporal, parietal, and frontal lobes, distinct secondary language sites are located in the posterior temporal and parietal regions [19]. Interestingly, a recent study using DCS in bilingual patients implicates the dominant posterior temporal area and SLF in mediating language switching [20].

## **4. Vision**

Visual information enters the retina and is transmitted via axons of the retinal ganglion cells in the optic nerves, which partially cross at the optic chiasm and continue as the optic tracts which synapse in the lateral geniculate nucleus (LGN) of the thalamus. Cells of the LGN then project to the primary visual cortex of the occipital lobe via a fan-like projection of fibers (optic radiations) that pass lateral and superior to the atria and temporal horn of the lateral ventricle.

The inferior optic radiations (Meyer's loop) project through the temporal lobe, carrying visual information from the contralateral superior visual field, with lesions causing a "pie in the sky" contralateral homonymous inferior quadrantanopia, though the ipsilateral eye is typically affected to a greater extent because of the lateral position relative to contralateral eye fibers in the optic radiations [21]. Superior optic radiations project through the inferior parietal lobe and carry visual information from the contralateral inferior quadrants. The primary visual cortex, corresponding to Brodmann Area 17, resides in the occipital lobe along both borders of the calcarine sulcus and is retinotopically organized.

## **5. Audition**

Peripheral perception of sound is relayed via cranial nerve VIII to the dorsal and ventral cochlear nuclei near the inferior cerebellar peduncle at the pontomedullary junction. Sound information then ascends bilaterally via a series of relays to the inferior colliculi, then medial geniculate nuclei, then the primary auditory cortex located along Heschl's (transverse temporal) gyrus and corresponding to Brodmann's areas 41/42. Heschl's gyrus lies just medial to the superior temporal gyrus deep within the Sylvian fissure posterior to the insula. Given the relatively deep location of the primary auditory cortex relative to most temporal gliomas, loss of hearing, and thus auditory mapping, is not common in glioma surgery. Higher order areas of audition include areas of the parietal and temporal lobes immediately surrounding Heschl's gyrus, including Wernicke's area.

## **6. Visuospatial Processing**

The nondominant parietal lobe has been implicated in visuospatial perception, with damage to this region in humans producing a clinical syndrome of unilateral neglect, where patients essentially ignore the left half of their visual field.

Recent studies using a line bisection task in awake patients undergoing direct cortical stimulation has further defined the regions responsible for spatial cognition as both the supramarginal gyrus and posterior-superior temporal lobe of the nondominant hemisphere [22]. Subcortically, both the optic radiations and nondominant SLF are involved in visuospatial information transmission.

## **7. Calculation**

Data from neuropsychological evaluation and functional imaging studies of patients with parietal lobe damage have suggested a localization of calculation within the inferior parietal lobule. Acalculia, defined as the inability to perform simple mathematical tasks, can be either isolated acalculia or in conjunction with other symptoms (e.g. Gerstmann's syndrome of agraphia without alexia, right/left disorientation, finger agnosia, and acalculia associated with dominant angular gyrus injury). An intriguing study performed by Duffau, et al. used direct cortical stimulation during awake surgery of a school teacher with a left parietal lobe lesion [23]. Interestingly, distinct sites for multiplication and subtraction were found in the anterior

angular gyrus, posterior to naming/counting language sites in the supramarginal gyrus and anterior to the tumor.

## 8. Judgment

The prefrontal cortex is known to have a role in planning and solving problems. Direct cortical stimulation of the prefrontal cortex in patients harboring frontal brain tumors was used to investigate the contribution to cross-modal judgment. Patients were given verbal and visual cues that referred to the same item (congruent) or in which the cues differed slightly with respect to semantic or phonemic content (incongruent). Patients were asked to determine if the verbal/visual pair was the same during stimulation. Incongruent judgment was impaired during stimulation of the left dorsolateral prefrontal cortex, suggesting a role for this area in judgment of complex scenarios.

## 9. Commissural Connections

The major connection between the right and left hemispheres is the corpus callosum, with the vast majority of cortical areas receiving at least some input from the contralateral hemisphere. Notable exceptions to this pattern are the primary visual cortex and hand areas of the rolandic cortices, which lack commissural connections [24]. Sectioning of the corpus callosum is in the standard repertoire for epilepsy surgeons and is an important part of surgical approaches to the ventricular system. Damage to the callosum can result in various disconnection syndromes, as described by Sperry [25]. With regard to intrinsic tumor surgery, callosal involvement is relatively common, particularly in tumors of the cingulate gyrus. Duffau et al. reported outcomes for a series of 32 patients with low-grade gliomas involving the corpus callosum. Direct stimulation of the callosum had no effect, and the resection of glioma-infiltrated callosum improved quality of resection without inducing long-term neurological deficits [26].

# TUMOR AND FUNCTION LOCALIZATION

## 1. Tumor Characteristics

Preoperative imaging of patients harboring gliomas begins with standard MRI sequences, typically T1-weighted images with gadolinium, as well as T2 or FLAIR imaging. Enhancement, when present, can demonstrate the tumor center and gross margin, which provides a target for resection, as tumor recurrence risk is greatest within 2 cm of the enhancing rim [27]. Also, enhancement is generally indicative of a higher grade tumor. T2 or FLAIR sequences demonstrate the extent of edema, although one cannot determine the relative contributions of tumor versus vasogenic edema. With that caveat, for the case of low-grade gliomas (LGG), FLAIR signal is generally considered the target of resection. MR

spectroscopy may be a useful adjunct in the setting of previously treated patients for which the diagnosis of recurrence versus treatment effect is equivocal.

## 2. Identifying Functional Brain Regions

### 2.1. Anatomic Localization of Functional Circuits with MRI and Diffusion Tensor Imaging (DTI)

In order to determine the relationship of the tumor to the Rolandic (somatomotor) cortices, it is imperative that the surgeon be able to localize the central sulcus (CS) from preoperative MRI. Multiple methods exist for identifying the CS. Most commonly, on high-vertex T2-weighted images, mirror-image transverse sulci that are nearly perpendicular to the midline represent the CS (Figure 2A). The “hand knob” omega-shaped region of the precentral gyrus can usually be seen anterior to the CS. Another method to identify Rolandic cortex is based on parasagittal MRI slices. The cingulate sulcus is identified and followed posteriorly and then superiorly where it terminates as the marginal sulcus. Immediately anterior to the marginal sulcus is the paracentral lobule, which is a medial extension of both the precentral (motor) and postcentral gyri of the lateral hemisphere (Figure 2B). A third method utilizes a sagittal MRI image through the insula. A triangle is drawn around the insula, and a line drawn perpendicular to the most posterior vertex of the triangle will approximate the location of the CS. These three methods (or combinations thereof) are highly reliable for identifying primary somatosensory cortex and its relationship to tumor. In addition to these topographic assessments of sensorimotor function, diffusion tensor imaging (DTI) is emerging as a routine part of preoperative evaluation of brain tumor patients as well as an important intraoperative adjunct to direct cortical stimulation. DTI fundamentally measures the degree of water diffusion within brain compartments, with more anisotropic regions such as white matter tracts having a higher degree of restricted directionality.

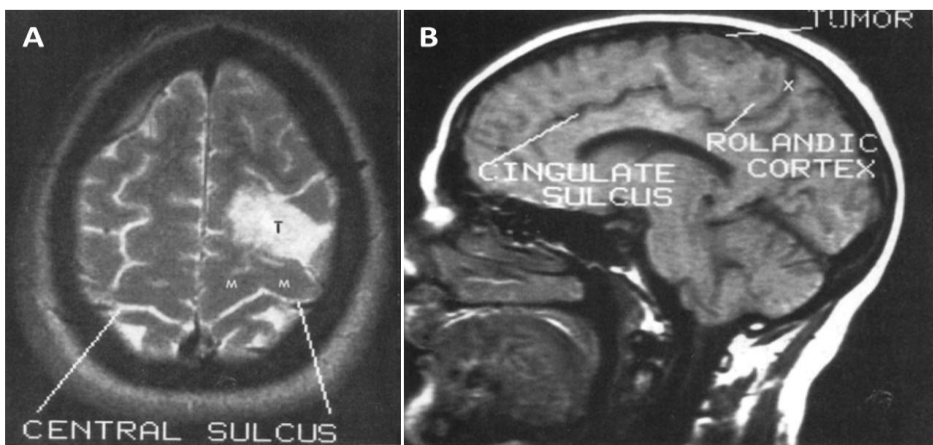


Figure 2. T2-weighted high vertex axial slice demonstrating the central sulcus perpendicular to the midline, along with the primary motor cortex in relation to a left posterior frontal glioma (A). T1-weighted image demonstrating the cingulate sulcus and its termination as the marginal sulcus (X), with the rolandic cortex just anterior to this sulcus, in a patient harboring a frontal tumor. M = primary motor cortex, T = tumor. *Adapted from Berger, et al. [44].*

The diffusion tensor can be evaluated for each MRI voxel of interest and a three-dimensional map is generated which accurately displays major white matter bundles. This map can be integrated into the presurgical scan to assess the relationship of the tumor to cortical and subcortical white matter pathways, most commonly the primary motor cortex and corticospinal tract. While the process is associated with some inherent error and must be verified with direct stimulation in the operating room, in our experience DTI is helpful in determining the mapping strategy for a given tumor, discussing expectations with patients, and improving the efficiency of mapping in the operative room. As a general rule, DTI maps tends to be larger than that observed by direct stimulation, suggesting that resection of a significant part of a DTI-defined tract can be well tolerated in terms of maintaining function and highlighting the importance of direct stimulation during resection. In contrast to preoperative localization of cortical and subcortical motor function, for which the standard MRI, DTI, and MSI findings are relatively consistent between patients, functional language circuits are more diffuse and variable, making preoperative anatomic localization more difficult. One particularly promising strategy is the use of preoperative DTI to localize language circuits. Recent data from a series of 230 patients undergoing glioma resection looking into the concordance of preoperative DTI with direct subcortical stimulation during surgery, demonstrated a high concordance rate for the corticospinal tract (motor), SLF (phonemic language), and occipitofrontal fasciculus (semantic language) [28]. Thus DTI is becoming an important preoperative and intraoperative adjunct to direct cortical mapping.

## **2.2. Functional Magnetic Resonance Imaging (fMRI)**

fMRI is based on the principle that neurons of more active brain regions receive increased blood flow to deliver glucose at a higher rate, which results in a localized increase of oxyhemoglobin (diamagnetic) relative to deoxyhemoglobin (paramagnetic). While localization of primary motor and sensory cortices by fMRI is accurate, several issues plague its widespread clinical use in brain tumor patients. First, at its most basic level, fMRI only demonstrates areas of increased blood flow, which is an indirect measure of neuronal activity. Also, the time lag between neuronal firing and diverted blood flow may be problematic for interpretation, particularly for complex actions involving several brain regions that have overlapping neural circuitry, such as language or higher-order cognitive function. fMRI at best illustrates which brain regions are involved in a particular task, but it does not demonstrate which regions are functionally necessary. In brain tumor surgery, given the clear data demonstrating that extent of resection correlates with outcome, we would ideally want to know the minimal circuit necessary for a given function so that all areas not required or which are normally used but can be adequately compensated for may be resected.

## **2.3. Magnetoencephalography**

Another methodology for detecting recruited neural circuitry during a given activity is magnetoencephalography (MEG). Unlike fMRI, MEG directly measures the magnetic field produced by electrical activity in the brain, specifically dendritic potentials. During a particular task, changes in the magnitude of  $\alpha$ ,  $\beta$ , and  $\gamma$  band cerebral oscillations can be detected, with higher and lower frequencies representing synchronization and desynchronization, respectively (29). For example, changes in the  $\gamma$  frequency band are associated with higher order cognitive processes such as language processing [30]. MSI, which refers to incorporating MEG-based functional data into intraoperative brain navigation

software, is being utilized in some centers. A recent study of 57 consecutive patients at our institution demonstrated a 100% negative predictive value and 64% positive predictive value for correlation of intraoperative stimulation of functional sites with MEG [31]. Thus MEG presents a promising technology for identifying function or absence of function near the tumor preoperatively, though additional validation with direct cortical stimulation and outcome data will be necessary before routine use is recommended.

#### ***2.4. Transcranial Magnetic Stimulation***

A more recent technology for mapping cortical function preoperatively is transcranial magnetic stimulation (TMS). TMS is a non-invasive technique that generates a precise, local magnetic field that induces an action potential in a small population of neurons. Through a simple algorithm, the patient's MRI data can be coregistered with the TMS software, allowing for precise delivery of the magnetic field to the cortical surface. A magnetic field is delivered to specific regions of the primary motor cortex using the MRI-based map, resulting in pyramidal neuron activation and subsequent movements of the relevant contralateral muscle groups, which can then be detected by EMG. Recent data indicates that TMS-based motor maps correlate well with direct cortical stimulation in the operating room [32]. TMS has also been investigated as a preoperative test for localization of language functions, though these data are somewhat less consistent compared to motor mapping [33]. Nonetheless, TMS represents a very promising modality not only for preoperative mapping of various cerebral functions but also a tool to study cortical plasticity before and after tumor resection, an important consideration for surgical strategies following initial resection.

## **PREOPERATIVE CONSIDERATIONS**

After initial imaging studies to evaluate the tumor characteristics, its location, and its relationship to known functional pathways and a careful neurological examination, it is important to consider the goals of surgery. For patients with diffuse disease and/or poor performance status, a biopsy to establish diagnosis may be the most prudent course of action. For a patient with a tumor within the motor strip that has imaging characteristics of a low-grade glioma and minimal or no functional deficit, observation with frequent serial MRI scans is reasonable, based on the concept that additional functional pathways may be recruited over time. In these patients, a worsening neurologic deficit or evidence of progression would be criteria for operative intervention. Once the decision is made to proceed with resection, the next decision is whether the patient needs direct cortical mapping. Some surgeons propose to do all cases with mapping or even all supratentorial tumor surgeries in the awake state if the patient is a reasonable candidate. Not only can basic motor and language function be mapped, but also individualized functions relevant to a particular patient's occupation and quality of life [34, 35]. For example, intraoperative visuospatial testing might be performed in an architect harboring a right parietal lesion, or calculation testing in a mathematics professor with a LGG of the dominant inferior parietal lobule. As intraoperative testing of higher order functions becomes more reliable, this strategy may become more popular. At present, however, the vast majority of glioma surgeries requiring mapping will be performed as either (a) awake craniotomy with language and motor mapping or (b) asleep craniotomy with motor



mapping. In the more rare circumstance of a glioma within the primary motor cortex or a patient with very poor but present motor function, an awake craniotomy with motor mapping may be considered, as the sensitivity is improved. Generally, patients with less than antigravity strength (0-2/5) will not have successful motor mapping.

Patients who qualify for awake craniotomy with language mapping include patients with dominant or co-dominant hemisphere tumors within or adjacent to known primary language cortical/subcortical circuits or any patient with language dysfunction concordant with the tumor location. There are a number of contraindications to awake mapping. Patients with tumors that have significant mass effect with midline shift noted on MRI are not good candidates, due to the potential for cerebral herniation after dural opening. One strategy in these cases is to treat with steroids and/or diuretics for 1-2 weeks, which can reduce the mass effect and allow for safer mapping. Another option is to perform an asleep biopsy followed by radiation to decrease the tumor burden and associated edema. Patients with significant preoperative language deficits (>25% error rate) that do not improve with short-course steroids may not be good candidates for awake mapping. Finally, pediatric patients or patients who may not be able to cooperate with the intraoperative language tasks should not be offered awake craniotomy. For these scenarios, the approach would be to perform a more conservative resection based on anatomic imaging and available preoperative functional data. Asleep craniotomy with motor mapping is generally considered for patients whose tumors are located within or adjacent to the Rolandic cortex and/or corticospinal tract. The major contraindication is poor baseline motor function (<3/5) that has not responded to a trial of steroids.

## INTRAOPERATIVE MAPPING TECHNIQUES

### 1. Patient positioning and incision

The main considerations for positioning the patient in the operating room are related to the location of the tumor. For tumors located in the frontal lobe, the patient is positioned one of two ways depending on the relationship of the tumor to the midline. For lateral frontal tumors (> 4 cm from the midline), the patient is placed supine with an ipsilateral shoulder roll and the head rotated 60 degrees towards the contralateral side (Figure 3A). A curvilinear incision extending from the zygomatic arch to just behind the hairline at the midline is used. For frontal tumors less than 4 cm from the midline (or Zone II cingulate tumors anterior to primary sensorimotor cortex, Figure 4A) [36], a similar incision is used and the patient is positioned supine with only a slight head rotation toward the contralateral side (Figure 3B). For temporal lobe tumors, the patient is positioned with the head turned 90 degrees so that it is parallel to the floor. Additionally, if there is a significant medial component, the head is flexed toward the floor by 10 degrees to improve access. For anterior temporal tumors, the incision starts at the zygoma, curves posteriorly above the pinna, and then anteriorly toward the hairline (Figure 5A). For posterior temporal tumors, a horseshoe incision starting at the zygoma and centered over the tumor is utilized (Figure 5B). Inferior/lateral parietal tumors are approached with the patient supine with the head turned 90 degrees and a horseshoe incision with the base centered over the pinna (Figure 6A). For superior/medial parietal

tumors, the patient is placed supine with the nose up and the head at 45 degrees from the horizontal (Figure 6B).

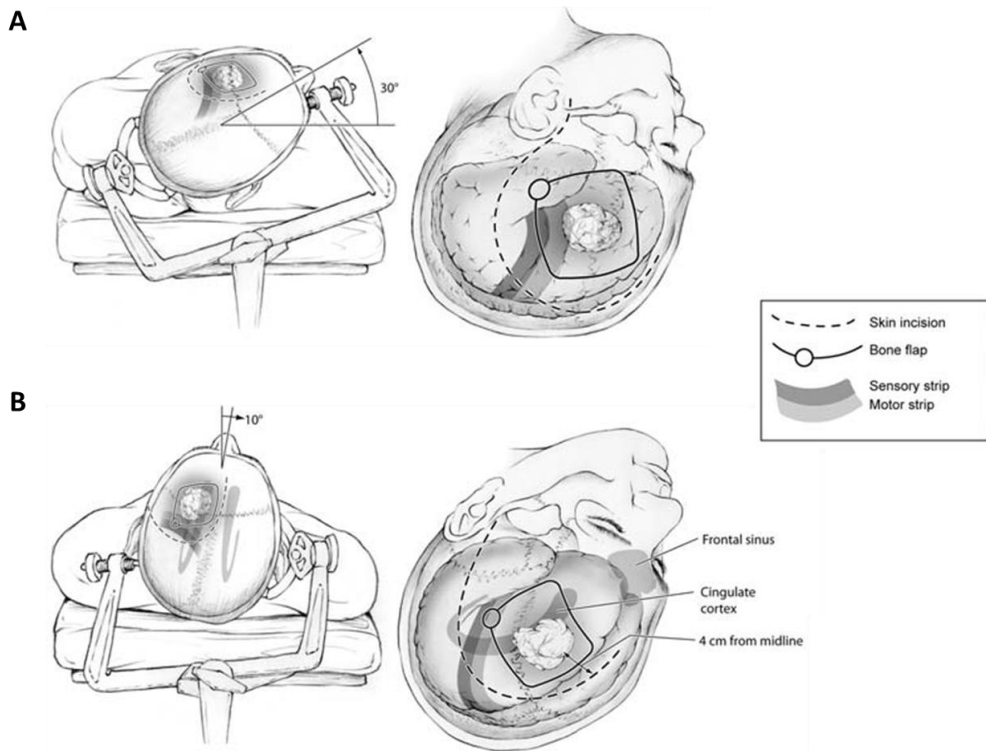


Figure 3. Illustration showing the surgical position and scalp incision for frontal tumors (A) within 4 cm of the midline and (B) lateral to 4 cm from midline. *Adapted from Berger, et al. [6].*

The incision starts at the midline just in front of the rolandic cortex, extends posteriorly behind the tumor, and then curves anteriorly and inferiorly toward zygoma. Zone III cingulate tumors can also be approached in this manner (Figure 4A) (36). For occipital tumors, the patient is placed in a lateral decubitus position with a protective axillary roll and the arm in a sling. The head is rotated to 30 degrees from the vertical and 15 degrees of lateral neck flexion toward the floor (Figure 6C). A horseshoe incision with its base centered above the ipsilateral transverse sinus and extending from behind the ear laterally to the midline medially is utilized. Finally, positioning for insular tumors is based on the relationship of the tumor to the sylvian fissure.

If the majority of the tumor is above the fissure (Berger-Sanai Zone I/II, Figure 4B), the patient is placed supine with the head rotated 60 degrees towards the contralateral shoulder and the head is angled upward 15 degrees from the horizontal. These maneuvers enable the surgeon to operate parallel to the insular vessels (Figure 7A). For insular lesions centered below the sylvian fissure (Zone III/IV) the head is turned parallel to the floor and angled 15 degrees inferiorly toward the floor, which provides adequate visualization of the inferior insula following superior temporal gyrus resection (Figure 7B). For insular tumors extending posteriorly (beyond the posterior limb of the internal capsule, Zone II/III), the head should be turned 60 degrees from the vertical instead of 90 degrees.

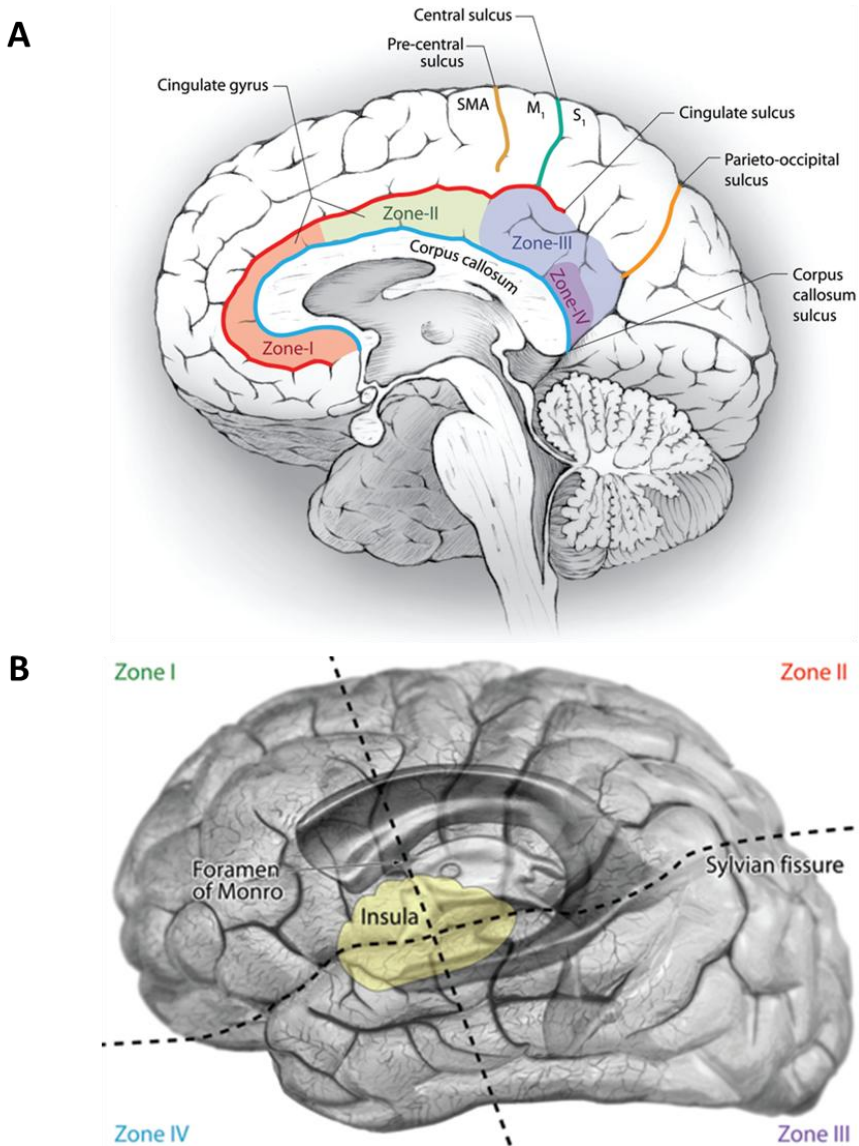


Figure 4. Anatomic illustration of naming schemes for (A) cingulate and (B) insular gliomas. Adapted from Tate, et al. [36] and Sanai, et al., [45] respectively.

The incision for insular tumors starts at the zygoma, curves posteriorly just above the ear, and then extends anteriorly toward the hairline. The aforementioned patient positioning details can be modified for lesions either encompassing multiple regions or located at boundaries of the supratentorial regions. Lastly, for all cases, local anesthetic is infiltrated along the incision site. Additionally, for awake craniotomy cases, the injections are placed circumferentially around the skin incision (Figure 8). After appropriate positioning, all pressure points are padded. Importantly, the contralateral body must remain free of lines, blood pressure cuff, etc. so that it can be readily visualized during mapping. A heating blanket is placed to ensure temperature  $> 36^{\circ}\text{C}$ . Intravenous steroids (4 mg dexamethasone) and preoperative antibiotics (1-2 grams cefazolin) are administered.

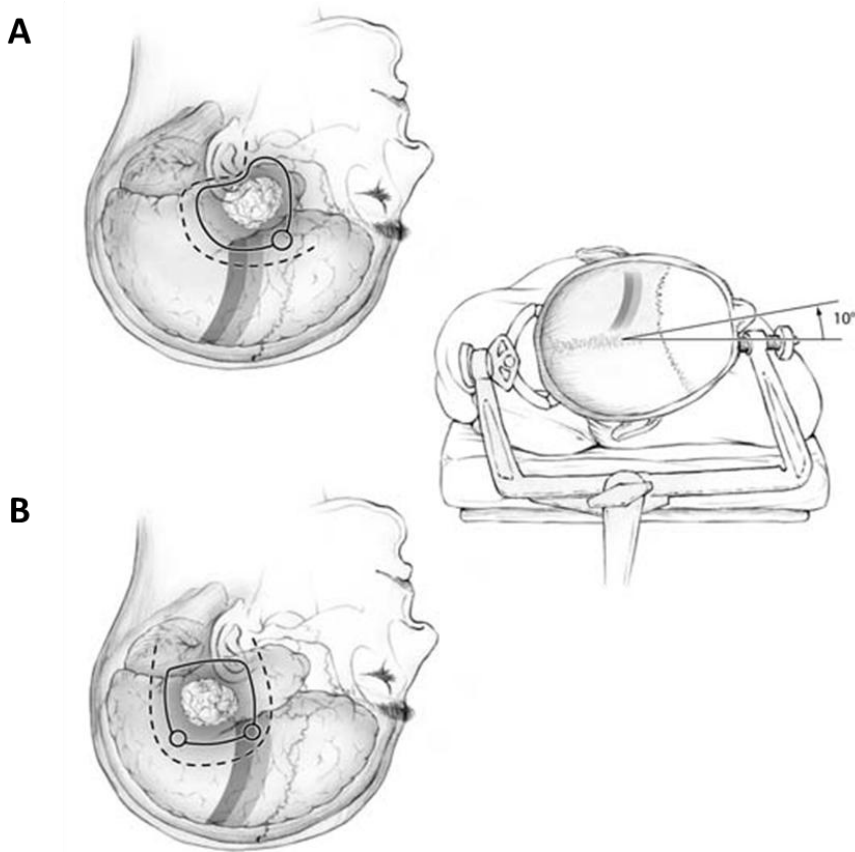


Figure 5. Illustration showing the surgical position and scalp incision for anterior (A) and (B) posterior temporal lobe tumors. *Adapted from Berger, et al. [6].*

Anticonvulsants are also administered, either the patient's home regimen or a dilantin load (15 mg/kg) if not previously on anticonvulsants. If elevated ICP is a concern, mannitol (1 gram/kg) can be administered. Finally, a time-out procedure verifying patient characteristics, tumor side, surgical plan, expected blood loss, and details of the proposed mapping strategy is absolutely essential.

## 2. Awake Craniotomy

Ultimately, the success of awake craniotomy and language mapping relies on a cooperative patient. Given that the length of surgery may be several hours, it is important to have the patient comfortably sedated for the portions of the operation when mapping is not performed. While there are multiple options in the neuroanesthesia literature, we prefer propofol ( $\leq 100 \mu\text{g/kg/min}$ ) and remifentanyl ( $\geq 0.05 \mu\text{g/kg/min}$ ). It is important to ensure adequate ventilation and thus not to over sedate. For mild obstruction, a nasal airway can be helpful. Propofol is administered at the time of foley administration and just prior to Mayfield pin application. Lidocaine/bupivacaine local anesthetic is also administered at the three pin sites.

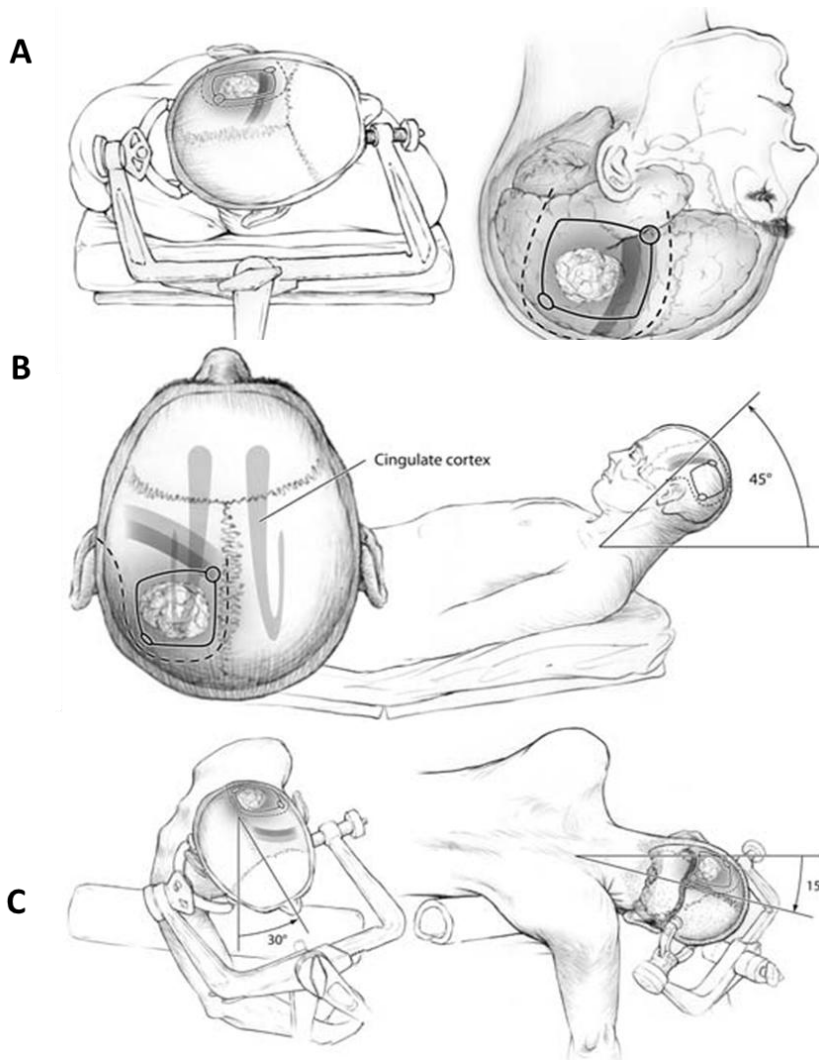


Figure 6. Diagram of patient positioning, scalp incision, and bone flap for (A) inferior parietal (B) superior/medial parietal and (C) occipital lobe tumors. *Adapted from Berger, et al. [6].*

The propofol/remifentanyl infusion is titrated during incision, muscle dissection, and craniotomy so that the patient remains comfortably sedated and breathing comfortably. After the bone flap is removed, all sedatives are discontinued and the patient is allowed to wake. During mapping, propofol should be within 6 inches of the IV line and ice cold lactated Ringer's solution should be available should seizures occur during stimulation.

Following mapping, either propofol/remifentanyl or remifentanyl ( $\geq 0.025$   $\mu\text{g/kg/min}$ )/dexmedetomidine (0.5  $\mu\text{g/kg}$  load, start at 0.5  $\mu\text{g/kg/hr}$  and increase by 0.1  $\mu\text{g/kg/hr}$  increments) infusions are used to slowly increase the level of sedation while avoiding respiratory depression. After positioning and pinning, the MRI-based surgical navigation system is registered. Using the trajectory views, a surface projection of the proposed surgical corridor to the tumor is drawn and a generous skin incision marked. Next, a bupivacaine (0.25%) and lidocaine (0.5%) local anesthetic mixture is prepared and injected circumferentially around the incision.

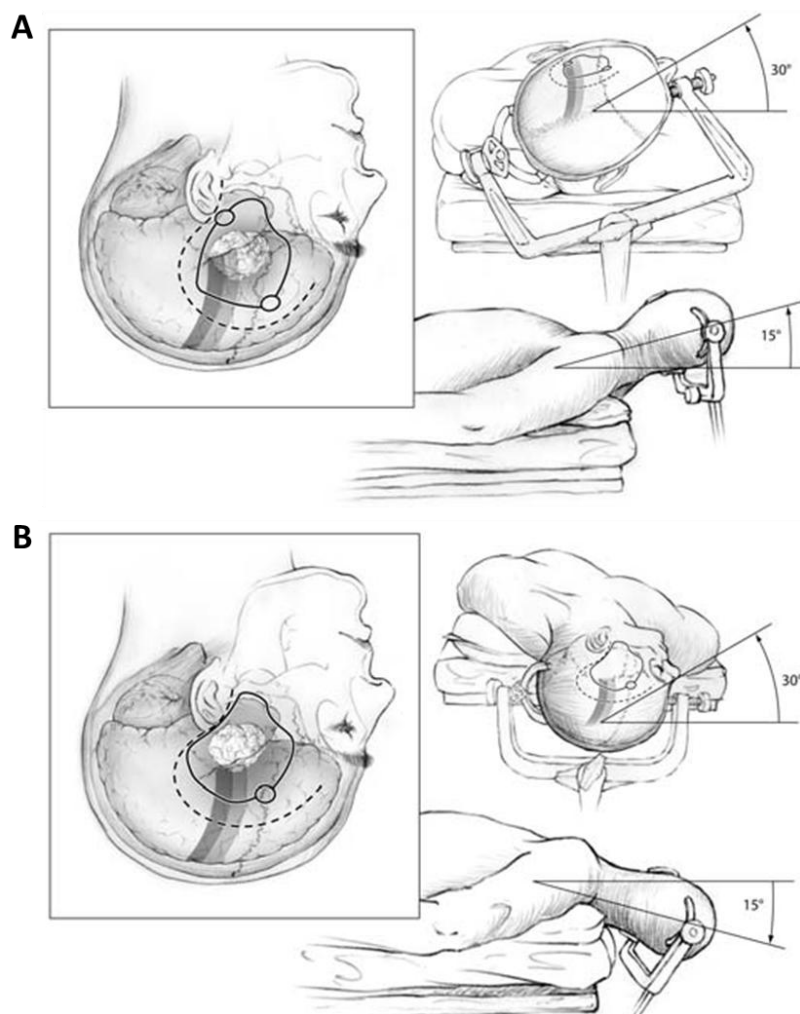


Figure 7. Operative positioning and scalp incision for insular lesions primary above (A) or below (B) the sylvian fissure. *Adapted from Berger, et al. [6].*

Specifically, the local block should address the territories of the supraorbital (above midpoint of orbital rim), auriculotemporal (1.5 cm anterior to tragus), zygomaticotemporal (midway between supraorbital ridge and the posterior margin of the zygoma), and lesser/greater occipital nerves (along line extending from inion to mastoid), depending on the location of the anterior-posterior extent of the scalp incision. Refer to Figure 8 for examples of local blocks tailored to tumors of various locations. At this point, the patient is prepped and draped in the usual sterile fashion while ensuring an adequate window to allow the patient to visualize the computer screen or slides during language mapping tasks. Following scalp incision, additional local anesthetic is infiltrated into the temporalis and/or occipital musculature. The intraoperative navigation system is again used to draw out a craniotomy flap, which will provide adequate exposure to the tumor. It should be noted that unlike motor mapping, which relies on confirmation of positive stimulation sites either on the exposed cortical surface or beneath the craniotomy edge with a strip electrode, there is strong data to support a limited craniotomy with reliance on negative data for language mapping [37].

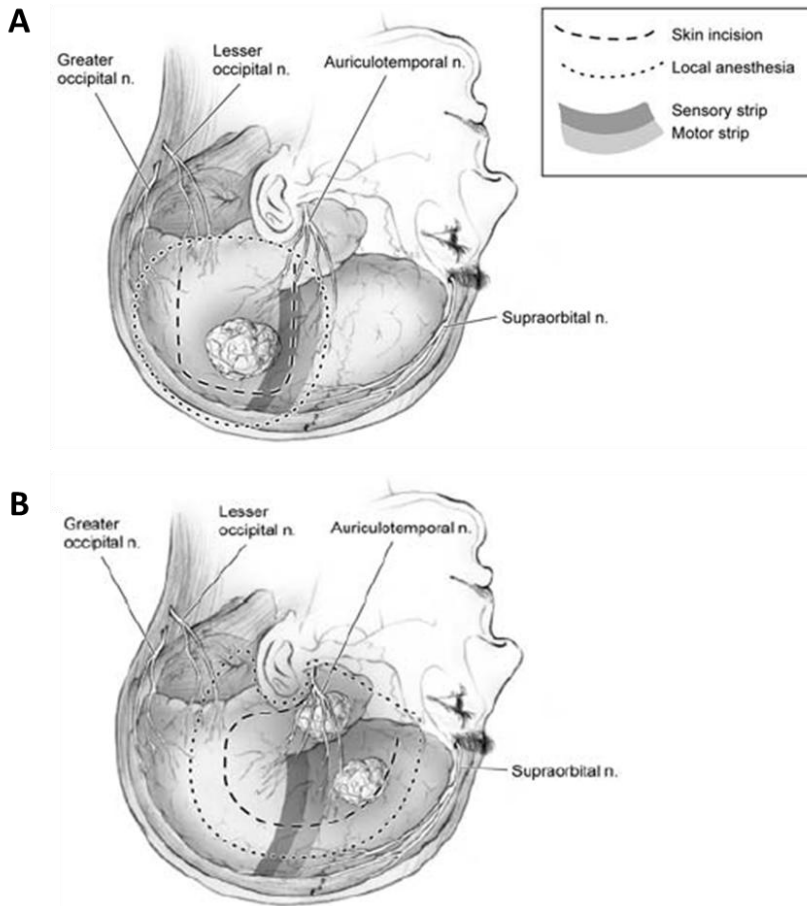


Figure 8. Illustrations showing the scalp incision and local anesthetic infusion sites relative to scalp nerve anatomy for parietal (A) and frontotemporoinsular (B) tumors. *Adapted from Berger, et al. [6].*

Following bone flap removal, a 30-gauge needle is used to administer local to the dura along the middle meningeal artery. Prior to dural opening, a small hole is drilled in the skull to anchor the electrocorticography crown post. The patient should be fully awake and cooperative prior to opening the dura, as emergence can otherwise cause coughing and brain herniation, particularly for large tumors or those with significant mass effect/edema.

### 3. Asleep Craniotomy

The patient is premedicated with midazolam and then brought to the operating room. Induction is performed using fentanyl and propofol. The patient is paralyzed prior to intubation and the blockade is reversed following skin incision. General anesthesia is maintained with nitrous oxide (70%), low-dose inhalational agent (typically <0.5 MAC isoflurane), and a fentanyl infusion (2  $\mu\text{g/kg/hr}$ ). After bone flap removal, mannitol and/or hyperventilation can be utilized if the brain appears full. Prior to initiation of motor mapping, the contralateral arm and leg should be uncovered, and adequate patient temperature ( $>36^{\circ}\text{C}$ )

and full reversal of neuromuscular blockade should be confirmed. As with awake mapping, first cold saline and then propofol can be used to abort seizure activity during mapping. Following completion of motor mapping, neuromuscular blockade can be resumed and is not reversed until after Mayfield pin removal. Fentanyl infusion is continued through scalp closure.

## 4. Mapping Details

### 4.1. Motor Mapping

Following dural opening, an Ojemann Cortical Stimulator (biphasic square wave, 60 Hz, 1 ms duration, current range 2-16 mA peak-peak) is brought into the field for motor mapping. Stimulation is performed by applying a bipolar electrode to the cortical surface for 2 seconds. For asleep motor mapping, a starting current of 4 mA (p-p) is applied at the primary motor cortex, and the current is increased in intervals of 2 mA until either a visual motor response or reproducible EMG activity in the muscle is noted. This latter EMG-based method is more sensitive than overt muscle contraction, allowing for decreased stimulation threshold and thus decreased risk of intraoperative seizure activity [38]. Positive cortical sites are labeled with sterile numbered paper squares. Typically, motor stimulation is elicited in the face and hand region of the motor strip. The leg motor cortex, which lies medially against the falx, may be stimulated via a subdural strip electrode. This blind insertion of the strip electrode is safe due to the lack of bridging veins between the leg motor cortex and falx. Similarly, if the craniotomy is anterior to the motor strip, a strip electrode can be passed posteriorly to stimulate the unexposed cortex. Following mapping of the primary motor area, language mapping may be performed as described below. If motor function is the only modality being mapped, then a barrier, usually a clear plastic sheet, is placed over the numbered positive stimulation sites for protection during the remainder of the case. Cortical window(s) are opened to provide adequate access to the intrinsic tumor below the surface. Subcortical motor mapping is performed once the resection nears the corticospinal tract system (descending corticospinal fibers, internal capsule, cerebral peduncle), and it is at this point that intraoperative motor DTI improves efficiency of stimulation. When bipolar stimulation is used, the current spread is minimal, approximately 2 mm. The implication of this limited spread is that (a) when performing resection near functional subcortical functional areas, stimulation should be repeated often (ideally after every 2 mm of tumor removal) and (b) the resection should be stopped when positive subcortical stimulation occurs. Prior to dural closure, a final stimulation at the cortical surface with preserved EMG activity distally provides confidence to the surgeon that the entire motor circuit is intact, whether subcortical stimulation was positive or not. Thus, even in the presence of a new postoperative motor deficit, the patient can be reassured that function will likely return, with only a 7% chance of a long-term deficit [39].

In addition to the traditional bipolar stimulation technique based on the methods of Penfield and Ojemann, another stimulation strategy that has been utilized by some groups employs a high frequency, short pulse train technique, referred to as dcMEP (direct cortical stimulation motor-evoked potential) because of the similarity to tcMEP (transcortical stimulation of motor evoked potentials) performed through the scalp [40]. Typical dcMEP stimulation parameters are 5-10 pulses delivered via a monopolar electrode at 250-500Hz and



5-25 mA intensity. dcMEP has been shown to reliably detect primary motor cortex, and intraoperative changes in dcMEP stimulation parameters correlate with postoperative deficits. Potential advantages of dcMEP over DCS include a decreased intraoperative seizure rate (0-4%) and the seamless conversion to continuous corticospinal tract monitoring during surgery, though the ability of monitoring to prevent neurologic deficit is not well established. Disadvantages of dcMEP include a broader charge spread relative to DCS and relative limited clinical experience.

#### ***4.2. Awake Language Mapping***

For patients undergoing awake language mapping, the sedation is discontinued prior to dural opening. After the dura is opened and prior to mapping, tongue protrusion and counting are checked to ensure a baseline level of function and cooperation. The motor pathways are identified as described for asleep motor mapping, with the exception that stimulation is started at a lower current (2 mA p-p) and increased in intervals of 1 mA. Following motor mapping, the electrocorticography crown is brought into the field and the electrodes are distributed throughout the exposed cortical surface. The cortical surface is then stimulated at multiple sites with a bipolar electrode. If afterdischarge (AD) potentials are noted on the electrical tracings, the current is lowered by 1-2 mA until no AD are noted, and this “ideal” stimulation current is utilized for the subsequent language mapping paradigms. Paper numbers are placed throughout the cortical surface to serve as a reference. Language mapping begins with a picture naming task where the patient looks at a computer screen and identifies simple objects presented at 4 second intervals.

All sources of extraneous noise are reduced to a minimum, and the patient is equipped with a microphone to ensure that the surgical team can hear all responses. The ability of the patient to correctly perform the task is verified prior to intra-task stimulation to ensure that the sedation is adequately reversed and that there are no anesthesia-related changes from baseline language function. Once reliable picture naming is established, cortical stimulation is performed at each numbered cortical site just prior to a picture change. The surgeon clearly announces which numbered site was stimulated, and the response is recorded as correct or incorrect (anomia/dysnomia). Each stimulation-accompanied picture change is followed by one without stimulation to allow for recovery to baseline if an error is made and serves as an internal control. The alternating sequence of picture naming with or without stimulation is carried out until each number site has been stimulated three times. A site is considered positive if stimulation-induced errors were present for at least 2 out of 3 trials. After picture naming is complete, a similar stimulation paradigm is employed as the patient is asked to read a series of words presented on the computer screen. Sites with stimulation-induced alexia/dyslexia are recorded. Following these receptive language tasks, expressive language sites are mapped. The patient is asked to count from 1 to 10 to ensure baseline performance and then from 1 to 50 with stimulation just prior to every other number. For this task, speech arrest is defined as interruption of counting without oropharyngeal movement (to distinguish from inability to speak due to stimulation of nearby face motor cortex). Generally, true speech arrest can be elicited just anterior to face motor cortex, within pars opercularis or pars triangularis. It is important to monitor for afterdischarges during stimulation to ensure that the perceived stimulation-induced language deficits are due to direct bipolar stimulation of a site and not current spread or ongoing depolarization from prior stimulation sites. After localizing cortical representation of language, these sites are covered for protection, and tumor resection

is initiated, leaving a 1 cm margin around any positive site., This is based on data that if the resection margin is greater than 1cm from the nearest language site, the likelihood of a permanent language deficit is low [6].

**Table 1. Direct cerebral stimulation defined cortical and subcortical substrates for major human brain functions. *Adapted from De Benedictis, et al.(34)***

| Function                      | Cortical Regions                                                                                           | Subcortical Pathways                                                                                         |
|-------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Motor                         | Primary motor cortex, SMA, premotor cortex                                                                 | Pyramidal system (centrum semiovale, corona radiata, internal capsule, cerebral peduncles)                   |
| Somatosensory                 | Primary/secondary somatosensory area, insula                                                               | Thalamocortical pathways                                                                                     |
| Oral language                 |                                                                                                            |                                                                                                              |
| Dorsal phonologic stream      | Posterior/superior temporal cortex, inferior frontal gyrus                                                 | Direct SLF (arcuate fasciculus)                                                                              |
| Ventral phonologic stream     | Posterior temporal, orbitofrontal and dorsolateral prefrontal areas                                        | Inferior occipitofrontal fasciculus                                                                          |
| Articulation                  | Supramarginal gyrus, ventral premotor cortex                                                               | Indirect SLF III                                                                                             |
| Speech production             | Dominant anterior insula (articulatory planning), ventral premotor cortex, primary mouth sensorimotor area | Operculo-insular fibers, pyramidal tract, lentiform nucleus, descending pathway from ventral premotor cortex |
| Writing                       | Inferior/superior parietal lobules, insula, SMA, 2nd/3rd frontal convolutions                              | SLF                                                                                                          |
| Reading                       | Visual cortex, visual object form area                                                                     | Inferior longitudinal fasciculus                                                                             |
| Visuospatial cognition        |                                                                                                            |                                                                                                              |
| Visual                        | Temporal-parietal-occipital junction, visual cortex                                                        | Optic radiations                                                                                             |
| Spatial awareness             | Right supramarginal and superior temporal cortex                                                           | Right SLF                                                                                                    |
| Vestibular                    | Right inferior parietal cortex, posterior insula, superior temporal cortex                                 | Right SLF                                                                                                    |
| Cognitive/executive functions |                                                                                                            |                                                                                                              |
| Language switching            | Left inferior frontal gyrus, posterosuperior temporal area                                                 | SLF                                                                                                          |
| Working memory                | Inferior frontal gyrus, dorsal premotor cortex, supramarginal gyrus                                        | SLF                                                                                                          |
| Syntactic processing          | Left inferior frontal gyrus and superior temporal gyrus                                                    | SLF                                                                                                          |
| Judgement/decision making     | Dominant prefrontal cortex and posterior temporal cortex                                                   | Inferior occipitofrontal fasciculus                                                                          |
| Attention/selection           | SMA, cingulum, frontal eye fields                                                                          | Subcallosal fasciculus, caudate                                                                              |

For the subset of patients with preoperative breakthrough seizures despite adequate anticonvulsant trials, electrocorticography can be used to guide resection of seizure foci in addition to the planned tumor resection. Such electrocorticography-guided glioma resections are particularly effective in reducing long-term seizure profiles in the pediatric population [41]. For multilingual patients, given that some representation of each language is distinct, it is important to map each language separately, starting with the patient's primary language [19]. In contrast to cortical language mapping, the utility and methodology of testing subcortical language circuits is less settled. The two primary methods of monitoring integrity of subcortical language circuits are (a) intermittent direct stimulation during naming/counting/reading paradigms as is done for cortical mapping (analogous to subcortical motor mapping) and (b) intermittent functional testing by simply asking to the patient to perform language tasks during active resection. For the latter, the ability to assess proximity to important circuits and thereby avoid damage, versus simply documenting that the pathway has been damaged, has not been well validated and is a general issue for functional monitoring strategies as a whole.

#### ***4.3. Neural Plasticity: Implications for Surgical Management of Gliomas***

Given the increasing evidence for “redistribution” of neural function in patients harboring gliomas, particularly in the case of low-grade gliomas (LGG), taking advantage of this property is becoming an important aspect of tumor management. For example, patients with LGG in the primary motor cortex may be observed over some period of time to allow for unmasking of latent or parallel circuits so that when resection becomes mandatory, due to tumor growth or worsened neurologic function, the primary motor circuits are farther away from the tumor center.

More recently, a number of groups have endorsed a strategy pioneered by Duffau which takes advantage of surgery-induced plasticity for LGG [42]. For patients with a limited first resection due to positive mapping findings within the planned resection field, the patient is allowed to recover and functional mapping is continued postoperatively via fMRI. Over the course of months or a few years, the eloquent function redistributes, presumably triggered by the initial surgery. Thus at a second surgery the cortical area once devoted to a functional pathway which has been redistributed can now be safely resected after proper confirmation of this functional shift with DCS. Importantly, this functional plasticity allowing for a more complete resection can occur over a relatively short time period relative to the expected time scale of tumor transformation to a higher grade.

## **CONCLUSIONS**

Direct cortical stimulation during resection of gliomas allows for accurate localization of functional circuits at both the cortical and subcortical levels. These maps enable the surgeon to maximize the extent of resection while decreasing the risk of permanent neurological morbidity. Recent developments such as magnetoencephalography, diffusion tensor imaging, and transcranial magnetic stimulation are becoming increasingly important for improving the accuracy and efficiency of intraoperative mapping protocols.

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## *Chapter 7*

# **PREOPERATIVE BRAIN MAPPING IN GLIOMA SURGERY**

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## **ABSTRACT**

Magnetic Resonance Imaging (MRI) currently represents the most elegant imaging method in differentiation of primary intrinsic brain tumors such as gliomas, thus offering valuable information prior to surgical resection. The last two decades have brought additional MRI-techniques such as Blood-Oxygenation-Level-Dependent functional-MRI (BOLD fMRI) and Diffusion Tensor Imaging (DTI)-Fibertracking, which may be used for mapping of eloquent brain in the cortical and sub-cortical planes. Together with conventional anatomical imaging, functional imaging with BOLD fMRI and DTI, tractography and magneto-encephalography may be helpful during planning and resection of brain lesions, as the goal of surgery is to maximize resection without inflicting new neurological deficits.

## **INTRODUCTION**

The human cerebral cortex alone contains 40 billion neurons crowded into 3 square meters of surface area. Each neuron makes thousands of synaptic contacts through which information rapidly flows from one neuron to another. Despite, the anatomical organization of the cerebral sulci and gyri as fundamental landmarks on the brain surface, the well-known variability in cortical function [72, 5, 9, 23, 80] calls for the aid of cortical mapping techniques to precisely identify specific sites related to cortical function. The white matter of the brain consists of complex three-dimensional architecture, known as fascicles or fiber tracts within the hemispheres and the brainstem. As a consequence, a detailed knowledge of

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the architectural anatomy of the white-matter tracts is paramount when dealing with intrinsic brain tumors such as gliomas.

Gliomas cause brain damage by infiltration into normal brain tissue and by the creation of a mass effect. Surgical treatment of gliomas in or adjacent to eloquent brain centers is associated with a high risk of postoperative deficits. There is growing evidence that more extensive surgical resection correlates to prolonged survival for both low- and high-grade gliomas and the National Comprehensive Cancer Network has acknowledged this in its recent guidelines. [83] Therefore, accurate localization of eloquent cortices, as well as the white matter tracts connected to these areas, is an essential adjunct to successful surgical excision in these patients. Evaluation of brain function is also important in the determination of whether a particular case is suitable for surgery, as well as the optimal surgical strategy for cases when surgery is indicated. Surgical selection of patients harboring infiltrative gliomas based on radiological criteria may be insufficient due to individual variability of cortical organization, [36, 72, 73] distortion of the cortical topography as a result of the mass effect of the tumor, functional reorganization due to plasticity and the fact that function can be preserved within infiltrated brain tissue. [74, 89, 102]

Functional mapping of the brain has, however, been pursued for decades and with the advent of new imaging techniques new ways of functional mapping have arisen. We discuss different brain mapping techniques (mainly preoperative) used in Glioma surgery for modifying the extent of resection and minimizing postoperative deficits.

## **METHODS OF BRAIN MAPPING: ADVANCES AND LIMITATIONS**

### **1. fMRI / Blood Oxygenation Level Dependent (BOLD) Mapping**

The primary motor cortex controls voluntary movements and is strongly somatotopically organized along the precentral gyrus. Damage to this region during surgery has severe consequences for the patients and has thus been subject to intra-operative mapping since the 1930's. [78] Blood-Oxygenation-Level-Dependent fMRI, hereafter referred to as fMRI, is a specialized MRI-technique using a T2 \*-weighted gradient echo sequence. Ogawa et al. first reported the basis for this technique in 1990. They discovered that this special MRI-sequence was sensitive to changes in the concentration of de-oxygenated hemoglobin (deoxy-Hb) in blood. [71] fMRI follows from the neurovascular coupling between neuronal electrical activity and cerebrovascular physiology that leads to three effects that can contribute to the fMRI signal: an increase in the blood flow velocity, blood volume and blood oxygenation level. The latter effect, gave the technique the name blood oxygenation level dependent (BOLD) fMRI. [42] With the advent of fMRI various motor tasks have been used for preoperative mapping, including finger tapping, hand clenching, elbow and shoulder movement, as well as tongue, lip, foot and toe movement. [95] fMRI has good concordance with other brain mapping techniques such as intraoperative motor evoked potentials and intraoperative electrocortical mapping. [8, 43, 50, 100]

Since the advent of functional magnetic resonance imaging (fMRI)-a noninvasive tool able to visualize brain function, several clinical applications have emerged. One of the major clinical uses is presurgical fMRI in patients with brain abnormalities. The goals of presurgical



fMRI are threefold: 1) assessing the risk of neurological deficit that follows a surgical procedure, 2) selecting patients for invasive intraoperative mapping, and 3) guiding of the surgical procedure itself. [90]

### ***Limitations of BOLD Mapping***

The challenges ahead lie in establishing standardized tasks and analysis for processing of these investigations, as well as carrying out prospective outcome studies or clinical randomized trials in order to produce evidence for the effect of presurgical BOLD fMRI on morbidity and mortality in patients with brain lesions.

For presurgical mapping, however, the statistical activation maps are individually thresholded for each subject and each task, in order to produce activation maps unequivocally delineating the cortical area of interest. This is a pragmatic way of analyzing fMRI for presurgical mapping. Another issue is that fMRI investigations do not necessarily differentiate between essential and participating brain areas unless the task has been specifically designed to do so. [90, 92] Thus, single subject fMRI analysis for pre-surgical mapping is inherently a user dependent process, heavily dependent on the user's competence, comprehension of functional neuroanatomy and interpretation of the activations. As stated previously, there are no standardized tasks, scanning procedures, administration of tasks, pre-processing or interpretation of activations for clinical applications of fMRI, something that is desirable and needed in order to obtain objective and comparative information among centers engaged in this activity. [90] This is one of the main challenges for clinical fMRI in the future.

## **2. Diffusion Tensor Imaging (DTI) and Fibertracking (FT)**

DTI is an advanced form of diffusion-weighted imaging which analyzes the Brownian movements of water molecules in cerebral tissue. In the gray matter, the movements of molecules of water are free, whereas in the white matter, the movements of molecules of water are in accordance with the direction of fibers (the so-called anisotropic movement). In areas of cerebral ischemia, the severe cytotoxic edema reduces the movement of molecules of water in both the gray and white matter. [26, 40, 47, 57, 64-66, 103] In DTI, the gradients are applied in at least three more directions (combinations of the X, Y and Z direction), making it possible to create a mathematical model of diffusion in three-dimensional space, known as the diffusion tensor. [40] From this diffusion tensor it is possible to calculate the direction of maximum diffusivity, which has been shown to coincide with the fiber orientation in the white matter of the brain. [59] The more directions the gradients are applied in, the more accurate the estimate of the tensor becomes. At least six directions are needed to accurately describe the tensor. Thus, by using DTI it is possible to map the anatomical location of neural tracts in the brain, e.g. the corticospinal tracts, the optic radiations of the optic tract, and the arcuate fasciculus. [1, 40, 96]

Several different parameters can be derived from the DTI images, such as maps of fractional anisotropy (FA maps; Figure 1) which is a measure of the magnitude of anisotropic diffusion ranging from 0 (isotropic diffusion i.e. non-directional) to 1 (anisotropic diffusion i.e. strongly directional). [1] Another frequently derived property of DTI is the direction of

maximum diffusivity modulated with the FA-value and represented as color coded maps, thus representing both the magnitude and direction of diffusion. [17] The accepted convention for color-coding is with blue representing the superior/inferior direction, red the left/right direction, and green the anterior/posterior direction. A third way of presenting the DTI images is through fiber tracking or tractography, which is a visualization technique for neural tracts in three dimensions. [40, 48, 57] (Figure 2)

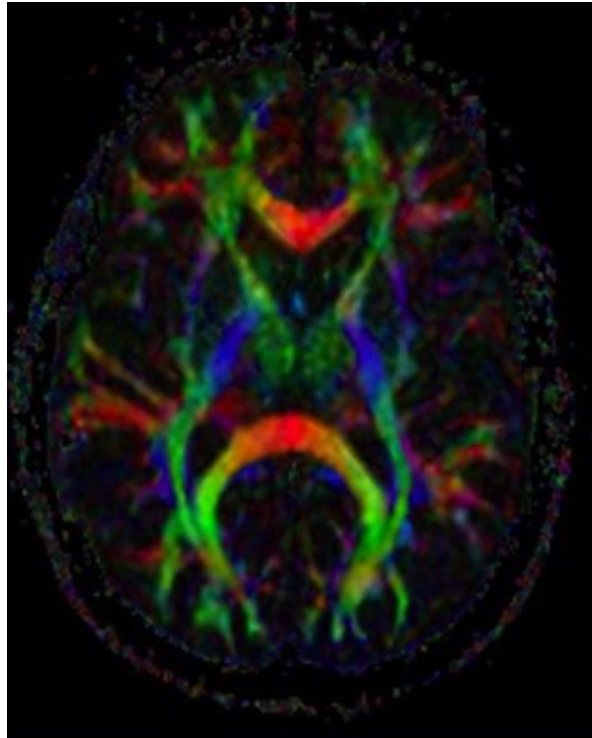


Figure 1. The fractional anisotropy (FA) map where bright colors correspond to high anisotropy displaying the preferred direction of the diffusion (red: LR; blue: CC; green: AP).

These algorithms automatically suggest tracts, which have to be virtually dissected using a region-of-interest tool to produce plausible tracts. Thus, if the corticospinal tracts are to be visualized one has to postulate that this tract runs through the posterior limb of the internal capsule and the precentral gyrus. A DTI study consists of several steps, but unlike fMRI, the subject being scanned is not required to perform a task and thus only needs to lie still, while the images are being acquired. After scanning, the images have to be processed using different algorithms before tractography is performed as a final step.

White matter regions where there are several fiber bundles with different orientations or where they “kiss”, cross, merge or diverge, are particularly troublesome for the tracking algorithms. In such regions the algorithm can either not track the fiber bundles present or track pathways that do not exist. [14, 57] Similar problems arise in the presence of tumor and edema, but some of these problems can be overcome using novel tracking algorithms or more advanced processing procedures. [2, 28] Nevertheless, the suggested tracts have to be manually and individually processed with a region-of-interest tool to virtually dissect

plausible tracts of interest, by choosing anatomical localization these tracts are known to run within. If one is to visualize for example the corticospinal tracts, some potential region-of-interests to use are the cerebral peduncles, posterior limb of the internal capsule, and the superior part of the precentral gyrus. This process heavily depends on the user performing the tractography where detailed knowledge of neuroanatomy is a prerequisite, thus making fibertracking (FT) a user dependent process.

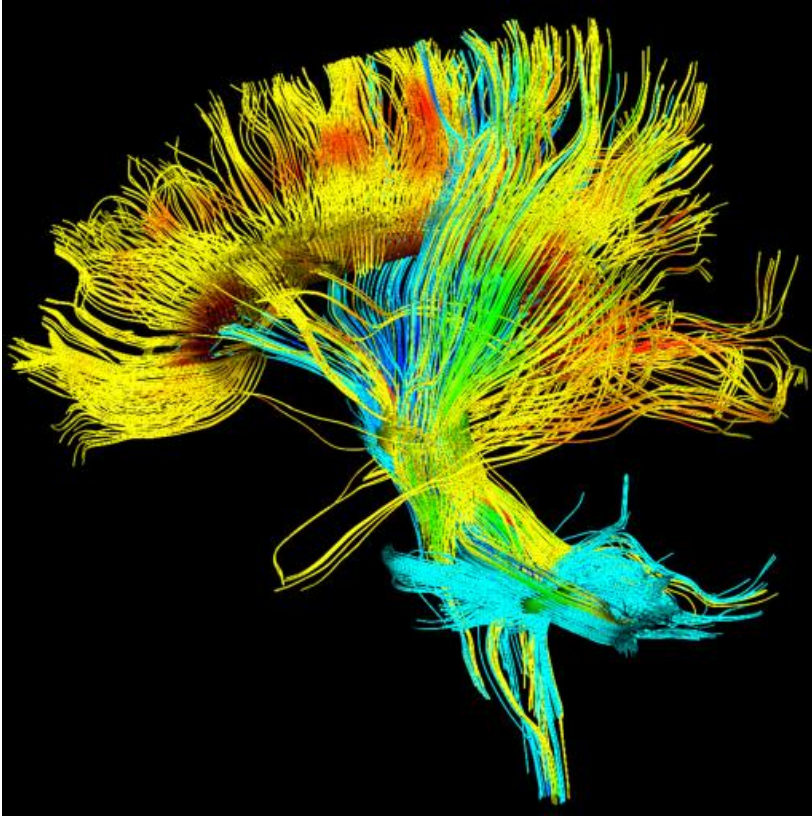


Figure 2. DTI-Fibertracking (tractography) analyzes white-matter anatomy and microstructural organization of tracts. The corpus callosum is used as an example. Virtual dissections allow reconstructing and studying the three-dimensional trajectories of the major divisions of the corpus callosum.

Fiber tracking algorithms compare local tensor field orientations measured by DTI from voxel to voxel and also allow delineation of white matter major tracts. With this technology, it is possible to obtain colored three-dimensional reconstruction of white matter fiber tracts with graphic resolution of relationships between tracts and gray matter. [65] (Figure 3) Therefore, DTI-based fiber tracking allows visualization, deformation, displacement, and infiltration of white matter fiber tracts caused by space-occupying lesions. The knowledge of positioning of white matter fiber bundles in relation to intracranial tumors may help to optimize surgical resection and to prevent new neurological deficits. [26, 40, 47, 57, 64-66, 101, 103]

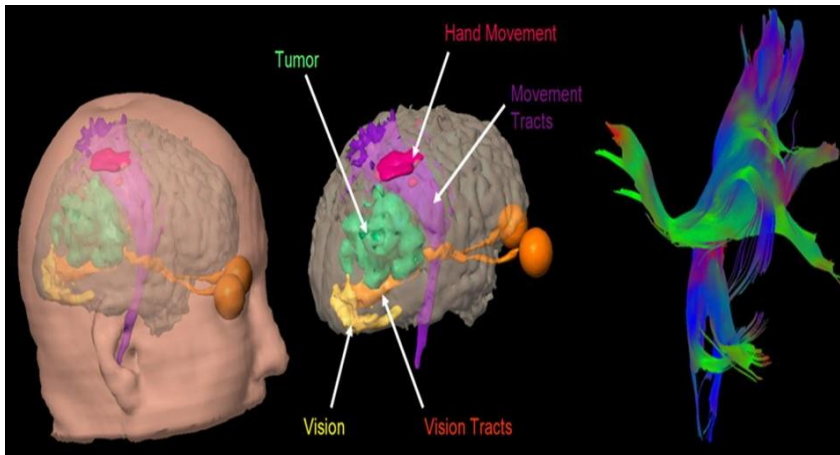


Figure 3. A) fMRI BOLD and DTI-Fibertracking integrated functional neuronavigation in image guided surgery outlining eloquent brain in the cortical and subcortical plane. B) 3D color-coded fibertracking isolating projection and association white matter fiber bundles.

The possible shifting of white matter fiber tracts has to be taken into account after major tumor parts are resected. Thus, the knowledge of tract position during surgery helps to prevent postoperative neurological deficits. DTI-MRI fiber tractography can provide unique quantitative and qualitative information to aid in visualizing and in studying fiber tract architecture in the brain. In particular, DTI-MRI with a fiber tracking algorithm is a method for preoperative and intraoperative localization of white matter fiber tracts, allowing delineation of white matter major tracts and showing their position after substantial tumor removal. [26, 40, 47, 57, 64-66, 103] The validation of DTI data with intraoperative electrical monitoring enhances the affordability for pyramidal tract localization. [65]

In conclusion, compared with the information provided by conventional MRI, DTI-MRI provides superior quantification and visualization of relationships between an intracranial tumor and white matter fiber tracts. In particular, preoperative and intraoperative DTI-MRI with fiber tracking (tractography) allows visualization of white matter fiber tracts, showing their position after removal of parts of an intracranial tumor, especially if guidance is needed for further resection. Three-dimensional visualization of white matter fibers such as corticospinal (pyramidal) tract, optic radiation, and arcuate fasciculus with relationship to intracranial tumors is helpful for preoperative and intraoperative evaluation of white matter fiber tract position, thus reducing surgical morbidity.

### ***Role of fMRI, DTI and Fibertracking in Preoperative Planning***

One of the first papers reporting the use of fMRI in a patient population is the one of Jack et al. in 1994, where they used both fMRI and invasive cortical mapping to map the motor cortex of two patients having epileptic seizures, due to neoplastic brain tumors located in close proximity to the primary motor cortex. [39] They found a correlation between the two techniques, and suggested that preoperative mapping was a potentially useful clinical application of fMRI. This was followed shortly thereafter by a dramatic increase in the use of fMRI in preoperative planning before neurosurgical procedures, focusing mostly on motor and language functions. [46, 58, 95] These reports have been followed by studies

demonstrating the use of DTI-fibertracking to visualize neural fibertracts in patients with neoplastic brain tumors. [15, 57, 99, 103]

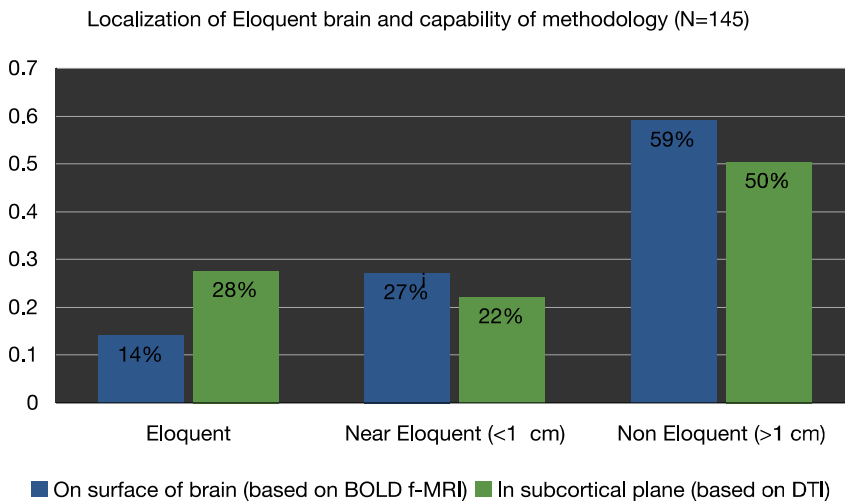


Figure 4. Combined role of both fMRI, and DTI-fibertracking techniques providing additional information on eloquent brain for the surgeon in the preoperative planning of the tumor resection that could not be extracted using either method alone. (Data presented to the AANS at the 78<sup>th</sup> Annual meeting, Philadelphia, PA, May 2010).

Since the brain consists of both white and grey matter, fMRI and DTI can together provide valuable information prior to surgery. Combined, both techniques provide additional information for the surgeon in the preoperative planning of the neurosurgical procedure that cannot be extracted using either method alone. [34, 90] (Figure 4) In 2004, Ulmer et al., demonstrated that twice as many functional systems were localized within 5 mm of the tumor borders when DTI and fMRI were combined compared to fMRI alone in 28 patients. [94] The functional systems taken into consideration were language, speech, vision, motor and pre-motor functions. Furthermore, only one out of the 24 operated patients (4%) encountered unplanned surgically induced deficit. Furthermore, the use of fibertracking seemed crucial and invaluable for visualization and understanding the complicated spatial relationship between the lesion, gray matter, and white matter fiber bundles.

### ***Role of fMRI, DTI and Fibertracking in Functional Neuronavigation***

During the last two decades several neuronavigation systems for performing frameless stereotactic neurosurgery have been developed, where anatomical data can be used intraoperatively to navigate in the brain during surgery. [29-31, 49, 76] (Figure 5) Thus the need for integration of functional information from fMRI and DTI into the navigation systems has emerged, in order to take advantage of this functional information during surgery. Some of the first papers on functional neuronavigation are the ones of Malidjan et al. and Shulder et al. in 1997, where they integrated fMRI into a neuronavigation system, allowing functional identification of eloquent cortex. [52, 88] This was followed by several others, either in larger number of patients or combined with other mapping techniques such as direct cortical stimulation. [63, 68-70, 81, 87]

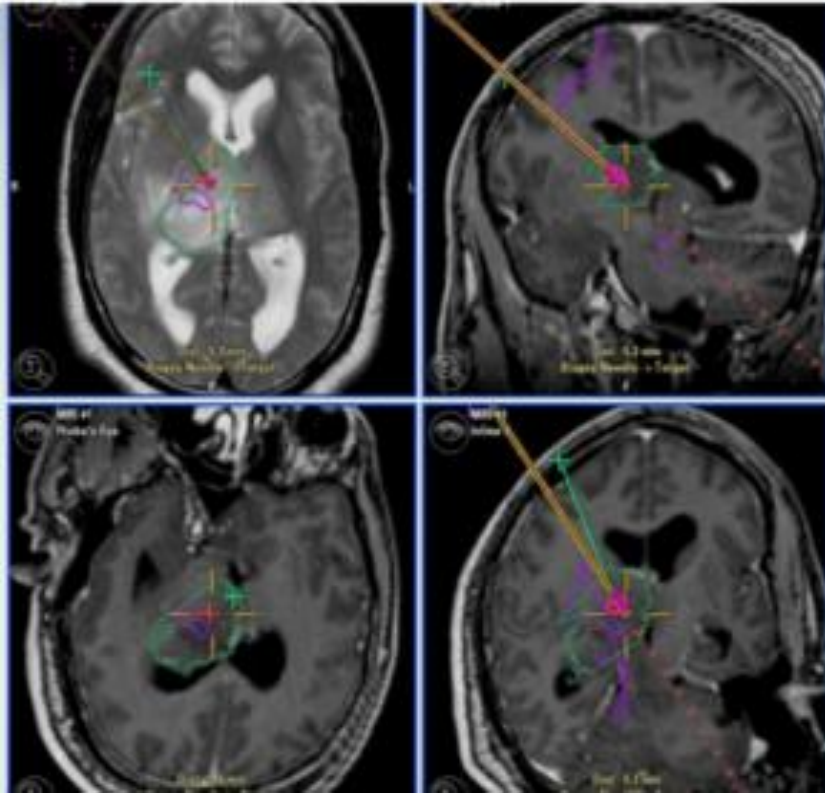


Figure 5. A frameless intraoperative stereotactic functional neuronavigation system display.

There have been a number of studies suggesting that the additional information provided by functional imaging (fMRI and/or DTI-Fibertracking) is highly valuable and enables safe resection [32, 38, 50, 53, 56, 62, 77, 103], especially when incorporated into a neuronavigation system. [45, 55, 63, 70, 82, 98] (Figure 6) In previous studies, either integrating fMRI data into neuronavigation systems [45, 104] or based on clinical demonstration of fMRI data and structural pathology [32] a minimal distance between 5 to 15 mm for feasible surgical resection has been suggested. A few papers have also focused on functional navigation using intraoperative 3D ultrasound to identify anatomical structures and dissection planes, correction of brain shift, as well as aid in the determination of the extent of resection. [16, 81] To our knowledge only one prospective randomized trial examining the clinical impact of DTI-data on degree of tumor resection, clinical outcome and survival has been published. Wu and colleagues demonstrated that high grade glioma patients benefit both in terms of increased tumor resection, improved postoperative outcome and time of survival when DTI-data were used to navigate during surgery, compared to those operated with conventional neuronavigation. [101] This is supported by other non-randomized uncontrolled studies. [82, 98]

Direct comparison of tractography of the motor pathways with intraoperative cortical electrostimulation shows that a number of factors can affect the reconstruction of the CST and generate pathways that have no direct correspondence with the exact location of the functional fibers. [56] Mikuni et al. emphasized that tractography and intraoperative cortical electrostimulation are complementary techniques [54] and can lead to better outcomes



compared to those obtained with cortical electrostimulation [13, 18, 44] or tractography alone. [4, 103] Furthermore, these authors suggest that tractography can be used to identify initial sites for cortical electrostimulation, allowing a more rapid localization of eloquent cortex during surgery. Tractography can be integrated into neuronavigation systems. [37, 66, 69, 75]

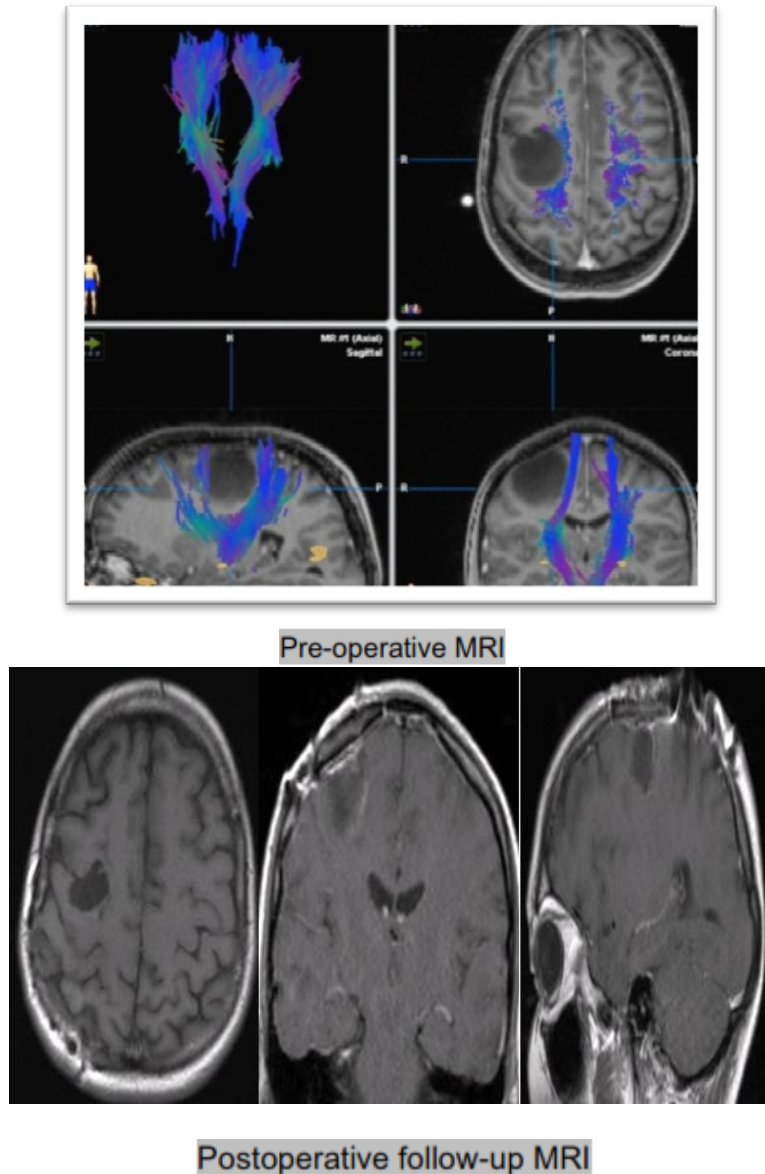


Figure 6. Case illustration: A 32-year-old right handed female presented with history of generalized tonic clonic seizures. Her preoperative MRI demonstrated right frontal non-enhancing tumor in the motor cortex. Fibertracking data was integrated and she underwent navigation-guided surgery. Fibertracking enhanced functional preservation and integrity of pyramidal pathways during surgery, her pathology showed a low-grade glioma. Postoperative MRI showed complete resection with no evidence of recurrence and patient had no neurological deficits.

Tractography has also been used to map language and visual pathways. The arcuate leaque connecting frontal, parietal and temporal regions is implicated in language function and the anatomy and organization of this structure has been studied using tractography. [12] Disruption by glioma of language pathways, notably the arcuate fascicle, as well as the inferior fronto-occipital fascicle (IFOF) (Figure 7) and uncinate fascicle, has recently been described. [3] Language DTI fiber tracking combined with intraoperative subcortical mapping aids in safe surgical removal of gliomas.

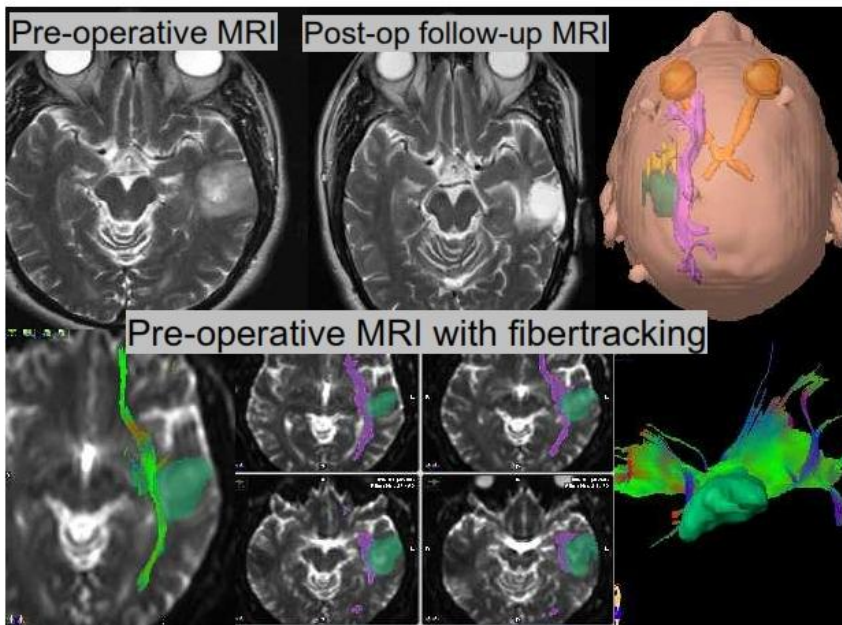


Figure 7. Case illustration: A 46-year-old right-handed male presents with history of generalized tonic clonic seizures and paraphasic errors. Preoperative MRI showed left temporal non-enhancing tumor infiltrating the inferior fronto-occipital fasciculus (IFOF) medially (association fiber bundle for speech). Fibertracking data was integrated and she underwent navigation guided surgery, pathology showed oligo-astrocytoma. Postoperatively his semantic speech was well preserved.

### ***Limitations of fMRI - DTI Based Functional Neuronavigation***

Even though the literature is full of illustrative case reports and studies demonstrating the usefulness of fMRI and DTI-fibertracking before and during neurosurgery, the two techniques have not yet reached the status of full clinical acceptance. There are two main reasons for this.

First, there is no consensus as to standardization of image acquisition, pre-processing, post-processing or interpretation of the statistical activation maps. [90] Second, to our knowledge there is only one prospective randomized outcome trial that definitively shows benefits to the final outcome, such as improved neurological status or survival of the patient, when applying fMRI and DTI preoperatively. [90] A big challenge for navigation based on preoperative images is the presence of brain shift during surgery. [60, 81] The dynamic changes of intracranial contents regularly occurring during the surgical procedures (the so-called brain shift) sometimes invalidates the preoperative surgical planning and, therefore, only intraoperatively acquired images allow us to perform what is the closest to real intraoperative image-guided neurosurgery presently available.



Nimsky et al. demonstrated that the intraoperative shifting of white-matter tracts, such as the internal capsule, can range from  $-8$  mm to  $+15$  mm, in patients undergoing glioma surgery. [66] The extent of brain shift needs to be compensated throughout the surgery. [65] Brain shift correction can be done mentally (i.e. in the surgeons mind), manually (i.e. landmark tracking) or preferably automatically, as the shift may be nonuniform and difficult to predict. Several methods have been suggested, such as intraoperative MRI [66] and 3D ultrasound. [16] It is feasible to update MRI data correcting for brain-shift using automatic co-registration of preoperative MRI with intra-operative ultrasound. [81] Another suggested solution is to use intraoperative MRI to acquire new anatomical and DTI images during surgery, when a marked brain shift necessitates updating the neuronavigation system. [65, 67] Intraoperative MRI does, however, require an MRI-compatible operating room and equipment, which is not needed for the ultrasound-based navigation system.

The navigation accuracy of the neuronavigation system is another potential source of error as there may be a registration error between the patient and the images. This has, however, been evaluated to be below 2 mm for the 3D ultrasound based neuronavigation system. [51]

In spite of this, fMRI and DTI-fibertracking have gained wide popularity, probably due to the promising prospects of further development and the usefulness of knowing where the functional cortices and their white matter connections are located. It is, however, important to remember that such imaging techniques are helpful adjuncts to the neurosurgeons toolbox and that good resection of pathological lesions eventually will depend on the operating surgeon, and his team. [41]

### 3. Magnetoencephalography (MEG)

MEG measures tiny magnetic fields outside the head that are generated by neural activity. Because it measures these fields directly, MEG offers excellent temporal resolution ( $<1$  ms). Furthermore, magnetic fields are unimpeded by biological tissue, so MEG recordings offer an undistorted signature of underlying neural activity. MEG scanners, however, are expensive and relatively rare, so MEG is less widespread than MR-based techniques. MEG studies are useful for localization of sensory, motor [61], and language regions. [24] They are also used to localize seizure foci, which is often helpful in the management of epileptogenic tumors such as oligodendrogliomas.

### 4. Transcranial Magnetic Stimulation (TMS)

A drawback with both PET and fMRI is that they do not differentiate between essential and participating brain areas when performing a task. This may, however, be accomplished with transcranial magnetic stimulation (TMS), a technique where a magnetic field is transmitted through the skull inducing an electric current within the cortex leading to activation or inhibition of the underlying cortical area, depending on the frequency used. By applying TMS to the primary motor cortex, movements will be produced, thus identifying the cortex activated as essential for production of movements. Similarly, TMS may be applied to language cortex and through inhibition produces temporary aphasia and identifies cortices

essential for language production. TMS may be especially useful in cases where it is not possible to perform fMRI but also in cases where the fMRI investigations are inconclusive. The usefulness of TMS for functional mapping has been demonstrated in brain tumor patients and preoperative TMS has been integrated into neuronavigation systems. TMS is frequently used in the management of peri-rolandic tumors, where the pyramidal tract is at highest risk of disruption from surgical resection.

## **5. Current Intraoperative Language and Motor Mapping Techniques in Glioma Surgery**

The use of direct electrical stimulation (DES) to perform intraoperative brain mapping in order to preserve eloquent areas during epilepsy surgery has been established as a reliable routine procedure by Penfield in 1937. [79] The technique has been further refined and adopted for the identification and preservation of language function and motor pathways. [6, 22, 84, 97]

### **5.1. Language Mapping**

Language mapping while the patient is awake is an intraoperative technique designed to minimize language deficits associated with brain tumor resection. Surgical treatment of gliomas in or adjacent to language areas is associated with a high risk of postoperative deficits. Even though LGGs and HGGs are distinct in their biological features, clinical behaviors, and outcomes, understanding the effect of surgery remains equally important for both. This is also true for lesions in areas of speech eloquence, where the proximity of critical language pathways, can present a significant challenge to standard operative strategies.

Language sites in the cortex of the brain vary among patients. Devices such as fMRI imaging, PET, and magnetoencephalography units may aid in the preoperative planning of the resection strategy, but these techniques remain too imprecise for complex functions such as language mapping: their sensitivity (PET, 75%; fMRI, 81%) and specificity (PET, 81%; fMRI, 53%) are suboptimal. [27, 35] Nevertheless, for the identification of functional language pathways and guidance of safe tumor removal, these diagnostic imaging tools are still only supplements, not substitutes, for direct intraoperative stimulation mapping.

### **Negative Language Mapping**

To preserve function during glioma resection in or near language areas, Sanai and colleagues [86] advocated negative mapping through intraoperative cortical stimulation (negative cortical stimulation) instead of traditional positive cortical stimulation. Positive cortical stimulation identifies cortical sites associated with and sites not associated with motor, somatosensory, or language function, whereas negative cortical stimulation typically identifies only sites not associated with such function. It is safe to use a minimal exposure of the tumor and resect based on a negative language mapping. SMA and opercular tumors allow radical resection with low morbidity whereas insular tumors remain a challenge even with the aid of this technique. [10, 85]

Negative language mapping, does not necessarily guarantee the absence of eloquent sites. Functional tissue can be found within the infiltrated brain and this must be considered in the

presurgical planning. Despite negative brain mapping, permanent postoperative neurological deficits have been reported. [91]

### **Impact of Language Mapping**

In 2008, Sanai et al. reported favorable functional language outcome after aggressive resection in a series of 250 patients with dominant-hemisphere gliomas. [86] In that series, using language mapping, only 1.6% (4 of 243 survivors) of all glioma patients developed a permanent postoperative language deficit. One explanation for this favorable postoperative language profile was strict adherence to the “one-centimeter rule,” first described by Haglund et al. [33], which demonstrated that, for temporal lobe tumors, a resection margin of 1 cm or more from a language site significantly reduces postoperative language deficits. Not surprisingly, it has been hypothesized that brain infiltration by gliomas leads to reshaping or local reorganization of functional networks as well as neosynaptogenesis. [20, 93] This would explain the frequent lack of clinical deficit despite glioma growth into eloquent brain areas, [19, 25, 89] as well as the transient nature of many postoperative deficits.

### **5.2. Cortical and Subcortical Motor Mapping**

Intraoperative motor mapping is a stimulation technique where a constant-current generator (Ojemann Cortical Stimulator; Radionics, Inc., Burlington, MA) is used to deliver a biphasic square-wave pulse to depolarize the cortical motor neurons or descending axonal pathways. Once the cortical motor region is stimulated and identified (Figure 8), the same current is used to localize descending motor pathways. [7] For patients with gliomas that are located within or adjacent to the rolandic cortex, and thus the descending motor tracts, stimulation mapping of cortical and sub-cortical motor pathways enables the surgeon to identify these descending motor pathways during tumor removal and to achieve an acceptable rate of permanent morbidity in these high-risk functional areas. [11, 21, 44]

When using stimulation mapping methods to identify subcortical pathways, the surgeon is able to achieve an acceptable risk of permanent motor deficits in this high-risk patient population, which consists of those with gliomas that are within or adjacent to motor tracts. In this setting, if both cortical and subcortical sites are found with stimulation mapping, one can expect a 7.6% rate of permanent motor deficits postoperatively. [44] In cases in which subcortical pathways could not be identified but in which the functionally intact status is confirmed by stimulation of the corresponding cortical sites, the incidence of permanent morbidity is less than 3%. [44] Results reported by Keles et al. indicate that these methods can be applied in patients whose tumors are located within or adjacent to functional motor pathways, and will result in an acceptable rate of postoperative morbidity that is mostly transient. [44] Specifically, any additional language deficit incurred as a result of the surgery will improve by 3 months or not at all. [84]

Throughout motor and language mapping, continuous electrocorticography should be used to monitor afterdischarge potentials, to eliminate the chance that speech or naming errors are caused by subclinical seizure activity.

## CONCLUSION

Gliomas located in eloquent territories can displace predicted fiber pathways in unpredictable conformations. The combination of advanced imaging paradigms in glioma surgery can best assure effective and safe resection with anatomic-functional mapping using tractography-integrated functional neuronavigation combined with direct cortical/subcortical electrical stimulation. Each of these modalities can offer valuable information in the pre-operative period, assisting in planning of the surgical approach, delineating the location of the tumor with respect to eloquent brain regions, and informing pre-operative discussions with patients regarding expected risks and post-operative course. Using this approach, and in conjunction with standardized neuroanesthesia and neuromonitoring, the postoperative motor and language resolution profiles following glioma resection may be predictable. As these modalities become increasingly refined and complex, a thorough understanding of their advantages and limitations will lead to better outcomes in the operative management of gliomas.

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## **SECTION II. CURRENT TREATMENT OPTIONS**



## **Chapter 8**

# **NEUROLOGICAL COMPLICATIONS OF GLIOMAS**

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## **ABSTRACT**

This chapter deals with those aspects of glioma management that are not directly related to tumor eradication. The diagnosis of a glioma is not always straightforward. Headaches, seizures and neurocognitive/behavioral changes are common patient complaints. Clinical expertise and an understanding of the underlying pathogenesis are needed. These problems continue to offer challenges at all subsequent stages of glioma management. Factors only indirectly related to the tumor may account for headaches. Hemorrhage into the tumor, hydrocephalus, postoperative complications, radiation damage, adverse drug reactions and emotional factors merit consideration. Seizure management may also become complex. Treatment controversies include seizure prophylaxis and the merits of various anticonvulsants. Neurocognitive/behavioral problems may stem from a variety of causes that need to be recognized in order to best manage patients. Treatment options are continually evolving. Newer aspects of gene regulation important in the pathogenesis of micro-hemorrhages, seizure related epileptogenesis, drug resistance and mechanisms of brain edema are altering our approach to glioma management. Finally, end of life issues and care-giver support are discussed.

## **I. INTRODUCTION**

Glial tumors can be difficult to diagnose in their early stages and, once diagnosed, pose challenging treatment and management decisions. This chapter will deal with those aspects of care that do not directly involve surgical and chemotherapeutic treatment. Prior to tumor

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diagnosis somatic complaints may be vague and poorly articulated. In retrospect most families can recall subtle alterations of behavior, clumsiness, vague headaches or lapses of attention weeks or months before diagnosis. Too often these complaints are dismissed by physicians unaware how subtle brain tumors can be. It is not unusual to send a patient with a ‘normal’ neurological exam and vague, possibly ‘psycho-somatic’ complaints for an imaging study only to find an unsuspected brain tumor. Other patients present with complaints that should immediately raise concerns about a brain mass. Complaints such as the new onset of headaches, seizures, personality changes or progressive neurological deficits are less likely to endure any delays in diagnosis.

This chapter focuses primarily on headaches, seizures and neurocognitive/behavioral complications of gliomas. Less attention is given to more obvious sensory, motor and language deficits.

Finally, looking after glioma patients and their families is a long and difficult journey. In the process, the physician has the privilege of becoming close to the patient and their families, helping them cope with a difficult roller coaster-like period in their lives. Most of the topics covered in this chapter directly impact patient care and hopefully play a part in improving their quality of life.

1. Headaches

Is there a characteristic glioma headache? Probably not. There are, however, warning signs that an experienced clinician comes to recognize as being suspicious of a brain tumor. This section will examine these statements and review some other interesting aspects of headache in glioma patients. Table 1 lists the common causes of headaches in glioma patients.

Table1. Causes of Headaches in Glioma Patients

|   |                                                                                                                                                                                         |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a | Tumor Expansion <ul style="list-style-type: none"><li>- Dural /vascular irritation</li><li>- Compartmental shifts</li><li>- Raised ICP</li></ul>                                        |
| b | Hemorrhage into the tumor <ul style="list-style-type: none"><li>- clinical presentation</li><li>- pathophysiology</li><li>- anticoagulation risks</li><li>- role of radiation</li></ul> |
| c | Hydrocephalus <ul style="list-style-type: none"><li>- communicating</li><li>- noncommunicating</li></ul>                                                                                |
| d | Post-operative headache <ul style="list-style-type: none"><li>- infection</li><li>- Low pressure headache</li></ul>                                                                     |
| e | Miscellaneous <ul style="list-style-type: none"><li>- radiation</li><li>- seizures</li><li>- medications</li><li>- vascular/tension</li></ul>                                           |



### ***a. Headache Related to Growth of the Glioma***

Headaches are common complaints associated with gliomas, but they are by no means universally present. At the time of diagnosis approximately 50% of patients will have complained of headaches [1, 2]. Only 20 % of patients with brain tumors have headache as the sole complaint at the time of presentation [1, 2]. For a busy practicing physician evaluating headache complaints on a daily basis, the challenge is to identify this small group. A brief anatomical review will help explain why there is no classic or characteristic headache of a brain tumor. Nevertheless, there are some clues that raise a warning flag.

Brain tissue itself is not pain sensitive and the majority of primary tumors do not initially produce headaches. The lack of brain sensitivity enabled pioneer neurosurgeons to interrogate lightly anesthetized patients and thereby determine pain sensitive structures. In the 1940s, Penfield and McNaughton at the Montreal Neurological Institute [3] and Ray and Wolff at Cornell [4] documented that pain sensitive structures were limited to the blood vessels at the base of the brain, the dura and the dural sinuses. These and other anatomical studies allow conclusions about intracranial nociceptive supply [5]. Supratentorial structures are supplied by the various meningeal branches of the trigeminal nerve. When these regions are stimulated, patients report ipsilateral steady aching pain in the corresponding somatic distribution of the trigeminal nerve. The pain will be described in the retro-orbital, frontal or temporal region. If the site of irritation is located away from midline structures the pain will more clearly localize to the affected side. The infratentorial dura and vessels receive their sensory innervations from the upper 3 cervical nerves, either directly or via their ascending contributions to the lower cranial nerves. Pain is felt in the back of the head and is usually bilateral.

How do brain tumors and specifically gliomas initiate pain responses from these nerves? As it grows, a tumor may directly or indirectly stimulate pain sensitive structures. A direct effect can be seen histologically as an intense mesenchymal reaction within the adjacent pia and dura [6]. Traction on these structures might cause a headache that is affected by head movement. The quality of the headaches might depend on how much inflammatory reaction occurs. It could be steady or pulsatile. The pulsatile nature of vascular headaches is often linked to mast cell degranulation [7] and it is likely that inflammatory reactive changes adjacent to the tumor could invoke similar effects. Brainstem gliomas usually feature multiple cranial nerve palsies, ataxia and corticospinal tract sign, and may produce headaches that can be confused with basilar migraines [8]. As the tumor and edema increase, traction, distortion and stretching of the dura and blood vessels will occur once again adding another type of headache pain. Unchecked, at some point, limitations in the size of the intracranial space cannot be compensated at the expense of the CSF and vascular compartments [9]. The result is a headache associated with raised intracranial pressure. Cyclical vasoconstriction is able for a time to temporarily decrease pressure and measurable plateau waves can be seen. These consist of elevated intracranial pressure (ICP) to 70-100 mmHg lasting several minutes followed by a rapid return to normal values of 5-10 mmHg [10]. They are associated with generalized headache, nausea and vomiting. Neck stiffness may suggest tonsillar herniation from a posterior fossa mass. Thus, a variety of headache patterns would not be unexpected.

An early morning headache that resolves with upright posture has been classically associated with brain tumors. The explanation for this temporal pattern has been that standing up increases venous return reducing intracranial pressure. This classic pattern, however, does appear to be very common [2]. In the greater context of headaches, early morning headaches

are more commonly due to sleep deprivation secondary to obstructive sleep apnea. Caffeine withdrawal, nocturnal seizures or migraines are other common causes.

Ultimately, there is no characteristic headache pattern of a brain tumor. Rather, it is the context of the headache presentation that tips off the experienced clinician. A new headache or change in the pattern of a chronic headache should raise suspicion. The stoic patient who minimizes his or her symptoms, or mentions headaches casually on leaving the office should cause concern. A subtle personality change, vague clumsiness or lapses of attention should cause further concern.

### ***b. Hemorrhage into the Glioma as a Cause of Headache***

Gliomas may sometimes present as an acute hemorrhagic event associated with headache. Headache due to hemorrhage can also occur at any stage in the illness. The gross pathology of high grade gliomas almost always features visible areas of necrosis with hemorrhages of different ages [6]. It is somewhat surprising, therefore, that clinically recognized hemorrhagic complications are not more common.

Hemorrhage into tumors can notoriously mimic other conditions. A small hemorrhage may be confused with a transient ischemic attack (TIA) or a seizure. A larger hemorrhage can present catastrophically with severe headache, nausea, vomiting, mental status changes and focal deficits. Hemorrhage may simulate a hemorrhagic infarct [11], hemorrhage into an arteriovenous malformation (AVM) [12], hypertensive hemorrhage [13, 14] or a subarachnoid hemorrhage (SAH) [15].

The pathogenesis of tumor related hemorrhage is of interest. Although hemorrhages are less common in low grade gliomas, Kato et al. [15] was able to accumulate reports of 15 pilocytic astrocytomas that bled. Rapidly growing high grade gliomas likely outstrip their vascular supply with resultant necrosis and hemorrhage, but this could not explain the occurrence in slow growing tumors. Other pathogenic theories also need to be considered [16]. These include abnormal angio-architecture with arteriovenous shunting, aneurysm formation, tumor invasion of adjacent blood vessels and tumor related coagulation defects.

A number of studies have looked at the vascular cytoarchitecture. Liwnicz et al. [17] examined 160 gliomas and found microscopic hemorrhages in 50% of glioblastomas and oligodendrogliomas. They examined the capillary structures of these tumors and found a correlation between primitive (net-like) retiform capillaries and hemorrhage. Uncontrolled vascular proliferation is a feature of malignant gliomas and recent treatments have focused on inhibiting the over-expression of vascular endothelial growth factor (VEGF). Cultured tumor cells expressing specific VEGF isoforms induced hemorrhages when injected into rat brains [18]. Inhibiting this action of VEGF with drugs such as bevacizumab might potentially reduce the risk of hemorrhage into a tumor. There is always concern about the risk of hemorrhage into a tumor when it becomes necessary to use anticoagulants. Could VEGF inhibitors alter the risk? The likelihood of deep venous thrombosis (DVT) in glioma patients is in the range of 30% [19, 20] so treatment decisions frequently arise. Even though there is a chance of bleeding into a tumor, several studies indicate that the risk benefit favors anticoagulation [19-21]. It will be of interest to see if VEGF inhibition makes anticoagulation safer.

Headaches can, of course, be part of the clinical presentation of an early or delayed radiation induced encephalopathy [22]. It is less well recognized that hemorrhage into gliomas may occur secondary to radiation therapy itself. Small vessel infarcts and micro

bleeds associated with radiation exposure have been recently reviewed [23]. MRI susceptibility weighted imaging (SWI) is able to detect small hemorrhages within areas of radiation induced infarction [24]. These smaller hemorrhages may not cause significant headaches; larger areas of bleeding will more likely cause headaches. In the absence of histological confirmation there will always be difficulty in distinguishing a hemorrhage into recurring tumor from an area of radiation induced necrosis. It is also not well recognized that radiation related hemorrhages may not occur until 5-10 years after the completion of treatment. [25-7]. One author has suggested that radiation induced vascular malformations may play a role in the late occurrence of hemorrhage [25].

### ***c. Hydrocephalus***

As a tumor enlarges, ventricular impingement may block cerebrospinal (CSF) flow resulting in obstructive hydrocephalus. Hydrocephalus may also result from leptomeningeal extension of the tumor itself or from shedding of cellular and proteinacious material limiting CSF reabsorption [28, 29]. If pressure builds up slowly, headache may not be a prominent feature but neurocognitive and personality changes predominate [30-31]. In children, a slow growing tectal glioma may for a while masquerade as congenital aqueductal stenosis. In adults the clinical presentation may resemble that of normal pressure hydrocephalus (NPH) with ataxia, urinary incontinence and impairment of higher intellectual function. Acute obstruction will feature severe generalized headache, nausea, vomiting and papilledema requiring prompt neurosurgical intervention.

### ***d. Post Operative Headache***

In the first few days following surgery more than half the patients report moderate to severe pain [32, 33]. Traction on dural structures may be part of the explanation but the extra cranial structures are also richly supplied with pain fibers. These also come from the trigeminal and upper cranial nerves [5]. In the course of the surgical procedures extracranial tissues are damaged, small nerve fibers severed and inflammatory processes activated. The pain in most patients will diminish steadily and cease to be a problem. Chronic headache may continue to be a problem for some [34, 35]. After 1 year up to 12% of patients still complain of headaches [32], and in 4% they cannot be adequately controlled medically.

When severe headaches occur or persist longer than expected, other possibilities need to be considered. Wound infection, parameningeal infection, sterile and infective meningitis are uncommon but of concern and warrant sampling of the CSF.

*Low pressure headaches* due to CSF leakage are uncommon but can be seen occasionally postoperatively. Headaches may only become apparent once the patient is mobilized. Patients complain of a generalized throbbing headache beginning 5-10 minutes after sitting up then gradually settling down with recumbancy. The headache is felt to result from irritation of the pain sensitive structures at the base of the brain due to a lack of buoyancy. This classical description is readily recognized and the leak surgically corrected. Not uncommonly, the presentation is less clear cut. After the headache has been present for several days positional changes may be less obvious [36]. The patient may be unable to function and become distraught. In these cases reviewing the history will reveal clues to the postural component. MRI scans will sometimes demonstrate a characteristic diffuse meningeal enhancement and brain “sagging” [37]. Thin section CT scans and CT cisternography may be of use [38].

### ***e. Miscellaneous***

Other causes of headache are also encountered. As mentioned above, headaches can be seen with early or delayed radiation induced encephalopathy [22]. Headaches are common after seizures. Post ictal headaches may not be suspected if the seizures occur at night. Many of the drugs used in glioma treatment may have headaches as a side effect. Finally, glioma patients are not immune to all the various headaches that plague the general population. These include migraines of various types, cluster headaches, muscle contraction headaches and headaches related to sleep deprivation. Obstructive sleep apnea is a common cause of headaches. Weight gain from steroids may predispose patients to this condition. There are a lot of stress factors and psychosocial issues that can lead to headaches as well.

## **2. Seizures**

The association between seizures and brain tumors was recognized over two hundred years ago and has been extensively written about [39-45]. The epidemiological data is extensive and varies with patient selection, tumor type, size and location. As a rough, easy to remember guideline it can be said that about 50% of patients with gliomas will have had a seizure by the time they come to surgery. Of those, half of them will have a seizure as the presenting complaint. Of the 50% who have not had a seizure by the time of surgery, half of those will have one at some subsequent time.

The diagnosis of a brain tumor may be missed for a while if the seizures are long standing and associated with a slow growing “benign” tumor. Seizures may be subtle and confused with psychiatric disturbances. Seizures from gliomas intuitively begin as localized events that become generalized. The history, however, may not readily reveal this sequence. Neither the type nor the duration of the seizures can exclude the diagnosis of a tumor. A change in the character of the seizures, drug resistance or a prolonged post ictal recovery are all more commonly seen with tumors [39] or with progression from a low grade to a high grade glioma.

As we explore the pathogenesis and treatment of seizures associated with tumors it is important to utilize the expertise gained by epileptologists [46]. Are drug side effects and drug resistance different in glioma patients? There are some healthy controversies surrounding seizure prophylaxis in the peri-operative period and in choosing anticonvulsants. New insights into gene expression have come to light.

### ***A. Pathogenesis***

For many years authors have emphasized the multifactorial nature of tumor related epileptogenesis [47]. Factors include location, tumor type, gene expression and the peritumoral milieu. Gene expression may play a role in drug resistance. Table 2 lists some of the topics involved in pathogenesis.

*Tumor location.* Tumors located primarily in the white matter are less likely to cause seizures early on than are those that initially invade the cortex. Infratentorial tumors are less likely to cause seizures than are supratentorial tumors. Among supratentorial tumors those involving the occipital regions are less likely to cause seizures than those involving the medial temporal regions and peri-Rolandic regions.

**Table 2. Pathogenesis of Glioma Related Seizures**

|                                                                    |
|--------------------------------------------------------------------|
| Regional differences in epileptogenicity                           |
| Tumor Type                                                         |
| Gene Expression<br>LGI1 alterations<br>Chromosome 1 and 19 changes |
| Peritumoral Environment<br>Ischemia and hemorrhage<br>Glutamate    |
| Drug Resistance                                                    |
| Peri-operative Seizures<br>Seizure prophylaxis                     |
| Additional Causes<br>Metabolic, Infectious and Toxic<br>Radiation  |

*Tumor Type.* Slower growing tumors, particularly those associated with primitive neuro-embryonal elements are universally recognized as being much more likely to feature seizures. In some of these subtypes such as dysembryoplastic neuroectodermal tumors (DNET) the seizure frequency may approach 100% and drug resistance is common. It has been suggested that the higher seizure frequency of slow growing tumors is in part related to a longer patient survival allowing more time for the adjacent tissue to become epileptogenic[43]. Another suggestion is that slow growing tumors may isolate regions of the brain, effectively causing an area of denervation hypersensitivity by de-afferentation [42,48]. For the most part, tumor tissue itself is electrically silent and the epileptic activity is felt to be generated in the peritumoral tissue [48]. This may not be true of all glioma subtypes. Primitive gliomas retain some neural elements that may potentially continue to generate electrical activity and release neurotransmitters and/or neuromodulators.

Different mechanisms likely come into play in more rapidly growing tumors. It would seem likely that ischemia induced hypoxia, acidosis and electrolyte disturbances would spill over into peritumoral tissues inducing seizures. Irritation from hemorrhagic blood products would add to epileptogenesis.

*Gene Expression.* The human Leucine rich Glioma Inactivated 1 (LGI1) gene on chromosome 10 was discovered in 1998 [49] and it has some interesting associations. These possibly link it to the pathogenesis of seizures. Disruption or loss of heterogeneity at this site, resulting in decreased expression of its polypeptide, is a common finding in malignant gliomas. It normally seems to function by keeping tumors in check. It is also felt to have an important role in suppressing tumor metastasis [50]. How is the gene related to epilepsy? Mutations at the same gene site are common in a rare form of familial epilepsy termed (ADLTE) Autosomal Dominant Lateral Temporal Epilepsy [51, 52]. The syndrome features focal seizures with a prominent auditory component. Mutations at this site have been found in up to 50% of patients [51]. Curiously, the changes differ in different brain regions. Furlan et al. [50] have found a preferential localization of specific isoforms of the gene polypeptide product (LGI1 Protein or Epitempin) to the medial temporal lobe. This suggests the possibility that regional differences in gene expression may play a role in regional differences

in seizure thresholds. The next logical question would be whether or not the incidence of gliomas in these families is unusually high or low. An initial sampling of families carrying mutations at the LGI1 site has not revealed an increased number of gliomas. It has been suggested that the situation is more complex and that changes of genetic expression at several sites might play a role [53]. The LGI1 story does not quite end yet.

There is another clinical scenario where this same gene pops up again. Antibodies to the LGI1 gene are one of the features of limbic encephalitis. Limbic encephalitis is characterized by the subacute onset of memory problems, personality changes, seizures and mesial temporal inflammation [54]. It occurs in both a non neoplastic setting and as a paraneoplastic presentation. It is thought to represent an autoimmune syndrome generally treated with immunosuppressive agents. Two recent case reports suggest a possible link to glioma. The first is a case of gliomatosis cerebri mimicking limbic encephalitis [55]. This patient had slight enhancement in the temporal lobes and improved on steroids. Three months later hemorrhage into the tumor bed led to the diagnosis. The second case [56] is that of a 41 year old woman with presumptive herpes encephalitis and characteristic changes in the temporal lobes. The patient improved and then two years later had radiographic features of a high grade glioma.

A closer look at the LGI1 protein also called Epitempin reveals it to contain leucine rich repeats and a unique tandem repeat (named EPTP) that defines a family of genes that may also be linked to seizure activity. [57]. Portions of the molecule bind actively to another superfamily of transmembrane proteins and in particular ADAM22 (A Disintegrin And Metalloproteinase) [58]. Among the many functions of this family of proteins is intimate involvement in cell to cell and cell to matrix adhesion, myogenesis, neurogenesis, transmembrane activity and regulation of synaptic transmission. This research is perhaps opening the window on a large number of genetic influences on epileptogenesis based on genetic discoveries.

The LGI1 gene is located on chromosome 10 and other members of this family of genes can be found in other chromosomal regions already associated with seizures. The closely related LGI4 gene maps to chromosome 19q. This location is adjacent to a voltage-gated sodium channel subunit [57], important in controlling membrane potentials. Oligodendrogliomas often feature deletions and loss of heterogeneity on chromosome 1p and 19q. A possible link of 19q alterations has been reported [59]. These studies taken together suggest that an interplay at a variety of genetic sites is involved in predisposing patients and likely specific tumor genetic fingerprints to an increased seizure occurrence.

*Peritumoral Environment.* Slower growing gliomas likely differ from high grade tumors in their effects on peritumoral regions. The more embryonic they are, the more likely they are to be associated with dysplastic brain tissue retaining neural elements. Gliomas grow by insinuating themselves into adjacent functioning brain tissues. Cellular elements in these regions retain the ability to generate epileptogenic activity. Higher grade gliomas feature ischemia, electrolyte changes and hemorrhage spilling over into adjacent normal tissues. Cellular injury and death, acidosis, membrane dysfunction, electrolyte imbalance and a breakdown of the blood brain barrier [43] within the tumor creates a population of hyper-excitable neurons in the peritumoral region.

*Glutamate.* Over the last few years glutamate and its receptors have been shown to have a role in tumor epileptogenesis [41, 44, 60-62]. Glutamate is a major excitatory neurotransmitter. Normally, glutamate and its receptor (*N*-methyl *D*-aspartate) NMDA are

localized to the synaptic region. The receptor functionally controls calcium entry into cells. Too much intracellular calcium can lead to toxicity and cell death. As gliomas grow they generate toxic amounts of glutamate. It has the effect of clearing the way for further tumor growth. The excess glutamate also plays a role in epileptogenesis. How do tumors generate excess glutamate? The transporter system for glial re-uptake of released glutamate (glutamate transporter – 1) is under expressed in glial tumor lines. Glioma cells actually release excess glutamate on their own. Normally this is carried out by exporting glutamate out of the cell in exchange for cystine. The process is carried out by a family of neurotransmitter transporters, the Excitatory Amino-Acid Transporters (EAATs) and specifically the X<sup>c</sup> cystine glutamate antiporter system. The system is strongly regulated up and over expressed in glioma cell lines. The exchange of glutamate for cystine also acts to protect the tumor cells. The increased intracellular cystine becomes available for the synthesis of glutathione (GTH). GTH helps protect the tumor cells from oxidative stress. Sulfasalazine is a drug that has the ability to inhibit this antiporter system and using sulfasalazine may have some potential as an adjuvant agent [62].

Other changes in the peritumoral region include complex alterations in the inhibitory neurotransmitter GABA (gamma amino butyric acid) and altered intercellular communication mediated by transmembrane junctional proteins. These and other mechanisms affecting the peri tumoral environment are reviewed by Shamji et al. [43]. It remains to be seen how these findings can be woven into an overall scheme of pathogenesis.

*Drug Resistance.* This discussion will include material important to management of both tumor related and non tumor related drug resistance. Most seizure patients regardless of the cause remain seizure free on a single agent and it does not seem to matter which drug is chosen. Another group requires a change of medication or dosage. A third group, perhaps 15-20%, is felt to have drug resistant or intractable epilepsy. Similar mechanisms underlie drug resistance in both tumor and non tumor related drug resistance. A number of recent in depth reviews [40, 44, 64-69] form the basis of the following discussion.

At the molecular level drug resistance reflects increased clearance of medications out of cells. Small lipid soluble molecules can get past the blood brain barrier into the brain. Once they enter brain tissue an energy dependent system designed to remove toxins is activated. This transport system consists of a family of proteins and their controlling genes. The controlling genes are termed ABC transporters. The name comes from ATP-Binding Cassette genes. Cassette is used in the sense of a group of genes of a similar function analogous to a music cassette containing the same genre of songs. The gene products make up part of an energy dependent complex pumping machine. The proteins expressed by the genes are numerous. The most common ones are the P (Permeability) glycoproteins P-gp, Multi resistant transporter proteins MRPs and multi drug resistant proteins MDRs. Over expression of these gene products accelerates drugs expulsion and limits their effectiveness. Increased expression has been demonstrated in both tumor and non tumor related drug resistance. Most anticonvulsants with the exception of valproate [70] and leviteracetam [71] are removed from the brain by this mechanism. Verapamil can inhibit P-glycoprotein and a few studies have shown encouraging results using verapamil in drug resistant patients [72-74]. Drug resistance remains an ongoing challenge to epileptologists.

*Peri-operative Seizures.* One of the best ways to limit seizure activity is clearly surgical removal of the tumor and adjacent epileptogenic material [41, 44, 46, 75]. Chemotherapy with temozolamide may also benefit seizure control [76]. Nevertheless 1/4 to 1/3 of patients

will continue to have some seizure activity following surgery [44]. This section will address a longstanding discussion as to the merits of prophylactic anticonvulsants in the perioperative period. Unfortunately, most studies do not meet rigid criteria making clear recommendations difficult. [77]. What would we need to know?

1. How often do seizures occur in a defined peri-operative period for specific tumor types, size and location? The range is wide and runs from 2% to 20% [39, 45].
2. How effective is each of the anticonvulsants and how much of an impact do side effects play in management? A number of articles suggest anticonvulsants lack effectiveness [78-82]. However, most of the studies were done with older enzyme inducing agents. Newer agents may fare better than the older ones. A more recent study suggests that seizure medications have a beneficial effect by impeding the transition from a focal seizure to secondary generalization [83]. Prior studies have not addressed this issue.
3. Do anticonvulsant interactions result in a worse outcome? A number of older anticonvulsants are metabolized by hepatic cytochrome P450 isoenzyme system and induce or inhibit enzymes shared with other anticonvulsants, chemotherapeutic agents and steroids [84].
4. Are side effects different or more common in brain tumor patients than regular patients? According to several reports, side effects, particularly adverse skin reactions and confusion appear to be more frequent [78-80].

In 2000 the American Academy of Neurology Subcommittee on Quality Standards released the results of their extensive literature review [80]. They concluded the literature did not establish that prophylactic anticonvulsant usage prevented perioperative seizures. Furthermore, the reported incidence of anticonvulsant side effects was high. They could not recommend short or long term prophylactic therapy. In 2009 another comprehensive review reached the same conclusions at least as applied to older anticonvulsants [82]. The authors of the subcommittee report stated that prophylactic use might be attractive if it could be shown that anticonvulsants were free of side effects. In the more than ten years since the recommendations came out, new drugs have emerged with more favorable side effect profiles. The recommendations, however, do not seem to have altered treatment practice.

Because seizures in the peri-operative period can be so devastating, many surgeons and neurologists continue to prescribe short term prophylaxis. Studies have shown that prophylactic treatment is actually the prevailing practice [85, 86]. Although it is important to continue to monitor the controversy, the more current question is which anticonvulsant is best. Some of the pros and cons of common anticonvulsants are described below. After surgery, decisions as to how long to continue prophylactic therapy are even less clear. They are generally made on a case by case basis. Factors include age, life expectancy, mobility, driving status, extent of resection and surrounding edema.

*Additional Causes of Seizures.* Electrolyte imbalance, alcohol, narcotic or sedative withdrawal, sleep deprivation and old ischemic foci can predispose glioma patients to seizures particularly if they have a lowered seizure threshold. Systemic and intracranial infections, commonly used antibiotics, antidepressants and anti-inflammatory agents may also lead to seizures. Chemotherapy related seizures are more commonly seen with agents used for other types of cancer than primary gliomas. A detailed review of medications, supplements,



medical history and habits is invaluable. Tramadol and venlafaxine are commonly used drugs that can lower the seizure threshold. Seizures may occur within hours of radiation treatment or as part of a delayed reaction to radiation therapy [22, 42].

### **B. Treatment Options**

The more commonly used anticonvulsants are briefly summarized below. It has been emphasized that “when treating patients harboring a brain tumor, it is mandatory to integrate the dogmas of epilepsy into a neuro-oncological viewpoint” [47]. Epileptologists and neurologists alter their prescribing practices over time as they gain more personal experience with newer agents. In order to sort out a new agent’s effectiveness and tolerability, patient feedback is the most critical information. Cost restrictions of newer non-generic medications also affect choices. Thus, it takes practitioners several years to assess the place of new drugs in the hierarchy of treatment.

*Levetiracetam:* Levetiracetam (Keppra) is emerging as the drug of choice for glioma patients for a number of reasons [46, 86-88]. Experience over the last 15 years has confirmed its effectiveness as a stand alone anticonvulsant. In a recent study, levetiracetam effectively decreased perioperative seizures [87-9] and seemed to reduce the progression of focal seizure activity to generalized seizures. A generic form is now available for both oral and intravenous usage. It is not metabolized by the cytochrome P450 (CYP450) enzyme system [89] or the Pgp/MRP transporter system [71]. Levetiracetam is not highly protein bound and does not have a lot of drug to drug interactions. The severe skin reactions associated with other anticonvulsants and radiotherapy do not occur with levetiracetam. [90]. It is for the most part well tolerated [91]. The most common side effect is occasional disorientation and agitation. A dose reduction will often dampen these difficulties. Finally, there is some recent information that it may even potentiate the antineoplastic action of temozolomide [92].

*Divalproex sodium:* Divalproex sodium (Depakote) and the intravenous formulation valproate sodium (Depacon) have been used for bipolar disease and vascular headaches as well as for seizure control. It augments the inhibitory action of GABA via an action on sodium and calcium flux. It is extremely effective in childhood epilepsies and has been a mainstay of adult seizure treatment for many years. Its metabolism is affected by the liver CYP450 enzyme system and thus drug levels need to be monitored particularly when combined with other anticonvulsants. Drug toxicity with somnolence and unsteadiness may occur with levels near or above the therapeutic range. Some patients, however, do not experience toxicity at these levels and actually require higher doses to achieve seizure freedom. Side effects such as weight gain and drowsiness are common, but rarely require discontinuing the drug. More severe adverse reactions include pancreatitis and elevated liver function tests. Sometimes, it is associated with high ammonia levels and encephalopathy. Occasionally, platelet dysfunction is also a problem [93, 94]. Most side effects resolve on lowering or discontinuing the drug. Some authors have suggested it as a first line drug [40, 41, 64], but this author reserves it for patients not responding well to levetiracetam. It requires vigilance in monitoring drug levels and routine chemistries. Some recent research suggests valproic acid may enhance the effect of temozolomide induced apoptosis in glioma cell lines [95]. Ongoing studies are assessing its clinical usefulness with some encouraging results [96].

*Phenytoin:* Phenytoin (Dilantin) remains a solid anticonvulsant choice both for treatment of Status Epilepticus as well as isolated generalized and focal seizures. It has been available for many years so its benefits and side effects are well known. It is now available in a water

soluble intravenous form (fosphenytoin) that is associated with less localized infusion site irritation. It stabilizes excitable membranes by modulating voltage gated sodium and calcium channels. It is also metabolized by the liver using the CYP450 enzyme pathway. It interferes with other anticonvulsants and may alter the effectiveness of chemotherapy protocols and steroids. It has a half-life of 22 hours so it takes several days for levels to plateau. Dilantin levels need to be adjusted with this in mind. Phenytoin is normally 90% protein bound and can displace or be displaced by other protein bound drugs. Low serum albumen levels produce low total phenytoin levels so measuring the free phenytoin level is necessary.

Although still widely used for epilepsy in general, it has lost favor amongst oncologists for a variety of reasons. Allergic skin rashes tend to be more severe in cancer patients. A severe localized exfoliative skin reaction has been observed in cancer patients who receive concomitant radiation therapy to the area involved [97, 98]. Other problems include lymphadenopathy, bone marrow suppression, folate deficiency, elevated liver function studies and simple toxicity manifested by ataxia, nystagmus and slurred speech. Regular vigilance in testing blood levels and blood parameters is needed. For these reasons, the author is increasingly using phenytoin as an add-on medication, particularly for patients experiencing breakthrough seizures on levetiracetam or valproic acid. Sometimes a patient will have been placed on phenytoin at an emergency room prior to assessment. In that case, if the drug is well tolerated and radiation is not planned, changing to another agent is probably not needed. Phenytoin continues to have a major role in managing status epilepticus.

*Carbamazepine (Tegretol) and Oxcarbazine (Trileptal).* These are highly effective oral anticonvulsants that are also used for a variety of neuritic pains. They are believed to act by blocking voltage sensitive sodium channels. Carbamazepine, and to a lesser extent oxcarbazine, are metabolized by the CYP450 system [99] and have abundant drug-drug interactions. The most troublesome side effects are hyponatremia, particularly in older patients, and mental confusion. Other less common side effects are elevated liver enzymes, lymphadenopathy and skin rashes [100].

*Lamotrigine (Lamictal)* is another effective oral agent. It is not a hepatic enzyme inducer [99]. Its major drawbacks are that the dosage has to be increased gradually, particularly when used with valproate, and that potentially serious skin reactions can occur. It is not yet available as a generic formulation. The author has used it more as a second line agent, but I suspect its use will increase, particularly as generic formulations become available.

*Topiramate (Topamax)* is widely used and, in lower doses, is effective for migraine prophylaxis. Its effectiveness as a stand alone anticonvulsant is less certain. Troublesome side effects are mental clouding in some patients and an increased incidence of kidney stones. It does have some impact on hepatic enzyme systems [99] and birth control pill effectiveness.

*Gabapentin (Neurontin).* Gabapentin is widely used for painful neuropathies and other neuropathic pains. Although structurally related to GABA, its mechanism of action is unrelated to these pathways. It is not metabolized in the body but may accumulate with renal insufficiency. It does not interact significantly with other medications or alter hepatic CYP450 activity. The major side effect is somnolence. One drawback is that it does not come in an intravenous form. There is also concern that it is not as effective as other stand alone anticonvulsants.

**Other Agents:** A variety of other agents are commonly used to control seizures. These include: pregabalin (lyrica), zonisamide (zonegran), and ethosuccimide (zarontin), phenobarbital, primidone (mysoline), tiagabine (gabitril), clonazepam (klonepin) and

lacosamide (vimpat). Lacosamide is also available in an intravenous formulation. At this time, some of the newer agents are not first line choices for glioma related seizure control, but their role may certainly change with time.

### ***C. Status Epilepticus***

Status epilepticus is variably defined. Any patient who has a seizure lasting more than a few minutes that has not responded to first line treatment with benzodiazepines or does not readily awaken may be considered as meriting treatment for status. The topic of status epilepsy is immense and cannot be done justice in a short summary. Treatment algorithms are constantly evolving and regularly being updated [101-103]. The reader is directed to these recent reviews. A list of the commonly used agents and their doses is provided in table 3.

The second line of treatment is the intravenous administration of one or more of the following agents: Phenytoin, Fosphenytoin, Valproic Acid, Levetiracetam, Phenobarbital and possibly Lacosamide. Refractory seizures mandate use of third line medications, with intubation, inducing general anesthesia with midazolam, propofol or pentobarbital. Bedside EEG monitoring is extremely useful. Physicians also need to be aware that propofol can, in high doses and with prolonged usage, induce metabolic acidosis, rhabdomyolysis and cardiac failure [104].

**Table 3. Drugs Used for Status Epilepticus**

| DRUG          | INITIAL DOSE    | INFUSION RATE                        | CAUTIONS                             |
|---------------|-----------------|--------------------------------------|--------------------------------------|
| Lorazepam     | 4mg IV          | Repeat 1-2 times<br>2 mg/min         | ↓BP, ↓Respiration                    |
| Diazepam      | 5-10 mg IV      | Repeat 1-2 times<br>5 mg /min        | ↓BP, ↓Respiration                    |
| Phenytoin     | 15-20 mg/Kg     | 50 mg/ min                           | ↓BP, arrhythmia                      |
| Fosphenytoin  | 15-20 mg PE/ Kg | 150 <b>mg</b> / min                  | ↓BB, arrhythmia<br>pruritis          |
| Levetiracetam | Up to 1500 mg   | Over 5-15 min<br>1500 mg twice daily | Sedation                             |
| Valproic Acid | 25-45 mg/Kg     | Up to 6 mg/Kg/min                    | Encephalopathy, ↑NH4                 |
| Propofol      | 2 mg/kg         | 25-75µgm/kg/min<br>Titrate to EEG    | ↓BP<br>Propofol Infusion<br>Syndrome |
| Midazolam     | 0.2 mg/kg       | 0.2-5 mg/kg/h<br>Titrate to EEG      | ↓BP                                  |
| Phenobarbitol | 10-20 mg/kg     | 100 mg/min                           | ↓BP                                  |

## **3. Neurocognitive and Behavioral Issues**

This section will include a variety of topics including mental status changes, memory loss, poor judgment, aggressiveness and agitation. At some point these issues impact every patient with a glioma. These changes can present even before the tumor is identified.

**Table 4. Neurocognitive and Behavioral Issues**

|    |                                                                                                                                                               |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a. | Presenting Challenges                                                                                                                                         |
| b. | Ongoing Issues<br>Seizures<br>Emotional stress<br>Surgery<br>Anticonvulsants<br>Radiation<br>Chemotherapy<br>Steroids<br>Psychiatric<br>Cognitive<br>Myopathy |
| c. | End of Life Issues                                                                                                                                            |

After diagnosis, neurocognitive and behavioral issues may persist as a manifestation of the tumor. They may also develop as a side effect of treatment or emerge due to the psychosocial stress of serious illness. I will discuss end of life issues separately.

**a. Presenting Challenges**

If tumors grow without involving critical motor, sensory or language structures, they can be associated with mental changes that are easily mistaken for primary psychological issues [105-108]. Depression, anorexia, anxiety, panic attacks, schizophrenia, and hallucinations may be the focus of attention. Subjectively, some patients may sense that something unexplainable is wrong with their thinking and families in retrospect will perceive personality alterations well before the tumor is discovered [108]. To some extent the behavioral changes depend on tumor location but some generalizations can be made.

*Posterior Fossa.* Patients with posterior fossa lesions usually present with gait deterioration, ataxia, vertigo, swallowing difficulties, oculomotor disorders, and sometimes crossed sensory findings. Neuro-cognitive features are generally lacking but some complaints such as vague dizziness may be misdiagnosed initially as functional. Later on, after the tumor progresses, hydrocephalus and raised intracranial pressure can affect mentation causing intellectual impairment, apathy or mutism.

*Frontal Lobes:* There is a curious picture that can be recognized with trauma, tumors and strokes affecting the frontal lobes. It is curious because its most notable characteristic is to make the examiner feel uncomfortable. Patients have a stern angry glare, humorless demeanor and terse verbal responses that are disquieting. Awareness of this sensation should prompt the examiner to suspect a frontal lobe dysfunction. Stubbornness, rigid concrete thinking and aggressiveness may be noted as well by the patient’s family.

*Parietal locations:* These areas may also be relatively silent, particularly if there are no motor or sensory deficits. Non-dominant hemisphere parietal lesions can cause spatial difficulties. Families misinterpret this as confusion, dementia or depression. Lesions of the dominant parietal lobes can cause a variety of language disorders that in their early stages,

may be hard to recognize. Again families may bring the patient to the physician assuming they are suffering from dementia or emotional problems.

*Occipital locations:* The patient may have a visual field loss and not be aware of it until they have an auto accident or report bumping into door openings. This cortical blindness is termed Anton's syndrome. Ocular hallucinations may occur and be suspected as a functional disorder.

### ***b. Ongoing Difficulties***

In the course of glioma patient treatment, neuropsychological problems almost always become issues that impact the patient's quality of life. This is particularly true with low-grade gliomas, where up to 90% of patients have neurocognitive issues [109]. These issues may reflect the continued growth of the tumor, but other causes need to be considered.

*Seizures.* Psychomotor seizures may feature stereotyped episodes including forced thinking, hallucinations, mood changes and automatisms that may superficially resemble emotional disturbances [105]. After a flurry of seizures it may take several days for the patient to return to baseline. The patient who manifests altered responsiveness for several hours may have nonconvulsive focal or generalized status. Subtle jerking of the eyes or eyelids may be the only observation to suggest seizure activity. Rapid awakening after IV ativan may at times be dramatic. An EEG helps confirm the diagnosis.

*Psychological Stress.* Patients faced with loss of motor function and mobility, lifestyle changes, uncertainty about the future and changes in family dynamics can be expected to react with depression and anxiety [110]. How the patient reacts will depend in part on his or her preexisting personality and ability for dealing with adversity. Antidepressant medications may assist and formal counseling or psychiatric treatment may be beneficial.

*Surgery.* In the early post operative period the benefit from reducing intracranial pressure or removing a seizure focus is balanced by the effects of surgery on surrounding normal brain tissue. A recent study addressed the question of neurocognitive changes after surgery and related it to tumor location [111]. Comparing pre-operative test scores to results obtained about one and a half months later revealed declines in neurocognitive measures that were most evident with tumors located in the insular region.

*Anticonvulsants.* As tumors progress, a decline in neurocognitive performance is expected. Patients on anticonvulsants seem to do worse [112]. The problem in ascribing neurocognitive difficulties solely to medications, however, can only be resolved by demonstrated improvement upon discontinuation of therapy. In the absence of a controlled prospective study it is difficult to exclude other factors.

*Radiation.* Neurocognitive studies showing an adverse effect of radiation on cognition suffer from a number of limitations [113]. These include the use of older techniques and higher doses [109, 114]. It is generally felt that any small risk of decline in neurocognitive function is overshadowed by deficits caused by unrestricted tumor growth[114].

*Chemotherapy.* A limited number of agents are used in glioma treatment. Henriksson et al. [115] have extensively reviewed the literature and concluded that Bevacizumab, if anything, actually improves cognitive function and that temozolomide does not adversely affect quality of life determinants either. Other agents such as procarbazine, lomustine and vincristine did have a negative impact on life quality.

*Steroids.* Dexamethasone began to be used in the early 1960's. Although effective in reducing peritumoral edema, high dose treatment is also associated with a variety of adverse

side effects such as: adrenal insufficiency, immunosuppression, peptic ulcer, osteoporosis, hyperglycemia and Cushing's syndrome [116, 117].

The mechanisms underlying the anti-vasogenic edema action of dexamethasone are starting to become understood [118-120]. It has been shown that they have direct effects on the endothelial/pericyte/glial unit that makes up the Blood Brain Barrier (BBB). An intact BBB depends on tight cell to cell junctions, a lack of fenestration and minimal pinocytotic action in endothelial cells. Glucocorticoids diffuse into the cells making up the BBB and activate a cytoplasmic glucocorticoid receptor complex. The receptor is normally held in check by a variety of proteins including heat shock proteins. These split off and the activated receptor becomes free to alter transcription at a number of gene sites. One of the repressed genes produces Vascular Endothelial Growth Factor (VEGF). VEGF is overexpressed in glioma cells and not only stimulates new vessel growth but also increases vascular permeability by disrupting the proteins that make up tight junctions. Another altered gene produces angiopoietin-1 [118] secreted by pericytes and astrocytes. Glucocorticoids upregulate its production. Angiopoietin-1 is the ligand or trigger for the endothelial tyrosine kinase receptor Tie-2 which when activated by angiopoietin-1 has antiedema activity of its own. This is part of the interplay between the endothelial cells and its supporting cells that comprises the BBB. The glucocorticoid receptor also results in up regulation of ATP sensitive potassium channels, increased production of tight junction proteins and down regulation of inflammatory cytokines. All of these actions work towards a reduction of vasogenic edema. The question of why glucocorticoids do not reverse cytotoxic edema as well as vasogenic edema may be linked to the glucocorticoid receptor system. Recently, increased destruction of the receptors by proteasomes in an animal model of cytotoxic edema has been shown to occur [121].

Clinically, the anti-edema effect can be appreciated within 6 to 12 hours, achieving a maximum improvement within two to three days. Currently the drug of choice is dexamethasone mainly because it has less mineralocorticoid effects than other glucocorticoids. Unfortunately, even after more than 40 years, no ideal validated dosage is available. A randomized study has suggested that doses of 4mg per day are as effective in reducing edema as higher doses [122]. Nevertheless, the usual approach seems to be that if a little is good, more must be better. A common dosage schedule for severe edema is often an initial dose of 10-12 mg given intravenously, followed by 4mg every six hours. There are also no recognized guidelines for doses during radiation therapy or how best to taper the dosage. Adverse side effects of steroids are dose related so having valid dosing guidelines is highly desirable. A few neuropsychiatric side effects deserve mention.

Approximately 5-10% of patients on steroids demonstrate personality changes [123]. The most common problems are insomnia, food seeking activity, irritability and mood swings. An increased sense of well being may manifest as euphoria, grandiosity and emotional lability. More severe manifestations include an acute 'steroid psychosis' that tends to occur early on in the course of treatment [123]. Symptoms gradually resolve in a few days. Subtle cognitive changes are even more commonly observed. Steroids given to normal volunteers have been shown to adversely affect cognitive function [124, 125]. Presumably similar effects occur in glioma patients.

Finally, a few other side effects on the neuromuscular system are of interest to neurologists. *Steroid Myopathy* characterized by proximal muscle weakness developing after weeks of treatment may not be appreciated at first. Patients usually report trouble getting up

from a chair or a squat. As symptoms progress shoulder girdle weakness can be demonstrated. The weakness probably reflects increased protein catabolism in muscle mediated by the ubiquitin-proteasome degradative pathway [126]. Steroids are also linked to *acute rhabdomyolysis* and *prolonged paralysis associated with neuromuscular blocking agents* [127-129].

Another condition of interest is *critical illness neuropathy/Myopathy*. There is some suggestion that steroids may play a role in this condition that is closely linked to septic encephalopathy and the systemic sepsis syndrome [130-131]. As critically ill patients start to awaken they have profound generalized weakness and respiratory difficulties. Some have findings suggestive of muscle dysfunction, others feature an axonal degenerating neuropathy and still others have features of both.

### **c. Declining Health**

At a certain point, active medical and surgical treatment become less intense. Patients and their care givers become faced with progressive illness and end of life issues. The most common problems are fatigue, drowsiness, uncertainty about the future, motor difficulties, communication difficulties and headaches [130]. As patients near the end of life, loss of consciousness, difficulty swallowing, incontinence, seizures and headache become increasingly problematic [131]. Rehabilitative efforts can achieve favorable results and cognitive therapy can benefit. [132]. Medications may play a role in controlling depression or agitation. Studies are underway to see if stimulants will benefit short term memory, energy level and quality of life. Finally, families and care givers can benefit from the support provided by specialist nurse support and social workers.

## **CONCLUSION**

This chapter has focused on neurological problems apart from treatment decisions about chemotherapy, surgery and radiation therapy. Rather, headaches, seizures and neurocognitive problems have been addressed in some detail. Additional attention has been given to the pathogenesis of each of these difficulties and some difficult clinical controversies have been discussed.

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*Chapter 9*

## **RADIATION THERAPY IN THE MANAGEMENT OF GLIOMA**

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### **ABSTRACT**

Radiation therapy is a standard treatment in patients with newly diagnosed high grade glioma. Involved field external beam radiation therapy to 60 Gy administered in 30 fractions is the current standard treatment regimen. Dose escalation, using various techniques such as 3-D conformal planning, stereotactic radiosurgery, or brachytherapy, has not been shown to improve the outcomes or survival of these patients. Concomitant temozolomide and involved field radiation therapy is the current standard for high grade glioma, and investigation in combined modality using concurrent standard radiation therapy and various chemotherapeutic agents is ongoing to improve survival. Stereotactic radiation therapy using standard fractionation or hypofractionation is well tolerated and may be considered in selected patients or in the recurrent setting. For patients with low grade glioma, external beam radiation therapy may improve progression-free survival but has no significant impact on the overall survival. As the result, radiation therapy is considered for selected patients with high risk features in the adjuvant setting and is often reserved for recurrent low grade glioma.

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## I. INTRODUCTION

Radiation therapy has been a mainstay of treatment for glioma for almost 40 years. After surgical debulking, it was the only therapy that provided any survival advantage for high grade gliomas until as recently as 2005 with the advent of temozolomide. In low grade glioma, while the timing of radiation therapy is still a matter of much debate, it nearly always becomes a part of each patient's therapy, either at recurrence after surgery or at the inevitable progression after chemotherapy. This chapter summarizes the use of radiation therapy in the treatment of high and low grade gliomas in the up-front and recurrent setting and reviews the relevant studies which established these recommendations.

The highly technical field of radiation therapy uses a fundamentally simple method to treat tumors. Modern external beam radiation therapy (EBRT) uses high energy x-rays generated by a linear accelerator (LINAC) and shaped by automated collimators in order to maximize dose to a 3-dimensional target delineated by interrogation of imaging, knowledge of anatomy and information gleaned from the surgeon and pathologist. Several modalities are employed by radiation oncologists to deliver radiation therapy to malignancies. EBRT refers to the use of beams of radiation, typically X-rays, to deliver radiation from a linear accelerator to a targeted area. 3-D conformal radiation therapy allows delivery of radiation dose precisely to the tumor using volumetric delineation of tumor and surrounding normal tissues to decrease amount of normal tissue irradiated. Intensity-modulated radiation therapy (IMRT) is an advanced method of delivering EBRT by modulating the intensity of radiation to achieve more precise field shaping and dose delivery to tumor targets while sparing adjacent radiosensitive tissues. These forms of radiotherapy are typically delivered over many fractions in small doses to make best use of the therapeutic window provided by cancer's dysregulated DNA repair mechanisms. On the other hand, stereotactic radiosurgery (SRS) or stereotactic radiotherapy (SRT) delivers highly precise high-dose radiation to the tumor by using many small beams of radiation converging on a single, often small, tumor target in a single or a few treatment sessions. Brachytherapy is fundamentally different and involves physically implanting one or many radioactive sources into the patient near or within the tumor and allows the source to decay for a specified period of time to deliver its dose.

The mechanism responsible for the majority of DNA damage from radiation therapy involves the formation of free radicals. The energy produced from the external source is transferred to water molecules in the cells located in the area of energy deposition causing breaks in the hydrogen bonds and creating free radicals.

These unstable species cause chain reactions which lead to double and single stranded breaks in the backbone of cellular DNA - called indirect damage. This structural damage ultimately leads to cellular death through a variety of mechanisms beyond the scope of this chapter. Another mechanism involves radiation causing direct damage to the DNA backbone, but this represents the minority of the damage inflicted by radiation.

The ability of inflicting damage to malignant cells while minimizing collateral damage is based on the ability to exploit four canonical mechanisms, colloquially referred to as 'the four R's' of radiobiology: repair, redistribution, reoxygenation and repopulation. Tumor cells often have a decreased capacity to repair themselves due to their mutations. It is this impaired cellular upkeep that partially accounts for radiation damaging cancerous cells while relatively

sparing noncancerous cells, which, having adequate self-repair capabilities, are able to survive. This difference in cellular repair helps keep adverse effects to a relative minimum.

Furthermore, it is this ‘therapeutic window’ between normal tissue and cancer repair that forms the basis of our modern fractionation schemes. Different tumors cells exist in all phases of the cell cycle at any given time. Particular times in the cell cycle, such as the M phase, are more susceptible to radiation than others. By dividing radiation therapy over a period of time, a technique called fractionation, allows malignant cells progress through the cell cycle.

This redistribution of phases ensures that cells receive radiation during all phases of the cell cycle, including those phases with increased susceptibility. Dividing treatment over time, in addition to aiding in redistribution, also allows treatment to the tumor during periods of sufficient oxygenation such that lethal damage can occur. Repopulation is the result of tissue repair; understanding of timing of repopulation can serve as a guide to designing rational fractionation schemes for normal tissue repair while minimizing potential tumor repopulation.

## II. RADIOTHERAPY FOR HIGH GRADE GLIOMAS

### Dose

The treatment of high grade gliomas (HGGs), WHO grade III-IV tumors, has always been particular challenging. In the late 1970s, Walker et al. performed a landmark randomized controlled study which provided conclusive evidence that adjuvant radiation therapy is superior to surgery alone. Whole brain radiation therapy (WBRT) was administered to patients with malignant gliomas after maximum surgical resection in doses varying from 5000-6000 rads<sup>1</sup>. Those that received adequate treatment saw a rise in median survival from 17 weeks with surgery alone (the previous standard of care) to 37.5 weeks with the addition of WBRT, a 120% rise in median life expectancy [59, 60].

Further studies verified the findings of Walker et al. by continuing to demonstrate the benefits of radiation therapy in the treatment of HGGs. The dose of 45 Gray (Gy) administered to the whole brain showed improvement in survival, although further dose escalation was limited by poor brain tissue tolerance [26]. Doses of <45, 50, 55 and 60 Gy were compared. Maximum survival was observed at doses of 60 Gy [61]. The addition of a 10 Gy boost failed to demonstrate any change in outcome [35]. Further dose escalation to 90 Gy using 3D conformal technique was feasible although failure was still predominantly secondary to local recurrence [6]. It is of note that the treatment for unresectable HGGs is similarly to that of resectable tumors with the obvious exception of surgery.

### Treatment Volume

Dose-dependent complications result from radiation delivered to non-cancerous tissue. Once the role of radiation was established in the adjuvant setting, researchers turned attention to providing a more focused delivery of radiation to gliomas. In 1989, a randomized trial

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<sup>1</sup> 1 rad = 0.01 Gy

compared whole brain radiation with a combination of whole brain radiation and a more focused, “coned down” radiation regimen. The established dose of 60 Gy was given either as WBRT or as 43 Gy dose of WBRT and a 17 Gy “coned down” dose. There was no statistically significant difference in outcomes between the two modes of delivery [46] establishing the possibility of field reduction without compromising survival. Concurrent and more recent studies have all demonstrated similar results with focused, “coned down” treatment of HGGs being equal to whole brain irradiation [13, 62]. Thus, involved field radiation therapy became the standard approach for radiotherapy treatment planning.

With the evidence supporting effective use of focal treatment, the next logical goal on the horizon for researchers was to establish appropriate margins of treatment such that the entirety of the tumor was included while continuing to minimize normal brain tissue irradiation. Advances in imaging allowed these studies to be performed. Comparison of antemortem CT images and postmortem histological slides showed that a margin of 1 cm beyond peritumoral edema on CT would have encompassed all malignant cells in more than 50% of the patients and a margin of 3 cm would have encompassed all malignancy in all patients studied [16]. Hochberg et al. reported that 90% of recurrence occurs within a 2-cm margin [20], and further studies continued to suggest an adequate margin of 2 cm [19]. Although the standard current protocol allows generous margins based on neuroimaging, future margins for radiation therapy in HGGs are still being questioned. European models of glioblastoma growth patterns posit microscopic invasions beyond conventional margins, potentially explaining the high failure rate [1].

In-depth analysis of imaging studies is required for radiotherapy treatment planning. Early research suggested that a CT-MRI composite provides a more conformal treatment volume, sparing the most noncancerous tissue [54]. Current planning is usually based on information gleaned from both T1 weighted MRI with contrast and T2 weighted MRI. There is active research attempting to determine whether more advanced imaging, such as MR diffusion, MR perfusion, MR spectroscopy or brain PET are better able to define tumor biology, but none of these modalities are in widespread use in the clinic to date. More sophisticated planning methods have been studied in an effort further the shift radiation therapy towards a more conformal treatment volume. IMRT, an elegant method which provides more focal treatment, has failed to improve clinical outcomes when compared to conventional 3D planning at equal doses [34], though it is still utilized in cases where critical normal structures would otherwise prevent delivery of the planned dose to the tumor volume with more conventional planning techniques.

A current planning guideline recommends 60 Gy of radiation given in 2 Gy per daily fraction over 30 fractions using a two-step design. The initial gross tumor volume (GTV) is defined by either T2 or the FLAIR changes on the MRI or CT (if the patients are unable to undergo MRI); this is expanded by a margin of 2 cm to define the initial clinical target volume (CTV). An additional margin of 5 mm is added to define the initial planning target volume (PTV), and this initial PTV is treated to 46 Gy in 23 fractions. Boost GTV is defined by contrast-enhanced T1 abnormality on the MRI or CT (if the patients are unable to undergo MRI) and this includes resection cavity and/or macroscopic tumor, and the same margins as the initial volumes are used to define boost CTV and PTV (2 cm and 0.5 cm, respectively). The last 14 Gy is given to the boost PTV over 7 daily fractions.

## Novel Radiotherapy Technologies

Novel radiotherapy treatment modalities have been studied extensively in an attempt to deliver higher doses to improve local control and survival in patients with high grade gliomas. Early single institution and retrospective studies had promising results with SRS using dose boosts from 8 to 28 Gy [36, 40, 49]. One study found that the main limiting factor to the success of SRS was peripheral recurrences [33]. A large, multi-institutional study found that the addition of SRS to existing treatment regimens was well-tolerated by patients, although it did not corroborate the findings that SRS increased survival [4]. A broad review of literature as well as a prospective study using SRS boosts of 15 to 24 Gy showed that the inclusion of SRS in the treatment of HGG did not provide any additional benefits but it did add increasing toxicity. Radiation Therapy Oncology Group (RTOG) 93-05 investigated the use of radiosurgery boost followed by conventional 60 Gy of radiotherapy for supratentorial glioblastoma and this phase III trial showed identical median survival of 13.5 months in the two treatment groups and failed to show improvement in overall survival and in failure patterns with addition of radiosurgery boost [52]. The American Society for Therapeutic Radiology and Oncology (ASTRO) has recommended against the use of SRS in the treatment of HGGs [55].

The addition of brachytherapy to existing treatment protocols for primary HGGs has shown some promise by using temporary and permanent low activity iodine-125 and temporary iridium-192 implants [25]. The adjunct use of brachytherapy added between 2.5 and 100 cGy/h to the baseline EBRT which resulted in total doses ranging from 40 to 80 Gy. Larger studies using iodine-125 have failed to show any improvement in outcomes [28, 45]. The addition of brachytherapy also resulted in an increased incidence of life threatening necrosis [51] and is not recommended in protocols at initial diagnosis.

The use of high-dose, accelerated hyperfractionated radiation therapy has been studied in attempts to shorten treatment, better facilitate chemotherapy and improve outcomes. Reports of a phase II trial suggested that improved outcomes without significantly limiting side effect profiles can be achieved through the use of hyperfractionated and accelerated hyperfractionated radiation therapy involving twice daily doses of 1.2 or 1.6 Gy and total doses ranging from 48 to 81.6 Gy [63]. These shortened treatment protocols have provided similar results while maintaining acceptable levels of toxicity [10]. Although these early studies suggested that hyperfractionation may be effective, these results were dismissed after a prospective phase III randomized trial failed to show improved outcomes with hyperfractionation. This study compared 70.4 Gy in 1.6 Gy per fraction given twice a day vs. 59.4 Gy in 1.8 Gy per fraction given once a day [39].

Hypofractionated radiotherapy delivers higher daily dose of radiotherapy in a shorter treatment course. Retrospective and prospective studies demonstrated that overall hypofractionated radiotherapy is safe and effective for elderly patients or patients with poor performance status. These studies used dose ranging from 30 Gy to 50 Gy with fractionation of 2.5 Gy/day to 7 Gy/day, and the results are non-inferior to conventional radiotherapy over 6 weeks. Further studies using a Phase III design is warranted to investigate this approach with or without chemotherapy [32].

Advances in chemotherapeutic agents as adjuncts to existing radiation protocols have been stagnant until recently. Bleomycin has not had any effect on HGGs [26]. A later study of 571 patients with malignant gliomas demonstrated no additional benefit from a multidrug

chemotherapy regimen including carmustine and procarbazine or carmustine, hydroxyurea and epipodophyllotoxin over carmustine alone [46]. No prolongation in survival was found with the addition of the PCV regimen (procarbazine, lomustine and vincristine) to existing radiation therapy protocols in patients with grade III/IV gliomas [3, 57].

In 2005, improved outcomes due to the addition of a chemotherapeutic agent were finally realized. A landmark study of patients with newly diagnosed glioblastoma randomized participants to receive either radiotherapy with temozolomide or radiotherapy alone. All patients underwent gross total resection. An increase of 2 year survival from 10.4 to 26.5% was observed after the addition of temozolomide to the standard radiation treatment protocol [53]. This regimen is the current standard of care and is used in the majority of contemporary treatment protocols.

## Prognosis

While advances in the treatment of HGGs have been made, the prognosis is still poor. Attempts to quantify the prognosis of HGGs have been challenging due to the rarity of the disease which has made powering studies to tease apart prognostic factors difficult.

The use of recursive partitioning analysis (RPA) - a powerful statistical tool to interpret data retrospectively - has identified several independent prognostic factors. The assertion and validation of age, histological subtype, mental status, Karnofsky performance status and duration of symptoms as independent prognostic factors has provided clinicians with the ability to provide a clearer prognosis for patients with HGGs and researchers with rational guidelines for clinical trial stratification. Specifically, the currently accepted RPA for glioma separates patients into six categories based on the above factors and separates them into statistically significant group with median survivals ranging from 4.7 to 58.6 months depending on the subgroup [11, 42].

Recently, a simplified model for prognosis has been proposed based on the original RPA. This model includes only four variables: age, performance status, extent of resection and neurological functioning. This results in three distinct prognostic groups which will be used for future RTOG studies on glioblastoma [30].

There has been a significant amount of recent interest in molecular stratification of glial tumors, but they have not yet become a standard part of the accepted RPAs. In spite of not being integrated into current RPAs, there are several specific markers that have been validated: the chromosomal loss of the 1p/19q in oligodendrogial tumors has been identified as a positive prognostic factor [57] and the methylation status of the O(6)methylguanine-DNA methyltransferase (MGMT) promoter gene has been associated with the prognosis of HGGs [14, 18].

Other studies have suggested that the MGMT promoter methylation status is significant only for being part of the larger, genome-wide methylation trend of the malignant cell line [58]. Another study suggests that mutations in the isocitrate dehydrogenase enzyme may have a positive effect on the prognosis of anaplastic gliomas greater than that of both MGMT promoter methylation status and 1p/19q deletion status [64] though the mechanism responsible for this change remains unknown [43].

## Recurrence

Despite multi-modality therapy, recurrence develops in the majority of patients with glioblastoma. The optimal management in this setting has not yet been conclusively defined. There is neither level 1 evidence nor randomized trial that compares treatment with supportive care in the setting of recurrent disease and we therefore rely on phase II data and comparison with historical controls. That being said, a number of studies using different radiotherapy techniques have given some hope to patients in the recurrent setting. Several single-institution studies reported modest survival benefit associated with the use of interstitial brachytherapy for recurrent malignant glioma [15, 29, 41, 50]. The results from a study of concurrent cisplatin and iodine-125 brachytherapy were within acceptable levels of toxicity [5]. More recently, stereotactic radiation therapy (SRT) using standard fraction or hypofractionation has been used to deliver more precise and potentially higher doses to patients with recurrent HGGs. Studies showed that this approach is well tolerated. Several SRT schemas have been reported by different institutions; a study by Combs et al. used a median dose of 35 Gy with a median fractionation of 5 x 2 Gy/wk [9]. At least one study showed hypofractionated SRT has maximum benefits in salvage therapy when recurrence occurs within 6 months of initial treatment when using a median dose of 35 Gy in 3.5 Gy fractions [12]. Maintaining quality of life and minimizing treatment-associated toxicity needs to be even more carefully evaluated in the setting of recurrent disease as radiation plays a primarily palliative role.

## III. RADIOTHERAPY FOR LOW GRADE GLIOMAS

The application of radiation therapy to the treatment of LGGs, or WHO grade II tumors, has been studied extensively. Definitive maximum surgical resection followed by close observation has remained the standard of care as the data for radiation in LGGs have not been overwhelmingly positive. The European Organization for Research and Treatment of Cancer (EORTC) 22845 trial randomized 314 patients to undergo immediate postoperative radiotherapy with 54 Gy or delay radiotherapy until the time of progression; the study showed a statistically significant increase in progression free survival in patients who received adjuvant radiation therapy compared to the control group which did not receive RT. However, the overall survival in the group of patients receiving radiation was the same as in the control group [56]. Two large randomized trials investigated potential benefit of higher radiation dose for LGG. EORTC 22844 included 343 patients who were randomized and assigned to 45 or 59.4 Gy adjuvant radiation therapy and failed to demonstrate a clear dose-response relationship for radiation therapy as adjuvant treatment for LGG. The overall survival and progression free survival were nearly identical, approximately 60 and 50%, respectively [23]. An intergroup study in North America included 203 patients with LGGs and randomized these patients into groups receiving either 50.4 or 64.4 Gy. The results of this study showed that patients receiving 64.4 Gy of radiation had a lower but not statistically significant overall survival rate (65 vs. 72%) [47]. Due to the lack of improvement in overall survival among patients receiving adjuvant radiotherapy for LGGs, radiation therapy is not currently

recommended in most standard protocols at initial presentation [56]. Radiation may be useful in recurrent disease or various subsets of patients with LGGs and is still being studied.

## Prognosis

While the vast majority of LGGs are cured by surgical resection, some exhibit more aggressive behavior. A comprehensive analysis of patients with LGGs who were enrolled in the EORTC 22844 and EORTC 22845 trials demonstrated several important negative prognostic factors including patient age over 40 at diagnosis, astrocytoma histology, preoperative neurological deficits, tumor diameter ( $\geq 6$ cm) and tumor crossing midline [37]. Patients who had 3 or more of these factors were considered high risk and had a median overall survival of 3.7 years compared to 7.8 years for those in the low risk group. The RTOG is sponsoring a multi-institutional phase II trial that is currently studying the effects of temozolomide in combination with radiation for patients with high-risk subtypes of LGGs as identified by Pignatti et al. This trial is closed to accrual and is awaiting maturation.

## IV. ADVERSE EFFECTS OF RADIATION THERAPY

The side effect profile for standard radiation therapy protocols to the brain is generally benign, although more serious adverse effects can occasionally be seen. The side effect burden is dependent on many factors including total dose, location of treatment, fractionation, total area irradiated and pretreatment level of functioning. The most severe adverse effects of radiation are generally seen in patients with prolonged survival as the adverse effects of radiation are cumulative [7] and the tissue of the central nervous system is considered a 'late responding tissue'. Because of this, prognosis has to be considered when trying to estimate the side effect profile of a possible therapeutic regimen. Taking an actuarial approach to the side effect burden of radiation makes treatment more patient-centered, and expectations more realistic. In patients receiving radiation therapy as palliative care, the patient's goals and overall comfort take precedence. In patients with HGGs that carry a poor prognosis, acute adverse effects are the most relevant. Because LGGs and some grade III/IV gliomas have a life expectancy measured in years, not months, the delayed and long term effects of radiation are most germane.

Acute side effects of radiation to the brain are usually self-limited and easily treated. Further, they rarely manifest before the third week of a standard course of radiation. Common topical side effects can include dermatitis, desquamation and alopecia, all of which are typically self-limited and occur only where the radiation is actually passing through the skin. Dose dependent nausea, vomiting, somnolence and fatigue may be transiently observed and are typically easily treated with standard medications. Demyelination and alterations in capillary permeability secondary to radiation may result in headache. Late effects of radiation can include ototoxicity, retinopathy, cataract formation, hypopituitarism, hypothyroidism and blindness. The most serious adverse effect of central nervous system radiation therapy is necrosis [17].



Radiation somnolence syndrome that is rare, poorly understood adverse effect of radiation therapy. This syndrome includes drowsiness or exhaustion, low-grade fever, nausea, vomiting and headache. This cluster of symptoms typically occurs one to six months following radiation therapy to the brain and is most commonly observed in children receiving whole brain irradiation [31].

One obvious concern associated with the use of radiation to treat brain tumors is the possibility of neurocognitive sequelae. Using the Folstein Mini-Mental Status Exam (MMSE) as a primary endpoint, patients treated for LGGs were assessed for cognitive deterioration. Patients with a baseline MMSE score within the normal range ( $>$  or  $=$  27) and without tumor progression failed to show deterioration in more than 90% of patients. In patients with abnormal baseline MMSE scores ( $<$ 27), the use of radiation therapy was associated with deterioration. Of note, dose did not play a statistically significant role in the change in cognition [2]. Additional research suggests that the neurocognitive effects secondary to radiation therapy are quite insignificant, particularly in light of the potential for defects secondary to the mass effect of the tumor itself [27]. The effects of SRS on neurocognition in the setting of metastatic brain disease showed that, in spite of a high incidence of baseline deficits, 5 of the 15 patients studied lived over 200 days. Of those 5 patients, 4 showed stable or improved neurocognitive functioning [8]. Overall, the use of radiation in the treatment of gliomas does not carry a high rate of cognitive sequelae.

## V. ADDITIONAL TOPICS

### **Radiosensitizers**

Radiosensitizers, designed to increase the susceptibility of tumor cells to radiation damage, have been studied in an effort to combat the protective features of hypoxia in malignant tissue. Nitroimidazoles, perfluorochemicals and halogenated pyrimidines have all been considered. A phase III trial involving bromodeoxyuridine demonstrated no change in survival among patients with anaplastic gliomas, excluding glioblastoma [38]; Difluoromethylornithine demonstrated no change in survival or progression-free survival in glioblastoma [39]; and misonidazole showed only modest benefits, 8% increase 1 year survival, when compared to controls [21]. Some chemotherapeutic agents may have radiosensitizing properties and are currently being investigated.

### **Treatment of the Elderly**

Patients over the age of 70 have largely been excluded from the randomized trials mentioned thus far in this chapter. Their treatment, therefore, has largely been at the discretion of the provider and has subsequently not been evidence based, or even standardized. Many practitioners choose to only offer palliative therapy, while others treat them just like their younger counterparts. The addition of radiation to the treatment of patients in this group with HGGs was tested in a trial which closed early due to the strong benefit demonstrated by therapy which showed an increase in median survival from 16.9 to 29.1

weeks. This increase in life expectancy was not associated with a decrease in quality of life or cognitive ability [24]. Further research demonstrated that, when adjusted for prognostic factors as described by the Curran et al. RPA, outcomes for patients over 70 years old are similar to outcomes of younger patients [48]. A retrospective analysis of 2,836 patients over the age of 70 with glioblastoma demonstrated that the use of radiation therapy alone resulted in longer median survival than surgery alone and that the combination of surgery and radiation leads to significantly improved outcomes when compared to radiation or surgery alone [44]. Other single institution studies have also shown the benefit of chemotherapy and surgery as well [22, 44]; though these data are retrospective, they suggest that patients in this age group should be treated no differently than their younger counterparts because of age alone.

## CONCLUSION

The diagnosis of HGG still bears a grim prognosis. While research into treatment with radiation and chemotherapy over the past several decades has led to more than a doubling of life expectancy in these patients, median survival remains poor, suggesting that there are fundamental pieces of the puzzle yet to be uncovered. The standard of care for HGG at this time includes maximal safe surgical resection followed by radiation of 60Gy to the involved field in 1.8-2Gy fractions. Temozolomide is given concurrently at a dose of 75mg/m<sup>2</sup> daily during radiation, followed by six months of 150-200mg/m<sup>2</sup> for 5 days at the beginning of each 28 day cycle.

The use of radiation therapy in the treatment of LGGs has not been extremely fruitful. Ongoing research is attempting to identify any potential benefits of adjuvant radiation in high risk subsets of patients with LGGs. To date, however, the standard of care remains maximal surgical resection followed by observation. Patients with high risk features may be considered for adjuvant radiotherapy. Radiation therapy is currently recommended in the recurrent setting. Treatment regimens include 50.4 to 54 Gy of involved field radiation therapy with 1 to 2 cm margin expansion around the surgical cavity and/or gross tumor.

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## *Chapter 10*

# **DOSE RATE MODULATION FOR THE SAFE DELIVERY OF RETREATMENT IRRADIATION IN RECURRENT GLIOMA PATIENTS**

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## **ABSTRACT**

In spite of the continued evolution of surgical and chemotherapeutic approaches to the treatment of low and high-grade glioma patients, most patients recur, at which time their therapeutic options are limited. Most recently the conditional approval of bevacizumab by the FDA for the recurrent glioblastoma patients has offered a significant improvement in progression free survival. Efficacy, however, regarding overall survival has been limited. One unique approach to such recurrent glioma patients is the use of dose rate modulated re-treatment irradiation. Radiotherapy (RT) delivered below standard dose-rates reduces normal tissue toxicity and can induce significant tumor regression in some tumor types including glioma. By reducing the effective dose-rate and increasing the treatment time, it becomes possible for repair processes to be active during irradiation. This reduction in dose-rate can result in a therapeutic advantage, as repair of sub-lethal damage in normal neural tissues is greater for late complications than for neoplastic cells; further tumor cells (that have received prior RT) can accumulate in a sensitive phase of the cell cycle, e.g., G<sub>2</sub>, thus leaving them at risk for radiation induced killing. In the chapter to follow, modeling for this approach, technological details, well as biophysical basis for its application is summarized.

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The published peer reviewed clinical literature on the treatment of recurrent glioma patients using this technique is reviewed in detail: This literature, which encompasses relatively large treatment volumes, is consistent with the concept that fractionated reduced dose-rate external beam RT can achieve an improved therapeutic index. Also outlined are the current planned prospective clinical studies for evaluation this new approach in the context of cooperative group clinical trials in recurrent glioblastoma patients.

## 1. INTRODUCTION

The use of combined modality therapy, including maximal safe resection, followed by adjuvant radiotherapy (RT) and chemotherapy, i.e., temozolomide, has become the standard treatment for glioblastoma (GB) [1] [A similar approach, i.e., the introduction of temozolomide into surgery/radiation approaches to grade 3 and low grade gliomas is currently being evaluated in prospective cooperative group studies. Unfortunately the gains in progression free survival (PFS) and overall survival (OS) for glioblastoma (GB) have been modest, and projected to be modest for lower grade gliomas. Thus, retreatment strategies for recurrent disease are critical to both palliate patients and provide survival benefit. In considering the management of recurrent glioma confounds to therapy relate to drug resistance, drug interactions (e.g., P450 induction or inhibition), as well as the blood–brain barrier with systemic drug therapies. The recent provisional approval of bevacizumab for recurrent GB by the Food and Drug Administration was based on a comparison to historical data [2, 3]; although PFS was dramatically improved (based on a historical control), OS benefit was modest [2, 3].

Similarly, reirradiation using conventional and conformal techniques has been associated with only modest palliative benefit. This in part relates to reduction of retreatment volumes or reduced doses to prevent toxicity, potentially reducing the efficacy. Thus, in an attempt to retreat relatively large volumes of recurrent disease (beyond the scope of radiosurgery) to projected efficacious doses, our group has explored a paradigm that allows for enhanced repair of radiation damage in normal tissues while maintaining a therapeutic dose to the tumor. The overall concept is to extend the period of time a give fraction of radiation is delivered, in order to exploit differences in repair kinetics between normal and malignant cells, allowing normal tissues to repair sublethal damage. Modeling suggests that the aforementioned dose-rate effect is most dramatic between 0.01 and 1 Gy/min [4]. From a practical standpoint, a reduced dose rate can be obtained by dividing a standard treatment fraction into a number of equal subfractions delivered in a pulsed manner separated by a fixed time interval (or alternatively by using continuous reduced-dose-rate RT, which can be achieved by employing a  $^{60}\text{Co}$  source whose age is more than eight half lives of  $^{60}\text{Co}$ ) [5]. The concept of fractionated reduced-dose-rate external beam RT has been shown to be feasible with an improved therapeutic index for oropharynx cancer [6] and breast cancer [7, 8]. The chapter to follow includes: i) a summary of the biological basis for dose rate modulation of RT; ii) the techniques for the clinical application reduced-dose-rate external beam RT; iii) the published clinical experience in applying this treatment approach to recurrent glioma patients with attention to its further evaluation in future prospective clinical trials.

## 2. THE BIOLOGICAL BASIS FOR DOSE RATE MODULATION

The general concept for RT dose rate modulation relates to exploiting the biological differences between normal neural tissue and glial neoplasms in terms of repair of radiation damage and the induction of radiation-induced lethal lesions. In this regard, it is of interest to note that low dose rate interstitial brachytherapy has been used for the treatment of recurrent glioma with reported improved median survivals [9]. The radio-biological properties of low dose-rate radiation have long been established relative to tumor/tissue sparing. Currently, a superior dose distribution is felt to be the reason for the efficacy and tolerance observed with brachytherapy as a treatment for malignant glioma. Beyond this, however, there is theoretical and experimental evidence relating to lower dose-rates potentially providing more efficient radiation damage. Indeed, there is increasing experimental evidence as well as empiric clinical observations suggesting that reduced dose-rate irradiation may be a biologically different and distinct process in contrast to acute high dose rate fractionated irradiation. Shultz and Geard [10] evaluated 2 human glioblastoma cell lines relative to the effects of dose rate on radiosensitivity. Dose rates ranged from 0.2 Gy/h to 5.4 Gy/h. An inverse dose rate effect, (i.e., a paradoxical increase in cell killing within a narrow range of decreasing dose-rate) was demonstrated at approximately 0.4 Gy/h [10]. This phenomenon is thought to be a consequence of the accumulation of cycling cells at the G2M checkpoint. During continuous low dose-rate irradiation, the G2M blocked cells are preferentially killed as they are relatively more radiosensitive than cells in G1 and S phase. In support of this concept, Shultz and Davies, performed cell cycle analysis which revealed maximal accumulation of cells at G2M checkpoint at a dose rate of 0.4 Gy/h [11, 12]. Based on these experimental observations, one can extrapolate to the hypothesis that actively proliferating glial tumors may be selectively more radioresensitive to continuously delivered low dose-rate irradiation than the quiescent surrounding normal brain tissue.

Beyond the aforementioned considerations, breaking each fraction of radiation into a number of subfractions takes advantage of a second intriguing radiation phenomenon known as low-dose hyper-radiosensitivity (LDHRS), (i.e., increased radiosensitivity to doses <0.3–0.5 Gy). This translates to a cell survival that is considerably lower than that which would be predicted from extrapolation of the survival curves derived from higher doses. Such data are available (i.e., low-dose responses) for many human cell lines [13], most of which have demonstrated LDHRS [14, 15]. As alluded to above, low dose hypersensitivity might be a product of radiation-damaged G2 cells entering mitosis prematurely [14]. That is to say there was inefficient cell cycle arrest of irradiated cells as a result of the inverse dose effect [14]. Short et al. [16] have hypothesized that low-dose hypersensitivity can be thought of as the constitutive response of cells without the enabling of repair mechanisms triggered by higher doses. Modeling of these phenomena has been accomplished by Short and colleagues [17]. Building on the above concepts, Tomé and Howard using modeling, developed a treatment strategy for gliomas that employs both a reduced dose-rate and pulsed treatment dose delivery [5]. Using five established glioma cell lines they demonstrated that a pulsed delivery scheme can yield a substantial increase in tumor control probability. They proposed a pulsed delivery scheme for the treatment of gliomas in which the daily treatment fraction is delivered using 0.20 Gy pulses, separated by three minutes for a time-averaged dose-rate of 0.0667 Gy/min. The dose per pulse of 0.2 Gy was near or below the transition dose observed *in vitro* for four

of the five glioma cell lines studied [5]. Although most evidence for low-dose hypersensitivity has been demonstrated *in vitro*, analysis of LDHRS in metastatic tumors (i.e., breast, melanoma, and sarcoma involving skin and subcutaneous tissue) has revealed a potential advantage to low-dose ultrafractionated therapy compared with more standard fraction sizes [15]. In the next section the technical aspects of applying these radio-biological concepts to clinical practice are described, as well as the clinical experience regarding their application to recurrent glioma patients.

### **3. TECHNIQUES FOR THE CLINICAL APPLICATION OF REDUCED - DOSE-RATE EXTERNAL BEAM RADIOTHERAPY**

The technique at the University of Wisconsin (UW) for pulsed reduced-dose-rate (PRDR) RT [8, 13, 18] is to generate 0.2-Gy pulses separated by 3-min intervals, thus creating an apparent average dose rate of 0.0667 Gy/min [5]. The dose rate of the linear accelerator is reduced to 1 Gy/min during each 0.2-Gy pulse. In a large series (n=103) of glioma patients [18] described in the next section, the median dose was 50 Gy (range, 20–60) delivered in 1.8–2.0-Gy fractions; with a mean treatment volume of  $403.5 \pm 189.4$  mL according to T2-weighted magnetic resonance imaging and a 2-cm margin. It may be of interest to the reader that the UW group has modeled the use of Tomotherapy: The method described in this paper involves a virtual grid style blocking scheme, where half of the beam angles are directionally blocked using 15 equally spaced segments surrounding the center of the image set [19]. Instead of employing the PRDR technique described above when using a Linear accelerator one can also opt for the delivery of reduced dose-rate RT using an attenuator that reduces dose rate to the desired level. Such an attenuator can be inserted into the wedge tray of the linear accelerator and the multi-leaf collimator can be employed for field shaping. As a matter of fact during the early phase of reduced dose rate radiotherapy at the UW just such an attenuator was constructed and employed, but its use was later on discontinued in favor of PRDR.

The attenuator consisted of a uniform brass slab of 55 mm thickness and yielded a dose rate of 5.38 Gy/hr in water at a depth of 1.5cm at a distance 101.5cm from the radiation source of the Linear accelerator. However, the use of such an attenuator changes the beam characteristics and necessitates the need to measure and model the beam in the treatment planning system before it can be used to treat patients. In addition, the handling of the attenuator is cumbersome and carries the risk of being left out prior to the delivery of reduced dose rate radiotherapy with dire consequences for the patient. The PRDR-RT, described above, does not suffer any of these draw backs and is therefore an inherently safer technique. Siker and colleagues [20] recently reported on their experience between 1993 and 1998 in recurrent glial tumors (the results of which are discussed below) using what now may be regarded as a somewhat impractical technique. “An Atomic Energy of Canada Limited Eldorado-6 modified cobalt-60 teletherapy unit was used to deliver SLDR. The desired dose rate of 40 to 50 cGy per hour was obtained by placing a custom metal attenuator in the beam. Radiotherapy was delivered 6 to 8 hours/day at a dose rate of 0.4 to 0.50 Gy/hour, delivering 30 to 35 Gy over 12 days [20].” In spite of the cumbersome nature of this approach, this

report involving recurrent glioma patients, as discussed below, does serve to support the underlying premise regarding low dose rate RT in terms of tolerance and efficacy.

## **4. CLINICAL EXPERIENCE FOR DOSE RATE MODULATION AS RECURRENT GLIOMA THERAPY**

### **1. Background**

The clinical feasibility of low dose therapy was established over 4 decades ago by Pierquin et al. [21]. As reviewed by others [18, 20-22] the idea of retreatment radiation for palliation is similarly not novel. However, by virtue of toxicity considerations median treatment volumes have generally been no greater than 74 mL [22, 23]. The large retreatment series using fractionated stereotactic RT from the University of Heidelberg reported a median planning target volume of 49.3 mL for 172 patients treated to a median dose of 36 Gy using 2-Gy fractions [21]. In this series, 1-cm expansion along T1-weighted MRI-enhancing tumor was used. It is in this context the clinical experience using dose rate modulation is described below. As the reader considers the efficacy data presented below, an appropriate yardstick may be useful. Although the outcome of recurrent low grade and grade 3 gliomas are often variable, the outcome for grade 4 astrocytoma tends to be generally uniform. An analysis by Wong et al. [23] had provided a useful comparator for survival at recurrence in GB. These investigators, reviewing eight phase II drug studies, observed a median survival of 25 weeks (95% confidence interval 21–28) for recurrent GB and 30 weeks (95% confidence interval 26–35) for the entire cohort of anaplastic astrocytoma and GB [24].

### **2. The Medical College Wisconsin (MCW) (Milwaukee) Experience (1993-1998)**

The MCW group using a dose rate modulation technique designated as semi-continuous low-dose-rate teletherapy (SLDR) (see section II above for technique) treated 20 recurrent patients: Grade 4 glioma (n=14), Grade 2 glioma (n=5), ependymoma (n=1) [20]. Overall, 18 of 20 patients were able to complete SLDR. The average field size was 8.4 - 9.25 centimeters (range 6 - 9 cm and 10 - 11 cm, respectively). The median radiotherapy dose was 60 Gy (range, 50.4–70.4 Gy). At 3 months there was no Grade 3 central nervous system toxicity (without radiographic evidence of tumor progression). Overall survival after SLDR was 56% at 6 months, 28% at 12 months, and 17% at 24 months. One patient survived >48 months, and 1 patient survived >60 months after SLDR treatment. Re-resection before SLDR treatment significantly improved 1-year overall survival for all patients and patients with Grade 4 glioma. Overall survival after SLDR was 56% at 6 months, 28% at 1 year, and 17% at 2 years. Additionally, 1 patient with a grade 2 glioma survived more than 48 months after SLDR. One long-term survivor had a recurrence that behaved clinically and radiographically like higher-grade glioma before SLDR. The longest survivor was a patient with WHO Grade 4 glioma at diagnosis who lived more than 60 months after SLDR. As might be expected Karnofsky performance score (KPS) was a prognosticator. Analysis of 14 patients, who had an initial

diagnosis of GB, demonstrated median survival after SLDR of 8 months for patients who underwent surgical resection before SLDR, and 4.5 months for patients who did not undergo resection. Treatment in this series was well tolerated. Only 3 patients required an increase in steroids during therapy. Two patients achieved long-term survival of greater than 4 years with stable neurological function. With the caveat that it is difficult to distinguish between necrosis and disease progression in the era in which this study was performed there were no gross therapy induced complications reported. In summary, this pioneering study provides results consistent with safety and efficacy.

### **3. The University of Wisconsin (Madison) Experience (2000-2007 and Beyond)**

A research group at the UW has recently reported on a series of 103 recurrent glioma patients using a dose modulation technique they describe as PRDR-RT described in section II above [18]. Histology's included: low-grade glioma (n=25); grade 3 oligodendroglioma (n=3); grade 3 oligoastrocytoma (n= 3); grade 3 astrocytoma (n=25), grade 4 oligoastrocytoma (n= 1), grade 4 astrocytoma (n= 44); brainstem glioma (n=1); (pineal tumor (n=1). Strikingly, the mean treatment volume was  $403.5 \pm 189.4 \text{ cm}^3$  according to T2-weighted magnetic resonance imaging and a 2-cm margin. The PRDR-RT regimen was well tolerated, and no patient discontinued treatment because of associated toxicity. Autopsy of the brain was performed in 15 patients, of which 4 had notable necrosis. In 2 of these 4 patients the PRDR dose was 50 Gy in 2 patients originally treated to a dose of 54 Gy and 59.4 Gy, with an interval of 12 and 40 months to PRDR-RT. In the other 2 of 4 patients, the PRDR dose was 54 Gy in 2 patients originally treated to 60 Gy, with an interval of 11 and 28 months to PRDR-RT. In addition, one of these patients received stereotactic radiosurgery to 12 Gy prescribed to the 50% isodose line 25 months after initial RT to 60 Gy. Other pathological findings only included punctuate necrosis in the presence of recurrent tumor, usually associated with mitoses or vascular endothelial proliferation. Follow-up MRI scans in this series revealed changes consistent with disease progression, pseudo-progression, or necrosis. Clinical deterioration was treated with increases in steroid dosage and supportive care. Three patients underwent post-PRDR-RT subtotal resection with the pathologic findings revealing progressive disease without treatment related necrosis. Although no formal endocrine, neuro-cognitive, or visual testing was performed, no patient manifested obvious blindness from treatment. Multivariate analysis revealed age at the initial diagnosis, initial low-grade disease, and KPS of  $\geq 80$  to be significant predictors of survival after initiation of PRDR. No significant survival difference (from the initiation of PRDR) was observed for patients who had previously been treated with no systemic therapy (including temozolomide), one previous regimen, two previous regimens, three previous regimens, or four or more previous regimens. Regarding GB patients, these authors found that time from initial RT ( $\geq 14$  months) was prognostic, and compared favorably relative to palliative chemotherapy [24]. Such PLDR-RT patients had a median survival of 28 weeks; patients reirradiated within 14 months of previous therapy had a median survival of only 21 weeks ( $p = 0.004$ ). It is noteworthy that 77% of the patients treated in the Phase II drug studies [24] had received previous chemotherapy, 25% of whom had received at least two regimens. In the UW series, 92% of patients had received previous chemotherapy, and 45% had received at least two regimens.

Of parenthetical interest, newly *de novo* GB patients were compared with those who were initially diagnosed with a lower grade glioma and transformed to grade 4 (confirmed by the biopsy); there was no significant difference in survival from the initiation of PRDR-RT (median survival 5.2 vs. 5.9 months,  $p = .12$ ). In considering histologic findings at the initial diagnosis, median survival for those with low-grade glioma from the initiation of PRDR-RT was 11.4 months (range, 1–33.8); for those with Grade 3 glioma: 5.6 months (range, 1.2–23.7); for those with GB: 5.1 months (range, 1–48.4). The 6-month and 1-year actuarial survival rate after retreatment with PRDR was 73.9% and 47.8% for those with low-grade glioma, 38.2% and 9.7% for those with Grade 3 glioma, and 34.8% and 4.4% for those with GB, respectively. For the entire cohort, the median survival from the initiation of PRDR-RT was 5.8 months (range, 1–48.4), and the 6-month and 1-year actuarial survival rate was 44.7% and 15.9%, respectively. For the reader's interest, a series of MRI's demonstrating a typical PRDR-RT response was published in 2007 as part of a case report [28]. In summary, PRDR as a reirradiation strategy was well tolerated, allowing for safe retreatment of larger target volumes to high doses (>100 Gy) with palliative benefit.

#### 4. Future Research Considerations

The recent approval of bevacizumab for the treatment of recurrent GB has yielded a new clinical cohort of patients with a uniquely poor prognosis, i.e., patients who have progressed after bevacizumab therapy [25, 26]. The 6 month progression free survival (PFS) for these patients is negligible. Relative to this, anecdotal experience at the UW (Howard and Robins, unpublished 2011) suggests that PRDR-RT does provide some palliation and even responses in this patient population.

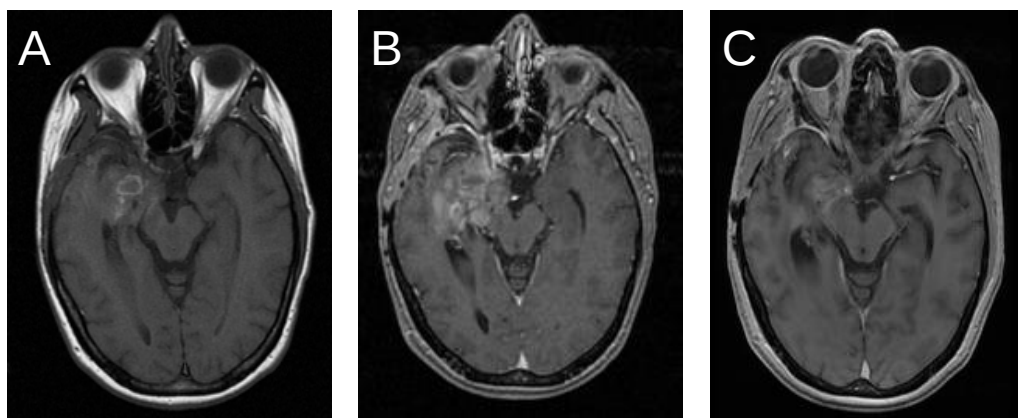


Figure 1. Axial T1-weighted MR image with gadolinium contrast of a patient with GB (MGMT non-methylated). The patient experienced rapid progression at the conclusion of RT and temozolomide. Bevacizumab was initiated with a dramatic improvement in the patient's MRI (pictured in Panel A) and a clearing of symptoms. Seven months later the patient became symptomatic requiring steroids and was found to have progressive disease (shown in Panel B). The patient received PRDR-RT (45Gy in 1.8Gy fractions) with bevacizumab. Symptoms improved obviating the need for steroids. Panel C obtained 4 months after the start of PRDR- RT demonstrates a response, which was durable for another 3 months.

Beyond this, our experience with continuing bevacizumab with PRDR RT after progression on bevacizumab may have utility, as illustrated in Figure 1. Additionally, adding bevacizumab to PRDR-RT in bevacizumab naïve patients may provide durable responses in excess of the typical PFS (i.e., ~ 4 months) seen with bevacizumab [2, 3]. These observations may in part relate to the potential for bevacizumab to prevent, and or treat, radiation induced necrosis [27]. Based on such observations a prospective clinical trial is now planned randomizing recurrent GB patients (both bevacizumab naïve, and bevacizumab progressed) to PRDR- RT with and without bevacizumab.

## CONCLUSION

Forty years have past since the first application of dose rate modulation for the treatment of neoplastic disease. Almost two decades have elapsed since the first clinical application of this concept to recurrent glial neoplasms, and a decade since the development and first testing of practical approach to the delivery dose rate modulated RT for recurrent glioma patients. It is noteworthy that the aforementioned clinical experience is grounded in preclinically tested biological principals. Thus, it is argued that there is now a unique opportunity to define the potential for this therapeutic approach in the context controlled prospective clinical trials. It is anticipated that results of such efforts will have implications beyond the scope of neuro-oncology.

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## **Chapter 11**

# **TARGETED THERAPIES FOR MALIGNANT GLIOMAS**

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## **ABSTRACT**

Malignant gliomas are the most lethal brain tumors, accounting for 13,140 deaths in the United States each year [1]. Despite significant advances in treatment, malignant gliomas remain incurable and carry a dismal prognosis. The median survival of patients with glioblastoma is only 15 months from diagnosis with standard treatment [2]. Until recently, the role of chemotherapy has been questionable but temozolomide has now become the standard of care for treatment of newly diagnosed glioblastoma concurrently with external beam radiation therapy. An increased understanding of the molecular pathogenesis of malignant gliomas has led to the development of a number of targeted therapies with encouraging results in early studies. Most of these agents are directed against angiogenesis or growth factor pathway ligands, their receptors, or intracellular second messenger pathways involved in signal transduction. Single agent bevacizumab, a humanized monoclonal antibody against the vascular endothelial growth factor (VEGF) ligand, was recently approved by the US Food and Drug Administration (FDA) for treatment of recurrent glioblastoma. Not all malignant glioma patients respond to targeted therapies and definitive predictive markers are lacking. Unfortunately, even for responding patients, the responses are generally short-lived and malignant gliomas uniformly recur. An improved understanding of the mechanisms of resistance and the role of glioma stem cells in tumorigenesis is necessary to develop more effective therapies.

## **INTRODUCTION**

The incidence of malignant gliomas has increased over the last three decades. Malignant gliomas are comprised of World Health Organization (WHO) grade III (anaplastic) and grade

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IV (glioblastoma) neoplasms. In 2010, approximately 22,020 primary brain and other nervous system neoplasms were diagnosed in the United States [1]. Glioblastomas (GBMs) and anaplastic astrocytomas (AAs) comprised the majority of these neoplasms, with GBM being the most common primary malignant brain tumor (60-70%). Most of the therapies discussed in this chapter have been evaluated in GBM, but are also utilized in AAs by standard neuro-oncology practice. Despite significant advances in surgical techniques, radiotherapy, and chemotherapy, the prognosis for malignant gliomas remains poor. Due to their locally invasive and infiltrative nature, malignant gliomas almost universally recur and remain incurable. Recent studies have increased our understanding of tumor biology and molecular pathways underlying the growth of malignant gliomas. Several novel therapeutic agents targeting these aberrant pathways are being investigated with encouraging results. This chapter focuses on the recent advances in the medical management of malignant gliomas with an emphasis on antiangiogenic and molecular targeted therapies.

## CHEMOTHERAPY

Surgery and radiation therapy have been the cornerstone of treatment for newly diagnosed GBM since the 1970's [3]. However, due to the limited efficacy of older cytotoxic chemotherapies for malignant gliomas, the utility of adjuvant cytotoxic chemotherapy has been debated until recently, when temozolomide was shown to have a definite survival advantage [2]. The central nervous system has traditionally been considered a pharmacologic sanctuary site due to the blood brain barrier (BBB), which limits drug delivery to the brain parenchyma [4]. Several alternative strategies have been attempted to maximize chemotherapy delivery and circumvent the BBB. These approaches have included direct intra-operative injection [5], placement of resection cavity catheters for convection-enhanced delivery of chemotherapy [6], intra-arterial delivery with or without disruption of the blood brain barrier using mannitol [7, 8], high dose chemotherapy followed by autologous stem cell rescue [9], and implantation of biodegradable wafers impregnated with chemotherapeutic agents [10]. With the exception of carmustine (BCNU)-impregnated wafers, all other strategies have been associated with increased toxicity and limited clinical benefit.

## ALKYLATING AGENTS

### Nitrosoureas

Nitrosoureas - carmustine (BCNU) and lomustine (CCNU) - are small lipophilic molecules that readily cross the BBB. Most of the early clinical trials utilized nitrosourea-based chemotherapy regimens. The survival benefit of these regimens is only marginal. In the largest randomized trial of adjuvant chemotherapy in newly diagnosed high-grade gliomas conducted by the Medical Research Council (MRC), 674 patients were randomized to either radiation alone or to radiation plus procarbazine, lomustine and vincristine (PCV). No survival advantage was seen with the addition of chemotherapy to radiation [11]. A modest survival advantage, however, was reported in two meta-analyses that reviewed data from

randomized trials of patients with malignant gliomas [12, 13]. The Glioma Meta-Analysis Trialists Group reviewed 12 trials involving 3004 patients and reported an absolute increase in 1-year survival from 40% to 46% with the addition of chemotherapy to radiation [13]. Nitrosoureas have also been used in recurrent GBM and until recently were considered the standard of care. Earlier literature had suggested response rates as high as 40 to 50%, but these studies were conducted prior to the widespread use of computed tomography (CT) and magnetic resonance imaging (MRI). Current data suggests limited efficacy in this setting, with response rates and 6-month progression-free survival (PFS) less than 20% [14]. These agents are associated with significant myelosuppression and pulmonary toxicity.

### **BCNU-Impregnated Wafers**

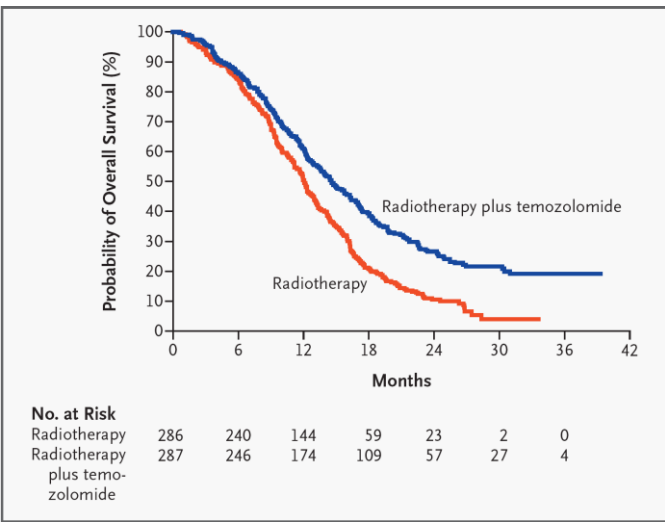
Biodegradable wafers (Gliadel) impregnated with carmustine (BCNU) can be used to line the surgical resection cavity to deliver continuous high dose chemotherapy without prohibitive systemic side effects. Brem et al. demonstrated an improvement in survival for patients with recurrent malignant gliomas (31 vs. 23 weeks) and BCNU wafers were approved by the Food and Drug Administration (FDA) for recurrent malignant gliomas [10]. Subsequently a phase III study in newly diagnosed malignant glioma patients also reported a statistically significant survival advantage compared to placebo (13.9 vs. 11.6 months) [15].

Based on this data and other studies, the FDA extended approval of BCNU wafers for use in the initial treatment of malignant gliomas. Gliadel therapy must be used judiciously, as it has been associated with delayed wound healing and infections [15]. Additionally, BCNU can potentially interact with other therapeutic agents and increase toxicity. The phase III study demonstrating improved efficacy of Gliadel wafers in newly diagnosed malignant glioma patients was completed prior to the establishment of the standard of care regimen for newly diagnosed GBMs. Thus, the additional efficacy of Gliadel wafers to concurrent temozolomide and radiation therapy is unknown. Also, Gliadel wafers may render patients ineligible for enrollment in clinical trials. Consequently, most centers do not advocate the routine use of BCNU-wafers in newly diagnosed GBM. Appropriate patient selection is imperative.

### **Temozolomide**

Temozolomide, an oral alkylating agent, became the standard of care for newly diagnosed GBM in 2005 when it was reported to confer a survival advantage. In a landmark phase III trial, 573 patients with newly diagnosed GBM were randomized to radiation therapy with or without temozolomide [2]. After maximal surgical resection, both groups received 60Gy of radiation in 2Gy fractions. The temozolomide cohort also received concurrent continuous daily temozolomide (75 mg per square meter of body-surface area per day) for the duration of radiotherapy, followed by six cycles of adjuvant temozolomide (150 to 200 mg per square meter for 5 days during each 28-day cycle). With the addition of temozolomide, the median overall survival (OS) improved from 12.1 to 14.6 months (Figure 1). More importantly the two year OS increased from 10% to 27%. This benefit was sustained at 5 years, with 10% of patients who received chemoradiation still alive versus 2% who received

radiation alone [16]. Patients with GBMs that had the promoter gene of the repair enzyme MGMT methylated were found to benefit the most from temozolomide [17].



The hazard ratio for death among patients treated with radiotherapy plus temozolomide, as compared with those who received radiotherapy alone, was 0.63 (95 percent confidence interval, 0.52 to 0.72;  $P<0.001$ ).

Figure 1. Kaplan-Meier Estimates of Overall Survival According to Treatment Group. Reprinted with permission from the New England Journal of Medicine. Stupp et al. Radiotherapy plus Concomitant and Adjuvant Temozolomide for Glioblastoma. N Engl J Med. 2005 Mar 10;352 (10):987-96. Copyright © 2005 Massachusetts Medical Society. All rights reserved.

Concurrent chemoradiation with temozolomide was fairly well tolerated with grade 3 or 4 myelotoxicity occurring in only 7% [2]. Although, the original trial used 6 cycles of adjuvant temozolomide after radiation, most of the ongoing trials incorporate 12 adjuvant cycles since serious adverse effects from prolonged use of temozolomide are uncommon. Recurrent malignant gliomas have a very poor outcome with most of the patients dying within 1 year [18]. If possible, surgical re-resection should be offered. Temozolomide has been used in recurrent GBM but the 6-month PFS is 18-21% and median OS only 5.4 months [19, 20]. Temozolomide is utilized in recurrent gliomas, particularly if the duration of remission from the last use of Temozolomide is 6 months or longer. Some groups have proposed the use of alternative schedules such as continuous low dose “metronomic” temozolomide [21]. Other treatment options such as bevacizumab are also available.

**MGMT Promoter Methylation**

The MGMT (O6-methylguanine–DNA methyltransferase) gene encodes a protein responsible for repairing deoxyribonucleic acid (DNA). Methylation of the MGMT promoter region leads to decreased expression of this protein and less repair by the removal of alkyl groups induced by chemotherapeutic agents. Tumor cells with a high level of MGMT protein are resistant to cytotoxic chemotherapy whereas cells lacking MGMT protein remain

susceptible [22, 23]. Epigenetic silencing of the MGMT gene by promoter methylation is associated with an increased overall survival in GBM patients receiving temozolomide and other alkylating agents [24, 25]. In the aforementioned EORTC-NCIC trial, MGMT promoter methylation status was determined retrospectively in 206 patients and was found to be the strongest predictor of survival [16]. Patients with a methylated MGMT promoter benefited the most from combined temozolomide and radiation (5yr OS of 14% versus 5% for those receiving radiation alone). MGMT gene promoter methylated patients who received concurrent chemoradiation had a median OS of 21.7 months, while those who received radiation only had a median OS of 15.3 months [17]. Among the unmethylated patients, benefit from concurrent chemoradiation versus radiation alone was smaller but still statistically significant. Patients with methylated MGMT promoter did better irrespective of whether they received concurrent chemoradiation or radiation alone. Hence MGMT promoter methylation is both a favorable prognostic factor for survival as well as a predictive factor for temozolomide sensitivity. Methylation status should be determined by polymerase chain reaction (PCR) in all newly diagnosed GBM patients and future clinical trials should stratify patients according to methylation status as this likely represents two distinct patient groups with different prognoses.

## **Resistance to Chemotherapy**

One of the proposed mechanisms of resistance to temozolomide chemotherapy in recurrent or refractory GBM is the increased expression of MGMT [26]. Tolcher et al. suggested that MGMT activity could be effectively suppressed by the continuous administration of temozolomide at a low dose, known as “metronomic chemotherapy.” [21]. One retrospective study reported that continuous temozolomide (50mg/m<sup>2</sup>/day) in patients with recurrent or progressive GBM after conventional 5-day temozolomide was well tolerated and effective [27]. This benefit was confirmed in another phase II clinical trial [28]. These studies were conducted prior to the approval of bevacizumab for recurrent GBM and this strategy seems to be ineffective after progression on bevacizumab [29]. Another approach has been to combine chemotherapy with MGMT inhibitors such as the pseudosubstrate O(6)-benzylguanine [26]. Unfortunately use of such inhibitors has been limited by the associated hematologic toxicity [30, 31]. Gene therapy to reduce MGMT activity is also being attempted [32]. The DNA repair enzyme poly ADP-ribose polymerase (PARP) may be partly responsible for chemotherapy resistance and various PARP inhibitors are currently under investigation [33].

## **MOLECULAR TARGETED THERAPIES**

By far, the most exciting advance has been our increased understanding of molecular pathways underlying the growth of GBM. These signaling pathways offer potential targets for the treatment of malignant gliomas using molecularly directed therapies. A wide variety of agents directed against angiogenesis, the growth factor pathways (EGF, PDGF etc.), their receptors, and downstream intracellular second messengers are being investigated. Most of

these targeted agents are either monoclonal antibodies or small-molecule tyrosine kinase inhibitors (TKIs). So far, the antiangiogenic therapies seem to be the most promising. In May 2009, bevacizumab, a humanized monoclonal antibody against the vascular endothelial growth factor (VEGF) ligand, received accelerated FDA approval for the treatment of recurrent GBM as a single agent.

## ANTIANGIOGENIC THERAPIES

### Rationale for Antiangiogenic Therapies in Gliomas

Glioblastomas are highly vascular tumors that require angiogenesis, formation of new blood vessels, for their growth [34]. These newly formed vessels are dysfunctional and leaky with abnormal basement membranes, leading to localized areas of hypoxia and necrosis [35, 36]. Hypoxia, in turn, drives further angiogenesis through the production of proangiogenic cytokines. These growth factors not only stimulate local endothelial cells but also recruit distant bone marrow derived and circulating endothelial cells to migrate to the tumor [37, 38]. VEGF is one of the most potent proangiogenic factors, but others are also involved, including hepatocyte growth factor (HGF), basic fibroblast growth factor (bFGF), interleukin 8 (IL-8), angiopoietins and platelet derived growth factor (PDGF). Higher grade gliomas express more VEGF, and increased levels of VEGF production directly correlates with a worse prognosis [39, 40]. VEGF binds to and stimulates the receptor tyrosine kinase, VEGFR-2, triggering a cascade of signaling events through the PI3K/Akt and Ras/MAPK pathways (Figure 2). Eventually, this leads to increased endothelial cell proliferation, migration, and survival.

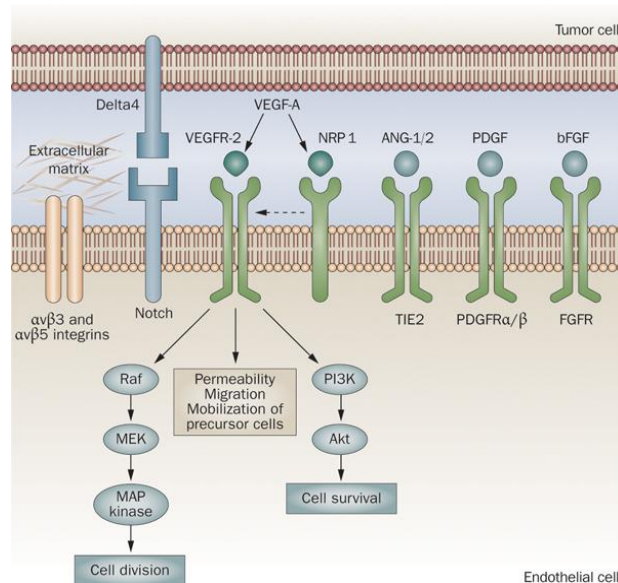


Figure 2. Angiogenic signaling pathways active in malignant gliomas. Reprinted by permission from Macmillan Publishers Ltd: Nat Rev Neurol. Norden AD, Drappatz J, Wen PY. Antiangiogenic therapies for high-grade glioma. 2009 Nov;5(11):610-20, copyright 2009.



Angiogenesis in malignant gliomas is also regulated by several other pathways, some of which are interlinked with VEGF. One such pathway is the angiopoietin-1/angiopoietin-2/tie-2 signaling pathway. Both angiopoietin-1 and angiopoietin-2 bind to the tie-2 tyrosine kinase receptor but have opposing effects on vascular stability and permeability. In the presence of VEGF, angiopoietin-2 destabilizes blood vessels to promote angiogenesis [41]. In preclinical studies angiopoietin-2 inhibitors have been shown to inhibit angiogenesis and may be of therapeutic value [42].

Similarly, the Notch-delta like ligand (Dll4) is a key regulator of angiogenesis [43]. VEGF upregulates the expression of Dll4, which in turn functions as a negative feedback regulator of VEGF by activating the Notch receptor on endothelial cells. Interestingly, inhibition of Dll4 promotes dysfunctional neovascularization with a paradoxical decrease in tumor growth due to reduced perfusion and hypoxia [44]. Other potential therapeutic targets include the transmembrane integrins  $\alpha v \beta 3$  and  $\alpha v \beta 5$ , which are involved in invasion as well as angiogenesis [45, 46].

## **Mechanism of Action of Anti-VEGF Therapy**

VEGF inhibitors are the most extensively studied antiangiogenic agents. Although still incompletely understood, these agents are thought to inhibit tumor growth by several mechanisms. Intuitively, anti-VEGF drugs inhibit neovascularization, thereby depriving tumors of essential blood supply, nutrients and oxygen. Another antitumor mechanism is “vascular normalization” of tumor vessels. Tumor vasculature differs significantly from normal blood vessels in that tumor vessels are more dilated, irregularly shaped, tortuous, leaky, and hemorrhagic. In preclinical models, VEGF inhibitors transiently “normalize” the structure and function of tumor vasculature, improving blood flow and oxygenation. This results in enhanced chemotherapy delivery and increased sensitivity to radiation [47]. VEGF is induced by ionizing radiation and seems to have a radioprotective effect on tumor vessels. By counteracting this surge in VEGF, antiangiogenic agents may increase the radiosensitivity of the tumor vasculature and potentiate effects of radiation [48]. Additionally, VEGF inhibitors can dramatically reduce tumor-associated vasogenic edema by decreasing the permeability of aberrant vessels [49]. Many patients on VEGF inhibitors are able to either completely stop or decrease the dose of corticosteroids, thus avoiding potential toxicities. There is also some evidence to suggest an improved survival with control of edema, even without an accompanying change in tumor growth [49]. Finally, anti-VEGF agents may have anti-glioma effects by targeting glioma stem cells. These highly chemo and radioresistant cells are self-renewing cells thought to be critical for tumorigenesis [50, 51]. Glioma stem cells are maintained by complex interactions with endothelial cells in “perivascular niches” [52]. By disrupting these perivascular niches, VEGF inhibitors can selectively target glioma stem cells and thus impact glioma growth [50].

## **Bevacizumab in Recurrent GBM**

After the success of bevacizumab and irinotecan in colon cancer, several investigators began to study the same combination in recurrent malignant gliomas. One of the initial

studies reported a remarkable radiographic response in 9 of 21 patients (43%) [53]. This was far superior to the historical 5-8 % response rate with temozolomide in recurrent GBM [19, 20].

Despite safety concerns, bevacizumab was well tolerated with only two treatment-related deaths; one due to intracranial hemorrhage and the other due to intestinal perforation. Other toxicities were similar to those observed in solid tumor patients receiving bevacizumab and were manageable. An additional advantage was the steroid sparing effect due to a decrease in peritumoral vasogenic edema. Several subsequent studies confirmed these results and set the stage for prospective studies. Vredenburgh et al. reported the first prospective phase II trial of bevacizumab and irinotecan in recurrent malignant gliomas [54]. They reported a 63% radiographic response in 32 patients (23 with GBMs and 9 with anaplastic gliomas) with a median PFS of 23 weeks for all patients. Of the 23 GBM patients, 14 (61%) had a response and the median PFS was 20 weeks, whereas 6 (67%) of the 9 anaplastic glioma patients had a response with a median PFS of 30 weeks. No patient experienced intracranial bleeding, although 4 patients developed thromboembolic complications and 2 had to discontinue therapy secondary to proteinuria. In another prospective phase II study, the same group reported a response in 20 of 35 (57%) recurrent GBM patients with a 6 month PFS of 46%. One patient developed intracranial hemorrhage and 4 had thromboembolic complications [55].

In 2009, the FDA provisionally approved bevacizumab monotherapy for the treatment of recurrent GBM based on the results of two phase II trials. The BRAIN study randomized 167 patients to receive either single agent bevacizumab 10mg/kg q2 weeks (n=85), or the combination of irinotecan and bevacizumab (n = 82) [56]. In the bevacizumab alone arm, the objective response rate (ORR) was 28% and the 6 month PFS 43%. For the combination arm, the ORR was 38% and 6 month PFS 50%. Median OS for the bevacizumab arm was 9.2 months, and 8.7 months for the combination arm. Updated results demonstrated that 11% and 16% of the patients were alive at 30 months in the two arms, respectively. Most of the patients were able to reduce corticosteroid doses by 50% or more. As expected, grade 3 or 4 toxicities were higher in the combination arm. This study was not powered to detect a difference between the two arms, so the FDA only utilized data from the bevacizumab arm for the purpose of approval.

The second trial was a single institution study at the NCI which evaluated bevacizumab as a single agent in recurrent GBM and reported an ORR of 35% with a 6 month PFS of 29% [57]. The exact contribution of irinotecan when used in combination with bevacizumab is unclear. Irinotecan was added to bevacizumab in GBM trials based on the extrapolation of metastatic colorectal cancer data, where the combination was found to have a survival advantage over bevacizumab alone. Albeit a non-comparative trial, the BRAIN study showed comparable OS rates for the two groups with increased toxicity for the combination. Currently, bevacizumab is approved for use in recurrent GBM as a single agent only. Several subsequent studies have demonstrated remarkable radiographic responses and improved PFS with bevacizumab; however, survival data from active phase II and phase III bevacizumab trials is still pending. Interestingly, bevacizumab is also being evaluated in combination with hypofractionated stereotactic re-irradiation for small recurrent gliomas. In a pilot study, a 50% response rate and 6 month PFS of 65% were reported with surprisingly minimal toxicity in 25 patients with  $\leq 3.5$ cm recurrent malignant gliomas [58].

## Bevacizumab in Newly Diagnosed GBM

Preclinical studies suggest that bevacizumab may have synergism with radiation [47]. In light of this data, bevacizumab is being studied in newly diagnosed GBM patients receiving concurrent chemoradiation. Early reports suggest that the addition of bevacizumab may significantly prolong PFS; however, final survival data is pending. In one phase II trial, the median PFS was 10 months longer for the bevacizumab plus radiation arm (17 versus 7 months) [59].

Risks must be carefully balanced with potential benefits for these patients, as bevacizumab is associated with delayed wound healing and bleeding. Additionally, there is concern that the use of antiangiogenic agents may promote a more invasive tumor pattern, and thus potentially shorten survival [5]. Two large placebo-controlled phase III trials looking at the use of bevacizumab in newly diagnosed GBM are underway in the US (RTOG 0825) and worldwide (AVAglio) [60]. Until results are available, this approach should be considered experimental and not used outside of clinical trials.

## Toxicity

Antiangiogenic therapies are generally well tolerated and most of the associated toxicities are transient. The incidence of serious adverse events is increased when bevacizumab is combined with chemotherapy. In the BRAIN study, the rate of  $\geq$  grade 3 events was 46% in the bevacizumab-alone arm versus 66% in the bevacizumab-irinotecan arm [56]. The most common adverse effects include hypertension, proteinuria, low-grade bleeding (epistaxis) and delayed wound healing. Most of these side effects are thought to be a direct result of inhibition of the VEGF pathway in normal tissues. For instance, hypertension is induced by decreased production of the potent vasodilator, nitric oxide. Under normal circumstances, VEGF enhances the production of nitric oxide by upregulating nitric oxide synthase in endothelial cells [61]. Bevacizumab blocks this process by depleting VEGF, thereby causing hypertension [62].

Serious adverse events have been reported, but are fortunately rare (<3%). Gastrointestinal perforation, reversible posterior leukoencephalopathy syndrome (RPLS), cardiac failure, stroke, and life-threatening hemorrhage are some of the dreaded complications. Gastrointestinal perforation is more commonly seen in patients with ovarian and colorectal cancer [63]. Although minor bleeding (e.g. epistaxis) is common, the incidence of life threatening intracranial hemorrhage is low. In one prospective study (n=167), only one of the five patients who experienced intracranial hemorrhages had a grade 4 bleed [56]. This is similar to the rate of spontaneous hemorrhage in patients with GBM not receiving bevacizumab [64]. Relatively high rates of thromboembolic complications (deep venous thrombosis, pulmonary embolism, arterial thrombosis) have been reported with the use of bevacizumab (11.4 to 12.7%) [54, 55]. However, this data must be cautiously interpreted because thrombotic complications are common in patients with malignant gliomas even without treatment [65]. The presence of GBM is a well-known risk factor for thrombosis. In conclusion, despite a small risk for life-threatening complications, bevacizumab has an acceptable safety profile in malignant glioma patients.

## Practical Considerations

Monitoring response to antiangiogenic therapy can be challenging. Malignant gliomas have dysfunctional blood vessels which leak contrast and enhance on CT and MRI. The Macdonald criteria, most commonly used for assessing response to treatment, utilize the extent of contrast enhancement in addition to corticosteroid use and neurological function [66]. Antiangiogenic agents decrease vascular permeability thereby reducing contrast enhancement on conventional imaging, often within days of starting therapy. This decrease in enhancement without a true anti-tumor effect has been referred to as a “pseudoresponse” [67].

Additionally patients progressing on anti-VEGF therapy can develop a more infiltrative, non-enhancing pattern of growth that is not readily detected by CT or contrast enhanced T1-weighted MRI, but appears as an area of increased T2 fluid attenuated inversion recovery (T2-FLAIR) signal abnormality (Figure 3) [68, 69]. In light of these shortcomings, new radiographic response criteria were recently proposed by the Response Assessment in Neuro-Oncology (RANO) Working Group. These criteria also incorporate the use of T2-FLAIR MRI sequences to evaluate response [69]. Alternative imaging modalities such as MRI perfusion imaging and positron emission tomography (PET) may be more useful to monitor radiographic responses in patients on antiangiogenic therapy [70-72].

## Mechanisms of Resistance

Despite remarkable responses with antiangiogenic therapy in some patients, benefits are transient and most patients eventually progress in a matter of months. Increasing evidence suggests that initially responsive malignant gliomas adapt to the presence of antiangiogenics and evolve mechanisms to evade angiogenic blockade [73]. Others have pre-existing resistance to antiangiogenic therapy and fail to respond from the outset. These modes of resistance have been referred to as “adaptive (evasive)” and “intrinsic” resistance, respectively [73]. At least four distinct mechanisms of resistance have been proposed: upregulation of alternative proangiogenic signaling pathways; recruitment of vascular progenitor cells from the bone marrow; protection of tumor vessels by increased pericyte coverage; and an enhanced invasiveness of tumor cells with co-option of existing normal vasculature [73].

In recurrent GBM patients, elevated levels of bFGF, SDF-1 $\alpha$ , and circulating endothelial cells have been noted at the time of progression on cediranib (a VEGFR TKI) [74]. It may be possible to overcome resistance by combining drugs (e.g. cilengitide and cediranib) that act on parallel pathways, or by novel agents that simultaneously inhibit the VEGF and alternative proangiogenic pathways (e.g. brivanib - an inhibitor of VEGFR and FGFR) [75]. Further studies to elucidate all mechanisms of resistance and strategies to overcome resistance are needed.

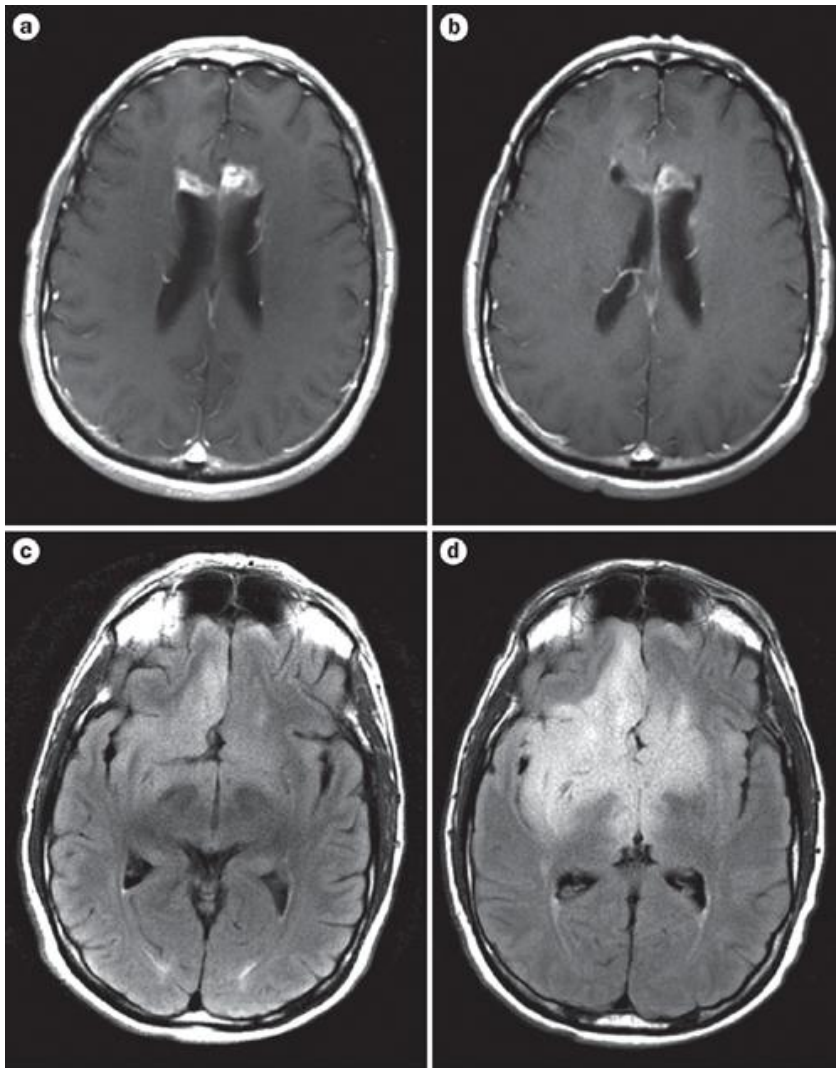


Figure 3. Post-gadolinium T1 MRI sequences obtained (a) immediately before and (b) 4 months after starting bevacizumab therapy. A partial response is evident, with marked reduction in size of the enhancing mass within the genu of the corpus callosum. (c), (d) Fluid-attenuated inversion recovery (FLAIR) MRI sequences obtained at the same respective time-points. A dramatic increase in the extent of FLAIR signal abnormality is observed, suggesting progression of non-enhancing tumor. Reprinted by permission from Macmillan Publishers Ltd: *Nat Rev Neurol*. Norden AD, Drappatz J, Wen PY. Antiangiogenic therapies for high-grade glioma. 2009 Nov;5(11):610-20, copyright 2009.

## Predictors of Response

At present there are no validated biomarkers to predict response to antiangiogenic therapy. Such biomarkers are essential not only to avoid potential toxicities in patients unlikely to benefit from antiangiogenic agents, but also to save valuable time and resources on ineffective therapies in patients who already have very limited survival. Several predictive markers have been proposed, including serum biomarkers (VEGF, b-FGF, collagen IV,

CD133+ circulating endothelial cells) [74, 76, 77]; tissue biomarkers (tumor levels of VEGF) [78]; and radiographic biomarkers (apparent diffusion coefficient (ADC) histogram analysis using MRI, [18F] fluorothymidine positron emission tomography (FL-PET)) [79, 80]. Some have even suggested incorporating serum and imaging biomarkers into a “vascular normalization index” to predict response [81]. This index utilizes changes in  $K^{trans}$  (a measure of vascular permeability/flow using MRI), micro-vessel volume, and serum collagen IV levels to predict response after a single dose of cediranib. The aforementioned biomarkers have been evaluated retrospectively and validation in prospective trials is necessary before their routine use. Furthermore, most of the proposed markers can only be utilized after initiating treatment with an antiangiogenic agent. A predictive biomarker is needed which can be measured prior to starting antiangiogenic therapy in an accurate, convenient, and reproducible fashion.

## Other Antiangiogenics

A variety of other antiangiogenic agents have been developed with encouraging results. These include aflibercept (VEGF-Trap), the tyrosine kinase inhibitors (cediranib, vandetanib, vatalanib, sorafenib, sunitinib and pazopanib) and cilengitide (an integrin inhibitor). A detailed discussion of each drug is beyond the scope of this chapter; however some of the more promising agents are highlighted and described below.

### Aflibercept (VEGF-Trap)

Aflibercept is a soluble VEGF receptor decoy fused with the immunoglobulin constant region (Fc) that scavenges both VEGF and placental growth factor (PlGF). With an affinity for the VEGF-A ligand several hundred times that of bevacizumab, it has shown potent antitumor effects in preclinical glioma models [82, 83]. Aflibercept monotherapy in recurrent malignant glioma was associated with a 30% response rate in one phase II trial. Unfortunately, this treatment was associated with significant toxicity and 25% patients had to discontinue therapy within 2 months. Further trials are underway with an emphasis on trying to identify biomarkers that could predict a response and select patients most likely to respond [84].

## Tyrosine Kinase Inhibitors

An alternative strategy of inhibiting the VEGF pathway is to target the VEGF receptor (VEGFR). Many tyrosine kinase inhibitors (TKIs) have been developed and studied in patients with malignant gliomas. Some TKIs are more selective for the VEGF receptor than others (e.g. cediranib). Currently, vatalanib, sorafenib, sunitinib and pazopanib have shown only limited anti-tumor activity in glioma patients [85-87].

## Cediranib (AZD2171)

Cediranib, an oral pan-VEGFR, PDGFR and c-KIT TKI, revealed significant efficacy in early studies. A phase II trial evaluating cediranib monotherapy in recurrent GBM, reported a 57% response rate and a 6 month PFS of 26% [88]. Therapy was well tolerated and the most common adverse reactions included hypertension, diarrhea, and fatigue.

There was a dramatic reduction in tumor-associated edema and all patients taking corticosteroids were able to either decrease or discontinue steroids altogether (Figure 4). A larger randomized phase III trial (REGAL) comparing three arms; cediranib monotherapy versus single agent lomustine versus the combination of cediranib and lomustine in recurrent GBM, was presented at the 2010 SNO annual meeting. Disappointingly no difference was noted between the three arms [89]. This lack of benefit has been attributed possibly to the lower dose of cediranib used in this trial compared to the phase II trials (30mg/day versus 45mg/day). Further trials of cediranib may be warranted before final conclusions regarding the role of cediranib in malignant glioma treatment are made.

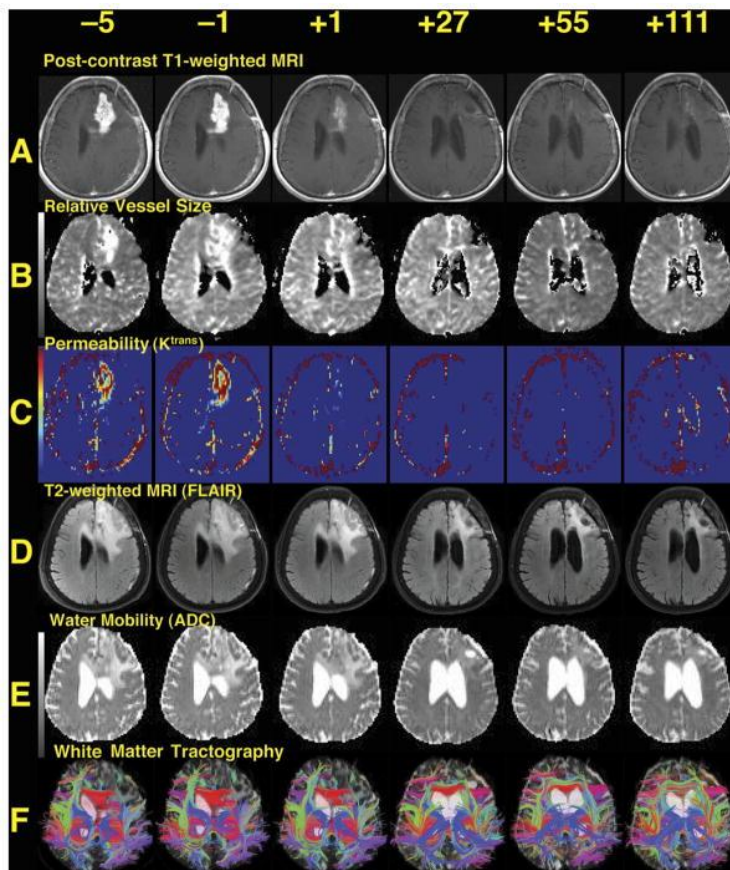


Figure 4. Imaging from a patient responding to cediranib. Reprinted with permission from Elsevier. Batchelor et al. AZD2171, a Pan-VEGF Receptor Tyrosine Kinase Inhibitor, Normalizes Tumor Vasculature and Alleviates Edema in Glioblastoma Patients. *Cancer Cell*. 2007 Jan;11(1):83-95. Copyright 2007.

## Cilengitide

Integrins are cell surface receptors widely expressed by glioma tumor cells and endothelial cells. Integrins promote angiogenesis and invasion by interacting with extracellular ligands [90]. Cilengitide is an inhibitor of the  $\alpha v \beta 3$  and  $\alpha v \beta 5$  integrins, and has been studied in GBM. Only a modest benefit was seen when used as monotherapy in recurrent GBM; however, significant synergistic activity was seen when combined with concurrent chemoradiation for newly diagnosed GBM [91, 92]. Similar to alkylating agents, patients with MGMT promoter methylation seemed to benefit most from cilengitide. A phase III trial comparing standard concurrent chemoradiation (temozolomide/radiation) with or without cilengitide in newly diagnosed GBM (CENTRIC) is currently accruing.

## OTHER MOLECULAR TARGETED THERAPIES

In addition to the angiogenesis pathway, a number of growth factor pathways are upregulated in GBM. This activation is accomplished by overexpression of growth factor receptors on malignant cells or by the amplification of growth factor receptor genes (EGFR, PDGFR, MET). Some cells harbor genetic mutations (EGFRvIII) that lead to constitutively activated receptors independent of ligands and uncontrolled growth in the absence of stimulating growth factors.

Alternatively, the downstream second messenger pathways may be constitutively activated when the negative regulators are lost by genetic mutations (PTEN tumor suppressor gene) or when positive signaling molecules acquire activating mutations (PIK3A). Therapeutic inhibitors of many of these activated growth factor pathways are available, and have shown promising results in preclinical models.

## EPIDERMAL GROWTH FACTOR RECEPTOR (EGFR) INHIBITORS

EGFR is a transmembrane tyrosine kinase receptor essential for normal growth, proliferation, and cell survival [93]. It is a logical potential therapeutic target, as EGFR is amplified and overexpressed in approximately 40-50% of primary GBMs [94]. Nearly half of the amplified tumors also harbor an EGFR mutation (most commonly EGFRvIII) resulting in ligand-independent constitutive activation with continuous downstream signaling and uncontrolled growth (Figure 5) [95]. Gefitinib and erlotinib are oral TKIs that reversibly block EGFR and have proven efficacy in non-small cell lung cancer [96, 97]. Both are well tolerated in patients with malignant gliomas, but responses rates to monotherapy are low. PFS and OS from these agents are no better than historical controls in unselected patients [98-102]. Attempts to identify predictive biomarkers and select patients likely to respond have been made, but EGFR expression, amplification, and mutation status do not seem to predict sensitivity to EGFR TKIs in GBM [99, 103].

One study suggested that tumors co-expressing EGFRvIII and PTEN were likely to respond [104], but a subsequent randomized phase II study by the European Organization for Research and Treatment of Cancer (EORTC) Brain Tumor Group failed to confirm these



findings [105]. Lapatinib, an oral EGFR/Her2 inhibitor approved for Her2 expressing breast cancer, has also been studied in GBM. Similar to other EGFR TKIs, it is also largely ineffective in malignant gliomas [106]. Preclinical studies suggested significant anti-tumor activity of cetuximab, an anti-EGFR monoclonal antibody [107]. Unfortunately in phase II clinical trials cetuximab only showed limited benefit [108], possibly due to its inability to penetrate the BBB. Current trials are focusing on novel inhibitors such as BIBW2992 (irreversible EGFR inhibitor), vandetanib (dual EGFR and VEGFR inhibitor), and nimotuzumab (monoclonal antibody against EGFR).

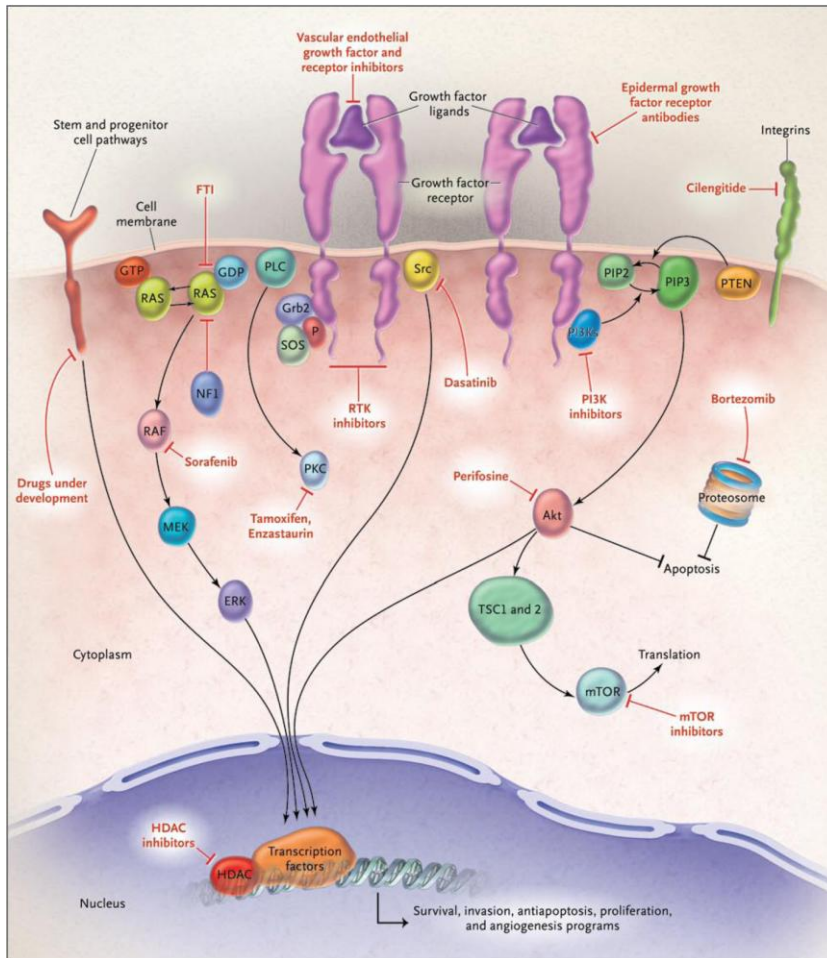


Figure 5. Growth factor and intracellular second messenger pathways upregulated in malignant gliomas. Reprinted with permission from the New England Journal of Medicine. Wen PY, Kesari S. Malignant Gliomas in Adults. 2008 Jul 31;359(5):492-507. Copyright © 2008 Massachusetts Medical Society. All rights reserved.

Other trials are combining EGFR inhibitors with mTOR inhibitors to simultaneously inhibit multiple targets. Given its radiosensitizing properties, cetuximab is currently being evaluated in combination with temozolomide and radiation in newly diagnosed GBM (GERT Study).

## **PLATELET DERIVED GROWTH FACTOR RECEPTOR (PDGFR) INHIBITORS**

Human malignant glioma cells almost universally co-express the PDGF ligand and its receptor (PDGFR), [109, 110] creating autocrine and paracrine loops that promote tumor cell proliferation and angiogenesis [110-112]. Imatinib mesylate, an oral PDGFR inhibitor, was reported to have significant antitumor activity in preclinical studies [113]. Unfortunately, clinical trials failed to show efficacy with imatinib monotherapy in unselected recurrent malignant glioma patients. In a phase I/II study, the North American Brain Tumor Consortium (NABTC) reported a 6-month PFS of only 3% in patients with GBM and 10% in those with anaplastic glioma. Five patients developed intracranial hemorrhages on imatinib 800mg/day [114]. Another phase II trial by the EORTC also reported limited antitumor activity with a 6-month PFS of 16% in recurrent GBM patients [115]. Single-agent imatinib has minimal anti-glioma activity; however some studies suggested significant activity when combined with the ribonucleoside diphosphate reductase inhibitor, hydroxyurea [116].

Hematologic toxicity was the most frequently observed adverse effect but the combination was well tolerated. A 6-month PFS of 27% and a radiographic response rate of 9% were reported in a phase II study of patients with recurrent GBM [117]. In one of the very few randomized phase III trials conducted in recurrent GBM, combination therapy with imatinib and hydroxyurea was compared with hydroxyurea monotherapy in 240 patients [118]. Unfortunately, no clinically meaningful differences were seen between the two treatment arms and the study failed to meet its primary endpoint. The median PFS was only 6 weeks for both arms ( $p=0.56$ ) and the 6-month PFS was also similar in both groups (5% versus 7% in the monotherapy arm). Although larger studies have failed to confirm significant activity in unselected patients with GBM, anecdotal evidence of durable response in some patients suggests a role in select patients. To date no biomarker has been identified that reliably predicts response to imatinib therapy, although c-KIT polysomy or amplification status may be potential candidates [119]. The number of GBM patients with c-KIT overexpression is low and the predictive value of c-KIT has yet to be validated in larger prospective studies. Other less selective PDGFR inhibitors including dasatinib (also a SRC inhibitor), sorafenib, and sunitinib are also being evaluated in malignant glioma clinical trials [120, 121].

## **INTRACELLULAR SIGNALING PATHWAY INHIBITORS**

Ligand binding activates cell-surface growth factor receptors which transduce signals through downstream intracellular signaling pathways such as the PI3K/Akt/mTOR and RAS/RAF/MAPK kinase. Intracellular signaling pathways are critical regulators of tumor growth, metabolism, proliferation, and invasion. These pathways are common to multiple growth factor receptors and are thus potentially suitable targets for molecularly targeted agents.

## PI3K/Akt/mTOR

One of the key second messengers activated by growth factor receptors is phosphatidylinositol 3-kinase (PI3K), which leads to activation of Akt and other downstream targets. These mediators ultimately activate mammalian target of rapamycin (mTOR), a crucial kinase that regulates protein synthesis (Figure 5). PI3K/Akt/mTOR signaling is frequently increased in GBM due to upstream growth factor overactivity (EGFR, PDGFR and MET), oncogenic mutations of PI3K, or loss of PTEN. Under normal circumstances the tumor suppressor PTEN inhibits Akt activation by PI3K. PTEN is mutated in almost 40% of GBMs [122]. Therapeutic drugs targeting several steps of this pathway have been developed and an oral Akt inhibitor, perifosine, has shown some activity in preclinical studies [123]. Direct PI3K inhibitors are being evaluated in clinical trials. PKI-587, a dual PI3K/mTOR kinase inhibitor has shown promising efficacy within *in vitro* and xenograft models [124]. Another dual PI3K/mTOR inhibitor, XL765, is also being evaluated [125]. The mTOR inhibitors are the most extensively studied PI3K/Akt pathway inhibitors. Temsirolimus and everolimus have been approved for a variety of malignancies; however, mTOR inhibitors have shown minimal efficacy as monotherapy in phase II trials of unselected recurrent malignant gliomas [126, 127]. Given the low toxicity profile and *in vitro* evidence of synergism between upstream growth factor inhibitors and downstream mTOR inhibitors, such combinations have been studied [128]. Although simultaneous administration of everolimus and gefitinib in recurrent GBM patients was well tolerated in one pilot study, the response rate was only 14% and no significant clinical benefit was seen [129].

## RAS/RAF/MAP Kinase

Another second messenger pathway employed by EGFR and PDGFR is the RAS pathway (Figure 5). RAS oncogene is mutated and constitutively activated in a number of solid tumors but such mutations are uncommon in malignant gliomas [130]. Nonetheless, the RAS pathway is activated in malignant gliomas due to increased upstream signaling from growth factor receptors and is a potential therapeutic target. Farnesylation is an essential step prior to activation of RAS. This allows RAS to translocate to the cell membrane and activate downstream effectors. Tipifarnib is a farnesyl transferase inhibitor with antitumor activity in a variety of malignancies [131]. Unfortunately, it has only shown modest activity as a single agent in patients with recurrent GBM and trials combining it with other agents as well as radiation are ongoing [132]. Inhibitors of RAF and MEK, which lie downstream of RAS, are also being evaluated in patients with malignant gliomas.

## Protein Kinase C (PKC)

Protein kinase C signaling pathway activity is also increased in malignant glioma and levels correlate with the growth rate *in vitro* [133]. Preclinical studies have shown that PKC inhibitors are able to block glioma proliferation and potentiate the effects of chemotherapy [133, 134]. Tamoxifen is an antiestrogen that inhibits PKC at higher doses. Tamoxifen

inhibits glioma tumor growth *in vitro* [135], but clinical studies have not shown a clear benefit in recurrent malignant gliomas even when combined with chemotherapy [136, 137].

Although well tolerated, high dose tamoxifen given concurrently with radiation in newly diagnosed GBM did not improve overall survival [138]. Enzastaurin, an inhibitor of PKC and the PI3K/Akt pathway, induces apoptosis and suppresses proliferation of glioma cells both *in vitro* and in xenograft models [139]. Based on promising phase II data in recurrent malignant gliomas (ORR 18%) [140], a phase III study comparing enzastaurin versus lomustine in recurrent GBM was initiated. Unfortunately, the study was closed after accruing 266 patients when a planned interim analysis demonstrated lack of efficacy [141].

## HISTONE DEACETYLASE (HDAC) INHIBITORS

Histone deacetylases (HDAC) are enzymes that modify chromatin and contribute to aberrant gene expression in cancer (Figure 5). Inhibitors of HDAC are promising anti-neoplastic agents. They inhibit growth, promote differentiation, and induce apoptosis in glioma cells [142]. Vorinostat, an HDAC inhibitor, has shown modest efficacy as monotherapy in recurrent GBM. In one phase II trial, the 6-month PFS was 15.2%; a subgroup of patients had stable disease for a prolonged duration [143]. Efforts to identify predictors of response are ongoing. Valproic acid is an antiepileptic drug that also inhibits HDAC.

A recent retrospective study has suggested a survival advantage for newly diagnosed GBM patients receiving valproic acid and temozolomide-based chemoradiotherapy [144]. Given the ability of HDAC inhibitors to sensitize glioma cells to temozolomide and radiation [145], current trials are evaluating the combination of temozolomide, radiation, and HDAC inhibitors (vorinostat and valproic acid). The novel HDAC inhibitor, panobinostat, has also shown some activity in combination with bevacizumab [146]. Other classes of drugs with diverse mechanisms of action including proteasome inhibitors (e.g. bortezomib), CD95 ligands, and novel ligand-toxin conjugates are also under investigation.

## RESISTANCE TO TARGETED THERAPY AND THE ROLE OF STEM CELLS

Despite promising results in preclinical studies, so far clinical trials with nonangiogenic targeted agents have not been as encouraging. Most drugs have limited efficacy when used as monotherapy in the recurrent setting. One plausible explanation is the lack of patient selection and inclusion of patient populations that were too heterogeneous at a molecular level. It is increasingly apparent that although the histopathologic features may be similar for GBMs, there are different molecular subgroups, each with unique outcomes and responses to treatment. Using gene expression profile studies, GBM patients can be molecularly subtyped on the basis of genes which are under or over-expressed within each group [147]. By defining a common pathobiology within each subgroup, this molecular classification may allow us to tailor therapies against specific aberrant pathways and expect more homogeneous and predictable responses to treatment. To this end, predictive biomarkers to help guide therapy

within each subgroup are also needed. Anecdotal reports of response suggest a role, although identification and selection of likely responders is essential.

Accumulating evidence suggests that there is co-activation of multiple parallel pathways in GBM [148]. This limits the efficacy of agents blocking one pathway and provides a rationale for combining agents acting on different pathways. Alternatively, novel agents inhibiting multiple pathways can be employed. Indeed, trials evaluating various combinations of inhibitors and multitargeted agents are ongoing, and results are eagerly anticipated. It is also evident that there is significant redundancy and crosstalk between the intracellular signaling pathways, leading to downstream signaling despite inhibition of some components of a particular pathway. One strategy to overcome resistance is to combine inhibitors that act at different points in the same pathway, simultaneously targeting upstream and downstream mediators. A major hurdle in combining drugs is overlapping and often prohibitive additive toxicity. Both EGFR inhibitors and mTOR inhibitors inhibit the PI3K/Akt pathway and are well tolerated when given as single agents; however, a high incidence of rash and mucositis were seen when erlotinib was given concurrently with temsirolimus [149]. In another trial, erlotinib in combination with sirolimus was well tolerated, but had negligible activity among unselected recurrent GBM patients [150]. Few trials have actually studied the levels of molecularly targeted drugs in the brain tissue. Inability to achieve sufficient drug levels across the BBB may be an important factor for the apparent inefficacy in clinical trials despite activity in preclinical studies. Even though the BBB is only partially intact in the tumor itself, malignant gliomas are considered diseases of the entire brain. It is hypothesized that some invasive glioma cells are able to escape and hide in sanctuaries within the brain. These cells are shielded by intact BBB and eventually lead to relapses [151]. It is crucial to analyze tumor samples to assess drug levels and to identify ways to improve drug delivery across the BBB. This will likely improve clinical responses in future trials.

A major advance in oncology has been the recent recognition of cancer stem cells in a variety of tumors including GBM. Glioma stem cells are multipotent, undifferentiated, self-renewing cells that resemble normal neural stem cells; however, an important distinction is that glioma stem cells are transformed and lack regulation of growth and differentiation. Although present in small numbers within malignant gliomas, stem cells are critical for tumorigenesis. When transplanted into mice, they result in highly invasive tumors resembling human GBM [152]. The exact origins and identity of these stem cells are still under debate. Some studies suggest they are derived from neural stem cells whereas others suggest they originate from de-differentiation of more mature brain cells [153]. Multiple identifying markers including CD133/Prominin, Nestin, Sox-2 and Musashi-1 have been proposed; yet, there is no consensus and the exact phenotype remains obscure. Evidence suggests that the highly chemo and radioresistant glioma stem cells may be responsible for therapeutic resistance and recurrences in malignant gliomas. High levels of the DNA repair protein MGMT, increased expression of multidrug resistance proteins, and inhibition of apoptosis all contribute to the chemoresistance of glioma stem cells [154]. Radioresistance is promoted by activation of the DNA damage repair and response pathways [51]. Therapies directed against glioma stem cells may result in a long term cure in malignant gliomas.

## TARGETING GLIOMA STEM CELLS

Glioma stem cells are dependent upon a number of dysregulated pathways (e.g. Notch, Sonic Hedgehog, Wnt- $\beta$ -catenin, PI3K) for their growth, indefinite regenerative capacity, and therapeutic resistance. By identifying the underlying molecular mechanisms and inhibiting them, we may be able to overcome resistance and develop highly effective therapies. One potential pathway is the Notch pathway. Through interactions with its ligands (Jagged and Delta-like ligands), the Notch signaling pathway maintains normal neural stem cells, promotes self-renewal, and suppresses differentiation [155]. Aberrant Notch activity is seen in glioma stem cells and is at least partly responsible for resistance to chemotherapy and radiation [156]. Notch inhibitors (GSIs –  $\gamma$  secretase inhibitors) not only induce apoptosis and differentiation, but also sensitize glioma stem cells to chemotherapy and radiation [156, 157]. Combining Notch inhibitors with chemoradiotherapy can potentially improve the efficacy of current treatment modalities and induce long term remissions. Another pathway of particular interest is the sonic hedgehog (SHH) pathway, which is required for sustained glioma growth and survival.

Hedgehog (HH) ligands bind to Patched (PTCH1) and release Smoothened (SMO), which in turn activates the GLI1 and GLI2 transcription factors to promote cellular proliferation (Figure 6). Although activating mutations of SHH are rarely seen in GBM, hedgehog signaling is increased in glioma stem cells [158]. Activated SHH contributes to chemoresistance of glioma stem cells [157]. Naturally occurring hedgehog inhibitors such as cyclopamine inhibit the tumorigenicity of glioma stem cells and enhance their sensitivity to chemotherapy [157, 159]. With the availability of synthetic hedgehog inhibitors (e.g. *vismodegib*), which are more stable and potent than cyclopamine, the SHH pathway offers a new therapeutic target for malignant gliomas. Dramatic responses in medulloblastoma have been seen in early clinical trials and further trials evaluating SHH inhibitors in GBM and other gliomas are pending. A rather unique strategy of targeting glioma stem cells is through promotion of differentiation. Similar to their normal counterparts, glioma stem cells retain their capacity to undergo differentiation when exposed to bone morphogenic proteins (BMPs). Exposure to BMP4 in particular, leads to a reduction in glioma stem cells *in vitro* and abolishes their capacity to generate tumors in transplanted mice [160]. Through activation of Smad signaling, BMP4 induces differentiation in glioma stem cells, thereby causing them to lose their tumorigenic potential and differentiate into committed mature cells unable to replicate. This effect has been seen in mouse models where BMP4 reduced tumor growth and prolonged survival [160].

Such an approach offers a less toxic and potentially more effective therapy to prevent growth and recurrence of malignant gliomas. Glioma stem cells are dependent upon their microenvironment for proper functioning. By disrupting these perivascular niches, antiangiogenic therapies form an integral part of the armamentarium against glioma stem cells [52]. Viruses may also play an important role in targeting glioma stem cells either by virotherapy or gene therapy. Virotherapy entails the identification of unique markers on glioma stem cells that can be preferentially targeted by therapeutic viruses, which in turn induce cell death via delivery of cytotoxic genes or direct virus-mediated-oncolysis [161]. Although it is not a form of chemotherapy, virotherapy is a potentially promising approach that is under investigation in early clinical trials.

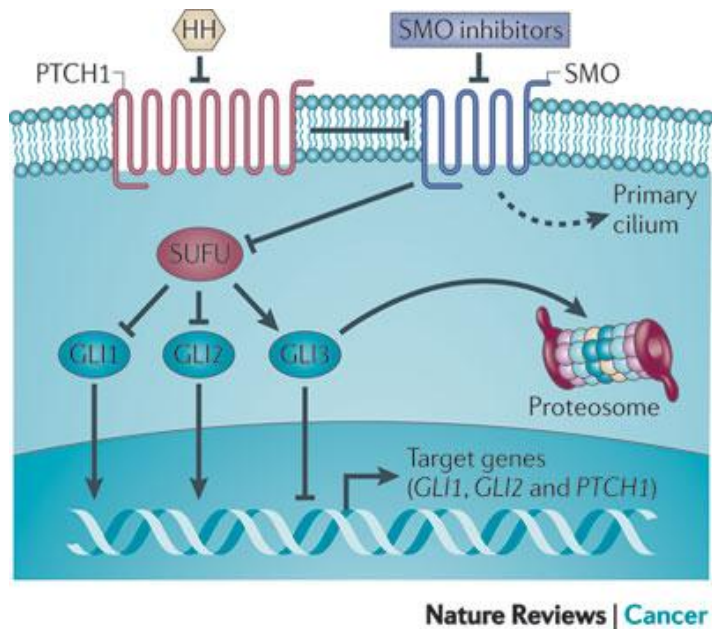


Figure 6. The Sonic hedgehog pathway. Reprinted by permission from Macmillan Publishers Ltd: Nat Rev Cancer. Ng JM, Curran T. The Hedgehog's tale: developing strategies for targeting cancer. 2011 May 26;11(7):493-501, copyright 2011.

## CONCLUSION

Over the last decade, there has been a tremendous increase in our knowledge and understanding of the complex molecular mechanisms underlying malignant gliomas. While this has spurred the development of a wide variety of novel and exciting targeted agents with activity against gliomas, no single agent is effective in all glioma patients. By identifying predictive molecular biomarkers and further developing genetic profiling of malignant gliomas, we may be able to personalize treatment and select more rational and effective drugs. With an improved understanding of the mechanisms of resistance and the role of glioma stem cells in tumorigenesis and recurrence, hopefully in the near future we will be able to develop more potent targeted therapies with a realistic chance of curing malignant gliomas.

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*Chapter 12*

## **ANTIANGIOGENIC THERAPIES FOR HIGH-GRADE GLIOMA**

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### **ABSTRACT**

Glioblastoma is among the most vascular of all neoplasms through stimulation of angiogenesis, the derivation of new blood vessels from existing vasculature. Tumor associated blood vessels are characterized by markedly abnormal structure and function making an attractive target for therapeutic intervention. Antiangiogenic agents, particularly those that target vascular endothelial growth factor (VEGF) and its receptors, have demonstrated efficacy and tolerability in recent years and have attracted significant interest as potential therapies for high-grade gliomas. Bevacizumab, a humanized monoclonal antibody against VEGF, received accelerated approval by the FDA in 2009 for treatment of recurrent glioblastoma. Bevacizumab prolongs progression free survival and decreases steroid dependence by means of controlling peritumoral edema. Many other antiangiogenic agents are currently under investigation. However, responses are rarely durable as resistance to antiangiogenic therapy develops and eventual tumor progression is nearly universal. Tumor progression on antiangiogenic agents has rapidly become a challenging problem as no effective salvage therapy has been identified.

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## INTRODUCTION

High-grade gliomas, in particular the most malignant subtype glioblastoma (GBM), are rapidly proliferative aggressive tumors with extensive local invasion, high vascularity and universal disease progression despite current therapies. Both microvascular proliferation and necrosis are pathologic hallmarks of GBM. Annually, an estimated 24,070 primary malignant brain tumors are diagnosed of which two-thirds are high-grade gliomas [1, 2]. Radiation therapy has long been the cornerstone of treatment for high-grade gliomas. The value of chemotherapy historically has been debated until 2005 when temozolomide was shown to confer a survival advantage in a pivotal phase III trial in patients with newly diagnosed GBM. The addition of temozolomide to radiation therapy has extended the median survival from 12.1 to 14.6 months [3]. Standard of care for patients with newly diagnosed GBM since this landmark trial has consisted of maximum safe resection followed by radiation therapy with concurrent and adjuvant temozolomide. Unfortunately, despite current optimal therapy, it is nearly universal for these tumors to recur necessitating subsequent therapy in the recurrent setting. Fewer than 10% of patients with glioblastoma are alive beyond 5 years after diagnosis and more than 70% do not survive beyond 2 years [4]. Patients with the less common anaplastic glioma have a more favorable median survival of 2-5 years, yet these tumors are also incurable [5-7]. When these tumors eventually recur, survival outcomes are often dismal as most patients with recurrent disease die within one year [8]. At time of recurrent disease in glioblastoma, only 21% of patients achieve a 6-month progression-free survival (PFS6) following chemotherapy with temozolomide and in patients with anaplastic glioma the PFS6 is 45% [9, 10]. Historically, after progression, salvage therapies for GBM have achieved unfavorable radiographic response of 5% and PFS6 of 9-15% [8, 11, 12]. The lack of favorable response rates at time of recurrence has left ample room for implementation of novel treatments to further improve outcomes. Angiogenesis has since emerged as an attractive therapeutic target.

Inhibition of angiogenesis has long been hypothesized to be an attractive means of curtailing tumor growth and has recently become a mainstay of therapeutic strategy for several types of cancer [13-18]. Antiangiogenic agents, particularly drugs that target the vascular endothelial growth factor (VEGF) pathway, are aimed at exploiting the highly vascularized nature of high-grade gliomas and have increasingly become incorporated as second-line therapy for recurrent disease in GBM. Bevacizumab (Avastin®; Genetech, South San Francisco, CA), a humanized monoclonal antibody directed against VEGF-A, has previously been approved by the FDA for use with cytotoxic chemotherapy in metastatic colorectal cancer, advanced or metastatic non-small cell lung cancer (NSCLC) and metastatic renal cell carcinoma. The initial accelerated indication for metastatic breast cancer was later revoked in 2011 by the FDA given lack of survival benefit based on subsequent clinical trial data and continues to be strongly debated [19-21]. In May 2009, the FDA granted accelerated approval of bevacizumab for the treatment of recurrent GBM based upon encouraging rates of radiographic response and progression-free survival in several retrospective and prospective studies [22-28]. This approach in the recurrent setting has been widely utilized along with clinical trial therapeutics yet has not demonstrated an overall survival benefit and has introduced new challenges. Initial enthusiasm has since been tempered given modest improvements in overall survival, the invariable resistance that develops in response to

antiangiogenic therapies, difficulties assessing radiographic response and tumor infiltration, and inability to identify effective therapy after bevacizumab failure as well as biomarkers to guide use of such agents. In this chapter, we review the current status and challenges of antiangiogenic therapies for patients with high-grade gliomas.

## ANGIOGENESIS IN HIGH-GRADE GLIOMAS

Glioblastoma (GBM) is the most aggressive of the high-grade gliomas and is amongst the most angiogenic of malignancies; in fact microvascular proliferation is one of the pathologic hallmarks of GBM. Angiogenesis, a key regulator of growth in GBM, is the process of new blood vessel formation from existing vasculature by endothelial cell migration and proliferation. While required for natural physiologic processes, angiogenesis is critical for tumor growth given the constraints of oxygen and nutrient diffusion and has since 1971, been hypothesized by Folkman and others to be an attractive therapeutic target [15]. Significant data since has supported the concept that the growth of malignant gliomas is dependent on angiogenesis [29]. These tumors stimulate the process of new blood vessels through processes driven largely by vascular endothelial growth factor (VEGF)[30], yet are structurally and functionally abnormal. Glioblastoma vessels are tortuous, disorganized, are highly permeable and characterized by abnormalities in their endothelial cell wall and basement membrane [29, 31]. Cumulatively, these changes result in markedly abnormal vasculature and a blood-brain barrier that is significantly more leaky and permeable. This in turn, generates abnormal blood flow resulting in hypoxia, acidosis, necrosis and increased interstitial pressure within the tumor microenvironment. In response to hypoxia, new vasculature is generated through production of a variety of proangiogenic factors, most notably VEGF [30]. Other factors produced include basic fibroblast growth factor (bFGF), angiopoietins, platelet-derived growth factor (PDGF), interleukin 8, hypoxia-inducible factor 1 $\alpha$  (HIF1 $\alpha$ ), and hepatocyte growth factor (HGF) among others [32-37]. Other pathways promoting GBM growth include PTEN/PI3-kinase/Akt, MAPK/ERK, and Notch-Delta-like ligand 4 (DLL4) [17, 38-40]. VEGF, the most prominent angiogenic growth factor, is robustly expressed in high-grade gliomas and the level of production has been shown to correlate with the degree of malignancy, high-grade tumors producing more than 10 times as much as low-grade tumors [37]. Poor outcome in patients with GBM has also been associated with increased VEGF levels [41, 42]. Binding of VEGF to VEGF receptors (VEGFRs) on tumor blood vessels enhances permeability and activates endothelial cell proliferation, survival and migration [43]. VEGFRs are additionally expressed in solid tumors including GBM, as well as tumor endothelium, and are thought to function in an autocrine manner to promote tumor growth [44]. Most antiangiogenic therapies that have been evaluated to date target VEGF signaling. Angiogenic agents targeting such pathways were initially hypothesized to work by preventing the formation of new blood vessels and by pruning of existing blood vessels, depriving tumors of necessary oxygen and nutrients [16]. The process of vascular normalization has been proposed by Jain as a mechanism of antiangiogenic agents, transforming abnormal tumor vasculature to a normalized state by restoring the balance between pro- and anti-angiogenic signaling [29, 45, 46]. Normalization of vasculature may subsequently improve tumor perfusion and consequently delivery and efficacy of cytotoxic therapies.

Antiangiogenic therapy has also been proposed to selectively target glioma stem-like cells, which have been demonstrated to be an important promoter of tumor angiogenesis through expression of VEGF [47-49]. While controversial, growing evidence has linked tumor angiogenesis to cancer stem cell biology. Glioblastoma stem cells are thought to give rise to tumors and are felt to be both refractory to traditional therapeutics and capable of regeneration following treatment [50]. The self-renewal of tumor stem cells is thought to underlie the remarkable resistance of gliomas to radiotherapy and chemotherapy. Recently, there has been growing evidence that tumor stem-like cells additionally arise from the tumor itself and contribute to the endothelial cells that form tumor vascularization [51, 52]. This concept of vasculogenic mimicry explains how some tumor cells acquire characteristics of vascular endothelial cells and line tumor blood vessels [53, 54]. These stem-like cells also thrive in a microenvironment in close association with tumor vasculature, referred to as the ‘vascular niche’ [55, 56]. The tumor microenvironment comprises a complex orchestra of signaling molecules and pathways influencing angiogenesis [57]. Disruption of tumor vasculature within this niche by antiangiogenic therapy may make these progenitor stem cells particularly sensitive to cytotoxic chemotherapy.

## ANTIANGIOGENIC THERAPIES

Multiple approaches to disrupting angiogenesis in tumors are in clinical development. These include monoclonal antibodies, tyrosine kinase inhibitors, small-molecule inhibitors, and transcription inhibitors. The majority targets the VEGF pathway and as newer agents are being evaluated, other proangiogenic molecules are being exploited. Currently, monoclonal antibodies and tyrosine kinase inhibitors are the most common drug classes to be investigated in clinical trials.

## TARGETING VEGF SIGNALING

### VEGF Inhibitors

#### ***Bevacizumab (Avastin)***

Bevacizumab (Avastin; Genetech Inc., South San Francisco, CA) is the most commonly used antiangiogenic therapy for GBM, having received accelerated approval by the Food and Drug Administration for patients with recurrent glioblastoma in 2009. It is a recombinant, humanized monoclonal antibody that targets VEGF, and was initially approved for use in combination with irinotecan and 5-FU-based chemotherapy in metastatic colorectal cancer [58]. It is additionally used in metastatic non-small cell lung cancer (NSCLC), and metastatic renal cell carcinoma. Bevacizumab exhibits linear pharmacokinetics with a half-life of approximately 20 days (range 11-50 days) and is administered every two to three weeks. Initial studies with bevacizumab in recurrent GBM demonstrated radiographic response rates of 28% to 40% and PFS6 of 40% to 50% [22–23, 25–27]. Prior studies in recurrent GBM have comparatively demonstrated a less favorable PFS6 of only 15% [8]. These dramatic results prompted two subsequent prospective phase II studies of bevacizumab and irinotecan

in recurrent malignant gliomas and a second in recurrent glioblastoma [25, 26]. These trials demonstrated a similar safety profile compared to other populations and confirmed the marked antitumoral effect achieved with bevacizumab and irinotecan. Radiographic response rates based on Macdonald criteria [59] was observed in 60% of patients and the PFS6 rates were 38% to 46% with an overall median survival of 40 to 42 weeks. Historic meta-analyses of salvage therapy in the recurrent setting in contrast is reported radiographic response rates of 5% to 10% and PFS6 rates of 9% to 15% with an overall survival of 22 to 26 weeks [8, 11, 12].

Based on these preliminary results, two subsequent phase II studies of patients with recurrent GBM were completed, prompting accelerated approval by the FDA of single-agent bevacizumab for recurrent glioblastoma in May 2009 [22, 27]. The first was a randomized trial of bevacizumab with or without irinotecan in recurrent GBM in 167 patients [22] and the second, a single-arm evaluation of bevacizumab in 48 patients [27]. The primary end point was PFS6 and both studies showed significantly better outcomes than historically reported with salvage therapy. Notably, the majority of patients were able to reduce their corticosteroid doses by 50% or more in both studies. Other VEGF inhibitors have also demonstrated the ability to decrease cerebral edema and minimize adverse effects of corticosteroids [60]. The FDA accelerated approval of bevacizumab was based upon the clinically meaningful and durable objective tumor response through independent radiographic review. The European Medicines Agency, however, did not approve bevacizumab primarily due to lack of a standard control arm in these studies and the failure to demonstrate a survival benefit [61].

Antiangiogenic agents may additionally enhance the efficacy of radiation therapy based on preclinical data [62]. From this suggestion, two large, placebo-controlled, phase III clinical trials are ongoing that compare radiation therapy, temozolomide and bevacizumab with conventional radiation therapy and temozolomide alone in newly diagnosed patients with GBM. The results of these trials, which are not yet available, will provide further data regarding the efficacy of bevacizumab in newly diagnosed patients with GBM. Despite improved response rates and progression-free survival with bevacizumab, recurrence is nearly universal. This has prompted studies evaluating a variety of bevacizumab regimens with combination chemotherapy including erlotinib, cetuximab, etoposide, histone deacetylase inhibitors and many others. Outcomes of combination regimens thus far have been comparable to that achieved with single-agent bevacizumab alone and many clinical trials of bevacizumab with and without other therapeutics are currently ongoing in patients with recurrent GBM [63].

## Other VEGF-Targeting Drugs

### *Aflibercept (VEGF-Trap)*

Aflibercept, also known as VEGF-Trap, is a soluble decoy VEGFR that is fused to an immunoglobulin constant region. Aflibercept binds to VEGF-A with a hundred-fold greater affinity than bevacizumab [64 65]. It additionally binds VEGF-B and placental growth factor (PIGF), which have both been implicated in tumor angiogenesis [66]. Data from a phase 2 trial of aflibercept in recurrent malignant gliomas demonstrated an overall radiographic response rate of 24% with minimal evidence for single-agent activity and significant toxicity for which notably 25% of patients discontinued therapy [67]. The main treatment related

toxicities included fatigue, hypertension, lymphopenia, CNS ischemia and systemic hemorrhage. Subsequent studies of aflibercept are ongoing to evaluate efficacy in newly diagnosed malignant gliomas with radiation therapy and temozolomide.

## **VEGF Receptor Inhibitors**

In addition to VEGF inhibitors, small molecule inhibitors of VEGFR have been evaluated in recurrent malignant gliomas. Cediranib (AZD2171) inhibits all subtypes of VEGFR and has been evaluated in a phase 2 trial of patients with recurrent GBM. Response rates were less than those reported for bevacizumab with a response rate of 26% and PFS6 of 26% [68]. All patients were able to reduce dosages of corticosteroids consistent with a striking steroid-sparing effect. Cediranib was well tolerated with hypertension, diarrhea and fatigue being the most commonly observed adverse effects. Imaging with dynamic contrast-enhanced and susceptibility MRI, reduced blood vessel size and permeability was reported during therapy with Cediranib. These data further support the concept that antiangiogenic therapy normalizes abnormal tumor vasculature [29]. Normalization of vasculature may result in improved chemotherapy delivery and reduced hypoxia supporting the concurrent use of antiangiogenic agents with cytotoxic chemotherapy and radiotherapy when they might be most effective [28]. However, a subsequent phase III study comparing cediranib to lomustine did not show improved efficacy [69].

Many other VEGFR inhibitors have been evaluated in recurrent malignant gliomas including CT-322, pazopanib, sorafenib, sunitinib, vandetanib, cabozantinib and E7080 [40, 63, 70-77]. In general the response rates and PFS6 have been inferior to bevacizumab with greater toxicity.

## **Other Antiangiogenic Approaches**

Vascular endothelial growth factor (VEGF) is one of the most important stimulants of angiogenesis in malignancy and the majority of angiogenic agents have been designed to target either VEGF or VEGFR. However, several other biologic pathways are involved in angiogenesis including angiopoietins, integrins, fibroblast growth factor (FGF), platelet-derived growth factors (PDGFs), scatter factor/hepatocyte growth factor and c-Met, and other proangiogenic factors have been considered as potential therapeutic targets and are under current investigation in clinical trials [77]. Additional agents under investigation include those that target multiple angiogenic growth factor pathways simultaneously by inhibiting several receptors, potentially yielding greater efficacy when compared to agents targeting a single growth factor. These agents may target angiogenic as well as other tumor cell signaling pathways, and combinations of agents may increase effectiveness by inhibiting parallel angiogenic pathways or by inhibiting an angiogenic and glioma cell signaling pathway simultaneously.

## TOXICITY OF ANTIANGIOGENIC THERAPY

Antiangiogenic therapies are generally well tolerated yet have several uncommon but serious complications [78, 79]. Hypertension, proteinuria, poor wound healing and thromboembolic events are the most frequently observed adverse effects necessitating dose reduction in some cases and are reversible on discontinuation of treatment. The mechanism by which angiogenic therapies produce hypertension is uncertain but probably involves blockade of nitrous oxide. Nitric oxide synthase in endothelial cells is upregulated by VEGF resulting in an increase in synthesis of nitric oxide. Inhibition of VEGF signaling blocks this process by decreasing nitric oxide production resulting in vasoconstriction, increasing systemic vascular resistance and endothelial oxidative stress [80, 81]. The incidence of bevacizumab-associated hypertension in GBM is approximately 31% and in 5% of patients, grade 3 or higher hypertension occurs [22]. Management of hypertension with a single antihypertensive agent is generally sufficient. Angiotensin-converting enzyme (ACE) inhibitors are more quickly effective than calcium channel blockers and should be considered when rapid treatment is desired and are preferred when proteinuria coexists [78]. Blood pressure should be monitored weekly during the first bevacizumab cycle and subsequently with each cycle. Home monitoring may be encouraged in patients with essential hypertension. Direct effect of VEGF inhibition on the glomerulus also contributes to proteinuria resembling thrombotic microangiopathy [82]. Proteinuria is also thought to be provoked by bevacizumab-induced hypertension by increased glomerular pressure [83]. A meta-analysis of proteinuria in cancer patients receiving bevacizumab demonstrated 13% of patients with at least grade 1 proteinuria and 3.3% with grade 3 or higher proteinuria [84]. In patients with glioblastoma, the incidence rates reported in the BRAN study appeared to be lower, finding grade 1 proteinuria in only 4% and grade 3 proteinuria in less than 1% of patients [22]. Patients receiving bevacizumab should be monitored for proteinuria with serial urinalyses. According to the manufacturer, a urine dipstick greater than or equal to 2+ should be further evaluated with a 24-hour urine collection for protein. Bevacizumab should be discontinued permanently in the case of nephrotic syndrome and temporarily suspended for 24-hour urine protein exceeding 2 grams (<http://www.gene.com/gene/products/information/pdf/avastin-prescribing.pdf>). Vascular events including venous and arterial thrombosis are reported as a relatively frequent complication of antiangiogenic therapies and likewise venous thromboembolic events (VTE) are common in patients with malignant gliomas. Given this, it is difficult to determine if the reported incidence of VTE exceeds the anticipated rate associated with malignant gliomas in the absence of antiangiogenic therapies. The overall risk of VTE in patients with malignant gliomas is 24% and is greatest in the peri-operative period [85]. Retrospective review has revealed an overall incidence of VTE with bevacizumab of 7.5% [86]. Management of thrombotic complications requires careful assessment of the relation of therapy to the VTE and risk of treatment with anticoagulation. Most clinical trials permit continuation of therapy with resolving VTE. Recommended treatment includes use of standard anticoagulants and continuation of bevacizumab in the absence of hemorrhage. Given reports of improved efficacy and absence of drug-drug interactions there has been a trend towards use of low-molecular weight heparin compared to warfarin in patients with cancer [87]. The concomitant use of anticoagulation during treatment with bevacizumab is not uncommon given the increased risk of VTE in this patient population. The hemorrhagic

risk associated with pairing anticoagulation with bevacizumab has been evaluated recently in a retrospective study of 284 patients treated with bevacizumab, 64 of who received concurrent anticoagulant therapy [88]. 20% of patients experienced hemorrhage and in 6% of patients, hemorrhages were serious (grade  $\geq 3$ ). 11% of patients experienced intracranial hemorrhage (ICH) of which most (8%) were grade 1. Anticoagulant use during bevacizumab therapy may increase the risk of hemorrhage, although is generally well tolerated. Independent of anticoagulant use, bevacizumab has also been associated with an increased risk of both systemic and intracerebral hemorrhage [89]. High-grade (grades 3-5) hemorrhages occurred in 3.5% of patients [22, 78]. The most common type of hemorrhage is epistaxis but more serious hemorrhage may occur including hemoptysis, gastrointestinal bleeding, intracerebral hemorrhage and intratumoral hemorrhage. Data for risk of intracerebral hemorrhage in patients with metastatic brain tumors treated with anti-VEGF therapy demonstrate this risk is low (less than 3%) [90, 91]. Antiangiogenic agents may additionally interfere with wound healing as angiogenesis is a necessary step in effective wound repair. Wound breakdown has been reported to occur in 0% to 6% of patients treated with bevacizumab and has been noted to occur significantly more frequently in patients receiving pre-operative bevacizumab [78]. Wound healing complications were notably more frequent with subsequent craniotomies and in patients with a short delay between bevacizumab and surgery. Based on these results, repeated craniotomy should be delayed for a minimum of 28 days after the last dose of bevacizumab whenever possible [92]. Wound dehiscence is also observed in patients receiving bevacizumab with an incidence greatest within 10 days of port placement [93]. Ideally, use of bevacizumab should be avoided four weeks prior to and following surgery. For smaller procedures, bevacizumab should be avoided for at least two weeks when possible. Bowel perforation may occur with use of antiangiogenic agents and may occur by several mechanisms including exacerbation of diverticular disease and peptic ulceration, obstruction and constipation, arterial thromboembolic events and use of corticosteroids. In a meta-analysis of oncologic patients, the incidence of gastrointestinal perforation was 0.9% [94] and in clinical trials with bevacizumab in patients with GBM, the reported incidence ranges from 0% to 2.9% [25-27, 95, 96]. Bowel perforation is associated with a high mortality and requires emergent surgical assessment. Surgical intervention while sometimes necessary is complicated by difficulties with wound healing. Conservative management involves bowel rest, intravenous fluids and broad-spectrum antibiotics [97]. Reversible posterior leukoencephalopathy syndrome (RPLS), largely associated with hypertension, has additionally been described with a number of pharmaceutical agents including VEGF and VEGFR targeting agents. Clinically RPLS manifests as acute onset of headaches, confusion, cortical blindness and seizures in the setting of marked hypertension. MRI classically reveals T2/FLAIR hyperintensities in the white matter, most prominent in the posterior cerebral hemispheres. Principally, it is thought to arise due to failure of cerebral vasomotor autoregulation secondary to hypertension and endothelial damage. Both bevacizumab and VEGFR tyrosine kinase inhibitors have been implicated in RPLS which typically resolves with treatment of hypertension and discontinuation of the offending agent [98-100].



## CURRENT CHALLENGES

### Mechanisms of Resistance to Antiangiogenic Therapy

Although antiangiogenic therapy benefits most patients with recurrent GBM, progression is inevitable and identification of suitable agents after progression on bevacizumab has remained a prominent challenge in management [101]. Two major methods of resistance to antiangiogenic therapy have been proposed [102, 103]. The majority of patients derive benefit from antiangiogenic therapies, however the benefits are transient and are followed by restoration of tumor growth and progression by means of an evasive, adaptive resistance. Uncommonly, patients exhibit an intrinsic or pre-existing resistance to antiangiogenic therapy representing a subset of patients who do not respond to bevacizumab. Several mechanisms of acquired resistance have been described by Bergers and Hanahan including upregulation of alternative proangiogenic growth factors, recruitment of vascular progenitor cells and bone marrow-derived endothelial precursor cells, and adaptations increasing invasion, migration and survival in a hostile microenvironment [102]. The exact mechanisms underlying resistance to bevacizumab remain poorly defined. However, preclinical data suggest an increase in proangiogenic factors including fibroblast growth factor and angiopoietins (Ang), especially Ang2, associated with VEGF/VEGFR targeting therapeutics [28, 104-106]. Other proposed mechanisms of resistance involve increased PDGF signaling promoting stabilization of the neovasculature by recruitment of pericytes [107]. Potential strategies to overcome progression on bevacizumab have thus considered targeting multiple angiogenic mediators simultaneously.

### INFILTRATIVE PATTERNS FOLLOWING BEVACIZUMAB

Emergence of a more infiltrative phenotype in response to anti-VEGF therapy in some patients with GBM has become increasingly recognized as an area of concern in the era of bevacizumab [23, 108-112]. Inhibition of angiogenesis may lead to an infiltrative growth pattern with co-option of existing cerebral blood vessels [36, 113-116]. Bevacizumab may thus suppress enhancing tumor more effectively than nonenhancing infiltrative tumor growth, represented by T2/FLAIR hyperintense areas on MRI. Several clinical studies have suggested that increased infiltrative growth is evident on MRI scans from bevacizumab treated patients. While not pathologically confirmed, this phenomenon is consistent with the observation that some patients continue to show progressive neurologic disability despite stability of contrast enhancing tumor. This continues to be an active area of debate with multiple reports arguing against promotion of tumor infiltration by bevacizumab [117-120].

### EVALUATING RESPONSE TO ANTI-ANGIOGENIC THERAPY

Antiangiogenic agents such as bevacizumab and cediranib can rapidly decrease enhancement after initiation of treatment resulting in marked radiographic response rates compared to historic controls [28]. The radiographic improvement results from rapid

normalization of abnormal tumor vascularity, restoring the integrity of the blood brain barrier and thus contrast leakage into the tumor.

Thus, antiangiogenic agents more effectively alter imaging characteristics of enhancing tumor than that of nonenhancing disease. Rapid decrease in enhancement after initiation of treatment is often more impressive than the clinical benefit derived from therapy, thus the term ‘pseudoresponse’ has been used to describe this paradoxical effect that may not adequately represent tumor response.

Therefore, new radiographic criteria have been proposed to assess treatment response in the setting of antiangiogenic therapy [121-124]. The new Response Assessment in Neuro-Oncology (RANO) criteria incorporates change in FLAIR hyperintensity as a measure of tumor response. Prior to this, the Macdonald criteria were the most widely used method to determine response in clinical trials, determined by measurements of contrast enhancing disease.

## BIOMARKERS OF ANGIOGENESIS

No validated biomarkers currently exist to appropriately evaluate response to anti-angiogenic therapies to predict efficacy and identify progression during treatment.

Potential biomarkers under current investigation to predict benefit from anti-angiogenic therapies include several novel imaging techniques, circulating factors, or factors expressed by tumor samples [125-128]. Radiographic response, specifically diffusion-weighted imaging has shown promise as a predictive and prognostic biomarker in gliomas and has the potential to serve as an imaging biomarker for overall survival [112, 129]. MRI has also been utilized to assess tumor blood perfusion which has been associated with longer survival [130].

Levels of circulating endothelial progenitor cells have also been shown to correlate with tumor angiogenic activity and the efficacy of bevacizumab regimens [131, 132]. Prospective validation will be required to determine predictive biomarkers that may be used to select patients for antiangiogenic therapies and to assess response.

## CONCLUSION

Angiogenesis is a complex process in glioblastoma and is driven primarily by VEGF signaling. The addition of antiangiogenic therapies has been an important advance in the treatment of recurrent high-grade gliomas. Studies of bevacizumab and similar agents have demonstrated encouraging rates of radiographic response and progression-free survival, although benefits in overall survival remain unproven. Salvage therapies when bevacizumab is no longer effective are desperately needed as resistance to therapy invariably develops. Studies evaluating bevacizumab in combination with cytotoxic drugs and other targeted molecular agents as well as regimens targeting multiple pathways are currently under evaluation. The development of a predictive array of biomarkers for measuring treatment response may help identify subgroups of patients that benefit from these therapies.

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*Chapter 13*

## **CSF DISSEMINATION OF GLIOMAS AND PRIMARY BRAIN TUMORS**

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### **ABSTRACT**

CSF dissemination of primary brain tumors is uncommon. Recognition of CSF dissemination may be challenging in a patient with pre-existent neurologic symptoms and signs. Leptomeningeal metastasis (LMD) occurring in the context of primary brain tumors may be seen at diagnosis or at time of disease recurrence. Treatment of LMD at initial diagnosis may be curative, whereas treatment at disease recurrence is nearly always palliative. Contemporary treatment of LMD is the subject of this review and is discussed below.

### **INTRODUCTION**

Cerebrospinal fluid (CSF) disseminated disease or leptomeningeal metastasis (LMD) is a common problem in neuro-oncology occurring in approximately 1-2% of all patients with primary brain tumors. LMD is a disease affecting the entire neuraxis and consequently clinical manifestations are pleomorphic and may affect the spine, cranial nerves or cerebral hemispheres. Because of craniospinal disease involvement, staging and treatment need encompass all cerebrospinal fluid (CSF) compartments in both brain and spine. Treatment of LMD utilizes radiotherapy to sites of bulky or symptomatic disease and frequently intra-CSF drug therapy. The inclusion of co-administered systemic therapy is often necessary in that the

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majority of patients with primary brain tumors have both recurrence of disease at the primary site and LMD. Further complicating treatment is the paucity of studies of LMD in patients with primary brain tumors. At present, intra-CSF drug therapy is confined to three chemotherapeutic agents (i.e. methotrexate, cytosine arabinoside and thio-TEPA) administered by a variety of schedules either by intralumbar or intraventricular intra-CSF drug delivery. Although treatment of LMD is palliative with an expected median patient survival of 2 to 6 months, it may afford stabilization and protection from further neurologic deterioration in patients with LMD and primary brain tumors.

## GENERAL OVERVIEW

### Clinical Presentation

LMD classically presents with clinical manifestations encompassing symptoms and signs in three domains of neurological function: 1) the cerebral hemispheres; 2) the cranial nerves; and 3) the spinal cord and associated roots. Neurologic signs on examination generally exceed patient reported symptoms. The most common manifestations of cerebral hemisphere dysfunction are headache and mental status changes. Other signs include confusion, dementia, seizures and hemiparesis. Diplopia is the most common symptom of cranial nerve dysfunction with the VI cranial nerve being the most frequently affected, followed by cranial nerve III and IV. Spinal signs and symptoms include weakness (lower extremities more often than upper), dermatomal or segmental sensory loss and pain in the neck, back, or following radicular patterns. Nuchal rigidity is unusual and only present in 15% of cases [1-3]. A high clinical index of suspicion needs to be entertained in order to make the diagnosis of LMD. The finding of multifocal neuraxis disease in a patient with a primary brain tumor is strongly suggestive of LMD, but it is also common for patients with LMD to present with isolated syndromes such as symptoms of raised intracranial pressure, cauda equina syndrome or cranial mononeuropathy. New neurological signs and symptoms may represent progression of LMD but need to be distinguished from the manifestations of parenchymal disease or from treatment-related side effects of chemotherapy or radiation.

### Diagnosis

#### *CSF Examination*

The most useful laboratory test in the diagnosis of LMD is the CSF exam. Abnormalities include increased opening pressure, increased leukocytes, elevated protein or decreased glucose, which though suggestive of LMD are not diagnostic. The presence of malignant cells in the CSF is diagnostic of LMD but in general, as is true for most cytological analysis, assignment to a particular tumor is not possible [4]. In patients with positive CSF cytology, up to 45% will be cytologically negative on initial examination [5]. The yield is increased to 80% with a second CSF examination but little benefit is obtained from repeat lumbar punctures after two lumbar punctures [2]. Notably however this data is obtained from other solid tumors and may not be reflective of cytology yield in patients with primary brain

tumors. Site of CSF sampling also likely affects results of CSF cytology. In one study, ventricular and lumbar cytologies obtained simultaneously were discordant in 30% of cases [6]. Not obtaining CSF from a site of symptomatic or radiographically demonstrated disease was found to correlate with false negative cytology results in a prospective evaluation of 39 patients, as did withdrawing small CSF volumes (<10.5ml), delayed processing of specimens and obtaining less than two samples (7). Even after correcting for these factors there is remains a substantial group of patients with LMD and persistently negative CSF cytology. Glass reported on a postmortem evaluation evaluating the value of premortem CSF cytology and demonstrated 40% of patients with clinically suspected LMD proven at time of autopsy are cytologically negative [5].

The low sensitivity of CSF cytology makes it difficult not only to diagnose LMD, but also to assess the response to treatment. Biochemical markers, immunohistochemistry and molecular biology techniques applied to CSF have been explored in an attempt to find a reliable biological marker of disease. Numerous biochemical markers have been evaluated but in general, their use has been limited by poor sensitivity and specificity [8]. Use of monoclonal antibodies for immunohistochemical analysis in LMD does not significantly increase the sensitivity of cytology alone [9]. However, in the case of leukemia and lymphoma, antibodies against surface markers can be used to distinguish between reactive and neoplastic lymphocytes in the CSF [10]. Cytogenetic studies have also been evaluated in an attempt to improve the diagnostic accuracy of LMD. Flow cytometry and DNA single cell cytometry, techniques that measure the chromosomal content of cells, and fluorescent *in situ* hybridization (FISH), that detects numerical and structural genetic aberrations as a sign of malignancy, can give additional diagnostic information and in patients with leukemia or lymphoma demonstrate increased sensitivity when compared to CSF cytology [11]. Polymerase-chain reaction (PCR) can establish a correct diagnosis when cytology is inconclusive, but the genetic alteration of the neoplasia must be known for it to be amplified with this technique, and this is generally not the case, particularly in solid tumors [12]. In cases where there is no evidence of systemic cancer and CSF exams remain inconclusive, a meningeal biopsy may be diagnostic. The yield of this test increases if the biopsy is taken from an enhancing region on MRI (see below) and if posterior fossa or pterional approaches are used [13].

### **Neuroradiographic Studies**

Magnetic resonance imaging with gadolinium enhancement (MR-Gd) is the technique of choice to evaluate patients with suspected leptomeningeal metastasis [14]. Because LMD involves the entire neuraxis, imaging of the entire CNS is required in patients considered for further treatment. T1-weighted sequences, with and without contrast, combined with fat suppression T2-weighted sequences constitute the standard examination [15]. MRI has been shown to have a higher sensitivity than cranial contrast enhanced computed tomography (CE-CT) in several series, and is similar to computerized tomographic myelography (CT-M) for the evaluation of the spine, but significantly better tolerated [16]. Any irritation of the leptomeninges (i.e. inflammatory, blood, infection, cancer) will result in their enhancement on MRI, which is seen as a fine signal-intense layer that follows the gyri and superficial sulci. Subependymal involvement of the ventricles often results in ventricular enhancement. Some changes such as cranial nerve enhancement on cranial imaging and intradural extramedullary enhancing nodules on spinal MR (most frequently seen in the cauda equina) can be

considered diagnostic of LMD in patients with cancer. Lumbar puncture itself can rarely cause a meningeal reaction leading to dural-arachnoidal enhancement so imaging should be obtained preferably prior to the procedure. MR-Gd still has a >40% incidence of false negative results so that a normal study does not exclude the diagnosis of LMD. On the other hand, in cases with a typical clinical presentation, abnormal MR-Gd alone is adequate to establish the diagnosis of LMD [14]. Radionuclide studies using either  $^{111}\text{In}$  diethylenetriamine-pentaacetic acid or  $^{99}\text{Tc}$  macro-aggregated albumin, constitute the technique of choice to evaluate CSF flow dynamics [17]. Abnormal CSF circulation has been demonstrated in 30 to 70% of patients with LMD, with blocks commonly occurring at the skull base, the spinal canal and over the cerebral convexities. Patients with interruption of CSF flow demonstrated by radionuclide ventriculography have been shown in three clinical series to have decreased survival when compared to those with normal CSF flow [18]. Involved field radiotherapy to the site of CSF flow obstruction, restores flow in 30% of patients with spinal disease and 50% of patients with intracranial disease [18]. Re-establishment of CSF flow with involved-field radiotherapy followed by intrathecal chemotherapy led to longer survival, lower rates of treatment related morbidity and lower rate of death from progressive LMD, compared to the group that had persistent CSF blocks [28, 30]. These findings may reflect that CSF flow abnormalities prevent homogenous distribution of intrathecal chemotherapy, resulting in 1) protected sites where tumor can progress, and 2) in accumulation of drug at other sites leading to neurotoxicity and systemic toxicity. Based on this, many authors recommend that intrathecal chemotherapy be preceded by a radionuclide flow study and if a block is found, that radiotherapy be administered in an attempt to re-establish normal flow.

## PRIMARY BRAIN TUMORS

### Glioma

The management of patients with gliomas and leptomeningeal gliomatosis (LG) is particularly challenging in that the majority of patients have compromised neurological function resulting both from the treatment and topography of their primary tumor [19-21].

The majority of studies regarding LG are primarily autopsy-based. Erlich and Davis performed an autopsy study of 25 patients with glioblastoma (GBM), amongst whom 20 spinal cord examinations were performed [22]. Five of these patients (25%) had evidence of LG. Yung subsequently reported on 52 patients with HGG who underwent autopsy in whom 11 (21%) had evidence of LG [23].

In an antemortem series where the diagnosis of LG was made based on neuroradiographic imaging but not CSF cytology, Vertosick and Selker reported an incidence of LG of 2% [19]. Awad reported on 13 patients seen at the Cleveland Clinic with HGG and LG in whom 8 had premortem symptoms consistent with LG [24]. In all patients, the appearance of LG was a preterminal event. Onda performed autopsies on 51 patients who died of GBM and demonstrated evidence of LG in 14 (27%) [20]. Subsequently Grant reported on a series of 11 patients with HGG and LG all diagnosed antemortem [25]. In 3

patients, LG was found at time of initial tumor presentation. Amongst patients with previously diagnosed and treated anaplastic astrocytoma (AA) or GBM, treatment of LG had little effect. In summary, these various reports suggest that autopsy evidence of LG is ten fold greater than clinically evident LG in patients with HGG. The reason for this antemortem/postmortem discrepancy is enigmatic. Most recently, Witham reported on 14 patients (GBM: n=9; AA: n=5) with LG and treated with intra-CSF thio-TEPA [26]. In 3 patients LG was the presenting symptom of the HGG. In only 5 patients were clinical symptoms and signs compatible with LG. CSF cytology were not performed in 6 patients and amongst the 8 patients in whom CSF was examined, it was positive in only 2. The neuroradiographic criteria permitted subependymal disease (seen in 8 patients) to be sufficient to diagnose LG. Of note, overall median survival was 10 months (10 months for GBM; 19 months for AA).

Another recent report suggests that patients with HGG and LG defined by clinical symptoms and signs, positive CSF cytology and neuroradiographic findings compatible with LG respond poorly to an aggressive multi-modal therapy [21]. The median survival following diagnosis of LG in this report is similar to that of LMD secondary to systemic solid cancers. Like the majority of patients with LMD, LG in patients with HGG is a late event in the natural history of their cancer and predicts for limited survival. At present there is very little evidence suggesting an advantage for any particular intra-CSF chemotherapy relative to another or when used in combination for the treatment of LMD [27-30]. Furthermore the majority of patients in the above-mentioned study died of complications secondary to LG and not as a result of progression of the primary brain tumor [21]. The therapy for leptomeningeal metastasis is necessarily multi-modal as approximately 60% of patients require radiotherapy to treat symptomatic or neuroradiographically bulky subarachnoid disease, and approximately one third receive systemic chemotherapy for primary tumor progression [21]. Therefore, standard treatment of LG utilizes clinically appropriate treatments (involved-field radiotherapy, systemic chemotherapy or intra-CSF chemotherapy) as the disease affects the entire neuraxis in a variable and pleomorphic manner. Treatment results consequently reflect a composite of therapies not unlike the treatment of high-grade gliomas, which creates difficulties when attempting to allocate benefit to any specific therapy as for example intra-CSF chemotherapy. In conclusion, notwithstanding the modest treatment-related toxicity observed in various studies, the poor response to treatment and limited survival in patients with HGG and LG suggests that a less aggressive approach may be justified. Palliative treatment with radiotherapy directed to symptomatic site(s) of disease perhaps in conjunction with simple systemic chemotherapy regimens, may prove to be as effective as an aggressive multimodal approach and much less intrusive to the patient. This statement however is not meant to exclude the enrollment of such patients from clinical trials designed to explore new approaches to the treatment of LG, a disease for which new and effective therapies would be welcomed.

## MENINGIOMA

Meningiomas are extra-axial brain tumors of middle to late adult life and have a female predominance. Overall, 90% of meningiomas are benign, 6% atypical and 2% are malignant

[45-51]. Most patients diagnosed with a meningioma proceed with surgical removal and are advised to do so based on their neurological symptoms and tumor size and location [31]. Complete surgical resection may be curative. For incompletely resected or recurrent tumors not previously irradiated, radiotherapy is administered. Radiotherapy may be administered as either conventional external beam irradiation or stereotactically. Stereotactic radiotherapy (SRT) either as LINAC or gamma knife radiosurgery is increasingly utilized. When the meningioma is unresectable or all other treatments (surgery, radiotherapy) have failed, immuno-chemotherapy may be considered. Rarely meningiomas metastasize and when seen, most often to extraneural sites (cervical lymph nodes, pulmonary) [32]. Metastatic meningioma is most often associated with either aggressive (WHO Grade 2) or malignant meningioma (WHO Grade 3) with a range of occurrence of 10-25%. Not clear from the literature however is the risk of CSF dissemination in patients with meningioma [32]. In a small series (arguably biased by referral to a neuro-oncology clinic) 4% of 200 consecutively seen patients with recurrent meningioma manifested CSF spread of disease defined both radiographically and cytologically. In addition to CSF spread of disease, the majority of these patients (5 of 8; 63%) also had evidence of extraneural metastasis. Furthermore, all patients had extensive prior therapy comprised of surgery (often multiple), radiotherapy (often both external beam and stereotactic) and chemotherapy (all had received prior hydroxyurea and progressed). Aside from hydroxyurea, alpha-interferon and somatostatin analogues, there is a paucity of chemotherapy agents with demonstrated activity against recurrent meningiomas complicating the management of these already complicated patients. Similar to patients with meningeal gliomatosis, these patients were treated in a multimodal manner including both systemic and regional (intraventricular) chemotherapy and involved-field radiotherapy to sites of symptomatic spine disease (3 patients). Despite this aggressive approach, median survival was only 5.5 months with all but three patients dying of progressive disease. In the three patients alive, all have residual disease, and two continue to receive therapy. Furthermore, no patient demonstrated a treatment response (best response was stable disease) as assessed by neuroradiography. Based on the results of this study, two conclusions seem justified. One that meningiomas do metastasize both hematologically and by CSF dissemination and secondly, that there is a need for more effective chemotherapy for recurrent surgical and radiotherapy refractory meningiomas.

## **Primary CNS lymphoma (PCNSL)**

PCNSL is a disease that primarily involves the brain, both as the source of origination and in patients who relapse [33,34]. The leptomeninges may be involved both at disease presentation and at recurrence. Notwithstanding that isolated disease of the leptomeninges is rare in patients with PCNSL regardless if one considers patients at presentation or recurrence, simultaneous brain and leptomeningeal disease is common. In a study by of 86 patients with PCNSL, all patients underwent a complete CSF/leptomeningeal evaluation [34]. 26% had a positive cytology and neuropathology and neuroradiography demonstrated disease consistent with leptomeningeal spread in another 15%. In studies that define lymphomatous meningitis as a positive CSF cytopathology the results are similar. Balmaceda and Lawler each reported a prevalence of 26% while other studies, including a larger series by Ferreri reported prevalence of 12-16% [58,64,74]. Other studies using different modalities, including PCR for



a component of IgG heavy chain or contrast-enhanced brain and spinal imaging alone reported lower prevalence figures for lymphomatous meningitis, 13% and 12.5% respectively. In an autopsy study of patients with PCNSL, 77% of patients (21 of 26) had evidence of lymphomatous meningitis.

A similar figure (40% incidence of PCNS-LM) was found in another autopsy study. Both of these studies are perhaps most relevant to patients with PCNSL at relapse and suggest combined brain and leptomeningeal disease is seen in the majority of such patients. Taken together, the above-mentioned antemortem and postmortem studies suggest that PCNSL is a disease of the neuraxis with frequent evidence of lymphomatous meningitis. A large multicenter study conducted by Ferreri of 370 patients with PCNSL evaluated the benefit of adjuvant intrathecal methotrexate [33]. Ferreri compared patients who received high-dose methotrexate based regimens with or without intrathecal chemotherapy and were unable to demonstrate a survival benefit. Furthermore, notwithstanding that some patients were treated with less than  $3.0 \text{ gm/m}^2$  of methotrexate (MTX), intrathecal chemotherapy had no impact regarding meningeal relapse or on survival, even in patients who had positive CSF cytology at diagnosis. It would appear therefore that even in patients with leptomeningeal disease at diagnosis, high-dose MTX systemic regimens provide sufficient CSF MTX levels to obviate the need for intra-CSF chemotherapy. Because lymphomatous meningitis is sufficiently common in PCNSL, leptomeningeal/CSF compartment therapy is indicated as part of the treatment regimen in patients with newly diagnosed PCNSL. Leptomeningeal/CSF compartment therapy can take several forms including intra-CSF chemotherapy, radiation (i.e. craniospinal irradiation) or high-dose systemic chemotherapy (MTX, cytarabine or thio-TEPA).

The most effective drug utilized in patients with newly diagnosed PCNSL is high-dose MTX. When this drug is administered in gram quantities (high-dose), cytotoxic CSF levels are achieved. In a study by Glantz of 16 solid tumor patients with leptomeningeal metastases, MTX was administered intravenously at  $8 \text{ gm/m}^2$  over four hours followed by serial sampling of CSF and blood [35]. In this study, there was in addition, a parallel group of patients who received intrathecal MTX at a standard dose and schedule. After a single intravenous dose, MTX levels of 1.0 micromolar are maintained in the CSF for, on average, 48 hours, and 0.1 micromolar for up to 93 hours. In the patients who received a single intrathecal MTX dose, levels of 1.0 micromolar are maintained for 35-48 hours, and 0.1 micromolar levels are maintained for about 57 hours.

Therefore, the duration of cytotoxic drug exposure in the CSF is similar in patients who receive intravenous or intrathecal MTX on these schedules and at these doses. In another study by Chamberlain, two groups of patients with recurrent PCNSL and isolated lymphomatous meningitis were compared [36]. One group was treated with intra-CSF chemotherapy whereas the other received high-dose ( $8 \text{ gm/m}^2$ ) systemic methotrexate. Outcome was similar in both groups (overall survival, progression free survival, lymphomatous meningitis response rates) and differences were seen only in toxicity and pharmacoeconomics (both favoring intra-CSF chemotherapy). These studies suggest that intra-CSF chemotherapy is not required in patients receiving systemic high-dose MTX however should be considered in patients receiving alternative therapies (i.e. PCV, CHOP, topotecan) and with evidence of CSF dissemination.

## EPENDYMOMAS

CSF dissemination occurs in 3-12% of all intracranial ependymomas and is most frequent in patients with infratentorial anaplastic ependymomas [37-44]. Because a small but measurable risk for CSF dissemination exists for all patients with newly diagnosed ependymoma, an extent of disease evaluation including both CSF cytology and craniospinal MRI is mandated following surgery. This staging permits stratification of patients into those with ( $M_1$ ) or without CSF metastasis ( $M_0$ ) and patients with or without residual disease following surgery, the two most important clinical parameters affecting outcome. Treatment of ependymomas is primarily surgical, as essentially all analyses have determined that completeness of surgical resection is the most important covariant affecting progression free and overall survival.

Radiotherapy represents the second most frequently utilized adjuvant treatment modality for ependymomas despite the lack of a randomized clinical trial showing benefit and the general consensus that ependymomas are radioresistant. Furthermore, there are no data regarding a dose response relationship in ependymomas and as such, total tumor dose has varied. Because of the possibility of CSF spread, one of the controversies regarding the radiotherapeutic management of ependymomas is the volume of brain that needs to be treated. Notwithstanding early enthusiasm for craniospinal irradiation (CSI), several recent studies support the application of limited-field radiotherapy for  $M_0$  tumors and reserve CSI for  $M_1$  tumors regardless of tumor histology and according to some authors for anaplastic ependymomas of the infratentorial compartment. Robertson reported a prospective trial involving 32 children with newly diagnosed intracranial ependymoma that examined the role of adjuvant chemotherapy (CCNU, vincristine, and prednisone versus so called "8 in 1" regimen) after initial surgery and CSI [45]. This multi-institutional Children's Cancer Group study demonstrated that both the extent of surgical resection and volume of residual disease on postoperative imaging predict for progression-free survival. Chemotherapy had no impact on progression-free survival with either chemotherapy drug regimen. Five-year progression free survival and overall survival rates were 50% and 64%, respectively. The majority of relapses were local treatment failures (71%) or concurrent local and distant CNS metastasis (21%) and isolated metastatic relapse was uncommon (7%) and occurred only in the setting of  $M_1$  disease at diagnosis. The study concluded that involved-field radiotherapy results in similar outcomes as compared to CSI and therefore CSI is appropriately reserved for disseminated neuraxis disease ( $M_1$ ). Whether intra-CSF chemotherapy could replace CSI in patients with  $M_1$  disease at presentation seems unlikely as adjuvant chemotherapy has not improved survival or are there studies comparing CSI to intra-CSF chemotherapy.

The management of recurrent ependymoma has not received much attention in the literature despite the fact that nearly 50% of patients will recur. There is consensus regarding an extent of disease evaluation (contrast-enhanced neuraxis MRI and CSF cytology) be performed in all patients with recurrent ependymoma as the discovery of disseminated disease changes management. Patients demonstrated to have disseminated disease ( $M_1$ ) are candidates for CSI (particularly if prior radiotherapy has not been administered), involved-field radiotherapy or intra-CSF chemotherapy. Goldwein reported on 36 patients with recurrent intracranial ependymoma in which 33 were treated with re-operation, 12 received conventional radiotherapy and all received chemotherapy [46]. Median time to recurrence

was 2.8 years and in the majority, relapse was either local (78%) or local with concomitant distant metastasis (14%). Twenty-nine (79%) of the initial cohorts had a second relapse in which a local component to the relapse was seen in 80%. Two-year overall survival and progression free survival were 29% and 23%, respectively. Considering only first relapse, 2-year actuarial survival was 39% and median survival was 17 months. Median progression free survival was 12 months. Among 36 evaluable patients and 37 chemotherapy regimens, there was one partial response (3%), seven stable disease patterns (20%), and 29 disease progressions (77%). In responding or stable disease patients, median duration of response was 9 months (range 3-23 months). Cisplatin was felt to be the most active agent amongst the four commonly used chemotherapeutics (cisplatin, procarbazine, CCNU, and vincristine). The role of intra-CSF chemotherapy for patients with recurrent  $M_1$  ependymoma is uncertain, as there are only case reports of patients with isolated CSF dissemination treated primarily with intra-CSF chemotherapy. In the author's opinion, intra-CSF chemotherapy should be offered to patients not otherwise treated with CSI for  $M_1$  ependymoma recognizing that the dominant variable affecting survival is disease recurrence at the primary site.

### Primitive Neuroectodermal Tumor (PNET)

PNET comprise both differentiated tumors (olfactory neuroblastoma, cerebral neuroblastoma, retinoblastoma and pineoblastoma) and undifferentiated tumors (medulloblastoma). Because of the propensity for PNET and in particular medulloblastoma to manifest CSF dissemination, standard initial treatment includes CSI regardless of extent of disease ( $M_0$  or  $M+$  disease). As a consequence there would appear to be no role for intra-CSF in the initial treatment of patients with CSF disseminated PNET as CSF based chemotherapy would be redundant to CSI. Furthermore, there are no adjuvant studies that in addition to CSI include intra-CSF chemotherapy for patients with  $M+$  disease however this has been proposed to permit further dose reduction of CSI (see below). An argument however could be made for intra-CSF chemotherapy in patients with positive CSF cytology following administration of CSI though the incidence of this clinical occurrence is not easily determined from the literature. Against this position are several studies that have demonstrated that  $M_1$  disease (positive CSF cytology only i.e. metastatic microscopic disease) does not predict for reduced survival raising the question whether there exists a compelling role for intra-CSF chemotherapy in the initial treatment of medulloblastoma. Lastly, a pilot study (CCG 9892) of average-risk medulloblastoma children ( $n=65$ ) suggest that reduced CSI dose (24 Gy) in conjunction with systemic chemotherapy (CCNU, cisplatin, vincristine) results in both excellent outcomes (5-year event free survival 79%) and no increased risk of neuraxis failures (i.e. CSF dissemination)[47].

Therefore the best indication for intra-CSF chemotherapy is seen in patients with recurrent disease and evidence of CSF dissemination. Data is most robust for medulloblastoma regarding patterns of relapse in which approximately one third are local, one third are local plus disseminated and one third metastatic only[48, 49]. Amongst the metastatic group approximately 20% will in addition have extraneural metastases and predominantly to bone. Overall, 50% of patients with recurrent medulloblastoma have evidence of CSF dissemination defined as either subarachnoid nodules (spinal or intracranial) seen neuroradiographically or by positive CSF cytology. In general however treatment of

recurrent and disseminated medulloblastoma is with systemic chemotherapy regardless of CSF spread of disease [48, 49]. The only exception (similar to ependymoma discussed above) is seen in case reports of patients with isolated CSF dissemination and medulloblastoma treated with intra-CSF chemotherapy. How to extrapolate from the limited data regarding intra-CSF chemotherapy for PNET is problematic and any statements regarding treatment must be considered in this context.

## **Germ Cell Tumor (GCT)**

GCT constitutes less than 1% of all primary brain tumors; occur primarily in the first two decades of life (60-70%) and principally in males (approximately 75%). GCT like PCNSL and ependymoma has a proclivity to CSF dissemination and as a consequence, staging of disease (CSF cytology and oncofetal protein analysis, contrast-enhanced neuraxis MR) is performed at diagnosis. CSF disseminated disease is seen in approximately 20% of germinomas and 35% of nongerminomatous GCT [50-53]. Among patients with GCT, those with germinoma (seminoma) can be cured by radiotherapy alone or by combined platinum-based chemotherapy and radiotherapy. Patients with mature teratoma are potentially curable by surgery alone. Patients with non-seminomatous GCT, though frequently recur, may be cured (approximately 65%) by radiotherapy alone, high-dose thio-TEPA based chemotherapy with autologous stem cell transplant or by combined platinum-based chemotherapy and radiotherapy.

Similar to ependymoma, a point of controversy in the treatment of GCT remains with respect to the appropriate radiation field (localized versus whole ventricle versus whole brain versus CSI) [50]. Increasingly, the volume of brain irradiated is determined by extent of disease and secondarily by tumor histology. In general, disseminated disease is treated with CSI whereas localized disease is treated by either involved-field or whole ventricle radiotherapy. Additionally, there is increasing interest in utilizing combined modality therapy for GCT (platinum-based chemotherapy and radiotherapy) permitting a reduction in both radiation dose and volume. The role of intra-CSF chemotherapy for the treatment of disseminated GCT is unclear, as essentially all studies have utilized systemic chemotherapy with radiotherapy. Patients with isolated CSF disease are rare as are reports of primary treatment with intra-CSF chemotherapy. Not unlike disseminated PNET, how to utilize intra-CSF chemotherapy in the treatment of disseminated GCT is uncertain and would reasonably be based on extrapolation of studies of carcinomatous meningitis.

## **TREATMENT**

In summary, the evaluation of treatment of CSF dissemination of primary brain tumors is complicated by the lack of standard treatments, the difficulty of determining response to treatment given the sub-optimal sensitivity of the diagnostic procedures and that most patients will die of their primary disease, and the fact that most studies are small, non-randomized and retrospective [54, 55]. Furthermore, it is not certain that intra-CSF chemotherapy treatment of LMD provides effective palliation as contrasted with therapy utilizing systemic chemotherapy

and CNS directed radiotherapy. Nonetheless if LMD-directed treatment is warranted, treatment requires the combination of surgery, radiation and chemotherapy in most cases. Surgery is used in the treatment of NM for the placement of 1) intraventricular catheter and subgaleal reservoir for administration of cytotoxic drugs and 2) ventriculoperitoneal shunt in patients with symptomatic hydrocephalus [54, 55]. Radiotherapy is used in the treatment of LMD for a) palliation of symptoms, such as a cauda equina syndrome, b) to decrease bulky disease such as co-existent parenchymal brain metastases, and c) to correct CSF flow abnormalities demonstrated by radionuclide ventriculography [54]. Radiotherapy of bulky disease is indicated as intra-CSF chemotherapy is limited by diffusion to 2 to 3 mm penetration into tumor nodules. In addition, involved-field radiation can correct CSF flow abnormalities and this has been shown to improve patient outcome as discussed above. Whole neuraxis radiation is rarely indicated in the treatment of LMD from primary brain tumors because it is associated with significant systemic toxicity (severe myelosuppression and mucositis among other complications) and is not curative. Chemotherapy is the only treatment modality that can treat the entire neuraxis. Chemotherapy may be administered systemically or intrathecally. Intrathecal (i.e. intralumbar) chemotherapy is the mainstay of treatment for LMD. Predominantly retrospective analyses suggest that the administration of chemotherapy to the CSF may improve the outcome of patients with LMD and solid tumors. Importantly and as indicated above, there are too few studies in patients with CSF disseminated primary brain tumors upon which an informed decision can be made regarding the value of intra-CSF chemotherapy. However, it is noted that most series will exclude patients that are too sick to receive any treatment, which may be up to one third of patients with LMD [54, 55].

Three agents are routinely used: methotrexate, cytarabine (including liposomal cytarabine or DepoCyt<sup>®</sup>) and thio-TEPA. No difference in response has been seen when comparing single agent methotrexate with thio-TEPA or when using multiple agents (methotrexate, thio-TEPA and cytarabine or methotrexate and cytarabine) versus single agent methotrexate in adult randomized studies of NM and solid tumors [29-30] (Table 1). Table 2 outlines the common treatment regimens for these drugs. A sustained-release liposomal form of cytarabine (DepoCyt<sup>®</sup>) results in cytotoxic cytarabine levels in the CSF for  $\geq 10$  days and when given bimonthly and compared to biweekly methotrexate, resulted in longer time to neurological progression in patients with LMD due to solid tumors [58]. Furthermore, quality of life and cause of death favored DepoCyt<sup>®</sup> over methotrexate. These findings were confirmed in a study of lymphomatous meningitis and in an open label study suggesting that DepoCyt<sup>®</sup> might be considered the drug of first choice in the treatment of LMD when experimental therapies are unavailable [27, 28, 57]. The rationale to give intrathecal chemotherapy is based on the presumption that most chemotherapeutic agents when given systemically have poor CSF penetration and do not reach therapeutic levels. Exceptions to this would be systemic high-dose methotrexate, cytarabine and thio-TEPA, all of which result in cytotoxic CSF levels. Their systemic administration however, is limited by systemic toxicity and the difficulty to integrate these regimens into other chemotherapeutic programs being used to manage the primary disease. Some authors argue that intrathecal chemotherapy does not add to improved outcome in the treatment of LMD, since systemic therapy can obtain access to the subarachnoid deposits through their own vascular supply [59-61].

**Table 1. Randomized Clinical Trials (Adapted from [54])**

| Study         | Design                                                                          | Response                                                                                                                                           | Toxicity                                                                                                                                                                                                                      |
|---------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Boogerd [56]  | N=35<br>Breast cancer<br>IT vs. no IT †                                         | IT vs. no IT:<br>Improvement or stabilization:<br>59% vs. 67%<br>TTP: 23 vs. 24 wk<br>Median survival: 18.3 vs. 30.3 wk                            | IT vs. no IT:<br>Neurological complications: 47% vs. 6%                                                                                                                                                                       |
| Shapiro [57]  | Solid tumors (n=103)<br>DepoCyt vs. MTX<br>Lymphoma (n=25)<br>DepoCyt vs. Ara-C | DepoCyt vs. MTX/Ara-C:<br>PFS*: 35 vs. 43 d<br>DepoCyt vs. MTX:<br>PFS: 35 vs. 37.5<br>DepoCyt vs. Ara-C: CR*: 33.3% vs. 16.7%<br>PFS: 34 vs. 50 d | DepoCyt vs. MTX/Ara-C:<br>Drug-related AEs: 48% vs. 60%<br>Serious AEs: 86% vs. 77%                                                                                                                                           |
| Glantz [27]   | N=28<br>Lymphoma<br>DepoCyt vs. Ara-C                                           | DepoCyt vs. Ara-C:<br>TTP*: 78.5 vs. 42 d<br>OS*: 99.5 vs. 63 d<br>RR: 71% vs. 15%                                                                 | DepoCyt vs. Ara-C:<br>Headache: 27% vs. 2%; nausea: 9% vs. 2%; fever: 8% vs. 4%; pain: 5% vs. 4%; confusion: 7% vs. 0%; somnolence: 8% vs. 4%                                                                                 |
| Glantz [58]   | N=61<br>Solid tumors<br>DepoCyt vs. MTX                                         | DepoCyt vs. MTX:<br>RR* 26% vs. 20%<br>OS* 105 vs. 78 d<br>TTP 58 vs. 30 d                                                                         | DepoCyt vs. MTX:<br>Sensory/motor: 4% vs. 10%; altered mental status: 5% vs. 2%; headache: 4% vs. 2%                                                                                                                          |
| Grossman [30] | N=59<br>Solid tumors and lymphoma (in 90%)<br>IT MTX vs. thiotepa               | IT MTX vs. thiotepa:<br>Neurological improvements: none<br>Median survival: 15.9 vs. 14.1 wk                                                       | IT MTX vs. thiotepa:<br>Serious toxicities similar between groups.<br>Mucositis and neurological complications more common in MTX group                                                                                       |
| Hitchins [29] | N=44<br>Solid tumors and lymphomas<br>IT MTX vs. MTX + Ara-C                    | IT MTX vs. MTX + Ara-C:<br>RR*: 61% vs. 45%<br>Median survival*: 12 vs. 7 wk                                                                       | IT MTX vs. MTX + Ara-C:<br>N/V: 36% vs. 50%; septicemia, neutropenia: 9% vs. 15%; mucositis: 14% vs. 10%; pancytopenia: 9% vs. 10%. AEs related to reservoir: blocked Ommaya: 17% vs. 0%; intracranial hemorrhage: 11% vs. 0% |

IT intra-CSF.

MTX methotrexate.

Ara-C cytarabine.

PFS progression free survival.

TTP time to tumor progression.

OS overall survival.

d day.

RR response rate.

wk weeks.

AE adverse event.

N/V nausea/vomiting.

**Table 2. Regional Chemotherapy for Neoplastic Meningitis. (Adapted from [54])**

| Drugs                | Induction Regimens                                 |                                                     | Consolidation Regimen                                               |                                                      | Maintenance regimen                                    |                                  |
|----------------------|----------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------|----------------------------------|
|                      | Bolus Regimen                                      | CxT Regimen                                         | Bolus Regimen                                                       | CxT Regimen                                          | Bolus Regimen                                          | CxT Regimen                      |
| Methotrexate         | 10-15mg twice weekly (Total 4 weeks)               | 2mg/day for 5 days every other week (Total 8 weeks) | 10-15mg once weekly (total 4 weeks)                                 | 2mg/day for 5 days every other week (total 4 weeks)  | 10-15mg once a month                                   | 2mg/day for 5 days once a month  |
| Cytarabine           | 25-100mg 2 or 3 times weekly (Total 4 weeks)       | 25mg/day for 3 days weekly (Total 4 weeks)          | 25-100mg once weekly (Total 4 weeks)                                | 25mg/day for 3 days every other week (Total 4 weeks) | 25-100mg once a month                                  | 25mg/day for 3 days once a month |
| DepoCyt®             | 50mg every 2 weeks (Total 8 weeks)                 |                                                     | 50mg every 4 weeks (Total 24 weeks)                                 |                                                      |                                                        |                                  |
| Thiotepa             | 10mg 2 or 3 times weekly (Total 4 weeks)           | 10mg/day for 3 days weekly (total 4 weeks)          | 10mg once weekly (Total 4 weeks)                                    | 10mg/day for 3 days every other week (Total 4 weeks) | 10mg once a month                                      | 10mg/day for 3 days once a month |
| $\alpha$ -Interferon | 1x10 <sup>6</sup> u 3 times weekly (Total 4 weeks) |                                                     | 1x10 <sup>6</sup> u 3 times weekly every other week (Total 4 weeks) |                                                      | 1x10 <sup>6</sup> u 3 times weekly one week per month) |                                  |
| Topotecan            | weekly (total 6 weeks)                             |                                                     | weekly (total 6 weeks)                                              |                                                      | once a month                                           |                                  |
| Etoposide            | 0.5mg twice weekly (total 6 weeks)                 |                                                     | 0.5mg once weekly (total 6 weeks)                                   |                                                      | 0.5mg once a month                                     |                                  |

In a retrospective comparison of patients treated with systemic chemotherapy and radiation to involved areas, plus or minus intrathecal chemotherapy, Bokstein et al. did not find significant differences in response rates, median survival or proportion of long term survivors amongst the two groups but, of course, the group that did not receive the intrathecal treatment was spared the complications of this modality [60]. Glantz et al. treated 16 patients with high dose intravenous methotrexate and compared their outcome with a reference group of 15 patients treated with intrathecal methotrexate [61]. They found response rates and survival were significantly better in the group treated with intravenous therapy. In addition, there are anecdotal reports indicating that systemic chemotherapy is effective for meningeal gliomatosis. Nonetheless, intra-CSF chemotherapy remains the preferred treatment route for LMD at this time. New drugs are being explored to try to improve the efficacy of treatment. These agents include; mafosphamide, diaziquone, topotecan, gemcitabine, interferon- $\alpha$ ,

trastuzumab and rituximab are some of the new drugs being evaluated for intrathecal administration [62].

Not all patients with NM are candidates for the aggressive treatment outlined above. Most authors agree that combined-modality therapy should be offered to patients with life expectancy greater than 3 months and a Karnofsky performance status of greater than 60%. Supportive care that should be offered to every patient, regardless of whether they receive LMD directed therapy; include anticonvulsants for seizure control (seen in 10-15% of patients with LMD), adequate analgesia with opioid drugs as needed as well as antidepressants and anxiolytics if necessary. Corticosteroids have a limited use in LMD-related neurological symptoms, but can be useful to treat vasogenic edema associated with intraparenchymal or epidural metastases, or for the symptomatic treatment of nausea and vomiting together with routine antiemetics. Decreased attention and somnolence secondary to whole brain radiation can be treated with psychostimulants.

## CONCLUSIONS

CSF disseminated disease secondary to primary brain tumors is a complicated disease for a variety of reasons. Firstly, most reports concerning LMD treat all subtypes as equivalent with respect to CNS staging, treatment and outcome. However, clinical trials in oncology are based on specific tumor histology. Comparing responses in patients with carcinomatous meningitis due to gliomas to patients with primary CNS lymphoma outside of investigational new drug trials may be misleading. A general consensus is that lymphomatous meningitis is inherently more chemosensitive than meningeal gliomatosis or CSF disseminated meningioma and therefore survival following chemotherapy is likely to be different. This observation has been substantiated in patients with primary disease recurrence though comparable data regarding CSF disseminated disease, and in particular LMD is meager. A second feature of LMD, which complicates therapy, is deciding whom to treat. Not all patients necessarily warrant aggressive CNS-directed therapy, however, few guidelines exist permitting appropriate choice of therapy. Based on the prognostic variables determined clinically and by evaluation of extent of disease, a sizable minority of patients will not be candidates for aggressive LMD-directed therapy.

Therefore supportive comfort care (radiotherapy to symptomatic disease, antiemetics, and narcotics) is reasonably offered to patients with LMD considered poor candidates for aggressive therapy as seen in Figure 1. Thirdly, optimal treatment of LMD remains poorly defined. Given these constraints, the treatment of LMD today is palliative and rarely curative with a median patient survival of several months based on data of the four prospective randomized trials (extrapolated to primary brain tumors). However, palliative therapy of LMD may afford the patient protection from further neurological deterioration and consequently an improved neurologic quality of life. No studies to date have attempted an economic assessment of the treatment of LMD and therefore no information is available regarding a cost-benefit analysis as has been performed for other cancer directed therapies. Finally, in patients with LMD, the response to treatment is primarily a function of CSF cytology and secondarily of clinical improvement of neurologic signs and symptoms.



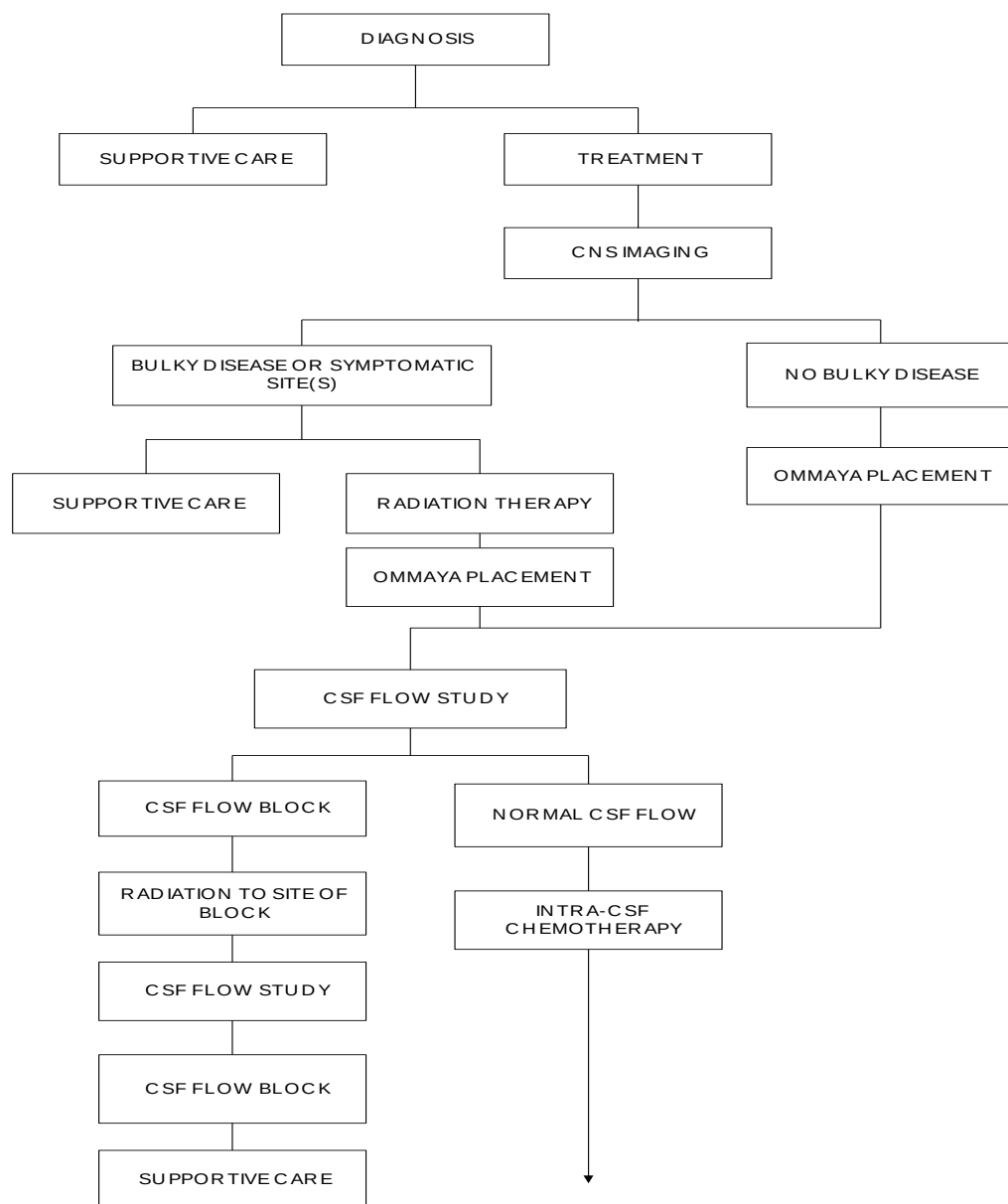


Figure 1. Treatment Algorithm of Neoplastic Meningitis (Adapted from [54]).

Aside from CSF cytology and perhaps biochemical markers, no other CSF parameters predict response. Furthermore, because CSF cytology may manifest a rostra-caudal disassociation, consecutive negative cytologist (defined as a complete response to treatment) requires confirmation by both ventricular and lumbar CSF cytologies. In general, only pain related neurologic symptoms improve with treatment. Neurologic signs such as confusion, cranial nerve deficit(s), ataxia and segmental weakness minimally improve or stabilize with successful treatment.

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### **SECTION III. FUTURE TREATMENT OPTIONS: NEW TARGETS & EXPERIMENTAL THERAPIES**





## *Chapter 14*

# **STEM CELL BIOLOGY AS IT RELATES TO GLIOMA PATHOGENESIS AND TREATMENT**

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## **ABSTRACT**

The discovery of stem-like cells in high-grade gliomas fundamentally altered the prevailing paradigm for the architecture of these tumors.

Once believed to consist solely of mutated somatic cells, with equal tumorigenic potential, malignant gliomas are now thought to contain a complex hierarchy of cells, ranging from brain tumor stem cells at the apex to fully differentiated progenitor cells. As explained in this chapter, brain tumor stem cells are hypothesized to drive tumorigenesis, confer treatment resistance, and have prognostic value.

In addition, these stem-like cells may represent novel therapeutic targets in the treatment of high-grade gliomas. In this chapter, we describe the current understanding of brain tumor stem cells and present some of the current controversies that remain in the field.

## **I. INTRODUCTION**

Our understanding of cancer has been vastly re-conceptualized in recent years. Tumors once thought to arise from the transformation of mature cells are now believed to arise instead from small populations of highly tumorigenic, self-renewing cancer stem cells that fuel tumor growth. This chapter highlights the implications of the cancer stem cell hypothesis as it relates to the pathogenesis and treatment of malignant glioma.

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## Overview: Stem Cell Biology

*Stem cells* are functionally defined as self-renewing, multipotent cells with broad developmental potential [1]. All stem cells maintain the capacity for both *self-renewal* - the ability to divide and sustain a pool of stem cells with equivalent developmental potential - and *differentiation* - the ability to generate specialized daughter cells under the appropriate conditions [1, 2].

Box 1: *Self-renewal*: the ability to divide and sustain a pool of stem cells with equivalent developmental potential to the parent cell

*Proliferation*: the ability to divide and generate progeny, which may have more restricted developmental potential than the parent cell

Over several generations, stem cells give rise to rapidly proliferating *transient amplifying cells*. These cells in turn give rise to lineage-committed progenitors that eventually differentiate to form mature somatic cells [3]. Empirically, the total variety of cell types into which a stem cell can differentiate is known as its *potency*. Embryonic stem cells are *pluripotent*, giving rise to every cell in the human body. Although pluripotent stem cells do not persist into adulthood, reservoirs of long-lived, tissue-specific multipotent somatic stem cells have been identified in various adult tissues, where they generate a restricted spectrum of daughter cells [4]. One particularly relevant example is *neural stem cells* (NSCs), which can differentiate to form neurons, astrocytes, and oligodendrocytes. While there is no definitive molecular signature for adult NSC populations, these cells tend to express the cell surface protein Notch, in combination with the RNA binding protein Musashi, the intermediate filament protein nestin, and glial fibrillary acidic protein (GFAP) [2]. None of these markers, however, is exclusive for the NSC population - Notch is expressed on mature neurons while Musashi and GFAP are expressed by mature astrocytes - reinforcing the need for functional criteria in the definition of adult stem cell populations [2]. Crucial to the maintenance of stem cell populations in adult tissues is the establishment of the *stem cell niche*, the environment that immediately surrounds and influences the stem cell population. In the adult CNS, NSCs reside predominantly in two niches: the sub-ventricular zone (SVZ) around the lateral ventricles and the dentate gyrus of the hippocampus [5]. Within these niches, various soluble factors are known to contribute to NSC proliferation, including epidermal and fibroblast growth factors (EGF and FGF). The observation that tightly regulated tissue-specific somatic stem cells persist into adulthood, both in the CNS and elsewhere, has opened exciting new avenues for scientific research. In particular, researchers have begun to investigate the consequences of dysregulation in the stem cell niche.

## II. CANCER STEM CELL HYPOTHESIS

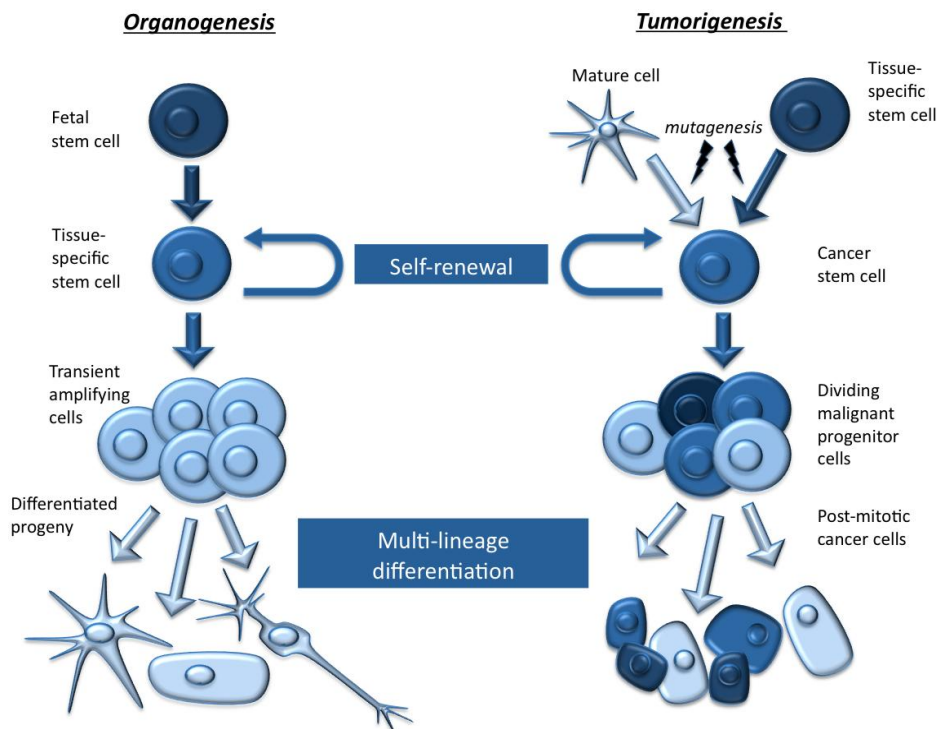
“A tumor can be viewed as an aberrant organ initiated by a tumorigenic cancer cell that acquired the capacity for indefinite proliferation through accumulated mutations. If one views

a tumor as an abnormal organ, then the principles of normal stem cell biology can be applied to understand better how tumors develop” [6]

Striking parallels have emerged between stem cells and transformed malignant cells (see *Figure 1*), providing the initial circumstantial evidence for an etiological link between stem cells and cancer [1, 6]. Both cell types have a significant capacity for self-renewal: highly regulated self-renewal is characteristic of somatic stem cells, whereas dysregulated self-renewal, in the form of replicative immortality, is a hallmark of cancer [7].

Furthermore, both cell types differentiate into morphologically and antigenically heterogeneous lineages: stem cells differentiate to form various mature tissue components, while cancer cells undergo aberrant differentiation to form phenotypically diverse progenitors [1].

These parallels have raised important questions about whether our understanding of tumor development and progression can be refined by conceptualizing cancer as a fundamentally stem cell-driven system.



**Figure 1.** *Organogenesis* occurs when tissue-specific somatic stem cells give rise to transient amplifying cells, which divide rapidly before differentiating to form mature tissue. The tissue-specific stem cell population is maintained through self-renewal. In *tumorigenesis*, cancer stem cells arise from mutagenic transformation of mature or stem cells. These cancer stem cells acquire or retain the ability to self-renew indefinitely, conferring the phenotype of uncontrolled replication. Like normal tissue-specific stem cells, cancer stem cells also differentiate over many generations, forming phenotypically heterogeneous tumors.

## Functional Definition of Cancer Stem Cells

The idea that cancer is initiated and maintained by its own unique set of abnormal stem cells, termed cancer stem cells (CSCs), is known as the *cancer stem cell hypothesis*. As the CSC hypothesis becomes more widely cited, considerable controversy continues within the literature regarding its definition, the requirements to test it, and its limitations. For this chapter, we use the definition established by a panel convened by the American Association for Cancer Research (AACR).

Cancer stem cell: a cell within a tumor that possesses the capacity to self-renew and to cause the heterogeneous lineages of cancer cells that comprise the tumor [8]

According to the AACR consensus, such cells form a self-sustaining reservoir that is crucial to continued tumor expansion. Based upon the AACR findings and the literature, we have summarized five central tenets that comprise the foundation of the CSC hypothesis (see *Figure 2*).

## Empirical Evidence for the CSC Hypothesis

The first evidence for the CSC hypothesis emerged from the study of leukemia and multiple myeloma.

### The Five Tenets of the Cancer Stem Cell Hypothesis:

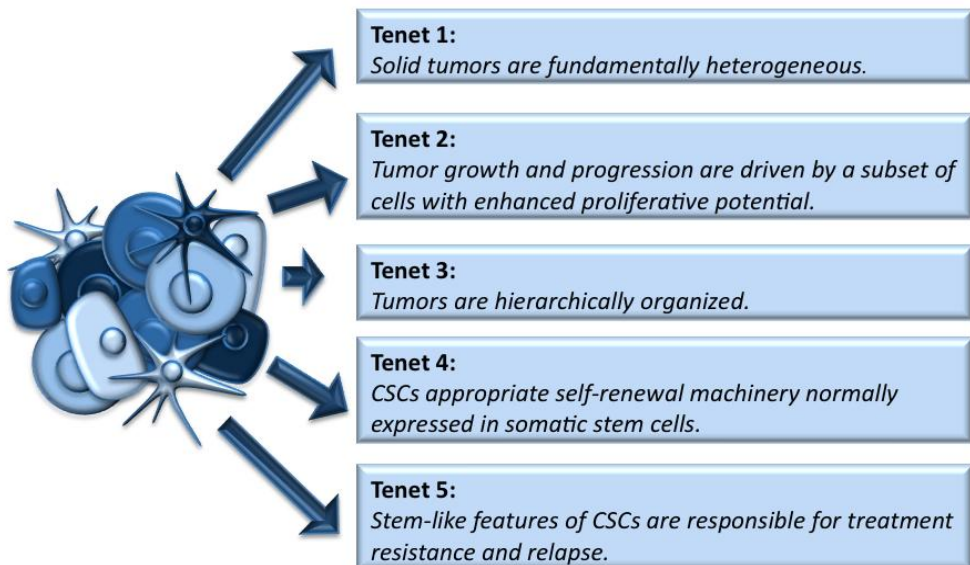


Figure 2. Based on the current literature, we have summarized five central tenets of the cancer stem cell hypothesis.

To demonstrate that a predictable subset of cells was capable of initiating disease, Dick et al. isolated CD34<sup>+</sup>CD38<sup>-</sup> cells from patients with acute myeloid leukemia (AML) and showed these cells had a highly enriched clonogenic capacity when engrafted in non-obese diabetic severe combined immunodeficient (NOD-SCID) mice [9]. Importantly, these cells, which occurred at a frequency of approximately one per million tumor cells, were capable of regenerating the phenotypic heterogeneity observed in the original patient samples.

Taken together, these observations constituted the first evidence that a human cancer (a) contained a distinct subset of stem-like cells with enriched cancer-initiating potential, and (b) these cells could differentiate into phenotypically distinct progenitors. With these experiments, the proof of principle was established for the CSC model. The next challenge was to demonstrate the existence of highly tumorigenic sub-populations in solid tumors. Al-Hajj et al. found that CD44<sup>+</sup> CD24<sup>-/low</sup> subpopulations isolated from human breast cancer specimens were highly tumorigenic when xenografted in NOD-SCID mice, forming tumors from as few as 100 cells. By contrast, cells of other antigenic phenotypes failed to form tumors even when tens of thousands of cells were injected [10].

### Empirical Criteria for Cancer Stem Cells

For a set of isolated cells to be considered CSCs, three observations are required [4].

- 1) Only a subset of cancer cells isolated from each tumor retain tumorigenic potential when xenografted into immunodeficient mice
- 2) These tumorigenic cells display distinctive stem cell surface markers
- 3) Tumors that arise from xenografts of these tumorigenic cells recapitulate the phenotypic heterogeneity of the original tumor

Using these criteria, CSC populations have been isolated from a number of solid tumors, including cancers of the prostate [11, 12], skin [13, 14], pancreas [15, 16], colon [17, 18], lung [19, 20], and brain [21, 22].

## III. BRAIN TUMOR STEM CELLS DISCOVERY

The first evidence for a CSC population in gliomas was found when Ignatova et al. first isolated stem-like cells from glial tumors utilizing growth conditions optimized for neural stem cells [23]. The investigators isolated stem-like cells from tumor samples that mimicked neural stem cells and formed non-adherent *neurospheres* in culture. The first *in vivo* evidence originated from the Dirks lab when Singh et al. isolated highly tumorigenic cells with the stem cell marker CD133 (also known as Prominin-1) from fresh glioblastoma patient samples (Figure 3) [22]. *In vitro*, these cells demonstrated significant capacity for both self-renewal and proliferation, forming neurospheres when cultured in serum-free medium supplemented with EGF/FGF.

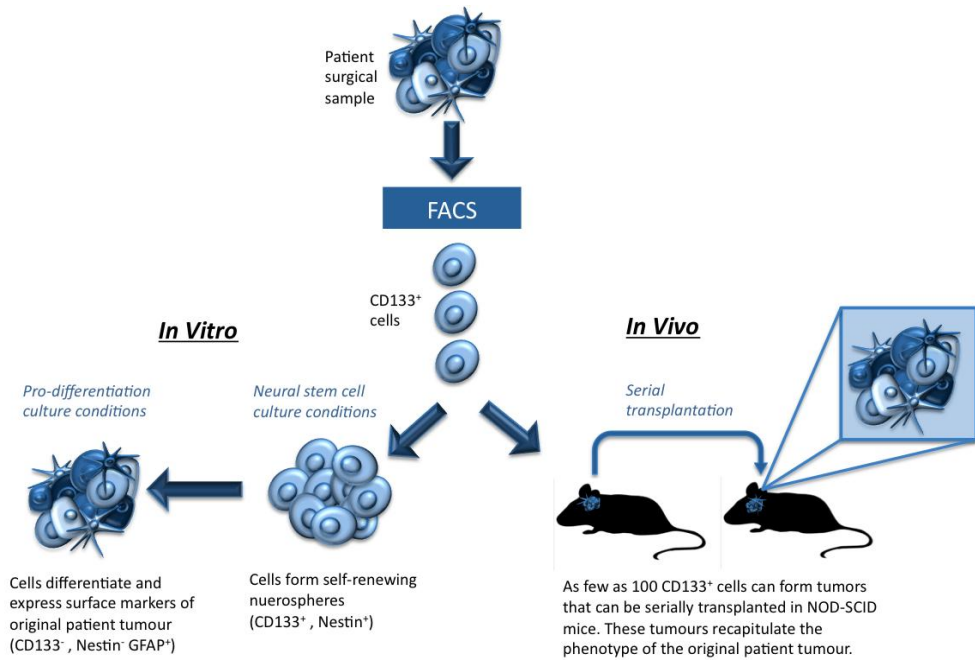


Figure 3. Schematic of isolation of brain tumor stem cells from patient surgical samples. CD133<sup>+</sup> cells were sorted from patient tumor samples via fluorescence-activated cell sorting (FACS). When grown under neural stem cell conditions (serum free, supplemented with epidermal and fibroblast growth factors), CD133<sup>+</sup> cells formed self-renewing neurospheres. When cultured in pro-differentiation conditions (serum, no growth factor supplements), the CD133<sup>+</sup> cells differentiated and expressed markers found in the original patient tumor (e.g. GFAP). *In vivo*, as few as 100 CD133<sup>+</sup> cells form tumors, while up to  $1 \times 10^4$  CD133<sup>+</sup> cells fail to form serially transplantable tumors. The tumors that arise from CD133<sup>+</sup> cells accurately recapitulate the phenotypic heterogeneity and architecture of the patient tumor.

*In vivo*, as few as 100 CD133<sup>+</sup> cells could regenerate tumors in immunocompromised mice, compared to  $10^5$  CD133<sup>+</sup> cells from the same sample [22]. Importantly, tumors derived from these CD133<sup>+</sup> cells recapitulated the phenotypic heterogeneity of the original patient tumor.

Through these experiments, the authors demonstrated that the putative brain tumor stem cells met all three functional criteria for CSCs. Subsequent groups explored alternative methods of isolating and enriching for CSCs in patient-derived glioma samples. Alternative surface markers emerged, including CD15/SSEA-1 [24, 25], L1CAM [26], A2B5 [27], and integrin  $\alpha 6$  [28]. Refined culture methods also demonstrated the existence of mitogen-independent brain tumor stem cells [29]. In virtually all studies, the phenotypic heterogeneity of the patient tumor was recapitulated in an orthotopic xenograft model from a small number of cells. As the field matures, clinical evidence that brain tumor stem cells exist and drive treatment resistance in patients remains lacking. As such, many experts in the field dispute the term “brain tumor stem cells” in favor of the less value-laden term *Brain Tumor Initiating Cells* (BTICs). This nomenclature emphasizes that such cells are isolated based on their ability to initiate tumors in xenograft models, and are not necessarily of neural stem cell origin.

## Key Questions

As apparent from the introduction above, there remain several important questions concerning BTICs and CSCs that we address below.

### ***Do Mutated Stem Cells Represent the Cell of Origin for Gliomas in Patients?***

To answer this question, we must first understand the developmental hierarchy of normal neural tissue. Replication-competent NSCs in the post-natal brain were first isolated under neurosphere conditions by Sam Weiss and Brent Reynolds in 1992 [30]. These endogenous NSCs persist into adulthood, poised at the apex of the brain's developmental hierarchy. Over time, NSCs undergo asymmetrical cell division in order to simultaneously maintain the stem cell population and give rise to transient amplifying progenitors, which proliferate and gradually lose their self-renewal abilities with each division, giving way to lineage-committed progenitors. These progenitor cells in turn give rise to mature astrocytes, oligodendrocytes, and neurons. To date, this model has been envisioned as being exclusively unidirectional, with multipotent cells gradually giving way to more committed lineages. However, it is possible that certain de-differentiation events may occur as part of normal neural tissue development. Neurogenic precursors have been shown experimentally to de-differentiate in response to EGF [31] and astrocytes have been shown to revert to a stem cell phenotype in response to transforming growth factor alpha [32], highlighting a possible role for de-differentiation events in neural development.

With the advent of the CSC hypothesis, it has been suggested that gliomas arise from actively dividing NSCs that mutate during the process of normal neural development. These mutant cells are thought to produce a renegade population of multipotent stem-like cells with dysregulated proliferative abilities, which eventually expand and generate the full spectrum of tumor cells. The expression of NSC-associated proteins such as nestin by certain BTIC populations highlights this possibility [33]. However, it is also possible that mature cells, such as astrocytes, may assume certain stem-like properties following oncogenic insult, and revert to a less differentiated phenotype. It stands to reason that different cells of origin are possible, perhaps at different levels of the hierarchy, and that the identity of the cell of origin may greatly influence the phenotype and classification (neural, proneural, mesenchymal, or classical) of the patient tumor. Many research groups have experimented with the generation of oncogenic mutations in specific cell types by genetic or anatomical targeting, with the objective of understanding which cells in the hierarchy are capable of acting as the cell of origin. For example, Holland et al. demonstrated that nestin-expressing NSCs could give rise to high-grade gliomas upon the activation of Ras and Akt [34]. In addition, GFAP-expressing NSCs and progenitor cells can give rise to high-grade gliomas upon GFAP-Cre-mediated inactivation of p53 and PTEN [35]. Similarly, it has been shown that inactivation of Nf1 and p53, or Nf1, p53, and PTEN, in nestin-expressing NSCs can give rise to malignant gliomas [36]. Jacques et al. also demonstrated that deletion of PTEN/p53 in adult SVZ stem cells can result in gliomagenesis, whereas mutation of the same tumor suppressors in mature parenchymal astrocytes did not [37]. Each of these studies demonstrated that NSCs or their progenitors are capable of oncogenic transformation. Importantly, however, these studies do not specifically address whether the mutant NSCs themselves, or their mutant progenitors, are ultimately responsible for tumor formation.

Recently, an influential paper by Liu et al. shed light on the cell of origin question using an elegant lineage-tracing technique termed mosaic analysis with double markers (MADM) [38]. The authors used a Cre-mediated recombination system in which p53/Nf1 inactivating mutations were initiated in dividing NSCs. Successful recombination events activated the expression of green fluorescent protein (GFP), while alternative recombination that did not inactivate p53/Nf1 activated red fluorescent protein (RFP) expression. Non-recombinant cells lacked both fluorescent tags. The authors observed that both wildtype (RFP) and mutant (GFP) NSCs could give rise to the normal repertoire of progenitors. Although a variety of different cell types bore the oncogenic mutation, including NSCs, oligodendrocyte precursor cells (OPCs), neurons, astrocytes, and oligodendrocytes, only OPCs underwent transformation and initiated tumor development. Importantly, these OPCs acquired the expression of NSC markers upon transformation, highlighting that certain NSC phenotypic traits, such as surface marker expression, can be acquired by an otherwise committed cell type. Furthermore, although NSCs may represent the cell of mutation, they are not necessarily the most likely candidates for the cell of origin, in this murine model of p53/Nf1-driven glioma.

### ***Do Current Experimental Models Reflect Stable Populations of Stem-Like Cells in Patient Tumors?***

A fundamental assumption in the field is that CSCs isolated from patient surgical samples accurately represent stem-like cells in the patient tumor. However, there is concern that the experimental strategy for isolating CSCs - using *in vitro* culture systems or xenografts to serially propagate stem-like cells - will select for populations that are not true CSCs. Once a cell reaches the stage of xenograft transplantation, it has survived various manipulations, including removal from its niche in the patient tumor, isolation from the surgical specimen, mechanical and enzymatic dissociation, sorting, and implantation into a series of foreign hosts. Each of these events exerts a unique selective pressure, confounding an assay that is intended to provide an unbiased read-out of tumorigenicity. Furthermore, no assay is available to interrogate the tumorigenicity of a given patient-derived BTIC population in a syngeneic host; all models, by definition, must cross the species barrier, which introduces an additional element of selective pressure [39]. Lastly, putative BTICs are transplanted from an immunocompetent host to an immunocompromised model, which may substantially influence the tumor microenvironment. In summary, the context in which tumorigenicity is measured differs widely from the context of the original tumor [39]. Our current experimental models select for cells that can survive in the murine microenvironment, rather than selecting for the most tumorigenic sub-population, making it difficult to extrapolate to patient tumors [40, 41].

Assuming a BTIC population does exist in patients, how stable is the stem-like phenotype? We must consider that the BTICs isolated from surgical specimens only provide a snapshot of the cellular hierarchies at the time of resection. Given the highly plastic nature of these cells, it is possible that their phenotypes are unstable, and may change over days, weeks, or the lifetime of the tumor. As described by the AACR panel:

The stem cells within an individual tumor may constitute a moving target, that is, the cells that drive growth at one point in time may not be identical to those doing so at another stage in tumor evolution...furthermore, the genetic and epigenetic instability that are



fundamental properties of tumor biology can induce cellular heterogeneity within the stem and non-stem cell populations of the tumor [8]

In addition to intrinsic cellular changes that may make patient BTIC populations unstable, one must also consider the evolution of the niche over the lifetime of the tumor. Various factors, such as the infiltration of immune cells and development of large areas of necrosis - as well as changes caused by surgery, chemotherapy, and radiation - profoundly alter the tumor microenvironment, which may impact the stability of BTIC populations. We must remember that BTICs are only an important therapeutic target if the phenotype in patients is both stable and reasonably similar to experimentally-isolated BTICs [42]. At present, there is no direct evidence in patients that untreated human gliomas maintain a stable subset of uniquely tumorigenic stem-like cells.

### ***Do BTICs Have Prognostic Relevance?***

The clinical data regarding the prognostic relevance of BTIC markers in high-grade gliomas remain somewhat conflicting. In 2008, Zhang et al. demonstrated that survival in glioma patients was positively correlated with low expression of the stem cell markers CD133 and nestin in tumor samples, as determined by immunohistochemistry (IHC) [43]. Similarly, expression of CD133 was identified as an independent prognostic marker in a study of 44 glioblastoma patients in Italy [44] and 95 glioma patients in Germany (45), with high expression being correlated with low survival. However, a recent clinical study examining the expression of three BTIC markers (CD133, CD15, and nestin) in 88 glioblastoma samples showed no correlation between BTIC marker expression and clinical outcome, suggesting that these markers may lack prognostic significance [46]. As more clinical data are accumulated, the prognostic relevance of BTIC markers will continue to be clarified.

### ***What Defines the BTIC Niche in the Tumor?***

It is well known that the adult NSC niche is finely tuned to support the balance between stem cell proliferation and quiescence. Similarly, it has been observed that BTICs preferentially occupy certain tumor microenvironments, particularly the nutrient-rich microvascular proliferating zone or *perivascular niche* (PVN) [47]. The PVN, which borders angiogenic microvascular structures within the tumor, is characterized by significant cellular heterogeneity - within this region can be found astrocytes, endothelial cells, macrophages, microglia, and post-mitotic tumor cells, in addition to BTICs [48]. In several types of brain cancer, including high-grade gliomas, the density of these PVN structures appears correlated to the number of BTICs in the tumor [47]. Disruption of the vasculature, and concomitantly the perivascular niche, has been shown to deplete the BTIC population and halt the growth of PDGF-driven gliomas [47]. Mechanistically, it has also been shown in a PDGF-induced glioma model that the endothelial cells, juxtaposed alongside BTICs, release nitric oxide (NO), a highly reactive free radical that activates the Notch pathway in BTICs and promotes self-renewal *in vitro* as well as tumor growth *in vivo* [49]. However, communication between BTICs and the vasculature is bidirectional; not only do endothelial cells enhance BTIC proliferation, but BTICs also enhance angiogenesis through the expression of vascular endothelial growth factor (VEGF) [50, 51]. It is believed that the expression of VEGF by BTICs is initiated by the stabilization of hypoxia-inducible factors (HIF) transcription factors, activated in response to hypoxic conditions within the tumor [52]. Interestingly, various genes

associated with the BTIC phenotype are transcriptionally regulated by HIF2 $\alpha$ , and hypoxic conditions have been shown to trigger non-stem cells to revert into a more stem-like state [52-54]. This demonstrates that BTICs can occupy both hypoxic and perivascular niches, and alter their gene expression according to their niche microenvironment.

### ***Are BTICs Resistant to Multiple Treatment Modalities?***

The current standard of care for high-grade gliomas includes surgical resection followed by radiotherapy combined with the DNA alkylating agent temozolomide. Despite this three-pronged approach, the prognosis remains dismal. The emergence of the CSC hypothesis provides one explanation for the failure of conventional glioma therapies: traditional approaches target the bulk of tumor cells but fail to eradicate the CSCs that fuel its growth. Similarly, small molecule inhibitors are designed based on the documented gene signatures of the bulk of tumor cells; these gene signatures may not be representative of the CSC population in these tumors, allowing them to escape eradication by targeted therapies. Over the past several years, BTICs have been shown to possess different mechanisms that specifically enhance resistance to both chemo- and radiotherapy. Resistance to radiotherapy may arise through both an enhanced capacity for DNA repair in response to genotoxic stress, and by resistance to radiation-induced apoptosis. Bao et al. demonstrated that CD133<sup>+</sup> BTICs have reduced sensitivity to ionizing radiation compared with their CD133<sup>-</sup> counterparts [55]. In fact, under the selective pressure of radiation, BTIC populations expand dramatically, a phenomenon enabled by their preferential activation of the DNA damage checkpoint response. A variety of molecules have been implicated in radioresistance, including Notch, SirT1, and Bmi1, suggesting that multiple overlapping mechanisms exist for maintaining a radioresistant state [56-58]. Slow transit time through the cell cycle of CD133<sup>+</sup> BTICs, compared to CD133<sup>-</sup> cells, combined with high basal expression of the checkpoint kinases Chk1 and Chk2, are also thought to contribute to radioresistance [59]. Furthermore, CD133<sup>+</sup> BTICs have shown heightened expression of a variety of anti-apoptotic proteins, including livin and survivin, which are thought to counteract the pro-apoptotic effect of genotoxic stress [60]. In addition to mechanisms for radioresistance, BTICs also have increased resistance to chemotherapeutics. In particular, they are known to express various ATP-binding cassette (ABC) transporters, which are transmembrane drug efflux pumps that confer reduced sensitivity to various chemotherapeutic agents. For instance, CD133<sup>+</sup> BTIC populations express high levels of ABCG2/BCRP1 [61, 62], a promiscuous transporter that can efflux both hydrophobic and hydrophilic chemotherapeutics, including topotecan, doxorubicin, irinotecan, imatinib, and methotrexate [63]. Some BTICs, though not all, are resistant to temozolomide (TMZ), though the mechanism of resistance has yet to be determined [64, 65]. It has been found, however, that inhibiting pathways unique to the BTIC phenotype, such as Notch and sonic hedgehog, can increase susceptibility to TMZ [66].

### ***Are BTICs a Viable Therapeutic Target?***

With the failure of conventional therapies to substantially improve patient prognosis, considerable emphasis has been placed on determining whether BTICs represent a valid therapeutic target in high-grade gliomas. As mentioned above, various surface markers have been identified for isolating BTIC populations, including CD133, CD15/SEA-1, L1CAM, A2B5, and integrin  $\alpha$ 6. Whether these proteins represent useful therapeutic targets, or serve

as viable prognostic markers, remains to be determined. Interestingly, a functional role has been shown for L1CAM and integrin  $\alpha 6$ , whereby inhibition or knockdown reduces tumorigenicity in xenograft models [26, 28].

In addition to cell surface proteins, various signaling cascades implicated in the maintenance of intra-tumoral BTIC populations may be targeted. For instance, the sonic hedgehog (SHH) pathway regulates BTIC proliferation, and high SHH expression correlates with reduced patient survival [67]. Abrogation of SHH signaling reduces BTIC tumorigenicity *in vivo* [67-69]. Similarly, Notch signaling is activated in BTICs and NSCs [70]. Forced expression of the Notch intracellular domain leads to the expression of nestin (a quintessential BTIC/NSC marker) and enhances the expression of resistance-conferring ABC drug transporters. Inhibiting Notch cleavage with gamma-secretase inhibitors reduces tumorigenicity drastically and prolongs survival *in vivo* [58, 70]. These results demonstrate the therapeutic potential of small molecule inhibitors targeting BTIC pathways. Another strategy to inhibit BTICs is the induction of differentiation to improve radio- and chemosensitivity. Piccirillo and colleagues demonstrated that bone morphogenetic proteins (BMPs) induce differentiation of BTICs *in vitro* and virtually eliminate tumorigenic potential in xenograft transplantation assays [71]. Similarly, Wurdak et al. demonstrated that knockdown of the adapter protein TRRAP induced BTIC differentiation *in vitro* and suppressed tumor formation in intracranial xenograft models [72]. In the clinic, differentiation strategies may prove to be safer, less toxic alternatives than treatments that generate massive cell death within the brain. CSC-targeting treatments are currently being piloted in the clinic for a variety of solid tumors. For instance, pilot studies have been established for the use of SHH pathway inhibitors to target CSC populations in pancreatic cancer (clinicaltrials.gov identifier: NCT01195415) and head and neck cancer (NCT01255800). Similarly, the gamma-secretase inhibitor MK-0752 is being tested in Phase I/II trials for its ability to eliminate breast cancer stem cell populations (NCT00645333). In gliomas, the gamma-secretase inhibitor RO4929097 is also being tested in various Phase I/II trials (NCT01269411; NCT01119599; NCT01189240). Similarly, the SHH inhibitor GDC-0449 is currently being tested in Phase I/II trials for gliomas and medulloblastoma (NCT00980343; NCT00939484; NCT00822458). Feasibility studies are also being conducted to optimize the clinical procedures required to rigorously test the efficacy of CSC-targeting therapies. For example, a feasibility study is underway at Stanford University to determine whether CSCs in head and neck cancers can be accurately quantified using presurgical patient tissue biopsies (NCT00610415). Such studies will also be necessary in high-grade gliomas before undertaking large-scale trials using BTIC-specific therapeutics.

## CONCLUSION

The discovery of stem-like cells in human gliomas has challenged our fundamental assumptions about brain cancer and offered a fresh therapeutic perspective to examine tumor heterogeneity and recurrence. New therapeutic targets have been elucidated and new treatment approaches suggested for high-grade gliomas; however, the initial rush of excitement has given way to prudent caution. Though the BTIC hypothesis offers much

promise to improve patient care, many vital questions still remain, and we must be ruthless, relentless, and rigorous in our pursuit of answers.

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*Chapter 15*

## **GLIOMA EXTRACELLULAR MATRIX MOLECULES AS THERAPEUTIC TARGETS**

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### **ABSTRACT**

Malignant gliomas which rarely metastasize, but are locally highly invasive into the extracellular matrix (ECM) microenvironment in the brain are difficult to treat. Despite years of research, current therapies including surgery, radiotherapy and chemotherapy are aimed at debulking of the brain tumor mass and targeting proliferating tumor cells. These therapeutic approaches do not take into consideration the ECM, its composition and the changes in ECM rigidity which can significantly affect glioma cell motility resulting in tumor cell dissemination through the brain and establishment of new, distant tumor foci. Some ECM molecules such as metalloproteinases (MMPs) which are essential in matrix remodeling have been targeted therapeutically.

However, due to the highly pleiotropic properties of MMPs, the outcomes of the clinical studies targeting these molecules have been disappointing and often resulted in increased tumor growth. In addition, current therapies do not address the quiescent population of the cancer stem cells which thrive in the ECM microenvironment of the vascular niche and are protected from therapeutic interventions. This chapter will focus on the review of the most recent information pertaining to the glioma extracellular microenvironment and identification of the possible biological targets within ECM for anti-glioma therapy.

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## 1. INTRODUCTION

Malignant gliomas are primary brain tumors, which are highly invasive. The median survival time of patients with glioma is only 6 months to 2 years depending on various patient, tumor and treatment parameters (Louis et al. 2007, reviewed by Wiranowska and Rojiani, 2011). Gliomas have highly aggressive character and glioblastoma multiforme also called glioblastoma (GBM) is the most aggressive subtype of gliomas. GBM is characterized by the diffuse infiltration into the normal brain parenchyma and interaction with the rich tumor microenvironment (Alves et al. 2011) containing extracellular matrix (ECM) components of the brain.

The infiltrative path of glioma into the normal brain parenchyma involves the basement membrane of blood vessels and myelinated nerve fibers of white matter tracts (Rao 2003, Lefranc et al. 2005). Despite, that GBM does not invade inside blood vessel walls or metastasize outside of the brain (Ananthanarayanan et al. 2011) the standard brain tumor therapies, such as surgery, chemotherapy and radiation are not effective in eradicating single glioma cells that migrated into the normal brain establishing new tumor foci. Glioma cells are locally invasive and when migrating through the ECM within several millimeters or centimeters from the main lesion they initiate recurrent tumors often distant to the primary lesion (Bolteus et al. 2001).

The pattern of glioma cell invasion is related to the unique composition of the cerebral ECM microenvironment, which is remodeled during invasion by activated matrix metalloproteinases (MMPs) (reviewed by Rojiani et al. 2011). In addition, new ECM molecules are secreted and receptor adhesion molecules are expressed by glioma promoting the glioma cell-ECM interaction and signaling. Some of the secreted ECM molecules such as tenascin-C are known to be associated with cell motility and angiogenesis which are both essential for tumor development. Another important microenvironment component affecting glioma development was found to be mechanical force determined by ECM rigidity. More rigid ECM promotes glioma migration and proliferation and lower rigidity of ECM (similar to that of normal brain) would have an opposite effect (Ulrich et al. 2009). The recent sequencing data presented by the Cancer Genome Atlas Research Network (2008) revealed genomic abnormalities in GBM that relate to several signaling pathways such as Epidermal Growth Factor Receptor (EGFR) /Ras /PI3K known to be associated with ECM-related signaling (Ulrich et al. 2009).

In addition, the recent integrated genomic analysis identified clinically relevant subtypes of GBM with its characteristic abnormalities in platelet derived growth factor receptor A (PDGFRA), isocitrate dehydrogenase 1 (IDH1), neurofibromin 1 (NF1), and confirmed EGFR mutations across all newly defined subtypes of GBM such as classical, proneural, neural and mesenchymal (Verhaak et al. 2010). The most recent studies by Katz et al. (2012) in PDGF-driven mouse models of proneural GBMs with a focus on the biology, therapeutic response and the complexity of the microenvironment showed that some of the genes found in mice are predictive of the survival of patients with this proneural subtype of GBM. Interestingly, many of these genes are expressed in the stroma of the tumor rather than by the tumor cells themselves.

## 2. EXTRACELLULAR MATRIX MOLECULES IN THE NORMAL BRAIN

In the central nervous system (CNS) approximately only 15-25% of the CNS volume is taken up by the extracellular space, while the majority of the CNS volume consists of cellular elements such as neurons, glia, astrocytic processes and blood vessels (Quirico-Santos et al. 2010).

The components found within the extracellular space include various ions, metabolites, neurohormones, peptides and ECM molecules produced by neurons and glia. The ECM environment of the normal brain contains high levels of spacefilling carbohydrate molecules unbound to proteins such as the large glycosaminoglycan (GAG) hyaluronan (HA). HA binds to specific cell surface receptors such as cluster determinant 44 (CD44) adhesion molecule and receptor for hyaluronate mediated motility (RHAMM) regulating properties of ECM and tissue, e.g., proliferation, adhesion, motility etc. Protein-bound carbohydrate molecules, which are present in the normal brain at high levels, include sulfated proteoglycans such as chondroitin sulfate proteoglycans (CSPGs) and heparan sulfate proteoglycans (HSPGs). In addition, fibrous proteins associated with the basement membranes of the brain's vasculature include collagens, fibronectin, and laminin (reviewed by Wiranowska and Plaas 2008). The levels of these fibrous proteins in the normal brain are low compared to the connective tissue outside the central nervous system (Bellail et al. 2004, Quirico-Santos et al. 2010). However, the ECM microenvironment of glioma differs from the normal brain and varies depending on the grade of glioma, e.g., with the highly aggressive GBM producing collagen, fibronectin or laminin (Mahesparan et al. 2003).

Several classes of ECM molecules that play an important role in the normal CNS development but have altered functions in glioma are reviewed below. The main classes of ECM components in the normal brain are GAG hyaluronan (HA), also called hyaluronic acid, and proteoglycans (PGs), which consist of a core protein attached to GAG chain. HA plays multiple roles in providing an organization of the pericellular matrix. There is a high diversity of PGs due to various core proteins, and variations in GAG side chains. Two classes of transmembrane PGs, glypicans and syndecans, which contain heparan sulphate (HS) side chains called HSPGs are found at high levels in the CNS. An important class of PGs are chondroitin sulfate PGs (CSPGs), which are expressed at high levels in the regions of the developing fetal brain and later in mature brain in astrocytes and neurons (Rao 2003, Quirico-Santos et al. 2010). CSPGs, and especially the subclass of lecticans, are one of the major families of HA binding matrix glycoproteins in the CNS. A second family of PGs that bind HA in the CNS are HA- and proteoglycan-link proteins (HAPLNs) also called "link proteins", which bind both HA and lecticans.

The PGs called lecticans contain lectin and HA-binding domains and within that group there are molecules such as aggrecan, versican, neurocan, and brain enriched hyaluronic acid binding protein/brevican (BEHAB/brevican) that act as linkers to ECM components. They bind to HA as well as to cell-surface receptors regulating many processes within the CNS during development, e.g., cell motility, axonal navigation etc. (Sim et al. 2009). Some of these molecules such as versican are known to be produced by glial cells and neural stem cells (Abaskharoun et al. 2010). Neuroglial protein-2 (NG2), also a CSPG proteoglycan, which is known as a characteristic marker of oligodendrocyte progenitor cells and pericytes in

developing vasculature, is expressed by many gliomas. The NG2 positive cells have been suggested to be the originating cells for glioma (Stallcup and Huang 2008).

Tenascins (C and R), a family of glycoproteins exist in the ECM as assemblies of several subunits expressed in zones of proliferation, migration, and morphogenesis and are known to play an important role in the developing CNS. For example, tenascin-C was found highly expressed in the subventricular zone and essential for neural stem cell development (reviewed by Wiranowska and Plaas 2008). Galectins (Gal), mannose-binding lectins, are glycan-binding proteins highly represented in the CNS. They take part in the development of neural networks, interact with other neural tissue derived glycoproteins and are expressed by astrocytes and endothelial cells (Quirico-Santos et al. 2010). The role of many ECM components of the normal brain described above, are altered dramatically in glioma. In the normal brain, the ECM complexes containing HA and PGs such as versican, brevican, neurocan, aggrecan, phosphacan and tenascin-C, tenascin-R, and link proteins form the ECM domains called perineuronal nets first described by Camillo Golgi in 1893. These perineuronal net aggregates enwrap the neuronal cell bodies and proximal dendrites of certain neurons and fill the space between neurons and glial processes. More recently, it was proposed that perineuronal nets within the brain are more heterogenous and include structures called “interstitial clefts” (Brightman 2002).

As described by Brightman (Brightman 2002), interstitial clefts comprised of astrocytic walls, basal lamina and ECM molecules may vary in size, shape and content depending on the brain region. In addition, the size and the content of interstitial clefts was found to be different in the mature brain by being narrower with limited capacity for cell movement compared to that in the fetal brain. Here, in the fetal brain, the size and the content of interstitial clefts permit cell migration and outgrowth of neurites while in the mature brain cell migration in the interstitial clefts could only occur after enzymatic degradation of the ECM (Brightman 2002).

## **2.1. Extracellular Matrix in the Brain as a Cytokine and Growth Factor Depot**

In the normal brain some regions are especially rich in ECM. These brain regions include subarachnoid space, subependymal pockets, circumventricular organs (CVOs) supplied by fenestrated capillaries without blood-brain barrier (BBB), and perivascular space around arterioles and venules.

These vessels are associated with stromal connective tissue space and lined by basal laminae containing heparan sulfate proteoglycans (HSPGs). HSPGs which are also components of ependymal, astroglial, and endothelial interfaces in the CNS (including interstitial clefts) have been suggested to serve as a storage site of growth factors and cytokines (Brightman and Kaya 2000).

A large number of growth factors, for example, insulin-like growth factor (IGF), transforming growth factor- $\beta$  (TGF- $\beta$ ), hepatocyte growth factor (HGF) were found to bind to HSPG (Folkman 1998). A similar observation was made for certain cytokines (reviewed by Wiranowska and Plaas 2008). It was suggested by Mercier et al. (2003) that cytokines and growth factors secreted by cells of connective tissue may accumulate in the basal lamina, interact with ECM proteins and affect biological processes including cytotogenesis of stem cells in the CNS.

## 2.2. Extracellular Matrix Stem Cell Niche

### 2.2.1. Normal Brain

In the CNS, the ability of normal stem cells to self-renew and to differentiate into specific cell types is controlled by the microenvironment of a CNS area in which these cells reside and which is called niche. Similarly, in other tissues and organs, stem cells are found in the protective microenvironment of niches, which are composed of ECM molecules and various differentiated cell types that release regulatory factors and provide direct contact with stem cells maintaining their quiescence.

The CNS microenvironment of the neural stem cells (NSC) niche is also called vascular niche, because stem cells concentrate near blood vessels. The NSC niche consists of several ECM components, and includes the basal lamina and endothelial cells of vasculature (Doetsch 2003). These mature, differentiated vascular endothelial cells have an intimate association with stem cells and play a regulatory role in the NSC niche through secreted soluble factors. These factors were shown to promote activation of Notch, a neural precursor receptor, resulting in self-renewal of neural stem cells (Shen et al. 2004). In addition, the basement membrane (also known as basal lamina) contributes to the microenvironment and provides a substrate for stem cells' movement. The subventricular zone, a highly neurogenic area in the CNS, contains transmembrane HSPGs bound to the supependymal basal lamina located in proximity to the stem cells.

As mentioned earlier (Section 2.1), HSPGs have the capacity to bind and to store a number of growth factors and cytokines thereby serving as a cytokine and growth factor depot. The growth factors and cytokines can diffuse quickly, and because of close proximity, they can reach high concentrations near the stem cells and regulate their development (Kearns et al. 2003). For example, EGF and basic fibroblast growth factor (bFGF) stored in the ECM of the subventricular zone can have a stimulatory effect on stem cells by enhancing their proliferation. The growth factors and cytokines can be stored in the ECM throughout life. As mentioned earlier, one of ECM molecules, tenascin-C, is highly expressed in the subventricular zone and essential for neural stem cell development (reviewed by Wiranowska and Plaas 2008). Tenascin-C plays a key role in the regulation of the developmental program of oligodendrocyte precursor cells (OPCs) and therefore confirming the importance of tenascin-C as an ECM component of the niche (Scadden 2006). Other ECM molecules, such as laminin and fibronectin, stimulate motility of stem/progenitor cells while CSPGs have an inhibitory effect (Kearns et al. 2003).

Another important ECM molecule is HA which may play a critical role in regulating NSC proliferation, differentiation and migration in the niche (Preston and Sherman 2012). It was also observed that upon activation of MMPs by proinflammatory cytokines, the neural progenitor cells were stimulated to migrate to the site of injury (Ben-Hur et al. 2006). In summary, many modulatory molecules were described within the ECM of the stem cell niche and, interestingly, many of them were found in glioma, but at higher levels than in the normal brain. In addition, not only the levels but also the functions of many of these molecules differ between the normal brain and glioma such as CSPGs, which are inhibitory for stem cells migration within the niche of the normal brain but stimulatory for glioma cell migration (Kearns et al. 2003; Sim et al. 2009).

### 2.2.2. Glioma

Recently, a new concept of cancer progenitor cells, also known as cancer-initiating cells or cancer stem cells (CSCs), was proposed. These self-renewing, multipotent CSCs are highly tumorigenic and resistant to conventional therapies (Lakka and Rao 2008). Glioblastoma CSCs resemble the normal NSC and express the markers Nestin+/- CD133+ found in the neural stem cell population.

Also, glioma CSCs, similar to NSCs, concentrate around blood vessels in the vascular niches with easy access to nutrients, signaling molecules, and the vasculature itself as a substrate for migration (Calabrese et al. 2007; Denysenko et al. 2010). However, CSCs differ from NSCs in their distribution in the brain and their capacity to proliferate. For example, the normal NSCs, which proliferate at a low rate, are found only in specific CNS regions such as hippocampus and subventricular zone.

In contrast, highly proliferative glioma CSCs can be found distributed across all regions of cerebrum and cerebellum within the tumors. It was proposed that the main difference between the normal NSCs and glioma CSCs may be the way in which these cells are modulated by the microenvironment of the vasculature within the niche (Calabrese et al. 2007). The vascular niche in brain tumors is abnormal in such that it contributes to the propagation of CSCs thereby enhancing tumor growth. Furthermore, the endothelial cells from this abnormal vascular niche can interact with brain tumor CSCs, as shown *in vitro*, providing certain extracellular regulatory factors and maintaining the self-renewal capability and undifferentiated state of these cells (Calabrese et al. 2007). In that way the glioma vasculature establishes a microenvironment of the niche in which CSCs can transmit and receive signals from the ECM.

For example, it was shown that upon stimulation of CSCs by ECM of the vascular niche, the CSCs can secrete vascular endothelial growth factor (VEGF) promoting angiogenesis and thereby enhancing tumor growth (Bao et al. 2006). In addition, the vascular niche was shown to interfere with radiation and chemotherapy by shielding the CSCs and contributing to the resistance to treatments (Denysenko et al. 2010). It was also suggested that the microenvironment of the niche may play a role in tumor initiation based on the observation that non-tumorigenic cell populations may become tumorigenic depending on a certain microenvironment (Rosen and Jordan 2009).

Recent reports show that GBM stem cells have similar capabilities as the normal NSCs and undergo differentiation into endothelial cells forming the majority of new blood vessels in gliomas (El Hallani et al. 2010; Ricci-Vitiani et al. 2010; Wang et al. 2010). Blocking VEGF or silencing VEGF receptor 2 inhibits the maturation of tumor endothelial progenitors into endothelium but does not stop the differentiation of CSCs (CD133+ cells) into endothelial cells. However, silencing of Notch (neural precursor receptor as mentioned in section 2.2) blocks the transition of CSCs (CD133+ cells) into endothelial progenitors (Wang et al. 2010). Further studies of the microenvironmental components within the brain tumor vascular niche could lead to new therapeutic targets for treatments of glioma.

### 3. EXTRACELLULAR MATRIX IN GLIOMA

#### 3.1. Extracellular Matrix and Mechanical Rigidity in Glioma

There are anatomic variations in stiffness in the normal brain parenchyma (Elkin et al. 2007) with basement membrane of blood vessels and the myelinated fiber tracts of white matter exhibiting a higher mechanical rigidity (Lefranc et al. 2005) and both serving as an infiltrative path for glioma invasion (Rao 2003; Ulrich et al. 2009). It was first observed *in vitro* that directed migration of fibroblasts occurs from soft to stiff areas of the ECM, a phenomenon named mechanotaxis (Lo et al. 2000). A similar observation was made for glioblastoma where changes in ECM rigidity can both increase and decrease cell motility and the extent of the effect was cell-type dependent (Thomas and DiMilla 2000). It was found that high ECM stiffness enhanced the expression of contractility-mediating proteins such as Rho (Paszek et al. 2005). A stimulatory effect of ECM proteins such as fibronectin, collagen, laminin and others on glioma cell migration were shown (Mahesparan et al. 2003). Ulrich et al. (2009) had shown that morphology, proliferation and motility of GBM cells cultured on two-dimensional (2-D) fibronectin-coated polymeric ECM with varied but defined mechanical rigidity, were regulated by the stiffness of the ECM.

The rigidity of the soft substrates used in that study was comparable to the ECM rigidity of normal brain parenchyma while an increased stiffness was characteristic for glioma and its surrounding stroma. More recently, the motility of glioma cells cultured on three-dimensional (3-D) HA-hydrogels, was similar to that observed in brain slices cultures (Ananthanarayanan et al. 2011). In addition, it was shown that increasing ECM rigidity resulted in increased cell spreading, motility and proliferation. It was suggested previously that glioma cells actively remodel their microenvironment changing it from normal brain ECM to rigid tumor-like ECM (Nakada et al. 2007). Therefore, it was suggested that glioma cells modify their ECM through proteolytic degradation of the normal brain matrix and secretion of new ECM components, thereby providing for a stiffer and more rigid microenvironment which in turn sends mechanobiological signals that support glioma cell invasion (Ulrich et al. 2009). By targeting either the signaling pathways for mechanotaxis or mechanical remodeling itself, new therapeutic approaches could be developed for the treatment of glioma which would affect glioma invasion and proliferation.

#### 3.2. Extracellular Matrix Molecules in Glioma

##### 3.2.1. Glycosaminoglycan Hyaluronan and CD44 Adhesion Molecule

Glioma cells constitutively produce HA and its production is increased during cell proliferation (Wiranowska and Naidu 1994; Wiranowska et al. 2010) promoting glioma invasion (Park et al. 2008). HA is synthesized at the plasma membrane by HA-synthases and the synthesis can be enhanced by various growth factors, e.g., epidermal growth factor (Knudson and Knudson 1993). Interestingly, the content of HA in glioma resembles that of embryonic brain cells (Delpech et al. 1993). HA binds to the HA-binding proteins called hyaladherins which include the CD44 surface receptor. CD44 is a transmembrane glycoprotein expressed by many cell types and by glioma. CD44 serves as a surface receptor

for ECM molecules such as HA and CSPGs (Ranuncolo et al. 2002). CD44 receptor is overexpressed in glioma cells *in vitro* (Wiranowska et al. 2000; Yu et al. 2010) and found *in vivo* at the leading edge of glioma at the brain-tumor interface (Wiranowska et al. 2006). The HA-CD44 interaction and CD44 shedding from the cell surface were found to be associated with glioma cell motility, migration, and infiltration into the normal brain parenchyma (Annabi et al. 2005). These authors also described that CD44 shedding was mediated by HA and accompanied by up-regulation of MT1-MMP expression.

After binding to the CD44 receptor, HA can be endocytosed, transported into lysosomes and degraded by hyaluronidases into small oligosaccharides shown to have glioma stimulatory activity (Novak et al. 1999). It was previously reported that while small HA fragments were found in the tumor tissues, native HA of high molecular mass was found in the normal and benign tissue (Rooney et al. 1995). Both the high levels of full length polymeric HA and its low molecular weight degradation products, HA fragments, known as oligosaccharides support glioma growth (Novak et al. 1999). The HA oligosaccharides, e.g., hexamer oligoHA-6 (HA-6) or decamer oligoHA-10 (HA-10) are able to displace full length HA via competition for CD44 receptor binding. HA can be effectively displaced by HA decasaccharides, such as HA-10, but not by HA oligosaccharides that are shorter than 10-mer (Tammi et al. 1998). It was observed that full length, large size HA had an antiangiogenic property, whereas smaller oligosaccharides after degradation (3-10 disaccharide units) were no longer anti-angiogenic (Deed et al. 1997). We found that small size oligosaccharide, decamer HA-10, exogenously added to the cell culture stimulated HA production by glioma cells (Wiranowska et al. 2010), as previously described for normal human fibroblasts (Luke and Prehm 1999). These authors found that displacement of nascent HA from the receptors by HA oligosaccharides led to stimulation of HA synthesis (Luke and Prehm 1999). Further studies of HA and the role of HA-CD44 interaction in glioma growth and invasiveness may provide new therapeutic targets for the treatment of glioma. Recent therapeutic approaches targeting HA-CD44 interactions are discussed below.

### **3.2.2. Chondroitin Sulfate Proteoglycans (CSPGs)**

CSPGs are expressed at elevated levels in the developing brain. In the normal brain, they are known for their inhibitory effect on stem cell migration. In glioma however, CSPGs are upregulated and stimulate glioma cell migration (Kearns et al. 2003; Sim et al. 2009). The two members of the CSPG subclass of lecticans such as versican and BEHAB/brevican are expressed at a higher level in glioma than in the normal brain tissue. In addition, versican and brevican can form complexes with mesenchymal matrix proteins found in the ECM of glioma, but not in the ECM of the normal brain (Sim et al. 2009). Gliomas of various grades, e.g., astrocytoma and GBM secrete high levels of BEHAB/brevican. The CSPG lectican has an N-terminal HA-binding domain that interacts with fibronectin, thereby further stimulating glioma progression (Viapiano and Matthews 2006). Several other ECM molecules such as HA, CD44, tenascin and transforming growth factor beta2 (TGFbeta2) also interact with versican and promote brain tumor cell invasion. Recently, the link proteins HAPLN4 and HAPLN2 were shown to be reduced in malignant gliomas and it was suggested that this reduction may be associated with matrix remodeling by glioma. Therefore, in contrast to the normal brain tissue where CSPGs lecticans associated to HAPLNs serve as inhibitors of cell motility, in glioma this stabilizing role of link proteins may be reduced or lost resulting in proinvasive activity of CSPGs in glioma (Sim et al. 2009). Another member of CSPGs



family, neuroglial protein 2 (NG2), is also overexpressed in glioma (Schrappe et al. 1991; Wiranowska et al. 2006). NG2 was first found to be expressed by oligodendrocyte progenitor cells (section 2). NG2 expressed by glioma cells has a strong association with ECM ligands such as collagen VI and cellular ligands such as CD44. It has been implicated in the invasive behavior of glioma and found to be expressed *in vitro* and *in vivo* by highly migratory glioma cells while not found in non migratory cells (Lin et al. 1996; Galli et al. 2004, Wiranowska et al. 2006, Stallcup and Huang 2008). NG2 is not only expressed by oligodendrocyte progenitor cells and glioma cells but also by pericytes, which are associated with microvasculature and may play a role in the development of glioma vasculature (Stallcup and Huang 2008). Therefore, NG2 may be considered as one of the main CSPGs involved in glioma progression.

### 3.2.3. Vasculature - Associated ECM Molecules Expressed by Gliomas

The basement membrane of the cerebral vasculature contains collagens (type IV and V), fibronectin, laminin, vitronectin and HSPGs such as glypicans and syndecans. Some HSPGs, e.g., syndecan-2 were reported to be increased in brain tumors (Theocharis et al. 2010).

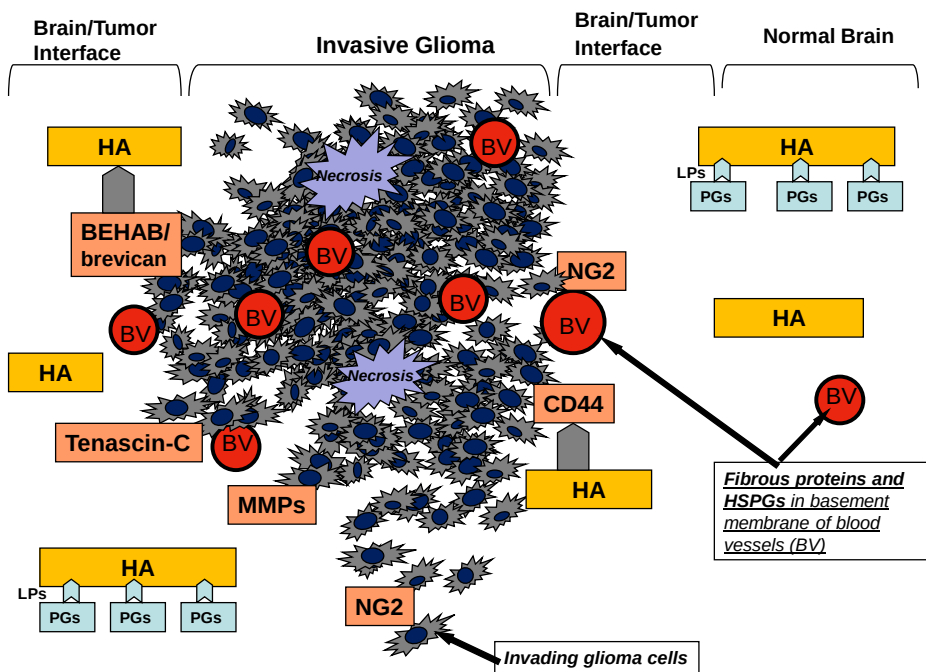


Figure 1. Schematic representation of the extracellular matrix (ECM) microenvironment of invasive glioma with necrotic centers and associated brain parenchyma. Hyaluronic acid (HA), a long space-filling molecule composed of a carbohydrate chain is shown either unbound or bound to proteoglycans (PGs) via link proteins (LPs) or bound to CD44 receptor. Also shown are two glioma associated ECM molecules (both chondroitin sulfate proteoglycans): brain enriched hyaluronic acid binding protein/brevican (BEHAB/brevican) and neuroglial protein-2 (NG2) with the latter expressed by glioma cells and pericytes of blood vessels. Also shown are glycoprotein tenascin-C, and matrix metalloproteinases (MMPs). Blood vessels (BV) shown in the glioma and the associated brain parenchyma contain fibrous proteins such as collagens, laminins etc. and heparan sulfate proteoglycans (HSPGs) associated with the BV basement membrane.

Table 1. ECM molecules and their functional significance

| Molecules                                                                                                                                                                                                                                                                                                   | Location                                              |                                                                            | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                             | Brain                                                 | Glioma                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Glycosaminoglycans (GAGs):</b><br>-Hyaluronan (HA)<br><br><b>-Proteoglycans (PGs):</b><br>• Chondroitin sulfate PGs (CSPGs)<br>.. lecticans:<br>aggrecan<br>neurocan<br>versican<br>BEHAB/brevican<br>.. NG2<br><br>• Link proteins (LPs)/ PG link<br>• Heparan sulfate PGs (HSPGs) (glypican, syndecan) | +<br><br><br>+<br><br>+<br>+<br>+<br>+<br>+<br>+<br>+ | +++<br><br><br>+++<br><br>?<br>++<br>+++<br>+++<br>+++<br>+ /- or ++<br>++ | <b>HA:</b> space filling molecule, binds to cell surface receptors: CD44 and RHAMM; regulation of ECM; regulates cell proliferation, adhesion, motility.<br><b>PGs:</b> HA binding proteins<br><b>CSPGs:</b> high levels in the developing fetal brain and glioma<br><b>Lecticans:</b> aggrecan (neurons), neurocan (neurons & glia), versican (glia), brevican (glia & neurons); bind to HA and cell surface receptors, regulate cell motility, axonal navigation; enhance migration of glioma cells; inhibit normal brain stem cell migration.<br><br><b>NG2:</b> expressed by CNS progenitor, pericytes, glioma cells; binds to collagen VI and CD44.<br><b>LPs:</b> bind to HA & lecticans; stabilize binding of core protein to HA.<br><b>HSPG:</b> bind to basal lamina of blood vessels, ependymal, astroglial, endothelial surfaces; storage site for cytokines and growth factors. |
| Tenascin:C & R (glycoproteins)                                                                                                                                                                                                                                                                              | +                                                     | +++                                                                        | <b>Tenascins:</b> expressed in zones of proliferation, migration and morphogenesis; Tenascin-C is over expressed in glioma.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Galectins, (lectins)                                                                                                                                                                                                                                                                                        | +                                                     | +++                                                                        | <b>Galectins:</b> (glia & endothelial cells), glycan binding proteins; neuronal and non-neuronal network development, angiogenesis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Fibrous Proteins</b><br>• collagens<br>• fibronectin<br>• laminins<br>• vitronectin                                                                                                                                                                                                                      | * In blood vessels only                               | +<br>+<br>+<br>+                                                           | * <b>Fibrous proteins:</b> structural elements of connective tissue in blood vessels' basement membrane in normal brain and glioma; in high grade glioma also expressed by glioma cells.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| MMPs, proteases                                                                                                                                                                                                                                                                                             | +                                                     | +++                                                                        | <b>MMPs:</b> tissue, ECM remodeling, angiogenesis. MMP-2,-9, -13 and MT1-MMP play a role in glioma progression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Laminin, collagen and fibronectin were also shown to be expressed by normal brain tissue bordering with glioma cells in spheroids (Knott et al. 1998). In addition, some of these molecules are also expressed by cells of highly aggressive gliomas. For example, it was found that fibronectin was expressed by GBM *in vitro* and in gliomesenchymal junctions in tumors and their blood vessels (Rao 2003). Laminins, which were found in blood vessels and in the glial limitans externa in glioma, were also shown to be expressed by human glioma cells positive for glial fibrillary astrocytic protein (GFAP) (Tysnes et al. 1999). An active site on laminin which was capable of binding to CD44 was identified (Hibino et al. 2004). In addition, Ljubimova et al. (2004) found that highly invasive GBMs overexpressed laminin-8, a member of the subset of laminins characterized by containing the alpha4 chain. Moreover, these authors also found that laminin-8 not only facilitated tumor invasion *in vitro*, but was involved in tumor regrowth after completion of a therapy. On the contrary, a different isoform, laminin-9, was found in lower grade gliomas, astrocytomas, and at low levels in benign brain tumors and in normal brain tissue. Therefore, many of these ECM molecules originally known to be associated with vasculature and now found at various levels expressed by glioma cells could be considered as biomarkers of glioma progression. Tenascin-C, a proteoglycan synthesized by glial and neural crest cells is highly expressed in the subventricular zone and essential for the development of neural stem cells. Tenascin-C, which is believed to be produced by endothelial cells, was found around blood vessels in

astrocytoma and its expression correlated with angiogenesis and tumor progression from grade II to grade III (Zagzag et al. 1995; Quirico- Santos et al. 2010).

Tenascin-C was found overexpressed in invasive glioma both *in vitro* and *in vivo* (Mahesparan et al. 2003). Also, galectins are upregulated in glioma and shown to be involved in glioma cell migration and angiogenesis thus confirming their significance as ECM molecules in glioma pathology. The schematic representation of ECM glioma microenvironment and the summary of representative ECM molecules and their functional significance are shown in Figure 1 and Table 1.

#### **4. MATRIX METALLOPROTEINASES (MMPs), THEIR INHIBITORS (TIMPs) AND REMODELING OF ECM IN GLIOMA**

Matrix Metalloproteinases (MMPs) are a class of enzymes known to be involved in normal tissue remodeling, but also produced by glioma cells (Wiranowska et al. 2000) and involved in modification of glioma ECM. In the past, MMPs were considered a potential target for anti-cancer therapies. The result of ECM degradation by MMPs is the release and diffusion of cytokines and growth factors stored in the ECM with subsequent further activation of MMPs by these factors (reviewed by Wiranowska and Plaas 2008). Since upregulation of MMPs was traditionally associated with inflammation and cancer progression, MMPs were considered a logical target for anti-cancer therapy. Supporting evidence was generated using transgenic mouse models overexpressing MMPs (Ha et al. 2001). Tissue inhibitors of metalloproteinases (TIMPs) are specific endogenous inhibitors of MMPs that have been correlated both positively and negatively with glioma invasion.

##### **4.1. MMPs in Glioma**

MMP up-regulation has been implicated in several broad disease categories including inflammation, vascular pathologies, and cancer. Analysis of MMP expression in cancer patients show strong correlation between increased expression of many MMPs and tumor progression in a wide range of malignancies including gliomas. Within the tumor, MMPs are secreted by tumor cells, as well as by stromal cells of the tumor (Rojiani et al. 2011). It appears that tumor cells produce a potent factor called extracellular matrix metalloproteinase inducer (EMMPRIN) a cell surface glycoprotein of the immunoglobulin superfamily. EMMPRIN stimulates MMP expression in stromal cells and also in tumor cells (Jodele et al. 2006). Several studies have documented overexpression of MMPs in gliomas compared to normal brain tissue.

However the MMPs involved in gliomas have almost exclusively been the gelatinases MMP-2 and MMP-9 (reviewed by Rao 2003) and strong gelatinase expression was correlated with tumor grade (Forsyth et al. 1999; Wang et al. 2003). Raithatha et al. (2000) found that MMP-2 expression was most prominent in tumor cells whereas MMP-9 expression was seen in tumor cell but was more strongly expressed in the vasculature. Recently Zhang et al. (2009) showed that knockdown of Akt2 resulted in decreased MMP-9 expression with concomitant decrease in glioma invasion *in vitro* and *in vivo*. It should be pointed out that

EMMPRIN levels have been shown to increase in glioma and correlate with tumor grade (Sameshima et al. 2000). EMMPRIN has also been shown to increase hyaluronan and to co-localize with its receptor CD44 (Toole and Slomiany 2008). Given that hyaluronan and CD44 are important players in the CNS and in gliomas, EMMPRIN may play a significant role in glioma invasion. Besides gelatinases, a number of studies document the role of Membrane-type (MT) MMPs such as MT1-MMP, as well as other MMPs involved in gliomas. Lampert et al. (1998) found increased levels of MT1-MMP and MT2-MMP in brain tumors along with gelatinases. Yamamoto et al. (1996) found that increased MT1-MMP expression is associated with the expression of activated form of MMP-2 which in turn correlated with malignant glioma progression *in vivo*. The gelatinases MMP-2 and MMP-9 as well as the membrane-type protease MT1-MMP have been well documented to play pivotal roles in invasion and angiogenesis (Handsley and Edwards 2005) and vascular basement membrane components are well recognized substrates of MMP-2 and MMP-9. MMPs are found at the invasive front of the tumor, in invadopodia, the actin-rich protrusions of tumor cells with proteolytic activity. The gelatinases and the MT1-MMP localize to or become activated at the invadopodia (Stylli et al. 2008). With regard to other MMPs, Lettau et al. (2010) have found that MMP-19 is strongly expressed in astroglial tumors and is also responsible for the invasion of glioma cells *in vitro*. Using the glioma cell line U251, Deng et al. (2010) found that MMP-26 promoted cell invasion *in vitro* and *in vivo*. In addition, Stojic et al. (2008) have shown enhanced expression of MMP-1, MMP-11 and MMP-19 in glioblastoma multiforme in comparison to low grade astrocytomas and normal brain.

Hence, it appears that MMPs play a pivotal role in glioma aggressiveness which would appear to make them potential targets for therapy. However, it should be pointed out that MMPs, by virtue of their degradation capacity also generate endogenous angiogenesis inhibitors. Proteolytic cleavage of plasminogen by several MMPs generates angiostatin whereas endostatin is generated from the C-terminal fragment of collagen type XVIII. MMP-9 is involved in the release of Tumstatin, another inhibitor of angiogenesis. MMPs as potential targets are further discussed below.

#### **4.2. Tissue Inhibitors of Matrix Metalloproteinases (TIMPs) in Glioma**

TIMPs have been known for their primary function of inhibiting MMPs, and it has now been widely recognized that TIMPs exhibit additional biological activities independent of their MMP inhibitory function. There are four TIMP members: TIMP-1, TIMP-2, TIMP-3 and TIMP-4, all of which inhibit MMP activity. TIMP-1, -2 and -4 are secreted, whereas TIMP-3 is associated with the extracellular matrix.

The MMP-independent activity of TIMPs, e.g. TIMP-1 and TIMP-2, includes promotion of cell growth as well as, apoptosis, angiogenesis and a role in cell signaling (reviewed by Rojiani et al. 2011). These roles of TIMPs surfaced when overexpression of these molecules gave conflicting results. Paradoxical tumor promoting and tumor inhibiting role of TIMPs may be extended to gliomas. Therefore, TIMPs deserve further consideration in the search for targeted therapies.

## 5. THERAPEUTIC TARGETING OF ECM MOLECULES IN GLIOMA

### 5.1. Targeting Metalloproteinases

MMPs play a crucial role in tumor growth, metastasis and angiogenesis and therefore have been the targets of antitumor therapy. Due to highly pleiotropic activities of MMPs the outcomes of the clinical studies targeting these molecules have been disappointing, resulting often in increased tumor growth (reviewed by Rojiani et al. 2011). The initial use of broad spectrum MMP inhibitors interfering with the function of many of these enzymes in clinical trials had unforeseen consequences and resulted in early termination of these studies (reviewed by Coussens et al. 2002). These disappointing results led to the realization that the experimental data had to be reevaluated. For example, in animal studies the inhibitors were administered in early or intermediate stages of cancer whereas in humans they were administered in advanced stages. Besides, it has now been well documented that a number of MMPs play a protective role and their elimination can have adverse consequences (Martin and Matrisian 2007).

However, given their significant contribution to tumor progression, MMPs still remain strong potential target candidates for therapeutic interventions. Therefore, it is not surprising that there are a number of MMP inhibitors (MMPI) in clinical trials (reviewed by Roy et al. 2009). Their effectiveness has yet to be proven. Also, the timing of delivery has been reconsidered since drugs given at earlier stages of cancer appear to be more effective than when given in advanced stages (Roy et al. 2009). Despite the adaptation of clinical trials and because of the pleiotropic activities of MMPs, a prediction of outcome is still difficult. Therefore, attention is now also given to other ECM molecules in the glioma microenvironment. New classes of targets need to be identified, eg. ECM molecules found within the cancer stem cell niche. Examples of other ECM targets are discussed below.

### 5.2. Targeting HA and CD44 Adhesion Molecule

We showed previously that blocking CD44 and interfering with HA-CD44 ligand-receptor interaction resulted in inhibition of glioma cell invasion, decreased HA production and led to glioma cell apoptosis (Wiranowska et al. 1998; Wiranowska et al. 2010). In addition, recent data by Xu et al. (2010) showed that CD44 attenuates activation of the Hippo signaling pathway and that knockdown of CD44 expression resulted in the inhibition of glioblastoma. It was suggested that CD44 is a prime therapeutic target for treatment of glioblastoma (Xu et al. 2010). Therefore, further studies of this promising biological target molecule in glioma are warranted. The soluble recombinant CD44-HA-binding domain (CD44-HABD) inhibited proliferation of endothelial cells *in vitro*, blocked angiogenesis *in vivo* and inhibited growth of various tumors (Pall et al. 2004) providing hope for a new therapeutic approach to glioma. In addition, HA was recently used *in vivo* as a delivery carrier for chemotherapeutic paclitaxel (HA-paclitaxel) targeting CD44 positive ovarian carcinoma (Auzenne et al. 2007). Also, recently cisplatin carrying HA nanoparticles were evaluated as potential treatment for cancer (Jeong et al. 2008).

## 6. FUTURE THERAPIES OF GLIOMA

### 6.1. ECM Therapeutic Targets: Hopes and Disappointments

Current therapies like surgery, radiotherapy and chemotherapy are aimed at debulking of the brain tumor mass as well as targeting and eradicating proliferating tumor cells. However, these therapies do not address the quiescent population of the cancer stem cells. These cells, which are nourished and protected from therapeutic interventions by the microenvironment of the CSCs vascular niche can repopulate and initiate new tumor foci in the brain (Denysenko et al. 2010). The modulation of the ECM microenvironment of the stem cell niche may be a promising approach especially in light of the finding that many of the genes and their products, prognostic for the fate of proneural GBM were identified in the stroma of the brain tumor but not in the brain tumor cells themselves (Katz et al. 2012).

Therapies targeting solely the brain tumors and their CSCs population may not be successful due to the high complexity and heterogeneity of brain tumors as demonstrated by the existence of various subtypes of GBMs. In addition, the genetic instability in expressing markers for undifferentiated cells within these tumors makes it difficult to assign the correct differentiation status of the tumor cells (Denysenko et al. 2010). Therefore, other therapeutic options should be pursued including therapies targeting the ECM of the CSCs vascular niches which would disrupt the microenvironment protective and supportive of the CSCs' self-renewal.

One of the stimulatory pro-angiogenic molecules released within the CSCs niche which enhances the survival of neural stem cells is VEGF (section 2.2.2). It has been shown that targeting VEGF and disruption of the niches by anti-VEGF treatment *in vivo* resulted in CSCs depletion and tumor growth inhibition (Calabrese et al. 2007). Some other studies however, showed that blocking VEGF *in vivo* resulted in growth of satellite tumors (Rubenstein et al. 2000). The early data obtained from clinical trials of GBM patients using VEGF-specific inhibitors such as bevacizumab combined with chemotherapeutic drug CPT-11, showed promising results (Calabrese et al. 2007). However, recent finding by Ricci-Vitiani et al. (2010) and Wang et al. (2010) showed that newly formed blood vessels originating from the GBM stem cells that differentiated into endothelial cells were not responsive to anti-VEGF therapy (section 2.2.2).

More recently, it has been shown (di Tomaso et al. 2011) that patients treated with anti-VEGF therapy still contained Nestin + cells, a characteristic marker of the CSCs, despite a decreased vascularization of the brain tumors. The phase II results from anti-VEGF therapy in combination with chemotherapeutic CPT-11 failed to show prolonged survival of the GBM patients (Lai et al. 2011). Similar results were obtained from phase III clinical studies with recurrent GBM patients treated with cediranib, an inhibitor of VEGF alone or in combination with chemotherapeutic lomustine (Batchelor, 2010). In addition, it was shown recently by Takano et al. (2010) that failure of bevacizumab treatment was associated with the high incidence of infiltrative tumors and MMP activity in the samples of urine. Therefore, based on the current state of knowledge, other approaches targeting ECM molecules of the vascular niche of CSCs need to be developed.

## 6.2. Glioma Possible New Prognostic Factors and New Targets

### 6.2.1. ECM Prognostic Factors

There is evidence that some ECM molecules could serve as prognostic markers of antiangiogenic therapy of glioma indicative of the fate of the therapy. For example, Takano et al. (2010) found that failure of bevacizumab (anti-VEGF antibody) treatment was associated with the high incidence of infiltrative tumors and levels of MMP activity in urine. Therefore, early detection and measurement of urine MMPs activity in samples from patients could be indicative of progressive disease. The detection of this biomarker could allow for earlier therapeutic intervention or alteration of a given anti-glioma therapy. Another ECM prognostic factor of potential therapeutic value could be presence of soluble collagen IV in the blood. It was shown by Sorensen et al. (2009) that in patients treated with anti-angiogenic therapy using cediranib (pan-VEGF receptor tyrosine kinase inhibitor) increased levels of collagen IV were found. This increase in circulating collagen IV was explained as the result of “blood vessels normalization” involving thinning of the abnormally thick tumor associated basement membrane of blood vessels. Increased levels of circulating collagen IV were positively associated with progression-free and overall patient survival.

### 6.2.2. New ECM Targets

Recent data by Inoue et al. (2010) showed that cancer stem-like cells of GBM express MMP-13 responsible for invasion and migration of these cells suggesting that, MMP-13 might be a potential new therapeutic target for glioblastomas. Blocking VEGF or silencing VEGF receptor 2 inhibits the maturation of tumor endothelial progenitors into endothelium but does not stop the differentiation of CSCs (CD133+ cells) into endothelial cells. However, silencing of Notch (neural precursor receptor as mentioned in section 2.2.1) blocks the transition of CSCs (CD133+ cells) into endothelial progenitors (Wang et al. 2010). Therefore, Notch may be another potential new target in the stem cell niche. Fujita et al. (2011) showed that non-steroidal anti-inflammatory drugs (NSAID) such as Cox-2 inhibitors suppressed gliomagenesis via inhibition of prostaglandin E2 mediated accumulation of myeloid derived suppressor cells. In addition, it was shown that the Cox-2 inhibitor, celecoxib, inhibited *in vitro* microvascular channel formation associated with VEGF down regulation (Basu et al. 2005). The effect was not only more pronounced but also prostaglandin E2-independent when evaluated in a highly invasive breast cancer cell line and compared to a less invasive cell line. Based on these findings, it could be hypothesized that the NSAID class may have potential application in highly invasive glioma by targeting of the inflammatory molecules and immune cells as well as microvascular channel formation in the ECM.

## SUMMARY AND CONCLUSION

Despite years of research, glioma therapy remains a challenge due to very limited therapeutic options and short survival of glioma patients. Until recently the focus of research evaluating possible target molecules for treatment was mainly on the cancer cell itself and its genetic characterization. Therefore, routine anti-glioma therapies such as chemotherapy or radiation are solely focused on targeting proliferating glioma cells by interfering with their

cell cycle and resulting in glioma cell death. Very little attention was given to evaluation and validation of therapeutic targets in the extracellular matrix in glioma, its vasculature and brain stroma outside the tumor. Recent clinical trials targeting glioma vasculature with anti-angiogenic molecules initially provided encouraging results but later failed to be effective in slowing glioma progression. On the contrary, some of the anti-angiogenic treatments despite achieving blood vessel “normalization” within the tumor later resulted in the development of new glioma foci in the brain parenchyma away from the main lesion. It became known recently that a population of glioma CSCs present within the tumors was capable of escaping this anti-angiogenic therapy giving rise to tumor endothelium which had no markers for anti-angiogenic therapy. Another attempt to stop glioma cell invasion into the normal brain parenchyma involved targeting various MMPs responsible for remodeling ECM during glioma progression. Although there were high expectations, the results of these studies were disappointing again. It was found that many of the targeted MMPs had anti-tumor as well as tumorigenic activities and blocking or eliminating their activity led to glioma regrowth. Clearly, the ECM and the surrounding stroma play an essential role in glioma progression but more studies need to be done in order to find proper targets within the ECM to slow or inhibit glioma growth. This is supported by most recent finding by Katz et al. (2012) that genes expressed in the microenvironment of the stroma rather than in the glioma itself may be predictive of glioma patient survival.

This chapter provides a review of recent information relating to ECM targets for anti-glioma therapy. Consideration was given to various ECM molecules within the normal brain, in glioma and in the vascular niche harboring glioma stem cells. Consideration was also given to ECM rigidity and its effect on glioma progression. In addition to discussing some pertinent ECM molecules in glioma progression, also new emerging ECM targets and new prognostic markers candidates were discussed. All in all, ECM molecules are of great importance in the development of new therapeutic strategies and the information compiled in this chapter summarizing their role should be suitable to give guidance for the search and the development of new ECM anti-glioma targets.

## FIGURE AND TABLE

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**Chapter 16**

## **CURRENT IMMUNOTHERAPEUTIC TARGETS IN GLIOMAS**

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### **ABSTRACT**

The prospect for high-grade astrocytic tumor patients continues to be dismal, even with advances in surgery, radiotherapy, and chemotherapy. There is a need for the development of novel therapies that would eliminate the heterogeneous and diffuse astrocytic tumor cells.

Innovative therapeutic approaches for the treatment of central nervous system neoplasia, such as antibody-mediated immunotherapeutics, cellular immunotherapeutics, and oncolytic viruses targeting tumor-associated antigens, have emerged during the past few years.

Herein we review several of the glioma-associated antigens and the current status of immunotherapeutics and oncolytic viruses targeting these antigens.

### **ABBREVIATIONS**

|      |                                  |
|------|----------------------------------|
| BTSC | brain tumor stem cell            |
| CTA  | cancer-testis antigen            |
| CARs | chimeric antigen receptors       |
| CNS  | central nervous system           |
| CSPG | chondroitin sulfate proteoglycan |
| CED  | convection-enhanced delivery     |

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|          |                                               |
|----------|-----------------------------------------------|
| CMV      | cytomegalovirus                               |
| DC       | dendritic cells                               |
| ECD      | extracellular ligand-binding domain           |
| EGF      | epidermal growth factor                       |
| EGFR     | epidermal growth factor receptor              |
| EGFRvIII | EGFR variant III                              |
| FNIII    | fibronectin type III                          |
| GAA      | glioma-associated antigen                     |
| GBM      | glioblastoma multiforme                       |
| GPNCMB   | glycoprotein nonmetastatic melanoma protein B |
| HSV-1    | herpes-simplex-virus type 1                   |
| HER2     | Human Epidermal Growth Factor Receptor2       |
| hTERT    | Human telomerase reverse transcriptase        |
| MHC      | major histocompatibility complex              |
| mAb      | monoclonal antibody                           |
| PDPN     | podoplanin                                    |
| PFS      | progression-free survival                     |
| PE       | <i>pseudomonas</i> exotoxin A                 |
| RTKs     | receptor tyrosine kinases                     |
| scFv     | single-chain variable fragment                |
| TAA      | tumor-associated antigens                     |
| TN       | tenascin                                      |
| WT1      | wilms' tumor 1                                |

## 1. INTRODUCTION

Glioblastoma multiforme (GBM) is the most common primary malignant brain tumor of the CNS in adults [1]. Malignant transformation is preceded by changes in cellular DNA and protein expression followed by uncontrolled proliferation of the altered cells. Altered gene expression in malignant cells often leads to the expression of tumor-associated antigens (TAAs).

TAAs allow for selective targeting of tumors, with minimal toxicity to the adjacent normal tissue [2]. A multitude of the TAAs are derived from overexpressed normal or mutated somatic genes and are therefore not strictly tumor specific. However, these antigens are highly overexpressed on tumors, often up to 10,000-fold the expression in normal tissue, and hence greatly accessible [3].

Initial experiments demonstrated the ability of lymphocytes from cancer patients to inhibit tumor cell growth *in vitro*, and antibodies obtained from certain patients were able to bind antigens expressed on tumor cells [4-9].

Many of the antigens were present on the cell surface and shared by tumors of different types [5, 6, 9], while others were patient specific [5, 8]. In recent years the identification of TAAs and using these TAAs in cancer treatment has accelerated. Several of the antigen targets for antibody based therapeutics, cellular therapeutics, and viral therapeutics are discussed below and summarized in Table 1.

**Table 1. List of glioma antigen targets for antibody, cellular, and viral immunotherapeutics**

| Antigen                                               | Therapy Mechanism |
|-------------------------------------------------------|-------------------|
| Tenascin-C                                            | Antibody          |
| Epidermal Growth Factor Receptor                      | Antibody          |
| Mutant Epidermal Growth Factor Receptor Variant III   | Antibody          |
| Melanoma-Associated Chondroitin Sulfate Proteoglycans | Antibody          |
| Glycoprotein Nonmetastatic Melanoma Protein B         | Antibody          |
| Podoplanin                                            | Antibody          |
| Gangliosides                                          | Antibody          |
| Survivin                                              | Cellular          |
| Telomerase                                            | Cellular          |
| Cancer-Testis Antigens                                | Cellular          |
| Human Cytomegalovirus                                 | Cellular          |
| HER2/neu                                              | Cellular          |
| EphA2                                                 | Cellular          |
| Wilms' tumor 1—WT-1                                   | Cellular          |
| Nectin-1                                              | HSV-1             |
| Nectin-like molecule 5                                | Poliovirus        |

## 2. ANTIGEN TARGET FOR ANTIBODY BASED THERAPY

A major challenge in the study of gliomas has been to identify glioma associated antigens (GAAs) which could aid in tumor cell destruction. It is difficult to identify GAAs from patient serum or animal antiserum owing to the presence of a large number of antibody contaminants that are challenging to remove before identifying TAA. The advent of monoclonal antibody (mAb) technology [10], opened a new dimension in the study of GAA. Using this technique Schnegg et al. [11] obtained three mAbs, BF7, GE2, and CG12. The mAbs BF7 and GE2 exclusively reacted with glioma cells, and their specificity was not altered by absorption with normal fetal and adult brain homogenate. The third mAb, CG12, exhibited extensive reactivity to glioma, melanoma, and neuroblastoma cell lines. Several GAAs have been described since then, and following is a description of some of these antigens.

### 2.1. Tenascin-C

Tenascin (TN) (also known as hexabrachion and cytotactin) [12], is a large disulfide-linked hexameric extracellular matrix glycoprotein. Each TN subunit has a molecular mass of 220-320-kDa and is composed of an amino-terminal portion consisting of a group of hydrophobic residues followed by eight cysteines, a linear segment of 14.5 epidermal growth factor (EGF)-like repeats, trailed by 8-15 fibronectin type III (FN-III)-like repeats, and a  $\beta$ - and  $\gamma$ -fibrinogen-like domain at the carboxy-terminus (Figure 1) [13, 14].

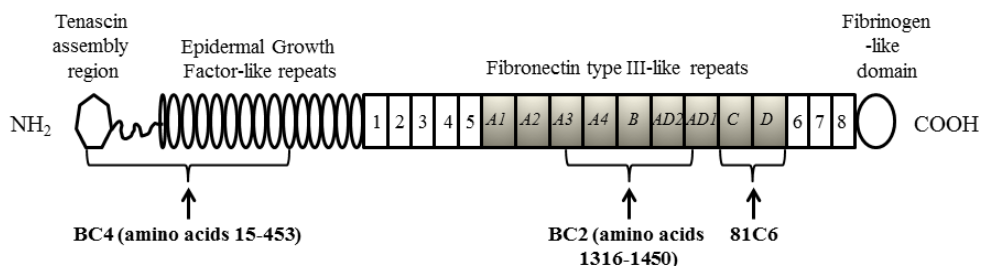


Figure 1. Model of a subunit of human tenascin-c. Structural domains include a tenascin assembly domain (heptad), epidermal growth factor (EGF)-like repeats (ovals), two types of fibronectin type III (FN-III)-like repeats (1) those conserved in all tenascin-C variants (open rectangles) and (2) those that are alternatively spliced (shaded rectangles) and the terminal fibrinogen globe (circle). Arrows indicate epitopes for BC-4, BC-2, and 81C6 monoclonal antibodies.

Different isoforms of TN (TN-C, TN-R, TN-W, TN-X, and TN-Y) generated by alternative RNA splicing have been reported [15]. TN-C, being the most extensively studied member of the family, has an unusually restricted expression in the developing embryo, and it is expressed extensively in the CNS and to a lesser extent in the peripheral nervous system [16]. TN-C induction is noted during neovascularization and wound healing in adults [17]. TN-C is expressed in a wide variety of human tumors, including glioma [12], osteosarcoma [18], melanoma [19], Wilms' tumor [20], mammary carcinoma [21], lung carcinoma [22], and oral squamous cell carcinoma [23]. TN-C was originally identified as a GAA with high levels of expression in glioma cell lines (10-625  $\mu\text{g}$ ) as opposed to normal brain (2-5  $\mu\text{g}$ ) [12, 24]. There is a direct correlation between TN-C expression and the degree of malignancy in gliomas, with more than 90% of GBM patient specimens demonstrating high levels of TN-C [12, 25]. Several anti-TN-C antibodies specific to various epitopes on TN-C have been generated with a view toward therapeutic applications and three of them BC-2, BC-4, and 81C6 have been introduced in clinical trials in Europe and the USA [26, 27].

## 2.2. Epidermal Growth Factor Receptor

The epidermal growth factor receptor (EGFR) is a member of the ErbB/EGFR family of receptor tyrosine kinases (RTKs). EGFR, also known as ErbB1 or HER1, is a 170-kDa, transmembrane glycoprotein and is composed of three functional domains – a 622 amino acid extracellular ligand-binding domain (ECD), a 22 amino acid hydrophobic transmembrane domain, and a 542 amino acid intracellular tyrosine kinase domain followed by a C-terminal regulatory domain [28] (Figure 2).

The C-terminal domain has several tyrosine residues that are phosphorylated specifically upon ligand binding. Binding of an EGFR ligand, such as EGF to its ECD induces receptor dimerization and activates its tyrosine kinase activity, in turn controlling cell growth and proliferation [29, 30].

The overexpression of EGFR is reported in brain tumors [31, 32] and many carcinomas. Up to 300-fold overexpression of EGFR is observed in glioma as opposed to normal control brain specimens [33]. GBM patients frequently exhibit *EGFR* gene amplification [34], while protein overexpression has also been demonstrated in the absence of gene amplification [35].

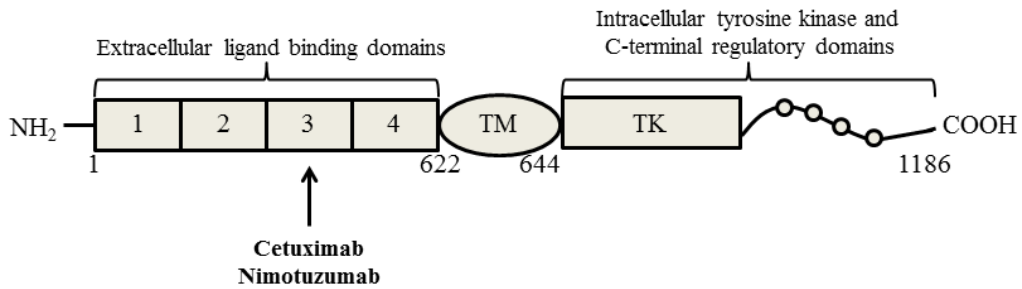


Figure 2. Pictorial depiction of wild-type epidermal growth factor receptor (EGFRwt). The amino terminus extracellular portion of EGFRwt consist of ligand binding domains 1-4, a transmembrane domain (TM), and the carboxyl terminus tyrosine kinase and regulatory domains. Arrows indicate the specificity of two monoclonal antibodies cetuximab and nimotuzumab.

In addition to protecting cells from apoptosis, EGFR activation induces proliferation, angiogenesis, and metastasis [36]. Several anti-EGFR mAbs for targeted therapy of EGFR-overexpressing tumors have been developed [37] and has been tested in GBM patients [38, 39].

### 2.3. Mutant Epidermal Growth Factor Receptor Variant III

*EGFRvIII* is the most common *EGFR* deletion mutant that is present in 20%-50% of GBMs with *EGFR* amplification [40]. The mutant *EGFRvIII* contains an in-frame deletion of 267 amino acids creating a novel glycine residue at position 6, between amino acids 5 and 274, generating a tumor-specific protein sequence [41]. *EGFRvIII* is a constitutively active RTK which is widely expressed in malignant gliomas [42], breast cancers [43], and head and neck cancers [44]. *EGFRvIII* overexpression induces resistance to common chemotherapeutics [45], which makes it an attractive target for therapy. A Phase I study with affinity-matured *EGFRvIII*-specific single-chain variable fragment antibody (scFv) MR1-1 fused to a truncated version of *pseudomonas* exotoxin A (PE), MR1-1-PE38KDEL, in malignant brain tumor patients is currently ongoing (ClinicalTrials.gov Identifier: NCT01009866).

### 2.4. Melanoma-Associated Chondroitin Sulfate Proteoglycans

Chondroitin sulfate proteoglycans (CSPGs) are extracellular matrix proteins composed of a 250-kDa core glycoprotein. The core glycoprotein is made up of a large ECD with numerous N-linked glycosylation sites, a hydrophobic transmembrane domain, and a short cytoplasmic domain with potential phosphorylation sites for protein kinase C [46]. The expression of CSPGs has been well documented in malignant gliomas and melanomas and is recognized by the mAbs 9.2.27 and Mel-14 [47-50].

Treatment with the 9.2.27-PE immunotoxin increased the survival from 30% to 90% in a rat glioma xenograft model [51]. Successful treatment of neoplastic meningitis with <sup>131</sup>I-labeled Mel-14 F(ab')<sub>2</sub> fragments was demonstrated in a 46-year-old female [52, 53].

## 2.5. Glycoprotein Nonmetastatic Melanoma Protein B

Glycoprotein nonmetastatic melanoma protein B (GPNMB) is a type I transmembrane glycoprotein composed of a long ECD, a transmembrane domain, and a short cytoplasmic domain [54]. In comparison to normal brain, elevated levels of GPNMB mRNA and protein are detected in patient samples of GBM [54]. A GPNMB-specific scFv (F6V) PE fusion protein (F6V-PE38) exhibited significant *in vitro* cytotoxicity and *in vivo* anti-tumor activity against glioma cells lines and xenograft models, respectively [55].

## 2.6. Podoplanin

Podoplanin (PDPN) is a 36-kDa, type I mucin-like transmembrane sialoglycoprotein that comprises a long ECD, a transmembrane domain, and a short cytoplasmic domain [56]. PDPN is extensively present on lymphatic endothelial cells and plays a major role in lymphatic vasculature formation [57]. Several anti-podoplanin mAbs have been generated, including NZ-1, D2-40, AB3, and 18H5 [58]. PDPN expression is upregulated in several cancers, including GBM [59]. In immunohistochemical analysis, 29 of 35 (82.9%) and 36 of 81 (44%) GBM patient specimens exhibited PDPN reactivity, which makes PDPN an ideal immunotherapeutic target for the treatment of GBM [59, 60].

## 2.7. Gangliosides

Gangliosides are sialic acid containing glycosphingolipids present on the plasma membrane outer surface of mammalian cells, with the hydrophilic carbohydrate group facing the external milieu and the lipophilic ceramide group localized within the lipid bilayer [61]. In mammals, the brain grey matter contains a high proportion of gangliosides, and their expression during nervous system development is tightly regulated [61]. Among the known gangliosides, the lacto-series gangliosides 3'-isoLM1 and 3'6'-isoLD1 are exclusively present in fetal brain, but undetectable in the normal adult brain. However, high expression of both 3'-isoLM1 and 3'6'-isoLD1 was reported in GBM patient and xenograft samples [62, 63]. Several mAbs of the IgM subclass against the GAAs 3'-isoLM1 and 3'6'-isoLD1 have been developed, and smaller scFv molecules from these mAbs are under development as glioma therapeutics [63, 64].

# 3. TARGETS FOR CELLULAR IMMUNOTHERAPY

Targets for cellular therapy consist of surface or intracellular antigens that can be recognized by the T lymphocytes of the immune system when processed and presented in major histocompatibility complex (MHC) molecules [65]. With the exception of EGFRvIII, few, if any, truly glioma-specific antigens have been identified that can serve as exclusive targets for cellular therapy in most cancers. Alternatively, most approaches have either targeted the unfractionated tumor antigen repertoire in the form of modified whole tumor cell

vaccines or tumor lysate-pulsed antigen-presenting cells, or targeted TAAs [66]. The potential advantages of targeting invasive malignant gliomas using cellular therapy include (1) the capacity to target intracellular and nuclear proteins, in addition to surface targets, due to the antigen-processing machinery which samples all cellular compartments [67], (2) the ability for activated lymphocytes to survey and eliminate invasive malignant tumor cells sequestered within normal brain without harm to eloquent cortex [68], and (3) the potential for the establishment of immunologic memory against TAAs that may allow for prolonged suppression of tumor regrowth [69].

### 3.1. Survivin

Survivin protein is overexpressed in a variety of human cancers, including but not limited to carcinomas of pancreas, lung, breast, colon, ovaries, liver, malignant melanoma, leukemia, and many CNS tumors, such as malignant glioma, medulloblastoma, neuroblastoma and neurofibroma [70-72]. The expression of survivin is found to correlate with pathologic grade of many tumors in the nervous system. In addition, survivin expression is found to be highly tumor specific, with only minimum expression detected in normal cells. Numerous published studies have demonstrated survivin as a potential candidate for immunotherapy [73], and using survivin as a cancer target still remains an active area of research. Studies have used DNA-transfected dendritic cells (DCs) and peptide vaccines to target survivin in murine tumor models, and a variety of studies have shown immunologic targeting of survivin in human cancers, such as carcinomas of prostate, breast, and lung, as well as gliomas [74-78]. However, there are relatively few published clinical trials evaluating immunotherapy directed against survivin in humans.

### 3.2. Telomerase

Human telomerase reverse transcriptase (hTERT) represents another potential target for cellular immunotherapy that is broadly expressed in multiple cancer types. hTERT is silent in normal tissues, but overexpressed in greater than 80% of human solid tumors [79]. Another attractive feature of hTERT as a tumor antigen is that telomerase activity is essential to the proliferative capacity of tumor cells [80], which reduces the likelihood that escape variants may occur due to immunologic selection of cells that have lost hTERT expression. Telomerase is frequently expressed in malignant gliomas (26-100%), and expression correlates with tumor grade [81]. Normal brain is devoid of telomerase expression. The immunogenicity of telomerase has been demonstrated both *in vitro* and *in vivo* in early-phase clinical trials [82-84]. To date, no clinical trials specifically targeting this protein in malignant gliomas have been reported.

### 3.3. Cancer-Testis Antigens

A number of cancer antigens identified as immunogenic targets in melanoma have a tissue-restricted expression with the testes and are referred to as cancer-testis antigens (CTAs)

[85-87]. CTAs include the melanocyte differentiation family of proteins including MAGE, CAGE, tyrosinase, tyrosinase-related protein 2 (TRP-2), and NY-ESO-1, all of which have been shown to be expressed in a subset of adult and pediatric malignant gliomas [88, 89]. The immunogenicity and potential for tumor regression in effective targeting of CTAs has been demonstrated in advanced melanoma [90], but exploration of the clinical utility of targeting CTAs in brain malignancies awaits further study.

### **3.4. Human Cytomegalovirus**

Recently, proteins from human cytomegalovirus (CMV) have been shown to be expressed in a high proportion of malignant gliomas [91-94] and may pose a novel and potentially immunogenic viral target for cellular immunotherapy within these tumors [95]. The potential relevance of CMV as an immunotherapeutic target in GBM was recently shown in a vaccine trial employing tumor lysate-pulsed autologous DCs, where a patient exhibiting a prolonged clinical response also demonstrated a robust *in vivo* expansion of CMV-specific T cells in response to vaccination [96]. Ongoing clinical trials are exploring the safety, immunogenicity, and potential efficacy of CMV-targeted immunotherapy in patients with GBM. A pilot study of DC vaccination targeting the immunodominant antigen pp65 in combination with regulatory T-cell depletion using daclizumab has shown promising immunologic and clinical responses [97].

### **3.5. HER2/neu**

HER2/neu, also known as HER2 and ErbB-2, is the human epidermal growth factor receptor 2, and is frequently overexpressed in malignant gliomas and breast cancers [98, 99]. HER2 has known immunogenic peptides and has been the target in many preclinical immunotherapy studies and several clinical trials [100-102]. While vaccination strategies targeting this protein have been safe, a more recent adoptive T-cell therapy trial using genetically engineered T cells that recognized HER2 through chimeric antigen receptors (CARs) resulted in fatal toxicity in a patient that was believed to be due to recognition of low levels of HER2 expression in normal tissues such as the lungs [103]. This extreme toxicity highlights the importance of immunologic specificity when employing high-affinity and high-avidity T cells against malignant tumors, and warrants caution in the future clinical application of increasingly potent immunologic therapies.

### **3.6. EphA2**

EphA2 is a tyrosine kinase receptor whose expression within tumors correlates inversely with survival in patients with GBM [104, 105]. Additionally, immunogenic epitopes specific to EphA2 have been identified and shown to elicit cytotoxic T-cell responses that recognize glial tumors *in vitro* [106, 107]. Due to the role of EphA2 in tumor cell proliferation, effective targeting of EphA2 with immunotherapy may reduce the propensity for immunologic escape or possibly select for less aggressive tumor cells.



### 3.7. Wilms' tumor 1—WT-1

WT1 is a zinc-finger transcription factor that functions as a critical factor in urogenital development and acts as an oncogene in a subset of cancers [108-110]. It is overexpressed in a number of cancers, including malignant gliomas, and has also been successfully targeted by using immunotherapeutic approaches in preclinical models. Ongoing clinical trials are evaluating the immunologic targeting of WT1 in patients with leukemia, lymphoma, breast cancer, and mesothelioma. A Phase II clinical trial of an HLA-A201 restricted peptide vaccine targeting WT1 in 20 patients with recurrent GBM demonstrated a clinical response in two patients (partial response) and median progression-free survival (PFS) of 20.0 weeks, and the 6-months (26-weeks) PFS rate of 33.3% [111]. Further studies exploring methods to enhance immunologic responses against WT1-positive gliomas are warranted.

## 4. ONCOLYTIC VIRAL IMMUNOTHERAPY

Recently, significant progress has been reported with clinical applications that elicit anti-tumor immunologic responses through administration of human viral pathogens designed to specifically target tumor cells (oncolytic viruses). For the treatment of primary CNS tumors, the natural neuropathogens herpes-simplex-virus type 1 (HSV-1) [112] and poliovirus [113] may have characteristics uniquely suited for use as oncolytic agents in the brain. Both recognize related cell-surface molecules of the nectin/nectin-like family of cell adhesion molecules as cellular receptors, nectin-1 [114] for HSV-1 and nectin-like molecule 5 for poliovirus [115]. Ectopic expression of nectin/nectin-like cell adhesion molecules is widespread in CNS tumors, which mediates natural susceptibility of such tumor cells to viruses recognizing them as receptors [116-120]. Also, these viruses possess the natural ability to spread within CNS tissue, which may be important to effectively target a locally diffuse process. Both HSV-1 and poliovirus have been genetically engineered to (i) eliminate their inherent neuropathogenic potential; (ii) prevent replication in normal CNS cells; and (iii) retain propagation and cytotoxicity in malignant cells [121, 122]. Accordingly, oncolytic HSV-1 and polioviruses efficiently target glioma cells expressing their receptors for destruction [113, 123]. This is reflected in prominent signs of direct tumor cell lysis and tumor destruction after virus inoculation in animal tumor models. An IND has been issued for a Phase I clinical trial with intracerebral inoculation of oncolytic polioviruses and will begin shortly. More importantly, oncovex, a multi-mutated HSV-1 construct carrying deletions of the coding regions for the ICP34.5 and ICP47 viral genes and an insertion of the GM-CSF gene with replication defects in normal, nonmalignant cells and potent immune-stimulatory properties, has demonstrated potent anti-tumor response in clinical trials [124, 125].

## CONCLUSION

Despite enormous efforts over the past four decades in the understanding of brain tumor biology, diagnosis, and treatment, the prognosis and mortality rate for CNS tumors essentially remains unchanged, as cancer treatment has traditionally been guided by the assumption that

cancer cell populations are homogenous. In contrast, research proves that brain tumors are complex, both at the cellular and the genetic level, and hence a better understanding of the phenotypic and molecular changes that underlie brain tumor growth could aid investigators in the development of improved therapies.

Studies by several groups have demonstrated the existence of a small proportion of cells designated as brain tumor stem cells (BTSCs), which drive tumor growth and confer resistance to conventional chemotherapy and radiation therapy [126, 127].

It is therefore essential to comprehend the role of BTSCs in tumor initiation and development, because this could aid in developing therapies targeting BTSCs. Alternatively, gene expression analysis in 22 human GBM samples by Parsons et al. identified novel genes, including isocitrate dehydrogenase 1 mutation in 12% of GBM patients, and signaling pathways previously not associated with GBM [128]. Further, work by The Cancer Genome Atlas group has classified GBM into proneural, neural, classical, and mesenchymal subtypes based on their gene expression profiles [129].

This study further identified classical and mesenchymal subtype GBM patients to be better responders to aggressive regimens of chemotherapy and radiation therapy. Such comprehensive genetic alteration studies could lead to the identification of novel targets and pathways which eventually will help us design personalized therapies for GBM patients.

Following the identification of potential targets on tumors in GBM patients, an ideal therapeutic regimen involving surgery, chemotherapy, radiation therapy combined with additional novel therapeutic approaches such as antibody-mediated targeted therapy, cellular immunotherapy, or viral immunotherapy could be explored for the treatment of brain tumors. Further, no single antigenic target is exclusively expressed on gliomas, and hence a combination therapy targeting multiple antigens or a combination of therapeutic approaches, such as antibody therapy with cellular immunotherapy, might be efficacious. Consequently, bispecific antibodies with specificity for antigens expressed on target cells and for CD3 receptors on T cells to harness the cytotoxic potential of T cells for glioma therapy are under development [130].

Next, the therapeutic agent has to be delivered to the tumor site. The challenging problem of conveying the therapeutic agent specifically into the brain parenchyma has been overcome by the locoregional technique of convection-enhanced delivery (CED), where the drug is infused through multiple catheters that are stereotactically inserted around the tumor resection cavity.

Since its initial testing 15 years ago, CED has come a long way with computer-generated models for desired volume of drug distribution and appropriate catheter positioning for optimal drug delivery personalized for individual GBM patients. To prevent leakage during CED imaging techniques to confirm adequate drug delivery to the tumor site are under development [131].

The treatment of GBM patients continues to be challenging due to the nonspecific nature of the therapy. According to the National Cancer Institute statistics, the incidence and mortality rates for CNS-related tumors has declined only slightly over the past 10 years. However, recent studies on gene expression profiling, signaling pathway characterization, and BTSC identification has led to the development of novel therapeutic strategies against GBM. These studies have stressed the importance of characterizing an individual patient's genetic and proteomic profile for a therapy to be specific. All of these exciting developments hold hope for brain tumor patients.

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*Chapter 17*

## **THE SIGNIFICANCE OF PTEN/AKT PATHWAY IN GLIOMA AND ITS RELATION TO PROLIFERATION, ADHESION AND APOPTOSIS OF GLIOMA CELLS**

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### **ABSTRACT**

Glioblastomas multiforme (GBM) is a highly invasive and aggressive primary brain tumor in adults, which occurs most frequently through loss of phosphatase and tensin homolog deleted on chromosome 10 (PTEN) function. The tumor-suppressive properties of PTEN inhibit the phosphatidylinositol 3-kinase (PI3K)/Akt signaling pathway through dephosphorylation of phosphatidylinositol 3,4,5-triphosphate (PtdIns(3,4,5)P<sub>3</sub>). PTEN/PI3K/Akt constitutes an important signaling pathway regulating various cellular responses including cell proliferation, cell growth, metabolism, apoptosis and migration. Loss of PTEN function causes accumulation of PtdIns(3,4,5)P<sub>3</sub> mimicking the effect of PI3K activation and triggering the activation of its downstream effectors. This unbalanced reaction leads to deregulation of PI3K/Akt signaling pathway, which is considered an essential driver in gliomagenesis.

### **INTRODUCTION**

Malignant glioblastoma is the most common primary brain tumor in adults. Despite advanced therapeutic approaches being developed, the prognosis of GBM patients remains poor. However, recent understanding of the molecular mechanisms in GBM pathogenesis has shed light on the “target therapy” approach. One of these targets is the PTEN/PI3K/Akt signaling pathway. There is increasing evidence demonstrating the crucial role of PTEN/PI3K/Akt signaling in the signal transduction system. Since this pathway links to various

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pivotal cellular functions, it is the most commonly activated signaling pathway in human cancers. This pathway therefore presents both an opportunity and a challenge for cancer therapy. In the present review, the contribution of PTEN/PI3K/Akt signaling in gliomagenesis will be discussed.

## GLIOBLASTOMA MULTIFORME

‘Glioblastoma multiforme’ (GBM), ‘Grade IV astrocytoma’, ‘glioblastoma’ are names for the same tumor. GBM is the most common and most aggressive of all primary brain tumors. According to the statistics of Central Brain Tumor Registry of the United States (CBTRUS), GBM accounts for 54% of all primary brain and central nervous system gliomas [1]. Generally, gliomas are classified into four major grades ranging from grade I to IV to describe their degree of malignancy under the World Health Organization (WHO) system [2, 3]. Grade I astrocytomas have slightly unusual looking cells, whereas the cells of grade IV astrocytomas are very abnormal in appearance.

The hallmarks of GBM include rapid growth with extensive blood vessel formation and central necrosis. Histologically, GBM has variable cytoplasmic and nuclear pleomorphism. It has octopus-like tentacles which facilitates its invasion into the surrounding normal brain tissue. The mortality rate of GBM is quite high, even though there have been major improvements in neuro-imaging, neurosurgery, radiation treatment techniques and supportive care. Prognosis for patients with these tumors is poor, with a median survival ranging from 9-12 months [4, 5]. Few patients with GBM can survive longer than 3 years. Due to its high mortality rate, considerable efforts have been put into understanding the pathogenesis of GBM.

Traditional surgical resection, radiotherapy and chemotherapy are still not effective in improving the prognosis of GBM patients. Generally, surgical resection is an important method for histological diagnosis and rapid cytoreduction [6, 7]. It benefits the patients with malignant situations by the rapid control of the tumor mass. However, aggressive surgery is nearly impossible to remove the entire tumor as the astrocytomas are highly infiltrative and they have no clear tumor margins. Moreover, there is an impact on the survival of a patient when an extensive tumor resection is performed [8]. Radiation therapy is one of the most effective treatments for gliomas. To date, radiation treatment of astrocytomas plus surgery is a standard treatment for patients [6, 7]. The effectiveness of chemotherapy is still questionable despite a number of chemotherapeutic agents have already been developed. One of the reasons is the passage of chemotherapeutic agents are often prohibited to enter the tumor tissues by the blood brain barrier. The latter highly restricts the passage of substances from the bloodstream into brain, thus only drugs with low molecular weights, high lipid solubility, low ionization and minimal protein binding can readily pass through [7, 9]. Another reason is the frequent hematological and organ-directed toxicity.

Due to the limitations of traditional treatments, many neurologists, neurosurgeons and neurobiologists have begun exploring new, more targeted and specific treatments for gliomas. Investigations on the molecular mechanisms of gliomagenesis are essential in understanding the biology of astrocytomas and for developing new targeted molecular drugs. GBM, like many cancers, has over-expression of proto-oncogenes, lost of tumor suppressor genes, as

well as uncontrolled intracellular signal transduction [6, 10]. Therefore, target inhibitors have been developed based on these genetic and signaling aberrations.

## GENETIC ABERRATIONS IN GBM

Advances in molecular biology, cellular biology and genomics have substantially improved our understanding of GBM pathology. These advances are being translated into targeted therapy and GBM is suitable for targeted molecular therapy because it has a variety of genetic aberrations and signaling pathway disruptions that present clear targets.

GBM may be *de novo* (primary glioblastoma) or developed through progression from low-grade or anaplastic astrocytoma (secondary glioblastoma). Primary GBM develops rapidly and without any clinical or histological evidence of an existing less malignant precursor lesion, whereas secondary glioblastomas progress slowly from preceding less malignant lesions [11, 12].

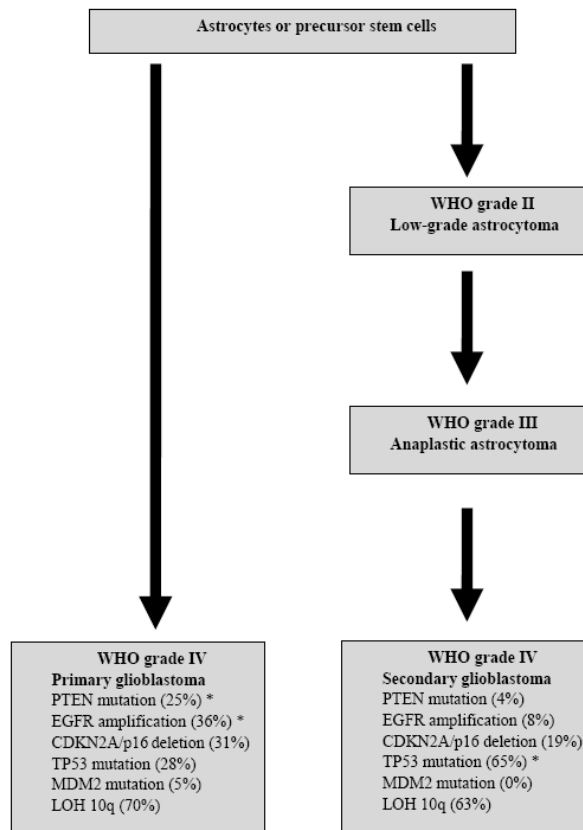


Figure 1. Different genetic alterations in primary (*de novo*) and secondary glioblastomas. PTEN mutation and EGFR amplification are typical genetic alterations in the pathway leading to primary glioblastoma, whereas TP53 mutations are frequent genetic alterations in secondary glioblastoma. The LOH 10q is commonly detected in both subtypes of GBM. \*Genetic alteration that is significantly different in frequency between primary and secondary glioblastomas. (Adapted and modified from Ohgaki et al., reference 13).

Although these subtypes of GBM are indistinguishable by histological assessment, they exhibit distinct clinical profiles and genetic alterations (Figure 1). Therefore, these subtypes of GBM may respond differently to therapy and can be considered as two different diseases. Certain population-based studies have shown that secondary glioblastoma is a rare disease which accounts for only 5% of all glioblastomas [13, 14]. Therefore, most clinical and molecular findings are obtained from primary glioblastomas. Genetically, GBM displays losses in part of chromosomes 6, 9, 10, 13, 17, 19 and 22 as well as gains or amplifications of chromosomes 1, 5, 7, 8, 11 and 12 [15]. Mutations of tumor suppressor genes such as phosphatase and tensin homolog deleted on chromosome 10 (PTEN) and TP53, deletion of CDKN2A/p16 gene, as well as overexpression of epidermal growth factor receptor (EGFR) and murine double minute 2 (MDM2) gene have been identified. Since PTEN alterations are more common in primary GBMs, this chapter will be discussed with an emphasis on the gliomagenesis based on PTEN and its related signaling pathways.

## PTEN SIGNALING IN GBM

The human PTEN gene, also known as mutated in multiple advanced cancers 1 (MMAC1) or transforming growth factor-regulated and epithelial cell enriched phosphatase (TEP1), maps to the chromosomal band 10q23.3 and was identified in 1997 as a tumor suppressor gene [16, 17]. Substantial biochemical and cell biological studies have found that PTEN is frequently disrupted in various types of malignant tumors, including glioblastomas as well as prostate, breast, and kidney carcinomas [18, 19]. PTEN mutations have been reported in 15-40% of glioblastomas, predominantly in primary GBM, whereas PTEN homozygous deletions occurred rarely (<2%) [19-21]. It is also mutated in 80% of the autosomal dominant cancer predisposition disorder, Cowden disease and Bannayan-Zonana syndrome [19, 22].

PTEN encodes a lipid phosphatase with an enlarged core catalytic domain that primarily binds phosphoinositides [19, 23]. As a tumor suppressor phosphatase, PTEN is the principal cellular antagonist of phosphoinositide 3-kinase (PI3K) signaling pathway. Numerous cellular processes are mediated by phosphatidylinositol 3,4,5 triphosphate (PtdIns(3,4,5)P<sub>3</sub>), a key second messenger in signaling cascades and, therefore its level needs to be strictly regulated. PTEN antagonises PI3K activity through its intrinsic lipid phosphatase activity by dephosphorylating PtdIns(3,4,5)P<sub>3</sub> to phosphatidylinositol 4,5 diphosphate (PtdIns(4,5)P<sub>2</sub>) [24, 25].

Normally, PtdIns(3,4,5)P<sub>3</sub> is maintained at low levels in cells containing the functional PTEN gene. However, in tumors where PTEN is mutated or deleted, PtdIns(3,4,5)P<sub>3</sub> is found at high levels [25], leading to excessive phosphorylation of downstream protein kinase B (PKB, also known as Akt) signaling cascades. Akt interacts with PtdIns(3,4,5)P<sub>3</sub> and translocates to the inner cytoplasmic membrane. This interaction causes conformational changes in Akt, exposing two main phosphorylation sites (threonine residue 308 and serine residue 473).

Activation of PKB/Akt in turn has been shown to phosphorylate numerous substrates, a majority of which are involved in cellular functions, for example, cell-cycle progression, cell migration and spreading, cell proliferation and apoptosis (Figure 2) [24, 25].



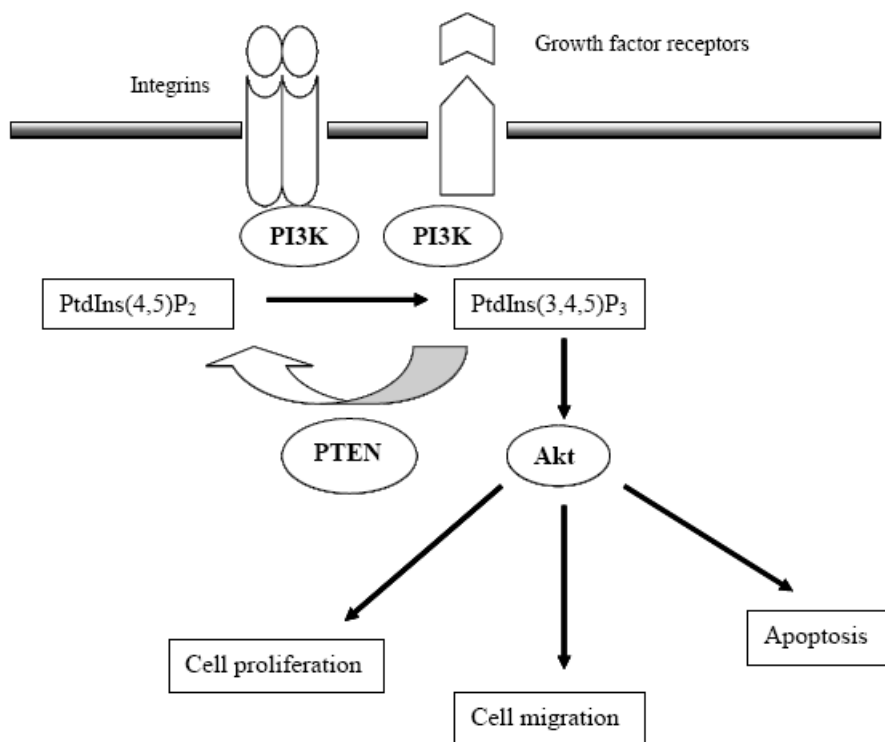


Figure 2. Schematic diagram showing the regulation of PI3K/Akt signaling pathway by PTEN. Upon ligand binding to the integrins or stimulation by growth factors, PI3K becomes activated and phosphorylates PtdIns(4,5)P<sub>2</sub> to PtdIns(3,4,5)P<sub>3</sub> in the membrane. PtdIns(3,4,5)P<sub>3</sub> then activates Akt, which elicits a broad range of downstream signaling events. PTEN negatively regulates this process by dephosphorylating PtdIns(3,4,5)P<sub>3</sub>.

Therefore, PTEN mutations may alter the PI3K/Akt signaling cascade, which in turn affect tumor cell proliferation and survival. Evidence shows that the presence of an elevated Akt activity level in a number of prostate and glioblastoma cell lines is correlated with the presence of PTEN mutations in those cells. In GBM patients, PTEN alterations have been associated with poor prognosis. In contrast, patients whose tumors with no PTEN mutation generally have a longer survival time [26, 27]. About two thirds of glioblastomas lack detectable PTEN protein expression [26] and most of them showed high levels of activated Akt [28]. Several studies using viral vectors have shown that wild-type PTEN suppresses tumorigenicity of glioma cells and inhibits the PI3K/Akt pathway [29-31]. Thus, the PTEN/PI3K/Akt pathway appears to be important in gliomagenesis.

## ROLES OF PTEN IN GLIOMAGENESIS

Numerous studies have implicated PTEN in the regulation of a number of cellular processes that are critical for tumorigenesis. The role of PTEN in the regulation of cell proliferation, cell migration and apoptosis in gliomagenesis will be discussed.

## PTEN REGULATES CELLULAR PROLIFERATION

Rapid cell proliferation, which depends critically upon the rate of protein synthesis, is one of the hallmarks in GBM [32]. Under normal physiological conditions, it is fundamentally important to regulate protein synthesis at the level of translation initiation for controlling the cell proliferation [33]. Conversely, alteration in the regulation of translation is a major contributing factor in tumorigenesis. The mammalian target of rapamycin (mTOR), also known as FK506 binding protein 12-rapamycin associated protein 1 (FRAP1), is one of the regulators for controlling translation.

mTOR is one of the important downstream targets of PI3K/Akt signaling. It regulates translation by phosphorylating ribosomal protein S6 kinase 70kDa polypeptide 1 (p70S6K) and eukaryotic translation initiation factor 4E binding protein 1 (eIF4EBP1 or 4E-BP1) [34]. The phosphorylated p70S6K regulates protein synthesis by activating 40S ribosomal protein S6. This promotes the translation rate of 5' terminal oligopyrimide (5'TOP) messenger RNA (mRNA) transcripts.

The 5'TOP mRNAs encode critical components of the cellular translational machinery, including ribosomal proteins and elongation factors, for promoting protein synthesis [35, 36]. The 4E-BP1 directly interacts with eukaryotic translation initiation factor 4E (eIF4E), which inhibits the 5'TOP mRNA translation. Upon phosphorylation of 4E-BP1 by mTOR, it dissociates from eIF4E and allows the initiation of mRNA translation [32, 37]. Together, p70S6K and 4E-BP1 contribute to the activity of the protein synthesis machinery which is necessary for controlling cell size and proliferation [38, 39].

There is substantial evidence indicating that the mTOR/p70S6K pathway is essential in oncogenesis mediated by PTEN/PI3K/Akt signaling. As mentioned before, PTEN loss leads to aberrant PI3K/Akt signaling, which is commonly found in GBM. Several studies have shown that tumors with PTEN missing (PTEN<sup>-/-</sup>) are highly correlated with activation of Akt, which leads to the phosphorylation of downstream effectors mTOR and p70S6K [40, 41]. Moreover, there is evidence confirming the role of PI3K/Akt and mTOR/p70S6 pathways in cell proliferation.

During the process of cell division, DNA and protein content duplicate to give rise to two daughter cells with conserved genetic materials and cell size. Cell proliferation is the final outcome of protein synthesis and DNA replication [38, 39]. When there is any alteration in the PI3K/Akt/mTOR/p70S6K signaling, in which mTOR and p70S6K are the regulators of protein synthesis in cell division, the rate of cell proliferation will be affected. Studies exploring the therapeutic potential of PTEN/PI3K and mTOR inhibitors in GBM and other solid tumors have shown that suppression of p70S6K phosphorylation inhibited cell proliferation [42, 43].

The mTOR inhibitor rapamycin reduces tumor cell proliferation and tumor growth in PTEN<sup>-/-</sup> tumors, but has no effect on PTEN<sup>+/+</sup> tumors, suggesting that mTOR is responsible for the oncogenic effect of PTEN loss [44].

Cell proliferation is tightly regulated by the activation and deactivation of a series of proteins that establish the cell cycle machinery. The aberrant cell proliferation in GBM also links to the alteration in the expression and activity of these proteins. The cell cycle consists of four distinct stages: (i) Gap phase 1 (G1), where a cell prepares to undergo DNA replication; (ii) DNA synthesis (S), where the cell undergoes DNA synthesis and replication;

(iii) Gap phase 2 (G2), where the cell prepares for mitosis and finally, (iv) Mitosis phase (M), where the newly replicated DNA condenses into sister chromatids which segregate to opposite ends of the cell, and then splits into two new daughter cells each containing an identical set of chromosomes [45, 46]. The cell cycle progression is strictly regulated by three major groups of regulatory molecules: cyclins, cyclin-dependent kinases (CDKs) and CDK inhibitors (CKIs).

The cyclins and CDKs are able to phosphorylate target proteins and promote cell cycle progression when they form complexes. The synthesis of cyclins fluctuates at specific stages in the cell cycle, whereas the CKI molecules are expressed constantly throughout. Different cyclins have different affinities for distinct CDKs. For example, D-type cyclins (cyclin D1, D2 and D3) bind preferentially to CDK4 and CDK6, cyclin E binds primarily to CDK2, cyclin A to CDK2 or CDK1 and cyclin B binds to CDK1. Different cyclins function in different phases of the cell cycle.

The D-type cyclins and cyclin E take part in G1-phase, cyclin A functions in S-phase and cyclin B plays its role in M-phase [45, 46]. On the other hand, CKI molecules antagonize the action of cyclin-CDK complex, and thus inhibit the progression of cell cycle. CKIs have two distinct families: (i) CDK2-interacting protein/CDK-inhibitory protein (Cip/Kip) family consists of p21<sup>Waf1/Cip1</sup>, p27<sup>Kip1</sup> and p57<sup>Kip2</sup>, and (ii) Inhibitor of kinase 4/Alternative Reading Frame (INK4/ARF) family consists of p15<sup>INK4b</sup>, p16<sup>INK4a</sup>, p18<sup>INK4c</sup> and p19<sup>INK4d</sup>.

When quiescent cells enter G1 phase of the cycle, D-type cyclins form active complexes with CDK4 or 6 and cyclin E associates with CDK2 to phosphorylate the retinoblastoma protein (Rb).

Rb is a nuclear phosphoprotein that causes cycle arrest at G1 phase by repressing transcription of genes required for the G1 to S phase transition. Inactivation of Rb by phosphorylation results in the dissociation of E2F transcription factor, which regulates expression of S phase genes [47, 48].

Cyclin A and CDK2 complex is activated in S phase and regulates DNA synthesis. In G2, the cyclin B-CDK1 complex promotes mitosis. To prevent unscheduled entry into the cell cycle, CKI is important in controlling the activity of CDKs. The INK4/ARF family members bind to D-type cyclin-CDK4/6 complexes while the Cip/Kip family members bind cyclin-CDK complexes containing cyclin D, E, A and B to inhibit CDK activity [45, 46].

Apart from regulating the cell growth through mTOR and p70S6K, PTEN/PI3K also regulates the cell cycle by controlling the stability of CDK regulatory proteins [49-51].

Therefore, those cell cycle regulatory proteins are 'hot' targets for studying tumorigenesis. The expression and contributions of these proteins involved in PTEN/PI3K signaling pathways have been extensively investigated. Several studies reported that restoration of wild-type PTEN in PTEN-deficient glioblastoma cells causes cell growth suppression *in vitro* as well as reduced tumorigenicity in nude mice.

The growth suppressive effect of wild-type PTEN is found to be associated with an accumulation of the cells in G1-phase and a decreased progression into S-phase through inhibition of the PI3K/Akt signal transduction [31, 52, 53].

This is accompanied by increasing the CKI p27<sup>Kip1</sup> expression, while the expression of p21<sup>Waf1/Cip1</sup> remained unaffected. Upregulation of p27<sup>Kip1</sup> is dependent on the phosphatase activity of PTEN, and is associated with a decrease in the G1-S cyclin-CDK associated kinase activities (cyclin A, cyclin E and CDK2) [31]. It was demonstrated that p27<sup>Kip1</sup> upregulation occurs following a PTEN-mediated inhibition of Akt.

## PTEN REGULATES CELL MIGRATION AND INVASION

Extensive infiltration of glioma cells into the surrounding normal brain tissue is a characteristic feature of malignant gliomas and a major reason for poor prognosis in GBM patients. Migration and invasion are two important prerequisites for the infiltrative and destructive growth patterns of malignant gliomas.

Both cell migration and invasion are complex processes which involve the coordination of several signal transduction pathways that allow cancer cells to remodel their surrounding microenvironment, migrate and invade into the surrounding tissues.

Cell migration is a multistep process involving (i) the protrusion of plasma membrane, lamellipodia and filopodia extension, at the leading edge of the cell, (ii) the formation of new adhesion sites to the extracellular matrix (ECM) at the front with concomitant release of the old adhesion sites at the back, and (iii) contraction of actomyosin-based cytoskeletal filaments in the cell body [54, 55]. During cell invasion, cells also release proteases that degrade and remodel the ECM to promote their passage through the stroma and entrance into surrounding tissue [56].

PTEN/PI3K has been reported to play an important role in regulating glioma cell migration and invasion [57, 58]. A study reported that overexpression of PTEN in NIH3T3 fibroblasts and U-87MG glioma cells results in reduced cell spreading and migration on integrin substrates such as fibronectin and vitronectin [56]. PTEN/PI3K most likely contributes by interacting with several intracellular signaling molecules including focal adhesion kinase (FAK) and Rho family of GTPases. FAK is a non-receptor tyrosine kinase that serves as a major mediator for cell migration. It functions as part of a cytoskeletal-associated network of intracellular signaling proteins that are activated by transmembrane integrin receptor clustering. It is expressed at elevated levels in GBM when compared with normal brain tissue [59]. FAK was found highly expressed at the protein level in astrocytomas and glioblastomas, however, FAK gene amplification has never been detected in glioblastomas [19].

Therefore, the link between PTEN/PI3K and FAK has been extensively studied in tumorigenesis. A study reported that overexpression of PTEN is associated with declined levels of phosphorylation of FAK and its putative downstream target p130cas (Crk-associated substrate), as well as reduced number of focal adhesion and actin stress fibers [22]. In contrast, the inactive PTEN does not reduce the number of focal adhesion nor the cell spreading capacity.

PTEN regulates FAK signaling through PI3K pathway. An activated FAK leads to activation of PI3K, PTEN can inhibit this signal transduction by antagonizing PI3K through dephosphorylation of  $\text{PtdIns}(3,4,5)\text{P}_3$ . Moreover, a study indicated that actin reorganization in various human cancer cells is through FAK/PI3K activation, suggesting the association of FAK and PI3K regulates the cell motility process [60].

Taken together, the data indicate that PTEN can regulate cell migration and focal adhesion formation by acting as a negative regulatory molecule in the integrin-mediated signaling cascade.

In the Rho family of small GTPases, Rho, cdc42 and Rac facilitate the migratory and invasive behaviour of tumor cells by regulating actin and microtubule cytoskeleton organization [61]. Most of the available data on the function of Rho family proteins in

malignant glioma transformation are based on dominant-negative mutants and RNA interference-mediated depletion of these GTPases. Strong correlations between the expression of Rho GTPase proteins and the invasive behaviour of glioma cells have been reported [62, 63].

A recent study also reported that radiation can enhance the invasiveness of primary glioblastoma cells through the activation of the Rho signaling pathway via PI3K [64]. Further evidence suggested that PTEN may suppress cell motility by down-regulating Rac1-GTP and cdc42-GTP [65]. Immortalized fibroblasts from PTEN-deficient mice showed increased cell motility when compared with wild-type controls cells. This phenomenon is associated with strongly elevated Rac1-GTP and cdc42-GTP levels without alteration in FAK phosphorylation.

## PTEN REGULATES APOPTOSIS

Malignant glioma is characterized by an extremely unfavorable prognosis. Active proliferation and extensive infiltration are the basic factors limiting the effect of standard therapy. Another significant problem is the reduced susceptibility to apoptotic stimuli in glioma cells, which frequently lack sensitivity to radiotherapy and/or chemotherapy.

PTEN plays an important role in maintaining the cellular susceptibility to apoptotic stimuli. Restoration of wild-type PTEN gene into a PTEN-deficient glioma cell line renders the cells susceptible to apoptosis [19, 66].

Moreover, PTEN gene transfer sensitized glioma cells to apoptosis when irradiated or treated with Fas ligand. Numerous studies show that PTEN regulates apoptosis is dependent on PI3K/Akt signaling. Akt, which is a serine/threonine kinase, can phosphorylate several apoptosis-associated proteins such as pro-apoptotic factor Bad [19, 22].

Upon phosphorylation, Bad can no longer function as a pro-apoptotic molecule and dissociates itself from anti-apoptotic proteins. This leads to a relative increase in the level of anti-apoptotic proteins and prevents apoptosis. In addition, Akt is shown to phosphorylate pro-caspase-9 and inhibits its proteolytic activation. These data illustrate that Akt regulates apoptosis. Therefore, inhibition of Akt by reducing the accumulation of PtdIns(3,4,5)P<sub>3</sub> constitute a potential mechanism of tumor suppression by PTEN.

## CONCLUSION

PTEN has been identified as a putative tumor suppressor, and substantial evidence has confirmed its tumor suppressor role both *in vitro* and *in vivo*. With regard to malignant glioma, aberration of PTEN signaling or other genes involved in the PI3K/Akt signaling pathway, appears to be crucial in gliomagenesis. Any alterations or aberrations in PTEN/PI3K/Akt signal transduction may contribute to various cellular responses in glioblastoma, such as loss of cell-cycle control, active cell proliferation, escape from apoptosis and extensive infiltration. From a diagnostic point of view, the understanding of PTEN signaling in gliomagenesis is important for the development of more specific and potent therapeutic agents for patients suffering from this fatal brain tumor.

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*Chapter 18*

## **THE THIOL REDOX SYSTEM IN GLIOMA BIOLOGY: CLINICAL TARGET AND SIGNIFICANCE IN RESISTANCE TO GLIOMA CHEMOTHERAPY**

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### **ABSTRACT**

Sulfur-containing compounds play an essential role in maintaining redox balance in glioma cells. Chief amongst these are the reduced and oxidised (disulfide) forms of cysteine (cysteine/cystine) and glutathione (GSH /GSSG) as well as thioredoxin and glutaredoxin, members of the thiol-disulfide oxidoreductases. GSH is also important as an antioxidant, as a 'safe' storage form of cysteine and for detoxification reactions involving the GSH sulfur transferase family of enzymes.

Glioma cells contain a high concentration of GSH, compared to normal astrocytes, which renders these cells particularly resistant to chemotherapeutic agents. The rate of synthesis of GSH is controlled by the availability of cysteine, which is imported in its oxidised form, cystine, through specialized channels in the plasma membrane. These channels, known as the cystine-glutamate  $x_c^-$  exchanger, operate by taking up cystine in exchange for glutamate, which is released into the extracellular medium. Glioma cells export large quantities of glutamate by this mechanism, which, if unchecked, causes damage and eventual death of surrounding neurones. Thus, GSH synthesis in glioma cells fuels resistance to chemotherapy and, at the same time, kills off surrounding neurones thereby providing space for tumor cell growth.

Much is now known of the molecular mechanism of cystine import and of GSH synthesis in glioma cells. In particular, a number of therapeutic strategies that target the cystine-glutamate exchanger have been proposed on the basis that this would inhibit synthesis of GSH and remove the source of glutamate release.

This review describes new developments in the field of glioma cell redox balance and evaluates the potential for clinical intervention at the level of cysteine and GSH biosynthesis that may prove effective in combating brain tumor growth and development.

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**Keywords:** Glioma, astrocyte, cysteine, GSH, oxidoreductase,  $x_c^-$  cystine-glutamate exchanger, transsulfuration

## 1. INTRODUCTION

Regulation of thiol redox balance is crucially important for multiple metabolic, signalling and transcriptional processes in mammalian cells. Thiol groups, whether in proteins or small molecules, are highly reactive and susceptible to oxidation that may cause significant loss of biological activity. In proteins, oxidation of the thiol group of cysteinyl residues produces modifications that, depending on the location of the cysteine(s), will impact on the structure, catalytic activity or ability to engage in protein-protein interactions. Such modifications include formation of inter- or intra-molecular disulfides between protein thiols (Protein (P)-S-S-P) or small molecular weight thiols such as GSH (P-S-SG), or successive oxidation to sulfenic (P-SOH), sulfinic (P-SO<sub>2</sub>H) or sulfonic (P-SO<sub>3</sub>H) acids and nitrosylation (P-S-NO). Cell-based thiol redox buffering systems are designed to protect thiol groups from oxidation and to repair those that may have become oxidised as a result of either normal or aberrant cellular metabolism. The key components of the thiol redox system are the cystine (Cys)/cystine (CySS) and glutathione (GSH) / glutathione disulfide (GSSG) redox pairs, and the thiol disulfide oxidoreductases.

The brain is highly susceptible to oxidative damage, due to its high metabolic load and dependency on oxidative metabolism. Astrocytes play a central role in the brain's defence from oxidative stress, particularly in the case of thiol-based antioxidants, and are the primary source of antioxidant molecules for neurones. Astrocytes synthesise the major cytosolic sulfur-based antioxidant, GSH, and actively export cysteine and GSH to manage extracellular redox balance and to supply neurons with cysteine for neuronal GSH synthesis.

Glioma cells are noted for their resistance to oxidative stress and apoptosis. In many instances, this relates to upregulation of several components of the thiol redox buffering systems, with the consequence that standard anti-cancer therapies are frequently ineffective in the treatment of gliomas. This is particularly true in the case of glioblastoma multiforme (GBM), which is the most common and highly aggressive primary malignant brain tumour in adults. Survival time of patients with GBM is still only about one year and therapeutic approaches that have a high success rate in other cancers are largely ineffective in GBM. Not only is accessibility limited, due to the blood brain permeability barrier, but GBM are unresponsive to all but a small number of drugs and CNS-related side effects are common [1]. Improved and effective treatment strategies are urgently required. Amongst these, the thiol-based redox buffering system is seen as having the potential for development of more effective and specific therapies. Significant differences in this system in gliomas compared to normal astrocytes signals the potential for therapies directed exclusively at glioma cells.

This review introduces the biochemistry and biological activity of the major classes of thiol-based antioxidants in astrocytes and describes the abnormal activity of the thiol-based redox system in glioma cells. It provides an account of the basis of chemotherapeutic resistance that is characteristic of glioma cell biology and describes new approaches to targeting the thiol-based redox system in cancer therapy.

## 2. THE COMPONENTS OF THE THIOL REDOX SYSTEM

### 2.1. Cysteine

The sulfur amino acid, cysteine, plays an important role in maintaining cellular redox potential and is the essential amino acid constituent in the antioxidant, GSH and in the functional (CXXC) motif of the major cellular antioxidant families that include the glutaredoxins, thioredoxins and peroxiredoxins [2]. Cysteine is highly reactive and readily oxidises to the disulfide form, cystine, with concomitant loss of electrons (Figure 1). Extracellular oxidising conditions favour cystine, whereas cysteine is the dominant form inside the cell. Even so, the intracellular concentration of free cysteine is kept in the low micromolar range, which minimises the risk of auto-oxidation to cystine and loss of redox balance.

Astrocytes accumulate cystine more readily than cysteine and, due to the relative abundance of cystine in the extracellular fluid, transport of cystine is the dominant mechanism for importing the amino acid. In astrocytes and glioma cells, the majority of cystine is taken up by the  $x_c^-$  cystine-glutamate exchanger. Lesser amounts can be imported as a substrate of the membrane-associated enzyme  $\gamma$ -glutamyltranspeptidase, which is a component of the  $\gamma$ -glutamyl cycle involved in GSH synthesis [3]. Experiments with cultured astrocytes and glioma cells have shown that cystine is a low affinity substrate for the high affinity glutamate transporters, but this is unlikely to make a significant contribution to intracellular cystine in physiological conditions, because of the considerably higher affinity of these transporters for glutamate [3-6].

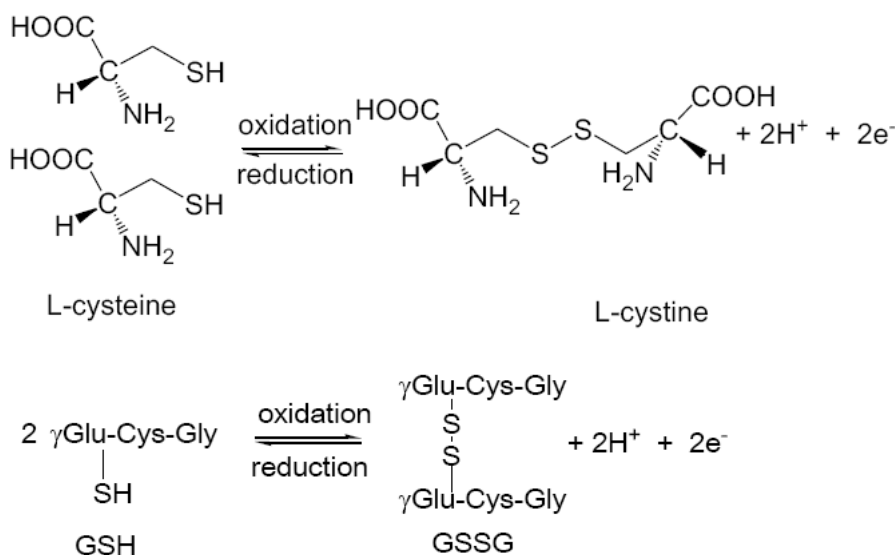


Figure 1. The cysteine / cystine and GSH/GSSG redox pairs.

As an alternative to direct transport of cystine from the extracellular medium, cysteine can also be derived from the indispensable amino acid, methionine, by transsulfuration via homocysteine. In mammalian liver, approximately 50% of cysteine in GSH is derived from

methionine via transsulfuration [7], but, until recently, this pathway was not thought to have any significance in astrocytes. It has lately been recognised that transsulfuration provides between one quarter and one third of cysteine for GSH synthesis in astrocytes. It is believed that this pathway is particularly important during oxidative stress or conditions in which the import of cysteine from the exchanger may be limited ([8, 9] and references therein).

### **2.1.1. The $x_c^-$ Cystine-Glutamate Exchanger**

The  $x_c^-$  cystine-glutamate exchanger is a sodium independent, chloride-dependent, electro-neutral antiporter that releases glutamate in exchange for cystine [6, 10-13]. The antiporter is composed of a heavy chain subunit, 4F2, that is common amongst amino acid transporters and a light chain, xCT, that confers substrate specificity [12, 13]. Several structural analogues of glutamate and/or cystine have been identified as substrate-inhibitors of the exchanger. These include L-homocysteate, L-quisqualate, L- $\alpha$ -aminoadipate and L- $\beta$ -N-oxalylamino-L-alanine [8, 14-16]. Non substrate inhibitors, such as 4-carboxyphenylglycine and sulfasalazine ((3E)-6-oxo-3-[[4-(pyridin-2-ylsulfamoyl)phenyl]hydrazinylidene]cyclohexa-1,4-diene-1-carboxylic acid) have also been identified [17].

The directionality of cystine-glutamate exchange, first identified by Bannai and co-workers in the 1980s, is governed by the respective concentration gradients of the transported species: intracellular glutamate is up to a thousandfold higher in the cytosol than outside the membrane. In contrast, the extracellular concentration of cystine is significantly higher than in the cytosol, because it is reduced to cysteine once inside the cell [10, 12, 13, 18]. In short, intracellular glutamate provides the driving force for cystine uptake by the exchanger and is continuously released into the medium.

In normal astrocytes, glutamate is re-cycled back into the cytosol by the high affinity sodium-dependent glutamate transporters (SLC 1 transporter family). This is essential for ensuring that the extracellular concentration of glutamate does not rise to a level that will cause excitotoxicity due to hyperactivation of NMDA receptors that would threaten the viability of surrounding neurones [19]. Furthermore, impaired glutamate uptake will result in an elevation in the extracellular concentration of glutamate that will compete with cystine for entry via the exchanger, resulting in a deficiency in cysteine for GSH synthesis, leading to oxidative stress [19].

Until now, it has been the assumption that glutamate was merely the counter-ion and had no physiological role beyond enabling cystine import by the exchanger. However, interesting new evidence suggests that extracellular glutamate from the exchanger has a specific function in the tonal regulation of extrasynaptic metabotropic glutamate receptors and the control of neurotransmitter release [20].

### **2.1.2. The Transsulfuration Pathway**

Synthesis of cysteine *in situ* from methionine provides an alternative to importation of cystine. This process begins with the adenosine-mediated methylation of methionine to homocysteine, which can either be re-methylated back to methionine (transmethylation), or directed towards transsulfuration via the intermediate cystathionine (Figure 2). The catalysts of the transsulfuration pathway are two pyridoxal-phosphate containing enzymes, cystathionine- $\beta$ -synthase (L-serine hydro-lyase (adding homocysteine); EC 4.2.1.22) and cystathionine- $\gamma$ -lyase (L-cystathionine cysteine lyase; EC 4.4.1.1) that catalyse the formation

and metabolism of cystathionine, respectively. Regulation of metabolic flux through the transsulfuration pathway occurs at several levels. S-adenosylmethionine is an allosteric activator of cystathionine- $\beta$ -synthase that channels excess sulfur towards cysteine when methionine levels are high. The enzyme is redox-sensitive and is two-fold less active in the reduced form, i.e., when the heme group is in the ferrous state [7]. Cysteine is a competitive inhibitor of cystathionine- $\gamma$ -lyase, thereby preventing an accumulation of cysteine that would otherwise threaten intracellular redox balance. Mild oxidative stress causes up-regulation of the enzyme in primary astrocytes and glioma cell cultures [8], whereas severe oxidative stress results in inhibition of the enzyme.

A deficiency in the transsulfuration pathway leads to excessive homocysteine production, loss of cellular redox homeostasis and reduced levels of GSH [21]. Genetic defects in cystathionine- $\beta$ -synthase are the most common hereditary causes of hyperhomocysteinemia in humans.

Cystathionine  $\beta$ -synthase is abundant throughout brain and is particularly highly expressed in Purkinje cells, the hippocampus, cerebellar Bergmann glia and astrocytes [22, 23]. Cystathionine- $\gamma$ -lyase is expressed much more restrictedly than cystathionine- $\beta$ -synthase, and until recently was thought to be entirely absent from brain, giving rise to the view that the transsulfuration pathway was not functional in this tissue [24]. It is now known that the enzyme localises to astrocytes, including retinal Müller cells, and microglia [8, 25-28]. Expression and functional activity of the enzyme has also been recorded in rat C6 glioma cells [8].

The functional significance of the transsulfuration pathway in astrocytic sulfur metabolism is only beginning to be understood, but there are indications that it operates as a reserve pathway that may be upregulated in response to mild oxidative stress [8]. It is hypothesised that, under conditions in which the availability of cystine via the exchanger is compromised, the transsulfuration pathway provides a crucial back-up system to ensure continuing supply of cysteine for synthesis of GSH [9]. Additional research is required for understanding the full significance of this pathway in glioma metabolism and to evaluate the extent of interplay between transsulfuration and cystine uptake in management of GSH.

## 2.2. Glutathione

GSH ( $\gamma$ -L-glutamyl-L-cysteinyl-glycine) is the major non-protein thiol-containing antioxidant in mammalian biology and has a crucial role in maintaining redox balance in the brain. GSH is a strong reducing agent, as the free sulfhydryl group of the cysteine residue readily oxidises forming an intermolecular disulfide bridge (GSSG; Figure 1) that is reduced back to GSH by GSH reductase using NADPH as electron donor. The cytosolic concentration of GSH is typically 2-3 mM, an order of magnitude higher than the cytosolic concentration of free cysteine (0.1-0.3 mM). Since GSH is non-toxic, it is an important storage and carrier form of cysteine. GSSG accounts for less than 1% of total GSH in normal cells, although the ratio shifts in favour of GSSG in aged animals and is accompanied by lower antioxidant efficiency [29]. Similarly, in oxidative stress, the GSH:GSSG ratio can fall to as low as 50. The bulk of GSH is localised to the cytosol, with small quantities in the mitochondria and

endoplasmic reticulum. Mitochondrial GSH makes up approximately 10% of the intracellular concentration of the tripeptide.

As an antioxidant, GSH is effective by a number of different mechanisms. It reacts non-enzymatically with reactive oxygen and nitrogen species, including superoxide ( $\text{O}_2^{\cdot -}$ ), nitric oxide ( $\text{NO}^{\cdot}$ ), hydroxyl radical ( $\text{HO}^{\cdot}$ ) and peroxynitrite ( $\text{ONOO}^{\cdot}$ ). GSH is more effective than cysteine in scavenging free radicals and is the only molecule able to detoxify hydroxyl radicals [30]. GSH is both cofactor and electron donor for GSH peroxidase that detoxifies hydrogen peroxide, with concomitant oxidation to GSSG. GSH also functions in the detoxification of xenobiotic compounds by GSH-S-transferases and in the protection of free sulfhydryl groups on proteins in association with the glutaredoxins.

### **2.2.1. The $\gamma$ -Glutamyl Cycle**

Synthesis of GSH occurs in the cytosol by two consecutive, ATP requiring reactions that form the first steps of the  $\gamma$ -glutamyl cycle first described by Meister and co-workers more than 40 years ago [31]. In the first step, which is catalysed by L- $\gamma$ -glutamate cysteine ligase ( $\gamma$ -glutamylcysteine synthase; EC 6.3.2.2;  $\gamma$ -GCL), the dipeptide  $\gamma$ -glutamylcysteine is formed from glutamate and cysteine (Figure 2). Thereafter, addition of glycine, catalysed by GSH synthase (EC 6.3.2.3), generates GSH. In the second half of the cycle, GSH is exported and the  $\gamma$ -glutamyl moiety transferred to an acceptor amino acid in a reaction catalysed by the membrane-bound enzyme,  $\gamma$ -glutamyltranspeptidase. A variety of amino acids can function as the acceptor, but cystine is one of the best [32]. In this case, the net result of the  $\gamma$ -glutamyltranspeptidase-catalysed reaction is the import of one cystine in exchange for cysteine, which equates to a gain of one cysteine per enzyme cycle. Overall, this reaction allows GSH to continuously provide a source of cysteine. The final steps of the  $\gamma$ -glutamyl cycle see glutamate and glycine regenerated via oxo-proline.

### **2.2.2. Factors that Regulate GSH Synthesis**

Cysteine is the rate-limiting precursor for GSH synthesis. The cytosolic concentration of this amino acid is close to the Michaelis constant ( $K_m$ ) value for  $\gamma$ GCL of 0.1–0.3 mM, whereas the intracellular glutamate concentration may be as much as an order of magnitude higher than its  $K_m$  value of 1.8 mM. Thus, regulation of GSH synthesis occurs at the level of  $\gamma$ GCL, which is also controlled by feedback inhibition by GSH.  $\gamma$ GCL is comprised of two subunits, each encoded by different genes. The heavy (catalytic) subunit,  $\gamma$ GCLc, has a relative molecular mass of 73,000 D, whereas the light (modifier) subunit,  $\gamma$ GCLm, has a relative molecular mass of 30,000 D.  $\gamma$ GCLm is inactive in catalysis, but increases the affinity of the catalytic subunit for glutamate and raises the  $K_i$  to approximately 2.3 mM for inhibition by GSH. GSH synthase is not a regulatory enzyme, indicating that  $\gamma$ -glutamylcysteine does not accumulate in cells.

Astrocytes are the site of GSH synthesis in the brain and the intracellular concentration of the tripeptide exceeds that of neurons [33]. Export of GSH from astrocytes can occur at a rate of up to 10% per h and follows the lines described above. Importantly, however, astrocytic release of GSH has the additional function of providing the component amino acids for GSH synthesis in neurons. Neurones do not readily take up cystine and preferentially transport cysteine as a substrate of the EAAC1/EAAT3 member of the SLC 1 high affinity glutamate



transporter family. Extensive analysis of the transfer of the components of GSH from astrocytes to neurones has been undertaken by Dringen and co-workers [26, 34], who identified that, whilst astrocytes in culture accumulate cystine in preference for cysteine, neurones have a reduced capacity for import of cystine, and rely on provision of cysteine from degradation of cys-gly that is a product of  $\gamma$ -glutamyltranspeptidase (Figure 2). Ultimately, therefore, neurones rely on GSH synthesised in astrocytes for protection from oxidative stress. In fact, astrocytes increase synthesis of GSH in response to oxidative stress [35].

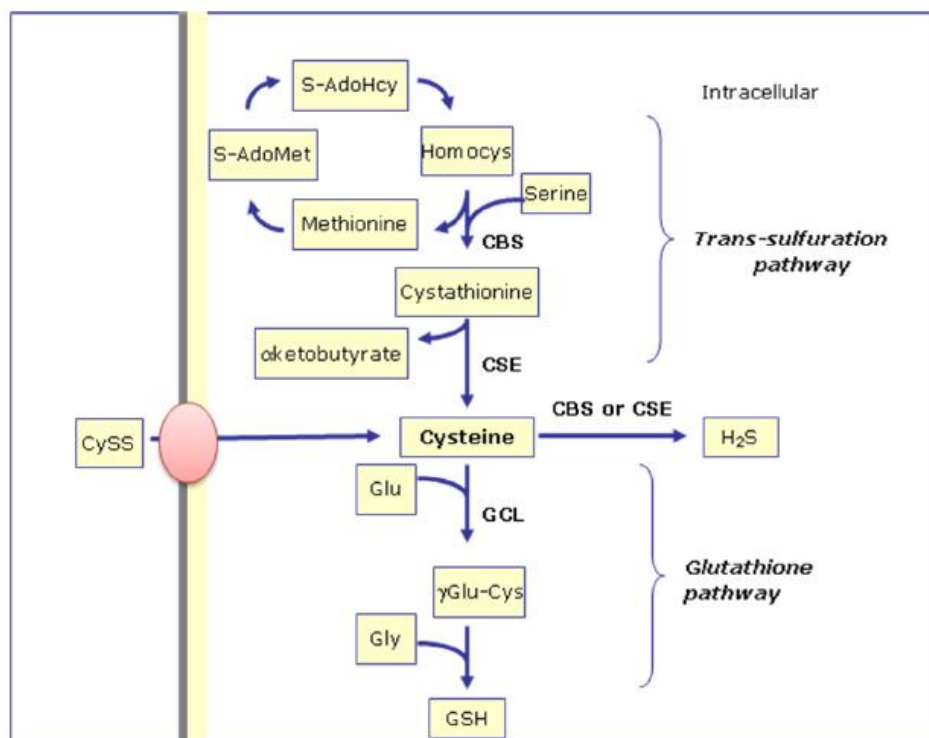


Figure 2. Cystine (CySS), taken up via the  $x_c^-$  cystine-glutamate exchanger is reduced to cysteine (Cys). Cys can also be synthesised intracellularly from methionine by the trans-sulfuration pathway, as shown. The rate-limiting step in GSH (GSH) synthesis is formation of  $\gamma$ -glutamylcysteine ( $\gamma$ -Glu-Cys), catalysed by  $\gamma$ -glutamate cysteine ligase (GCL). Conversion of methionine to cysteine in the transsulfuration pathway is catalysed by cystathionine- $\beta$ -synthase (CBS) and cystathionine- $\gamma$ -lyase (CSE). These enzymes also catalyse formation of hydrogen sulfide (H<sub>2</sub>S) from cysteine.

### 2.3. Thiol-Disulfide Oxidoreductases

The thiol-disulfide oxidoreductases (TDOR) include the thioredoxin (Trx) and glutaredoxin (Grx) systems that catalyse complementary reactions involved in protection of vulnerable sulfhydryl groups on proteins from oxidative damage. The peroxiredoxins (Prx) are a third member of the family that are involved in detoxification of H<sub>2</sub>O<sub>2</sub>. Thioredoxins catalyse reduction of intramolecular and intermolecular disulfides and sulfenic acids.

Glutaredoxin specifically reduces protein-GSH mixed disulfides. All of these enzymes share the common Cys-X-X-Cys motif, located at the end of a  $\beta$ -strand and beginning of an  $\alpha$ -helix in the protein.

Extensive immunohistochemical analysis of thioredoxin proteins in mouse brain has shown all members of the thioredoxin family localised in the hippocampus, substantia nigra, cerebral cortex, striatum, cerebellum and spinal cord. Subtypes of Trx and Grx showed strong staining in neurons, whereas specific subtypes of Prx (Prx3 and Prx6) were localised in astrocytes in all regions studied [36].

### 2.3.1. Thioredoxins

The thioredoxins are a small family of redox-active proteins (10-12 kDa) that mediate the reversible oxidation-reduction of sulfhydryl groups on proteins. The two principal forms of thioredoxin in mammalian cells are thioredoxin 1 (Trx-1) and thioredoxin 2 (Trx-2). Trx-1 (104 aa) is a cytosolic protein, whereas Trx-2 (166 aa) localizes to the nucleus. Thioredoxins have a number of additional roles, including acting as co-factor in DNA synthesis by providing electrons for reduction of ribose by ribonucleotide reductase and inhibition of apoptosis [37]. These proteins are effective antioxidants, in which the thiol groups operate as an electron source for scavenging reactive oxygen species (ROS), due to formation of homo – and hetero-dimers linked by disulfide bonds of the conserved cysteine residues. Thioredoxins increase the DNA-binding activity of a number of transcription factors, for example, Nf-kB, AP-1, AP-2 and nuclear receptors for oestrogen and glucocorticoid. They also bind (causing inactivation of) a number of kinases, including protein kinase C (PKC) and apoptosis signal regulating kinase 1 (ASK1) that regulates activation of the JNK and p38<sup>MAPK</sup> pathways. During oxidative stress, thioredoxin is oxidised by ROS, which causes its release from ASK-1, leading to activation of JNK and p38<sup>MAPK</sup> and a pro-apoptotic response [37], and references therein.

The redox activity of the thioredoxins is due to the presence of a highly-conserved catalytic site with the sequence Trp-Cys-Gly-Pro-Cys-Lys. The negative redox potential of the two cysteine residues provides a strong reducing environment that, with electrons provided by NADPH / glucose-6-phosphate dehydrogenase in the hexose monophosphate shunt, maintains the free sulfhydryl group of cytosolic proteins in the reduced state.

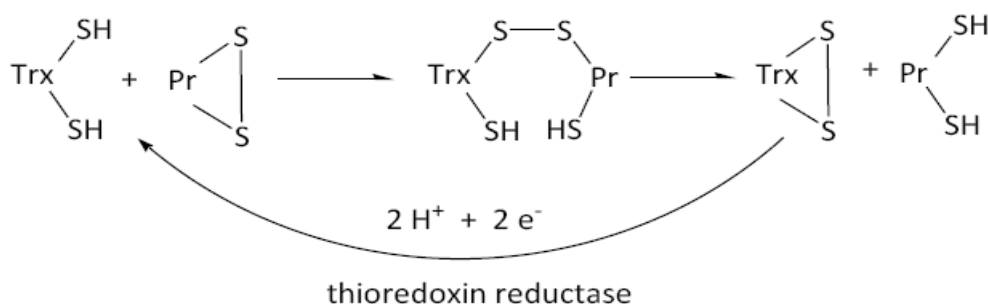


Figure 3. Schematic diagram of the thioredoxin (Trx)/ thioredoxin reductase-mediated reduction of oxidised sulphydryl groups in protein (Pr).

Recycling of oxidised Trx occurs via Trx reductase, by the mechanism illustrated in Figure 3. The mammalian Trx reductases are a family of selenium-containing flavoproteins with the conserved sequence Cys-Val-Asn-Val-Gly-Cys at the catalytic site. They exist as a homodimer and activity is regulated by NADPH. They function in conjunction with Trx in the transfer of electrons to oxidised proteins.

### **2.3.2. Glutaredoxins**

One of the major functions of GSH is S-glutathionylation, a process in which free sulfhydryl groups in proteins are reversibly oxidised by GSH as a protective mechanism to prevent damaging irreversible oxidation. This process is catalysed by a GSH-S-transferase family of enzymes. S-glutathionylation occurs in response to a number of external factors, including hormones and cytokines, and as an integral part of signal transduction cascades initiated by ROS and RNS. Glutaredoxins, on the other hand, catalyse the de-glutathionylation of proteins, which restores the protected cysteinyl residues to the reduced state.

There are several forms of glutaredoxin in mammalian cells: glutaredoxin 1 (Grx1) and glutaredoxin 2 (Grx2) are the principal forms, but others, Grx3 (also known as ‘protein kinase C-interacting cousin of thioredoxin’; PICOT), and Grx5 have also been identified [38]. Grx1 is the most abundant form, occurs in a micromolar concentration and is localised primarily in the cytosol. It contains the sequence Cys-Pro-Tyr-Cys at the catalytic site. The protein is also found in the mitochondrial inter-membrane space although its function here is unclear. Grx2 localises to the mitochondria and shares 30% sequence homology with Grx1 and has an active site motif of Cys-Ser-Tyr-Cys. To date, two human clones of Grx2 have been discovered: Grx2a and Grx2b that differ in their N-terminal sequences. The catalytic activity of Grx2 is approximately ten-fold lower than Grx1. Whilst glutathionylation of proteins is a protective mechanism that increases in response to oxidative stress, there are instances where de-glutathionylation has a regulatory role. For example, Grx-mediated de-glutathionylation of the small GTPase RAS is believed to be a factor in regulation of cell proliferation and hypertrophy [39].

In brain, most research on the glutaredoxins to date has been centered on investigating their neuroprotective function. For example, Grx2 is constitutively expressed in both neurons and glia in mouse and human brain [40].

It has a neuroprotective role in maintaining mitochondrial integrity in neurodegenerative diseases, such as Parkinson’s disease. In Alzheimer’s disease brains, an increase in Grx1 and a reduction in Trx1 has been observed [41], which is believed to be a direct effect of amyloid beta neurotoxicity and signals that deregulation of Grx1 and Trx1 antioxidant systems may be important events in the pathogenesis of AD. Increased de-glutathionylation (i.e., reduction) of DJ-1, a protein that functions as an antioxidant, transcriptional co-activator and molecular chaperone, may be the cause of decreased proteolysis of the protein and may contribute to the pathogenesis of sporadic Parkinson’s Disease [42]. Specific references to Grx activity in astrocytes and/or glioma cells are absent in the literature.

### **2.3.3. Peroxiredoxins**

The peroxiredoxins (Prx) represent a third branch of the Trx family of proteins, of which up to 6 subtypes have been identified (Prx1- Prx6). These enzymes specifically reduce

hydrogen peroxide and, based on their catalytic mechanism, are divided into three categories: (1), 2-cysteine Prxs, (2) atypical 2-cysteine Prxs and (3) 1-cysteine-Prxs.

During the peroxidase reaction, a sulfenic acid intermediate is formed on one of the cysteinyl residues close to the N-terminal region of the protein. A second cysteinyl residue, known as the resolving cysteine, which is located at the C-terminal region, reacts with the sulfenic intermediate, resulting in an inter-subunit disulfide bond that is subsequently reduced by Trx.

### **3. REDOX BUFFERING SYSTEMS IN GLIOMAS: RESISTANCE TO THERAPY AND OXIDATIVE STRESS**

#### **3.1. Cysteine and GSH in Glioma Cells**

Gliomas have elevated intracellular GSH, compared to normal astrocytes, which is considered largely responsible for their noted resistance to oxidative stress and chemotherapeutic agents [43]. Samples of primary human glioma show increased expression of the xCT subunit of the  $x_c^-$  exchanger, compared to peritumoural regions and human brain tissue without malignant transformation [44]. Correspondingly, experimental cell lines that are resistant to oxidative stress have a significantly up-regulated  $x_c^-$  exchanger, as indicated by increased mRNA encoding the xCT subunit and greater functional activity of the protein, and elevated GSH [19].

One of the most remarkable features of glioma cell biology is the failure to express functional high affinity glutamate transporters and consequent loss of 95% or more of glutamate uptake observed in normal astrocytes [17, 45]. Moreover, glioma cells actually release glutamate via the  $x_c^-$  exchanger, the complete opposite of the role of normal astrocytes in sequestering glutamate.

This is illustrated in Figure 4. Due to overexpression of the  $x_c^-$  exchanger combined with an absence of high affinity uptake, a significant and prolonged increase in the extracellular concentration of the amino acid has been recorded [17, 46]. Direct sampling of peritumoural extracellular fluid in Grade IV GBM patients by microdialysis revealed glutamate concentrations in excess of 100  $\mu$ M at the tumour margin, whereas non-malignant tissue had normal glutamate levels in the submicromolar range [47]. It is well established that hyperactivation of glutamate receptors, particularly the NMDA sub-type, triggers a sustained increase in intracellular calcium ions, that exceeds the intracellular calcium buffering capacity, leading ultimately to cell death.

This process, known as ‘excitotoxicity’ was first described 40 years ago and is now known to represent the final common pathway of neuronal cell death in a variety of neurodegenerative disorders [48, 49]. Thus, glutamate released from tumour cells causes widespread destruction of peritumoural neurones. Experts agree that this strategy allows the tumour to expand into the space vacated by dying neurones [17, 46]. Furthermore, evidence suggests that released glutamate acts as a trophic factor that promotes tumour invasion and formation of tumour metastasis [17, 50]. Overall, glioma cells display a remarkable adaptation that ensures a survival advantage over surrounding neurones and generates space for tumour expansion (Figure 4).

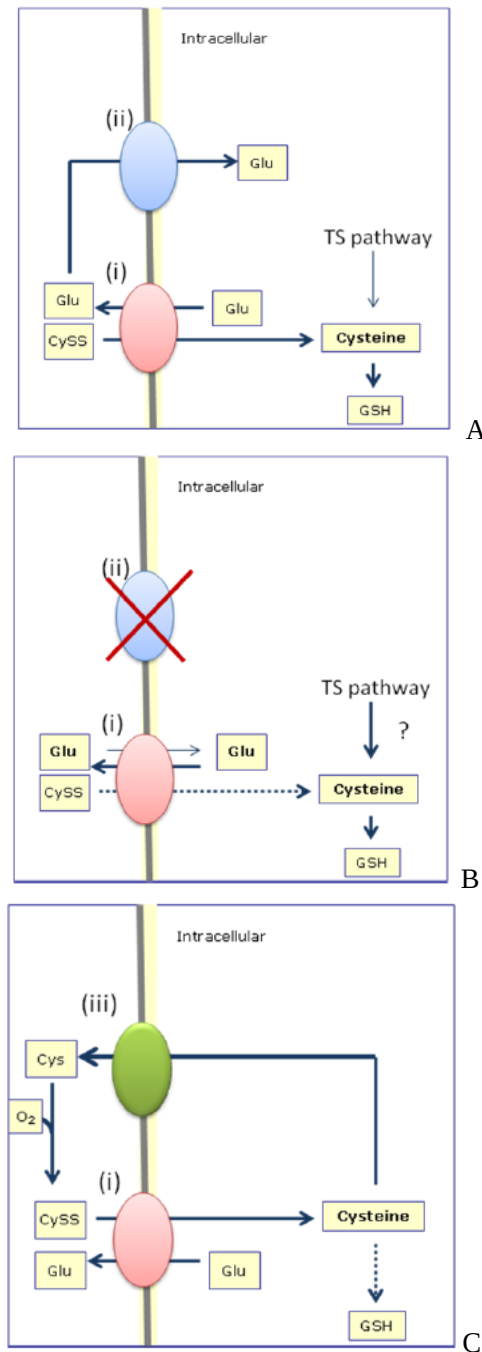


Figure 4. Transport of cysteine (Cys), cystine (CySS) and glutamate (Glu) in (A) normal astrocytes, (B) glioma cells and (C) in response to oxidative stress. CySS import via the  $x_c^-$  cystine glutamate exchanger (i) is matched by release of Glu that, in normal astrocytes, is taken up by the high affinity glutamate transporters (ii). In glioma cells, outflow of glutamate via the exchanger in the absence of high-affinity uptake increases the extracellular concentration of the amino acid that may compete with import of CySS and insufficient Cys for synthesis of GSH unless provided by an alternative source, such as the trans-sulfuration pathway (TS). Oxidative stress causes an increase in uptake of CySS and export of Cys (iii), potentially diverting Cys away from GSH synthesis.

Other disruptions in glutamate metabolism are evident in gliomas, compared to normal astrocytes. Gliomas do not express glutamine synthase, an enzyme that converts cytosolic glutamate to glutamine, which is then exported back to neurones to replenish the neurotransmitter pool of glutamate (the glutamate-glutamine cycle). Whilst not directly related to redox buffering systems, the absence of this enzyme highlights the far-reaching consequences that disruption of high affinity uptake may have on glutamate metabolism. In the absence of glutamine, neurones may contribute to glutamate-mediated neuronal dysfunction, which would exacerbate the effects of increased extracellular glutamate due to extrusion via the  $x_c^-$  exchanger. Glioma cells display heightened uptake of methionine, compared to normal astrocytes, that correlates with the proliferative activity of the tumours [51]. Whilst acknowledging that methionine is required to support DNA synthesis in rapidly proliferating cells, the question arises as to whether glioma cells upregulate the transsulfuration pathway as a mechanism to support heightened demand for GSH in growth and development and for enhanced ability to withstand oxidative stress. In particular, if one considers that gliomas must maintain an elevated concentration of GSH in face of reduced import of cystine due to an increased extracellular concentration of glutamate, then there must be an alternative source of cysteine for GSH synthesis. Clearly, there is a need for further experimentation to determine the full significance of this pathway in glioma cell biology.

### 3.2. Oxidoreductases in Glioma Cells

To date, few investigations on thio-oxidoreductases have been conducted in brain, particularly in the context of glioma biology. What reports there are tend to concentrate on the link between neurodegeneration and dysregulation of apoptosis in neurodegenerative disorders. For instance, there is accumulating evidence that dysregulation of apoptosis through enhanced glutathionylation of p53 may be linked to the inhibition of apoptosis that is observed in Alzheimer's disease [52]. In cultured astrocytes, carbon monoxide triggers glutathionylation of adenine nucleotide translocase, again resulting in inhibition of apoptosis [53]. Nonetheless, continued investigation into redox buffering in neurodegenerative disease may provide important insights into how the system could be targeted in glioma therapy. So far, there is a lack of consensus in the literature in regard to the link between expression of oxidoreductases and tumour growth and progression in human gliomas.

Increased levels of Trx and Trx reductase in glioblastoma/astrocytoma cells signify aggressive tumor growth, resistance to chemotherapy and poor patient prognosis. Normal cells, including astrocytes, have low levels of Trx, but overexpression has been noted in several malignancies. Increased Trx expression is linked to prevention of apoptosis in diffuse astrocytomas [54] and oligodendroglial tumors [55], which increases viability and the chance of survival of the tumour, but signifies poor patient prognosis. Trx overexpression increases that of HIF-1, which provides cancer cells with a survival advantage over normal cells. This is particularly important in the case of gliomas, as growth and expansion of brain tumors cannot occur unless space is provided as a result of death of surrounding neurons. Expression of other redox proteins, like superoxide dismutase, also increases in line with the severity of the tumour [56]. The opposite may be true of ependymomas, wherein decreased expression of antioxidant enzymes is associated with more aggressive tumors [57].

In a recent study on Prx expression in malignant glioma cells, Prx I and Prx II expression decreased significantly according to increasing tumour grade. Higher expression of these Prxs was associated with younger patient age and with a significantly better survival rate. Prx I expression was associated with a lower rate of tumour cell proliferation [58]. On the other hand, expression of Prx was increased in pilocytic astrocytomas, with Prx VI expression having a positive association with patient survival [59]. Human glioma cells overexpress Prx II, and lowering Prx II expression decreased intracellular GSH and resulted in a significant decline in GSH reductase activity, which may have accounted for the observed shift towards a more oxidising environment in these cells. It was suggested that decreasing Prx II expression in glioma cells might reduce clonogenic cell survival following exposure to ionising radiation and  $\text{H}_2\text{O}_2$  [60].

There is a scarcity of reports of changes in glutaredoxin activity in glioma cells, compared to normal astrocytes. However, this is an area ripe for investigation, since glutathionylation is anti-apoptotic and one would expect that there would be a greater level of protein glutathionylation in gliomas.

## 4. THE THIOL REDOX SYSTEM AS A THERAPEUTIC TARGET

### 4.1. Targetting the Cystine-Glutamate Exchanger

Several investigators have examined the hypothesis that targeting the  $\text{x}_c^-$  exchanger may provide an effective treatment for glioma, on the basis that limitation of incoming cystine would deplete GSH and render the cell more vulnerable to oxidative stress, leading to enhanced effectiveness of chemotherapeutic strategies. There is no doubt that targeting the  $\text{x}_c^-$  exchanger provides a mechanism for limiting glutamate efflux and improves patient outcome by lessening glutamate-mediated neurodegeneration. However, evidence for depletion of GSH following  $\text{x}_c^-$  exchanger knock-out is less clear-cut. Absence of a functional  $\text{x}_c^-$  exchanger has a bigger impact on extracellular glutamate than on intracellular GSH, suggesting that  $\text{x}_c^-$  inhibitors may have greater potential for treatment of CNS conditions linked to excessive release of glutamate than for achieving glioma cell death [61]. Research indicates that the  $\text{x}_c^-$  exchanger is geared towards facilitating a cystine-cysteine redox cycle, rather than feeding cystine into GSH for export of the tripeptide [61]. In this study, overexpression of xCT in Burkitt's lymphoma cells did not translate into an increase in GSH. Rather, the extracellular cystine concentration was elevated, from which it was concluded that the  $\text{x}_c^-$  exchanger was driving a highly efficient cysteine/cystine redox cycle, consisting of cystine import, intracellular reduction of cystine to cysteine, cysteine excretion and re-oxidation to cystine in the extracellular environment (Figure 4c). Additional evidence has come from the experiments of Mandal et al. [62], who showed that forced expression of the  $\text{x}_c^-$  exchanger in  $\gamma\text{GCL}$ -knock-out cells *in vitro* led to increased uptake of cystine and augmented intracellular and extracellular cysteine concentrations. More recent support for this view has come from experiments in which a mathematical model was used to examine the dynamics of cysteine, GSH and their disulfides in the medium of cultured astrocytes.

The indications are that astrocytes export cysteine, not GSH to provide antioxidant protection of neurones [63]. Mice lacking the xCT subunit of the exchanger do not have lower

GSH than normal controls and dysfunction of  $x_c^-$  does not induce oxidative stress *in vivo* [64]. The implication is that regulation of GSH and cystine-cysteine operate much more independently of each other than was previously supposed. Upregulation of the exchanger may drive resistance to oxidative stress because of increased cycling between cystine-cysteine. This information raises the question of how GSH levels are maintained (and up-regulated) if cystine import is channelled to cysteine export. The transsulfuration pathway is an obvious candidate for channelling cysteine to GSH and the demonstration that the transsulfuration pathway is upregulated in astrocytes and glioma cells in response to oxidative stress and inhibition of the  $x_c^-$  exchanger suggests that astrocytes have the capacity to adapt in favour of maintaining cysteine supply for GSH synthesis [8]. As stated, inhibition of glutamate efflux from gliomas is important for limiting tumour development. Sontheimer and co-workers make a compelling argument for targeting excess glutamate in development of treatments for glioma patients [17, 46]. Accordingly, a number of  $x_c^-$  inhibitors have been assessed for their efficacy as pharmacological agents to target gliomas [65, 66]. The search was necessarily confined to non-substrate inhibitors, as substrate inhibitors, whilst limiting GSH synthesis, would promote release of glutamate and contribute to peritumoural neuronal injury [67]. (S)-4-carboxyphenylglycine and sulfasalazine have been evaluated for their anti-tumour potential [17, 80, 65, 66]. As expected, the rate of tumour growth in mice was slowed by treatment with sulfasalazine [68, 69]. However, although already approved by the FDA, a recent Phase 1 clinical trial of sulfasalazine in glioma patients had to be curtailed due to investigator concerns about the efficacy of the drug [70]. Moreover, sulfasalazine is less than ideal, as its half-life in circulation is only approximately 80 min and its ability to cross the blood brain barrier is unknown. Nonetheless, there is a demand for sulfasalazine analogues with improved capacity to penetrate the brain and with greater effectiveness *in vivo*.

Other approaches to modification of  $x_c^-$  exchanger activity include ablation of xCT expression in glioma cells with small interfering RNAs (siRNAs), which produces a considerable reduction in glutamate secretion and delays neurodegeneration [44].

Similarly, rats implanted with xCT-silenced gliomas show a delayed onset and progression of neurological deficits compared to controls. In cystine/glutamate exchanger knock-out mice, there is a doubling in the plasma concentration of cystine and a drop in GSH. The plasma cystine/cysteine redox potential value (Eh) shows an oxidative shift from -100 mV in normal mice to -89 mV in xCT<sup>-/-</sup> mice [71]. The plasma concentration of cysteine is unchanged, but the GSH concentration falls by 50%.

Other approaches to glioma treatment that rely on targeting GSH have been proposed. *In vivo* experiments with mice have shown that elevated levels of GSSG in mouse brain induces neuronal cell death in a 12-lipoxygenase pathway-dependent manner, which is believed to be a result of increased glutathionylation of the enzyme. Based on these observations, it has been proposed that a strategy aimed at arresting cellular GSSG clearance may be effective in killing tumour cells [72].

## 4.2. Targetting Thioreductases

The Trx system has been heralded as an ‘attractive target for cancer therapy’ [73], particularly in cells that do not contain Grx reductase, as treatment would not require parallel depletion of GSH. Proposals include directly targeting Trx, or indirectly by inhibition of the



hexose monophosphate shunt and loss of reducing power in the form of NADPH [73]. Cisplatin, for example, is a metallodrug that irreversibly inhibits Trx reductase via covalent modification of the selenocysteine at the active site, but has no effect on glutaredoxin [37].

Second-generation platinum complexes (compounds with a platinum (II) and (IV) core) are also viewed as potential drug candidates and have been the subject of scientific and clinical investigations. These complexes, for example terpyridineplatinum(II) complexes (TPCs), target the C-terminally located Sel-Cys redox pair of thioredoxin reductase, causing specific and irreversible inhibition of the enzyme and are considerably more potent than cisplatin, with IC<sub>50</sub> values in the low nanomolar range.

It has been reported that TPCs inhibit proliferation of glioblastoma cells *in vitro* and inhibit tumor growth in C6-glioma-containing rats without producing severe side effects [1, 56]. Other drugs that target Trx/Trx reductase systems include carmustine (BCNU), dehydroepiandrosterone (DHEA) and motexafin gadolinium (MG) [37, 74]. However, none of these has proven suitable for use in glioma patients.

## CONCLUSION

In summary, the various components of the thiol-based redox system are closely associated with the resistance of glioma cells to oxidative stress and remain an attractive target for glioma cell therapy. However, much remains to be understood in relation to the interaction between the components of this system and how the cell adapts to resist oxidative stress before truly meaningful therapies can be designed. Indeed, Mandel et al. [75] caution that cancer therapies that are aimed at depletion of intracellular GSH and consequent cell death may not be effective due to cross-talk between different redox buffering systems.

GSH deficiency in cancer cells was rescued by upregulation of the Trx / Trx reductase system that is used by the cell to counteract the accumulation of excessive quantities of intracellular cystine. The upshot of these observations is that a combined strategy, targeting both the Trx / Trx reductase and GSH systems, may be required for effective treatment of cancer cells [62]. Glioma therapy has the additional challenge of limiting the neurotoxicity that arises from glutamate efflux. Targeting the x<sub>c</sub><sup>-</sup> exchanger remains an attractive option, although additional information is needed before we can fully understand the relationship between cysteine, GSH and antioxidant enzymes in glioma biology.

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*Chapter 19*

## **LOCO-REGIONAL AND TARGETED THERAPY OF GLIOMA WITH CHLOROTOXIN**

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### **ABSTRACT**

Glioma is the most common, as well as most lethal form, of primary brain tumor. Primary treatment strategies for glioma, including cytoreductive surgery, external beam radiation, and systemic chemotherapy have had generally disappointing results. Limitations of these approaches include tumor invasion into functional brain tissue, lack of chemo-sensitivity, and limited systemic delivery due to the blood-brain barrier.

Recent attention has focused on loco-regional strategies for treatment, as well as new methods for delivering therapy. Identification of tumor-specific surface targets, biological agents, and more sophisticated means to deliver macromolecules to the brain is offering new promise in the treatment of these tumors.

This chapter will review the current state of the art of available loco-regional therapies for glioma, with a particular focus on convection enhanced delivery, targeted toxin, and other biological strategies.

### **INTRODUCTION**

In the U.S., an estimated 22,000 new cases of brain tumor were diagnosed in 2011 [1], almost half of which were malignant gliomas. Gliomas, including glioblastoma multiforme (GBM), anaplastic astrocytoma (AA), astrocytoma, and oligodendroglioma, are amongst the most lethal forms of all tumors. Approximately 13,000 brain cancer deaths related to glioma-diagnosed disease occur annually in the US. The median survival for glioblastoma multiforme

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is approximately 12 months. For AA, the survival is approximately 3 years, and for low-grade (WHO grade II) astrocytomas 5-7 years [2]. The incidence of primary brain tumors appear to be on the rise, although it is unclear if this is attributable to better reporting, environmental, or genetic factors [2]. Despite over four decades of intensive research and a variety of chemotherapy, radiotherapy, and surgical approaches, the prognosis for these tumors has not changed significantly.

## **Standard Treatment Strategies for Glioma**

The mainstay treatment approaches for glioma include surgical resection, radiation therapy, and chemotherapy. Although each of these individual topics will be discussed in much greater detail in other chapters, they are briefly reviewed here as an introduction to a more extensive review of non-traditional loco-regional therapies which will constitute the bulk of this chapter.

### **Surgery for Primary Brain Tumors**

Open surgical removal of brain tumors has been a mainstay of glioma management for decades [3-6]. Unfortunately, total excision of glioma is rarely possible. Glioma cells are highly invasive, and are commonly present four or more centimeters away from the primary tumor mass [7]. Most of these cells are interdigitated with normal functioning brain parenchyma, resection of which can result in unacceptable neurological deficits. Further, while gliomas rarely metastasize outside the CNS, they frequently disseminate widely in both hemispheres of the brain. Tools to aid surgeons in differentiating normal tissue from glioma cells at the periphery of tumors can improve the extent of microscopic removal. Such methods include the use of tumor fluorescence [8, 9], and infrared imaging [10]. Gross total excision of low-grade gliomas, especially pilocytic tumors is curative in upwards of 80% of cases [5]. In contrast, gross total removal of all other gliomas and most primitive neuroectodermal tumors (PNET) is not clearly correlated with a higher cure rate. Cytoreductive surgery which removes at least 85% of the enhancing volume of high-grade glioma has been correlated with improved length of survival and improved quality of life [6, 11], with this effect especially evident for >98% resection [11, 12]. These results indicate that methods to improve the extent to which all macroscopic and microscopic tumors can be removed should result in better outcomes. Surgical resection remains the most effective single therapy for gliomas, especially when tumors recur or progress [11].

### **Radiation Therapy**

Gliomas are not particularly radiosensitive tumors. Radiation doses in excess of 50 Gy are effective in retarding glioma progression, but do not generally produce a cure or long-term control of disease [13]. Increasing doses of radiation have improved efficacy on higher-grade tumors, although most dose escalation studies have failed to demonstrate improved survival. Unfortunately, normal brain parenchyma is also sensitive to radiation effects. The



incidence of brain injury escalates quite rapidly above 70 Gy, effectively limiting total radiation doses to the 60 Gy range. The use of radiosensitizers has improved the effects of radiation or long-term outcome [14]. Furthermore, radiation injury and radiation induced cell death (radiation necrosis) is often as damaging as the primary tumor. Despite these limitations, radiation therapy is used almost uniformly in the treatment of gliomas, and is likely to remain a mainstay of therapy in the foreseeable future.

## **Chemotherapy**

Systemic chemotherapy has proven to be disappointing in the treatment of gliomas. This is partially attributable to the poor distribution of drug in the brain due to the blood brain barrier (BBB). Many agents, including BCNU (carmustine), CCNU (lomustine), Procarbazine, and temozolamide have demonstrated response against high-grade gliomas [15, 16]. However, these agents tend to have fairly limited or partial responses in both an upfront and recurrent setting. The combination of temozolamide with radiation therapy in patients with newly diagnosed GBM, followed by a course of temozolamide has been found to increase the median survival by 2 months when compared to radiation alone, with approximately 35% of patients surviving beyond 18 months [17]. Bevacizumab, a targeted VEGF receptor antibody, has demonstrated efficacy especially in recurrent or progressive disease, extending survival by about 4-12 months in 35% of patients, with up to 63% demonstrating some response [18, 19].

Delivery of chemotherapy via time-released polymer wafers is currently the only FDA approved form of loco-regional chemotherapy [20]. Polifeprosan 20 with carmustine (Gliadel®-Guilford Pharmaceuticals) is a synthetic biodegradable polymer wafer containing a 3.8% concentration of BCNU [21]. The wafers are implanted directly along the walls of a resected tumor at the time of surgery. Typically 6-8 wafers are implanted. Several animal and human studies indicate the effect mimics a 4-6 week infusion of BCNU. A phase III trial comparing Gliadel® to placebo wafers demonstrated a slight improvement in median survival of about 8 week [20].

Significant interest has developed around the use of convection enhanced delivery (CED) of chemotherapies to brain tumors. CED refers to the process of applying uniform positive pressure to overcome the natural resistance of the surrounding tissues, and essentially “push” the drug into the parenchyma. Using very slow infusion rates (0.5-4ul/min) over long time periods (3-5 days) with a positive pressure one can deliver relatively uniform concentrations of drug up to 4 centimeters away from the tip of the infusion catheter, with well-tolerated side effects and a reasonable risk profile [21].

## **Loco-Regional Treatment Strategies**

### ***Cellular Immunotherapy***

Immunotherapeutic strategies for the treatment of cancer have gained tremendous interest in recent years. The brain is generally considered to be an immune-protected organ, with less capability to invoke profound immunogenic or rejection responses than other organ systems.

furthermore, gliomas appear to actively inhibit immunologic responses such as delayed hypersensitivity reactions, and serum from glioma samples actively inhibit T-cell function *in vitro* [22]. Adoptive immunotherapy utilizing cytolytic T-cells is a loco-regional strategy of growing clinical potential [23]. This form of “passive” immunotherapy relies utilizes tumor-specific T-cell injected directly to the tumor microenvironment causing a local immune response. Initial efforts utilized non-activated peripheral lymphocytes injected directly into a tumor cavity. Subsequent attempts utilized lymphocyte activated killer (LAK) cells isolated from peripheral blood activated by exposure to IL-2. However, LAK cells do not have the capacity to migrate and several clinical trials with this therapy did not demonstrate any clinical benefit [24, 25].

Major advances in immunology and molecular biology have rejuvenated interest in this field with the promise of antigen specific T-cells. Current efforts aim to activate a select pool of T-cells embodied with tumor-specific attributes, and then infuse these T-cells along with IL-2 and/or other immune modulators to enhance their immune impact. The most notable study reported recently was a Phase I trial by Plautz et al. [26]. In this study patients underwent leukopheresis of T-cells from lymph nodes after the patient was exposed to a dose of irradiated tumor cells and GM-CSF. The T-cells were expanded *ex vivo* and then re-infused intravenously, along with IL-2 and anti-CD3 antibodies. Despite the several obstacles to this approach, three of ten patients demonstrated a partial radiographic response that lasted 6-13 months.

An alternate loco-regional approach is being explored by Jensen et al. [27]. In this approach CD8<sup>+</sup> T-cells are removed from patients via leukopheresis, and genetically modified to express chimeric mutant IL-13 antigen receptors, consisting of an extracellular single chain antibody (scFvFc) fused to the intracellular signal domain of the T-cell antigen receptor complex zeta chain.

This chimeric receptor provides antigen-specific targeted immunotherapy to glioma, in which tumor binding by the IL-13 antibody receptor activates the T-cells. Animal models of this method have demonstrated dramatic ability to eradicate xenografts. A Pilot feasibility study in which patient receive genetically modified T-cells via direct infusion is currently underway. CED of these cells may be an important method to increase the volume of distribution, pending the results of the initial study.

## Stem Cells

Neural stem cells demonstrate tremendous trophism towards gliomas in animal models [28]. The exact mechanism of this trophism is not well understood, and is under heavy investigation. Nonetheless, this trophic property has raised the possibility that stem cells may be an excellent mechanism for delivering loco-regional therapy to gliomas[29].

In contrast to viral therapy which relies upon the host cells to produce transgenes, stem cells may survive for long time periods, and do not depend on integration into host cells to exert their influences. Model systems have been proposed such as utilizing the transgene approach to have stem cells carry a prodrug converted to an active chemotherapeutic agent in the presence of a specific enzyme. For example, the drug 5-FC may be converted to the active chemotherapy drug 5-FU in the presence of cytosine deaminase (CD). Thus, stem cells producing CD provide a vehicle for site -specific targeted chemotherapy to brain tumors.

Local delivery of IL-2 for immunotherapy trials and other agents is also under consideration [30]. Whether these strategies prove viable it remains to be seen. More than likely however cells will be delivered by direct infusion into a tumor or surrounding parenchyma.

## Targeted Therapy

In light of the limits of the treatments described above, tremendous attention has focused on the development of tumor-specific targeted therapies. The broad goal of this technology platform is to directly bind a cytotoxic agent to tumor cells while avoiding or preventing any therapy delivery to normal brain cells. While this concept is broadly applicable to many malignancies, it is particularly important and challenging for gliomas due to the fact that these tumors so heavily infiltrate into and interdigitate with normal brain tissue. Unlike other solid organs, the destruction of the normal brain parenchyma results in specific loss of neurological functions. Therefore secondary damage to non-cancerous cells is poorly tolerated. Furthermore, the blood brain barrier functionally excludes the majority of large molecules from entering the brain, making the ability to actually deliver the therapy key to that therapy's success.

Targeted cancer therapies principally depend upon receptor mediated selective binding to tumor cells. The success of this approach requires high specificity or selectivity of binding to tumor cells or other tumor related cells such as tumor-induced neovascular blood vessels.

The targeting agent should take advantage of either the over-expression of receptors by tumor cells, or preferential expression of receptors not found on normal brain tissues. The targeting agents for these receptors are either 1) antibody or antibody-like ligands, 2) proteins that bind the receptor, 3) proteins that bind the cell and are internalized, or 4) large molecules with receptor-specific binding properties.

## Targeted Cytotoxin Therapy

Targeted therapy of glioma with tumor-specific cytotoxins has generated great interest lately. The basic strategy employed is to identify a cellular toxin, most commonly one derived from bacteria, modify that toxin to maximize anti-tumoral activity, and deliver the toxin directly to the tumor with a tumor-specific ligand acting as the carrier molecule[31]. One example of these is a modified variant of *Pseudomonas* exotoxin (PE) joined to Interlukin13 (IL13).

The IL13 receptor is preferentially expressed in many gliomas[32]. When this fusion protein binds to glioma cells, it is internalized and results in cellular death. Phase I and phase II studies with this agent were encouraging [32], but a phase III trial has indicated minimal survival benefit, with much toxicity.

An alternate strategy employs diphtheria toxin (DT), which has been fused to a fragment of the transferrin-receptor antibody and delivered via direct brain infusion (convection enhanced delivery) [33].

This agent is currently undergoing phase III clinical trials, but the ubiquitous nature of transferrin in the brain is a potential limiting factor. Fusion of DT to human plasminogen activator as well as IL13 has also been accomplished [34].

## Targeted Radioimmunotherapy and Radiopeptide Therapy

Antibodies to cell surface antigens have been most commonly employed in clinical settings, and several of these compounds are FDA approved for clinical use. Examples include trastuzumab (Herceptin®), an antibody to the Her-2/Neu receptor found in many breast cancer cells, and ibritumomab tiuxetan (Zevalin®), a radiolabelled antibody to the CD20 antigen on lymphoma cells. Examples of this approach in glioma therapy include targeting the epidermal growth factor receptor (EGFR) [35], fibronectin[36], and the extracellular matrix molecule tenascin with <sup>131</sup>Iodine radiolabeled ligands [37]. However, imaging studies demonstrate that the antibodies were too large to penetrate beyond a few millimeters into the surrounding brain parenchyma essentially neutralizing any potential targeting advantage [38]. Furthermore, tenascin is widely expressed throughout the nervous system and epidermal growth factor receptor, while upregulated in some gliomas, is also expressed by normal brain cells, limiting the specificity of these approaches. Both of these compounds are unlikely to have widespread impact on the disease or treatment strategies in the foreseeable future.

## Chlorotoxin- A Tumor-Specific Peptide for Targeted Therapy

Chlorotoxin (CTX) is a small neurotoxin isolated from the venom of the Giant Yellow Israeli scorpion *Leiurus Quinquestriatus*. Chlorotoxin is a 36 amino acid (Molecular Weight: 3950 Dalton) peptide containing a single tyrosine residue which is available for radioiodination, 8 cysteine residues, and 4 disulfide bonds, yielding a tightly folded tertiary structure.

In nature CTX functions as a paralytic agent for small insects and other invertebrates. The scorpion bites an insect which induces paralysis so the scorpion can either eat the insect or potentially escape a dangerous situation. When crayfish are injected with CTX they become completely immobilized for a period of approximately 2 minutes and then resume normal locomotion.

This response is dose dependent and highly reproducible, thus providing as an important biological assay for CTX and any engineered modifications of its structure. Human response to this venom in nature is generally bland and does not result in paralysis or death. The lack of a paralytic response in humans underlies the species selective nature of CTX binding, and the absence of a large number of binding targets in normal human tissues.

CTX was first identified as having tumor-specific targeting during patch clamp recordings of cells derived from cultures of normal and malignant human astrocytes (astrocyte= glial cell; malignant astrocytes=glioma) [39-42].

These studies demonstrated that CTX blocked a voltage-activated chloride channel in malignant glioma but not in normal glia, and the degree of response correlated to the degree of malignancy of the primary cell type.

This finding was subsequently confirmed in an extensive assortment of tissue explants obtained from human gliomas [39, 43, 44] and other tumors. Subsequent studies demonstrated that CTX facilitates the down-regulation of cell surface levels of CLC-3 chloride channel via internalization, and therefore the molecule was initially described as a glioma-specific chloride channel ligand [39].

## Tissue that Bind CTX

A host of immunostaining studies have been performed to better elucidate which cell types bind to CTX and to what degree [39, 43, 44] (and internal data, Transmolecular, Inc) . A summary of findings is listed in table 1. Basically all tumors of neuroepithelial origin such as glioma, meduloblastoma, PNET, and oligodendroglioma avidly bind CTX, with increasing receptor density as tumor grade increase. Similarly robust binding was observed in the overwhelming majority of malignancies tested, including melanoma and lung cancer. Minimal or absent binding was observed in normal brain and other solid organs. The only normal organ in which appreciable binding was detected was proliferating human vascular endothelial cells such as those in human umbilical vein (HUMVEC) [43].

## Identification of the Cell Surface Binding Partner for CTX

While *in vitro* cell culture and immunostaining data support the view that CTX binds a tumor cell surface receptor, the exact receptor has proven much more difficult to identify. Initial studies demonstrated that a human glioma cell line contained 1,300 high affinity binding sites per cell [39].

Given that electrophysiological experiments showed that CTX inhibits chloride ion fluxes across the plasma membrane of glioma cells, it was initially suggested that the receptor was associated with the chloride channel [39-41, 45]. Subsequent analysis using a recombinant polyhistidine-tagged version of CTX indicated the binding site may be a matrix metalloproteinase, MMP-2 [46]. MMP-2 is considered a chloride channel associated receptor (CCAR), and is believed to indirectly regulate the chloride channels expressed by gliomas but not normal human cells [39-41, 45]. MMP-2 stimulates breakdown of basement membrane and facilitates cellular migration of glioma cells and invasion of normal brain parenchyma. Application of CTX inhibits cellular migration *in vitro*, consistent with the idea that it interacts with MMP-2. Subsequent studies have shown that CTX more likely binds a macromolecular complex associated to a lipid raft on cell surfaces containing MMP-2, membrane type -1 MMP, transmembrane inhibitor of metalloproteinase-2 (TIMP-2) metalloprotease, and the CLC-3 chloride channel. The net result of this binding is internalization and subsequent down-regulation of both MMP-2 and CLC-3 [46, 47]. In this fashion CTX decreases MMP-2 enzyme activity, which is important for glial cell migration and a variety of other functions.

Demonstrating that CTX specifically interacts with MMP-2 as initially postulated has proven difficult. Using a fluorescent conjugate of CTX (CTX:Cy5.5), Veiseh et al. demonstrated that MCF-7 breast cancer cells that normally express small amounts of MMP-2 on their surface, exhibited marked increase in CTX binding when transfected with a plasmid encoding for MMP-2, re-enforcing the concept that MMP-2 mediates binding of CTX, either directly or indirectly. However, pulldown assays of recombinant MMP-2 incubated with either CTX:Cy5.5 or albumin and then anti- MMP-2 antibodies did not confirm MMP-2 specific binding by CTX [44]. Taken together these data support the view that while CTX cell binding modulates MMP-2 activity, MMP-2 is unlikely to be the direct binding partner.

Annexin A2 has been identified as a likely molecular target and binding partner for CTX [48]. Utilizing the pancreatic cell line PANC-1 and human umbilical vein endothelial cell

(HUVEC) cell lines, it was demonstrated that the presence of Annexin A2 on the cell surface was required for CTX mediated inhibition of angiogenesis mediated by vascular endothelial growth factor (VEGF) and several other growth factors. In the presence of normal or elevated levels of Annexin A2, CTX dramatically reduced tumor-induced neovascularity in the chick chorioallantoic membrane (CAM) assay. Reduction of Annexin A2 on cell surfaces using a single strand RNA (siRNA) knockdown technique results in a marked reduction of this effect. Further, CTX inhibition of PDGF-CC-induced U373-MG glioma and HUVEC cell migration was blocked with reduction of Annexin A2 levels. Irreversible cross-linking of cellular proteins followed by affinity pulldowns and mass spectrometry identified Annexin A2 as a binding partner to CTX. It is unknown whether other binding partners exist.

## Effects of CTX on Human Cells

### Inhibition of Migration

Using *in vitro* invasion assays, CTX has been shown to inhibit invasion in a dose-dependent fashion [46]. Several studies suggest that through its inhibitory affect on ion efflux, CTX prevents cell shrinkage thereby diminishing the ability of glioma cells to migrate through tight extracellular spaces in brain tissue. Further, by decreasing MMP-2 activity, CTX prevents proteolytic degradation of extracellular matrix (ECM) thus preventing the release of glioma cells from the constraints of cellular interactions with ECM [47].

### Anti-Angiogenesis

In a transwell migration assay, CTX profoundly inhibits agents that normally promote angiogenesis. CTX blocked invasion of proliferating HUVEC cells exposed to VEGF and bFGF by approximately 50%. Further, CTX blocked MMP-2 activation in response to bFGF. CTX inhibits formation of neovasculature by VEGF in a dose-dependent fashion. This effect has been observed for a wide range of angiogenesis promoting factors, including bFGF, EGF, IL-6, PDGF, TNF $\alpha$  and HGF. When CTX was co-administered with bevacizumab—an anti-VEGF antibody used in clinical cancer treatment—neovascularity was reduced by more than the effect observed by a 10-fold increase in bevacizumab dose alone. This observation suggests that CTX may inhibit more angiogenesis pathways than the VEGF-mediated one, and therefore provide an important synergistic effect in clinical treatment. As such, CTX may be an important anti-angiogenesis therapy, especially when used in conjunction with other treatments.

## Clinical Studies

Chlorotoxin was synthetically produced using solid phase synthesis methods by Transmolecular Industries, a biotechnology company that has subsequently been acquired by Morphotek, Inc. The product (TM-601) can be produced in large quantities for reasonable

costs, and the amino acid structure can be modified as needed. Mass spectrometry and HPLC assays indicate that the compound is identical to the naturally occurring compound in shape and structure. GMP grade TM-601 has received FDA approval for clinical testing. The crayfish biological assay shows identical response with the synthetic compound as with the naturally appearing one. Therefore all clinical testing in the past and anticipated future will likely be performed with TM-601 rather than naturally purified CTX. For purposes of this chapter, the terms CTX and TM-601 are used interchangeably.

Despite its selective binding properties and numerous cellular interactions noted above, CTX does not kill glioma cells *in vitro* when given as a single agent. Therefore TM-601 was initially developed clinically as a targeting agent for delivering a therapeutic payload. For initial studies the therapeutic payload utilized was <sup>131</sup>Iodine because it had been demonstrated to have efficacy against gliomas and had been utilized in previous radiolabeled antibody trials. Further, the use of a radioactive isotope as a tracer facilitates dosimetry and biodistribution studies.

A phase I study to evaluate the safety, tolerability, bio-distribution and dosimetry of intracavitary <sup>131</sup>I-TM-601 in adult patients with recurrent high-grade glioma has been completed [49]. In this trial, the dose of <sup>131</sup>I was kept fixed at 10 mCi, while the dose of CTX was escalated from 0.25 mg, to 0.50 mg, and 1.0 mg in three dosing panels containing 6 patients each.

The compound, <sup>131</sup>I-TM-601 was injected into the resection cavity of 18 patients with recurrent high-grade gliomas via an Ommaya reservoir two weeks after surgery. Radiation doses to normal organs were clinically insignificant. In contrast, mean radiation dose to within 2 cm of the cavity wall was 81 cGy/mCi (median 49) and ranged from 12 to 275 cGy/mCi. These values are substantially higher than doses to the whole body (mean 0.4 cGy/mCi) and to any other organs. Furthermore, the biological half-life of <sup>131</sup>I-TM-601 in the tumor cavity margin was longer than in any other organ including the normal brain, indicating long-term retention of the drug in and around the injection site. The median biological half-life in cavity margins was 70 (range 32-193) hours, 80 (range 25-86) hours, and 55 (range 41-62) hours for patients receiving 0.25, 0.50, and 1.0 mg of peptide, respectively.

Total body and brain SPECT imaging showed the <sup>131</sup>I-TM-601 localized to and remained primarily concentrated in and around the patient's surgical cavity for all five days imaged. Three patients had serious adverse events possibly or probably related to study medication reported within 22 days of administration. These included the following events: fever, chills, upper extremity paresthesias accompanied by mild cerebral edema on CT; progression in left-sided weakness; and infection of the tumor resection cavity and osteomyelitis. Additional serious adverse events reported by investigators to have possible attribution to the study agent beyond the initial 22-day observation period included one patient with generalized seizure and one case of headache, face droopiness, dysathria and instability.

Over the course of the 180-day observation period, there were a total of seven deaths. Two patients survived over 30 months. Median survival was 27.0 weeks from surgery for all three dosing groups. Histochemistry of the tumor tissue from all patients tested thus far stained intensely positive for TM-601.

An analysis of the imaging data obtained for a subset of the patients enrolled in this trial was also performed [50]. The purpose of this analysis was to determine if radiolabelled CTX might be a useful tool for imaging the extent of glioma invasion since it is tumor-specific and small enough to diffuse away from the primary tumor site. This study demonstrated that <sup>131</sup>I-

TM-601 SPECT scan estimates of tumor volume were midway between estimates on T2 weighted and T1 weighted contrast enhanced MRI scan, a result consistent with known glioma invasion patterns. This study provides promise that CTX may be useful for tumor imaging as well as tumor therapy.

A phase II trial of this agent was also completed for patients with recurrent glioma. The goals of this trial were to determine the safety of multiple dose administrations, at 40 mCi/0.8 mg per dose, and to test potential efficacy of this agent. The first portion of the study consisted of a dose escalation schema in which individual patients each received 3 doses of  $^{131}\text{I}$ -TM-601, spaced 1 week apart. The specific activity of the radiolabelled peptide was fixed with the dose varying between 20 mCi/0.4mg and 40mCi/0.8 mg. There were no major toxicities observed. In the later phase, patients were randomized to receive either 3 or 6 weekly intracavitary injections of 40 mCi/0.8mg  $^{131}\text{I}$ -TM-601. This trial was conducted at 17 centers in the US and completed accrual in 2007. Although the final results have not been published, preliminary data [A. O'Neil, personal communication] suggest improved survival for patients receiving 6 doses compared to those receiving 3 doses. As with the initial phase of the trial, no deaths or major toxicities directly attributable to the therapy have been reported.

Simultaneously an intravenous imaging trial was performed in patients with progressive cancer of any origin (CNS or systemic). The primary endpoint was the ability to visualize specific uptake of the agent on total body imaging. If so, this would suggest that  $^{131}\text{I}$ -TM-601 or variants could be used as both a cancer imaging agent and as a systemic targeted therapy.

Studies published in 2011 have supported further efforts to use CTX as a glioma targeting agent. The toxin has been used to target doxorubicin-loaded liposomes as well as gene-bearing nanoparticles to glioma in *in vitro* as well as animal studies[51-53]. Clinical application of these strategies could be promising.

## Other Uses of Chlorotoxin

### *In Vivo Tumor Imaging*

Chlortoxin demonstrates many of the properties ideally suited for use as a cancer imaging agent. Due to its small size it can cross the blood brain barrier, and penetrate deep into solid organs. It appears to demonstrate almost no binding to non-malignant human tissues, and is therefore highly tumor-selective. Further, it appears to be non-immunogenic and is rapidly cleared by the kidneys [50].

CTX has been bioconjugated to the fluorescent probe CY5.5, a flourophore that emits in the near infrared range [44, 54]. This compound, CTX:CY5.5, was demonstrated to bind multiple tumor cell lines *in vitro*, with the extent of binding significantly inhibited by blocking the MMP-2 . Subsequent *in vivo* biophotonic imaging in a subcutaneous rat 9L gliosarcoma model, and spontaneous medulloblastoma, prostate cancer, intestinal cancer, and sarcoma animal models documented accurate detection of the tumors. *Ex vivo* imaging of the intestines from an intestinal adenomatous familial polyposis mouse model also documented discrete and focal detection of tumor nodules and the ability to discriminate normal tissue from tumor [44]. These studies indicate that fluorescently tagged CTX may be useful as a “tumor paint” given prior to surgery and identified during surgery by infrared fluoresce imaging.



This method could aid in achieving more complete surgical resections and tumor negative margins, both of which have been correlated with improved survival for many cancers. One of the technical limitations of this work is the need to demonstrate the safety of this specific fluorophore for clinical use. Nonetheless, based on encouraging data that has been reported from the use of 5' ALA induced cellular fluorescence to detect tumor cells during glioma resection [9, 55], this technology appears promising. This year, the use of chlorotoxin-conjugated nanoprobe for direct visualization of tumors has been reported [56].

## Chlorotoxin Nanoparticles for MRI Based Tumor Detection

Chlorotoxin has also been bound to iron oxide magnetic nanoparticles for *in vivo* tumor detection via MR [54, 57]. In animal studies, nanoparticles coated with CTX targeted brain tumor cells with high affinity and were internalized despite an intact blood brain barrier [54, 58].

Subsequent studies in which CTX nanoparticles were also linked to green fluorescent protein (GFP) DNA and given to animals harboring tumors identified the presence of fluorescence in tumor cells removed several weeks later [59].

These studies demonstrated the feasibility of the concept that CTX coated nanoparticles would be internalized by tumor cells and lead to activation of a gene transcript, thereby suggesting that CTX nanoparticles may provide a novel delivery system for gene therapy. The tumor-specific targeting observed with CTX is robust and provides a novel platform for the development of a wide range of targeted based imaging and therapy for tumors.

## CONCLUSION

Loco-regional treatment strategies provide the greatest hope for effective control or eradication of glioma. The unique aspects of glioma management, including the invasive nature of glial tumors, and the functional organization of brain tissue are driving an increasingly sophisticated effort for developing, testing, and delivering loco-regional therapies. Novel delivery mechanisms such as time-released polymer implants and CED pumps allow neurosurgeons to deliver therapy with greater precision and reduced systemic side effects. Novel treatment strategies such as targeted toxins, antigen-specific T-cells, and more selective methods of radiotherapy are providing new promise in a field that has been quite frustrated in its efforts.

CTX represents a novel and exciting platform for cancer therapy and imaging. CTX meets many of the criteria of an ideal targeting agent. It is small and highly compact in structure. This means it could cross the blood brain barrier and is far more likely to diffuse deep into solid tumors than antibodies, mini-bodies, or other such larger molecule targeting agents. It is synthetically manufactured and can be easily modified by amino acid substitution and pre- or post-folding modification. It contains a single tyrosine residue for iodination and can be modified by covalent linkage for the conjugation of other imaging or therapy payloads. Because it is a naturally occurring biological peptide that was evolutionarily derived from an invertebrate with no human biology “design”, it fortuitously demonstrates binding to human

malignancies, but is otherwise alien to human tissues. Therefore it appears to have no innate toxicity in humans despite repeated administrations. It is likely that CTX has anti-tumoral and anti-angiogenic activity even in the absence of a therapeutic payload, and as such may be used as a stand-alone therapy or in conjunction with other cancer therapies as a treatment modulator. The ability of CTX to selectively bind to tumor cells may be exploited by linkage to a fluorescent or radioactive ligand for intra-operative tumor identification. Similarly, the agent may find use as a means to deliver target gene therapy or cellular toxin to tumor cells.

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*Chapter 20*

## **CYTOKINE-DIRECTED RECOMBINANT CYTOTOXINS FOR GLIOMA TREATMENT**

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### **ABSTRACT**

Single-chain recombinant proteins composed of tumor-targeting ligands and engineered isoforms of bacterial exotoxins, termed anti-cancer cytotoxins, exhibit highly potent killing of cancer cells *in vitro* and prominent anti-tumor effects *in vivo*. Several cytokine-directed cytotoxins have been employed in the experimental treatment of glioblastoma multiforme (GBM), a primary brain tumor of dismal prognosis.

These cytotoxins moved relatively rapidly from laboratory research to clinical investigations, including Phase III efficacy trials in patients with recurrent GBM. Interleukin 13 (IL-13)-based cytotoxin in particular demonstrated encouraging clinical effects throughout all stages of clinical evaluation. The IL-13 based cytotoxin extended the overall survival in a cohort of patients treated in centers more experienced with loco-regional delivery of the cytotoxin.

Thus, these trials have gathered important information for translational research and for further clinical development of these anti-cancer compounds. One aspect of paramount importance is the improvement of the method of drug delivery using convection-enhanced delivery (CED).

The other issue is a need for real-time monitoring of CED that is used for loco-regional administration of cytotoxins. Other crucial factors are related to target specificity and patients' inclusion criteria for treatment. A wealth of knowledge gained in both laboratory science and clinical application of cytokine-targeted recombinant cytotoxins promises to optimize this modality of GBM treatment in a near future.

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## INTRODUCTION

### Standard Treatment of GBM

Glioblastoma multiforme (GBM) is the most prevalent primary malignant brain tumor and it is classified as grade IV astrocytoma. The disease is of dismal prognosis, because two-year survival rate is less than 20% [1]. The mainstay of treatment of GBM relies on combination of surgery [2], radiation therapy [3], and chemotherapy [4]. Standard combinatorial treatment is surgery with radiation plus temozomolide, which offers ~14 months median survival [4]. It is safe to say that all patients with GBM are refractory to available treatments. The hope for significantly better outcomes is linked to novel modalities of treatment.

### Cytotoxic Fusion Proteins in Experimental GBM Treatment

Among a relatively large number of experimental anti-GBM therapies under investigation, a novel approach has emerged in a form of targeted recombinant chimera cytotoxic fusion proteins [5-8].

Cytotoxins are composed of targeting ligands and genetically-engineered derivatives of extremely potent bacterial exotoxins, mainly from the AB type group of toxins [9].

These recombinant single-chain fusion proteins are highly soluble and extremely toxic only to cells expressing targeted receptor [10]. The cytotoxins exert their killing effect independent of pro-, or anti-apoptotic cancer cell characteristics [10-13].

One of the most desirable pharmaceutical features is that they do not produce resistance readily, because bacterial exotoxins shut down *de novo* protein synthesis, a mechanism indispensable to all living cell. They do not have to rely on genetic changes like receptor mutations and kill cancer cells over-expressing targeted receptor independently of pathophysiological role of the receptor in cancerogenesis [10, 14]. Cytotoxins need to be administered using loco-regional delivery system termed convection enhanced delivery (CED) [8, 15]. This system of drug delivery is under continuous development and optimization [16, 17]. The CED offers bypassing of blood-brain barrier or blood-tumor barrier, the barriers that make treatment of the CNS disorders much more complex and difficult [8, 18]. Even though the potency of cytotoxins declines under hypoxic conditions frequently seen in GBM, they still retain cell killing ability at nanomolar concentrations [19]. Thus, there are multiple potential advantages in using cytotoxins as anti-GBM agents.

As mentioned above, cytotoxins are composed of two fundamental elements: a targeting ligand and a modified bacterial toxin. For targeting ligands, naturally occurring cytokines can be used like interleukin 13 (IL-13) or IL-4 [14, 20-22]. Targeting ligands are responsible for cell recognition function in cytotoxins. The cell killing role is assumed by bacterial exotoxins that are extremely toxic to eukaryotic cells. The most frequently employed for this purpose toxins are represented by *Pseudomonas* exotoxin A (PE) and *Diphtheria* toxin (DT). In the past, innovative genetic engineering generated a variety of PE derivatives suitable for fusion with targeting ligands [10, 23-25]. They are either deletion or substitution mutants, or both. This includes PE1E, PE4E, PE40, PE38, PE38QQR and DT390 or DT388, to mention some



of them. For example, PE40 is a derivative of PE, which lacks all of domain 1a of PE while PE38QQR is a form of PE that is missing all of domain 1a and additional 16 amino acids from domain 1b, and lysine residues were substituted for precise conjugation purpose at positions 590, 606, and 613 of domain 3 [21, 24, 26, 27]. The benefit of mutational changes in PE or DT is related to making toxins inactive on eukaryotic cells on their own, but deadly when attached to appropriate ligand.

For example, low concentrations of IL-13 based, PE or DT derivative-containing cytotoxin kill cancer cells, but neither IL-13 nor engineered toxins, individually or in mixture, kill those cells [22, 25, 28].

## **CYTOTOXINS TARGETED TO CYTOKINE RECEPTORS IN GBM**

For successful clinical implementation of targeted therapy using recombinant cytotoxins several critical points must be addressed. One of the most important ones is the specificity of the tumor target. The target should be expressed at sufficiently high levels and relatively homogenously in order to transmit effective GBM cell killing by cytotoxins, but the target must be practically absent in normal brain parenchyma. Furthermore, wild type cytokines chosen as targeted natural ligands in cytotoxins interact with various subtypes of their receptors causing undesirable activation of normal physiological systems [29-31].

We have discovered several unique targets over-expressed in GBM, but not in normal brain [e.g., 32, 33]. Most of these targets can be utilized for various clinical applications including imaging and treatment of GBM. Among cytokine receptors, we have found that IL-13R $\alpha$ 2, one of the two types of IL-13 receptor is frequently over-expressed in GBM cells [31]. Of interest, IL-13R $\alpha$ 2 is a gene target for both epidermal growth factor receptor (EGFR) and an EGFRvIII isoform which is constitutively activated [32, 34]. IL-13R $\alpha$ 2 is different from normal tissue physiological counterpart, because it is a monomer to which only IL-13 binds and not homologous IL-4. The physiological receptor for IL-13 on the other hand is a heterodimeric receptor complex to which also IL-4 binds to and signals through as well [35, 36]. IL-13R $\alpha$ 2 is over-expressed in a vast majority of patients with GBM, which has been documented in multiple studies and is indicative of IL-13R $\alpha$ 2 being a very attractive molecular target in GBM [14, 21, 22, 32, 37-43]. The laboratory work on IL-13R $\alpha$ 2 has been translated into multiple clinical applications including targeted recombinant cytotoxins, re-targeted cytolytic T cells, viruses and a number of vaccination approaches [22, 43-48]. Thus, the discovery of IL-13R $\alpha$ 2 in gliomas spearheaded a variety of interesting translational developments with new efforts underway.

### **Delivery of Cytokine-Based Cytotoxins Using Convection-Enhanced Delivery (CED)**

Recombinant cytotoxins are proteins usually within the 50 to 200 kDa range of molecular weight [20, 22, 24]. Unless attempted to be engineered with an idea of crossing blood-tumor or blood-brain barrier (BTB and BBB, respectively), these compounds are not expected to

home easily to intracranial tumors when given systemically. Although long-term infusion of cytotoxins at non-toxic doses could in some cases result in considerable homing of proteinaceous cytotoxins to CNS tumors, CED is a clever way of bypassing BBB or BTB [8, 15, 18]. CED is now a relatively well-established way of delivering a variety of compounds locally through interstitial infusion into brain tumors [8, 18, 49-51]. There are number of factors weighing into successful delivery of drugs into malignant and non-malignant brain tissues. One main drawback in almost all initial clinical trials with the cytotoxins using CED was that the catheters employed had been designed for the purpose of delivering infusates into a liquid space like blood vessels[8]. They have one outlet (port) at the tip, which creates inherent danger of reflux along the walls of catheters. This unfortunately has been observed clinically and experimentally [16, 52]. In other words, there is much less drug delivered to the intended location like tumors and its distribution during the CED becomes uncontrollable. The other drawback with the use of CED has been only sporadic performance of real-time imaging, which is a continuous monitoring of drug's distribution [50].

Most information on treatment with cytokine-guided cytotoxins using CED is coming from studies on huIL13-PE38QQR, a cytotoxin composed of wild type human IL-13 and PE38QQR derivative of PE. This cytotoxin was conceived and produced in our laboratory for follow-up comparative studies with IL-14 based cytotoxins [20, 22]. This compound has traveled from the bench to the clinic culminating in an efficacy trial named Phase III Randomized Evaluation of Convection Enhanced Delivery of huIL13-PE38QQR with Survival Endpoint (PRECISE) [53]. huIL13-PE38QQR was evaluated in direct comparison to recurrent GBM standard care carmustine wafers, Gliadel Wafers. The sole reason for entering into Phase III was that interim Phase I clinical trials demonstrated encouraging clinical responses [54, 55]. Up to 60 centers around the world took part in the trial.

## **Cytokine-Targeted Cytotoxins in the Treatment of Patients with GBM**

The first cytotoxin delivered with CED was DT.CRM107-Tf, which is targeted by transferrin conjugated to a modified DT [15, 56]. The results of the Phase I and Phase II clinical trials were encouraging. However, Phase III efficacy trials DT.CRM107-Tf was not finished [57]. IL-4-PE38KDEL is a cytokine-directed cytotoxin, which was evaluated clinically [58]. The difference in IL-4 receptors expression between normal and malignant CNS is however not large [21, 59]. Initially, human IL-4 based cytotoxin was examined in mice with which human cytokine does not exhibit any cross-reactivity [20, 60]. Perhaps not surprisingly, the results of early phase clinical trials using this cytotoxin have not been encouraging and the chance for developing efficacy trial is rather slim [58].

The most significant development in the area of cytokine-directed cytotoxins has been related to targeting GBM with recombinant cytotoxins based on IL-13. The first generation of an IL13-based cytotoxin is represented by huIL13-PE38QQR, which however was not designed for specific targeting of IL-13R $\alpha$ 2 [20]. Newer, rationally designed cytotoxins based on mutated IL-13 are certainly better equipped for molecular targeting of IL-13R $\alpha$ 2, a GBM-associated receptor [31, 61]. huIL13-PE38QQR went through early phase up to efficacy, prospective Phase III clinical trials. They were clearly high expectations for the drug based on pre-clinical and initial clinical testing [54, 55]. huIL13-PE38QQR moved thus quickly to

Phase III, because the results of Phase I multi-center trials documented quite uncommon effects of the cytotoxin [62]. Phase III trial was designed to directly compare huIL13-PE38QQR efficacy in patients with recurrent GBM with standard of care, which is Gliadel Wafers [53]. The trial did not include imaging of drug delivery nor was status of the IL-13R $\alpha$ 2 expression taken into account as eligibility criterion. The catheters used were Medtronic® PS Medical (Goleta CA, Catalog #43209) and Vygon US LLC (Valley Forge, PA) with one opening at the tip of the catheter [8]. In summarizing the results of the trial, the overall survival was not found to be statistically significantly better from standard of care, in contrast to the results obtained in Phase I trials [53].

Several recent reports described retrospective analyses of the data from the PRECISE trial. More than ten percent of the treated patients did not fulfill the study inclusion criteria of gross total resection [63]. Only 53% of the treated patients had catheters placed fully according to the guidelines while 51% had adequate catheter positioning. However, catheter positioning did not correlate with progression free survival (PFS) or overall survival in the PRECISE trial while this was found to be the case in prior Phase I studies [63]. Furthermore, it has become apparent that the drug delivery was insufficient into areas of highest risk to recur [64]. Thus, even properly positioned catheters did not deliver adequate amount/volume of drug. The optimal catheter positioning was only achieved after catheters placement in at least four patients by one center. Importantly, the overall survival was statistically significantly higher in a group of patients treated by investigators considered experienced [64]. This further underlines high potential of IL-13 based cytotoxins for being efficacious in recurrent GBM when used properly. huIL13-PE38QQR appears to be one of only few agents showing documented clinical efficacy in patients with recurrent GBM during the last couple of decades [65]. This clinical efficacy of hIL13-PE38QQR is likely related to specific over-expression of IL-13R $\alpha$ 2 in a majority of patients with GBM and potency of the IL13-based cytotoxin.

## Novel Cytotoxins Targeted by Cytokines

Enzymatic domains of modified bacterial exotoxins are not the only effectors that can be utilized for targeted delivery of cytotoxic drugs to GBM cells. We have recently constructed a universal single-chain protein module that recognizes not only cancer cells, but also pre-determined intracellular compartments, such as nuclei [66]. The construct is composed of a (i) mutated IL-13, IL-13.E13K, binding more specifically to IL-13R $\alpha$ 2, (ii) genetic engineering-adapted “get cleave and exit endocytic compartment” domain D2 of PE, and (iii) a nuclear localization signal (NLS) from the SV40 T, the IL-13.E13K-D2-NLS vector [66]. We documented in a direct way the binding of the construct to plasma membranes of GBM cells, its subsequent presence in the endocytic compartment and perinuclear region and, most importantly, time-dependent readily detected accumulation in the nuclei. In contrast, none of the several control proteins could enter the nuclei. This work was a continuation of our original findings of an efficient intracytosolic delivery of peptides/proteins using modified PE [23].

In terms of delivering cytotoxicity to GBM cells, IL-13.E13K-D2-NLS can be used for conjugation to either chemotherapeutics or photosensitizers, or radiolabels. Standard

chemotherapy agents like doxorubicin, which intercalates in DNA and causes topoisomerase II-mediated strand breaks can be employed in this approach. Doxorubicin was successfully delivered to brain tumors by liposomes [67], but we envision higher efficacy and safety with delivery of the chemotherapeutic directly to the nuclei of GBM cells. Topotecan, another chemotherapeutic which inhibits DNA topoisomerase I would likely be more effective when delivered directly to the target. Furthermore, photosensitizers necessary for photodynamic therapy represent attractive agents for conjugation to IL-13.E13K-D2-NLS-type of delivery vectors. Photosensitizers generate singlet oxygen species active within the 40-nm radius [68, 69]. Moreover, radiolabels of low energy electrons like Auger electrons are potentially very useful for targeted tumor cell killing with the expected low bystander effects [70]. Thus, a multiple-specificity vector for nuclear delivery in IL-13R $\alpha$ 2-expressing cancers is now available. This vector can become a cytokine-targeted cytotoxin when labeled and/or conjugated to various labels/chemotherapeutics/photosensitizers.

## Monitoring Response to Therapy

In cancer therapy, determining responses to treatment are not well developed, but they are highly desirable. This is a particular challenge in the treatment of gliomas. For example, using traditional imaging techniques treatment with Avastin “improves” radiologic picture of tumors while for fact they become more invasive in the course of therapy [71, 72]. Clinical trials using cytokine-targeted cytotoxins employed standard imaging methods without possibility to define responders from non-responders early or determine the eligibility on the basis of target expression in tumors. To this end, we isolated peptides binding IL-13R $\alpha$ 2 from disulphide-constrained heptapeptide phage-display library [73]. One of the peptides, Pep-1, binds IL-13R $\alpha$ 2 interestingly at a site that is not competed for by a natural ligand, IL-13. The peptide internalizes upon binding to the receptor. Even more importantly, the peptide homes to and it is retained by subcutaneous and intracranial GBM tumors when given systemically [73]. Only few targeting agents have been documented to cross the BBB [74, 75]. Therefore, the isolated peptides may be useful for therapeutic targeting and/or for imaging of GBM. Considering that peptides do not compete for IL-13 binding site on IL-13R $\alpha$ 2, they can be utilized concomitantly with IL-13 based targeted imaging/therapy [66, 76, 77].

This provides a unique management scenario of concomitant molecular targeting using cytokine-directed cytotoxins and non-interfering with the therapy specific visualization of the status of the receptor expression using peptides. Thus, confirmation of eligibility to treatment and measurement of the response to therapy can be achieved. Another peptides’ application could be “tumor illumination” benefiting image-guided surgical resection [78].

## CONCLUSION

Clinical experience with recombinant cytotoxins speaks favorably to their further successful development for the treatment of GBM. Cytokine-directed cytotoxins have gone through the first full circle of bench to bedside translational research. These cytotoxins are protein-based hence their preferred way of administration is a loco-regional one. On one

hand, this makes it a minimally invasive approach to treatment. On the other hand, loco-regional delivery offers multiple advantages when performed properly. Initial clinical implementation of the cytotoxins was thus a complex endeavor, because of both the nature of the drug and the way of its delivery were technologically novel, and the two were combined into one clinical application. However, many lessons have been learned that are critically important for the successful design of this type of targeted therapy. The cytotoxins have already shown clinical efficacy, which is extremely difficult to obtain in patients with GBM. Therefore, the efforts should be undertaken for the optimization of the treatment with cytotoxins with reasonable expectations of improving patients' survival.

The optimization of treatment with the cytotoxins will need to encompass specific issues with loco-regional delivery of these drugs, CED, as well as the improvement in these drugs properties. Importantly, the science of CED has been progressing and together with now quite extensive clinical experience, the most important factors are quite clearly defined. Of outmost importance is monitoring of drugs delivery to tumor bed. This can be accomplished with standard imaging techniques like MRI [79, 80]. Drugs' delivery monitoring needs to be performed throughout the entire procedure of CED. There are catheters available now, especially reflux-preventive catheters that are designed for the pressurized infusion into solid tissue like malignant lesions of the CNS [16]. Most likely, the time of infusion need not to be several days but probably less than a day and repeated at least once weeks later. The possibility of installing platforms for repeated infusions will enable to deliver cytotoxins more than once, which may be decisive in determining how efficacious can the treatment be [79]. And finally, targeting tumor cells, especially the invading ones, and/or tumor environment, but not normal brain parenchyma can be accomplished. Cytokine receptor IL-13R $\alpha$ 2 has proven to be an excellent target for a variety of therapeutic approaches, such as active and passive immunotherapy. It is expressed in a majority of patients with GBM at a relatively homogenous, pharmaceutically adequate levels and it can be found in a majority of invading cells. Targeting solely this cytokine receptor however requires elimination of patients with low level or no receptor present meaning some 30% of them will not be eligible for treatment [32]. Combining cytotoxins targeting IL-13R $\alpha$ 2 with others targeting suitable targets will make such therapy applicable to all patients with GBM [81].

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*Chapter 21*

## **NANOPARTICLE BASED DELIVERY OF THERAPEUTICS TO GLIOMA**

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### **ABSTRACT**

Nanoparticle (NP) drug delivery for glioma is recently gaining attention due to the numerous advantages it possesses over conventional drug delivery systems. The blood-brain barrier (BBB) which presents an immense challenge for the delivery of therapeutics by conventional methods has been shown to be permeable to certain NP drug delivery systems.

These advancements have enabled the achievement of high enough drug concentration in the brain to show a therapeutic effect. Additionally, local enrichment, specific targeting, and controlled release kinetics of the encapsulated therapeutics provide significant advantages.

The stability of these nanoparticles (NPs) and the ease with which their surfaces can be modified for targeted delivery are other hallmarks of these systems. Further, NPs have unique optical, electronic, chemical and structural properties which are quite distinct from the larger micro or macro particles.

This review focuses on NPs as means to deliver therapeutics to brain tumors. Nanotherapy using magnetic NPs is currently in clinical trials whereas therapies using ZnO NPs, gold NPs, chlorotoxin conjugated NPs, and photosensitizer NPs are still in the pre-clinical phase.

Also discussed is a novel niosome-chitosan drug delivery technique which we are currently evaluating *in vitro* in glioma cells.

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## 1. INTRODUCTION

In the last half decade, there has been a significant increase in the amount of research on novel ways to both aggress and improve the treatment of glioma. To that effect, the number of publications in the area of NP based drug delivery therapies for the treatment of glioma increased three fold from 2007 to 2011.

In fact, the number of publications in the first quarter of 2012 has almost reached the total number of papers in this research area as in year 2011. [1] The successful treatment of malignant gliomas has been hindered by their perplexing physiology and pathology, and because of their intracerebral location.

Therefore, nanotechnology has been utilized to design effective delivery of therapeutics to the central nervous system (CNS) and to gain tumor access through the blood-brain barrier (BBB) overcoming spatial and temporal constraints that other methodologies may encounter. [2-4] The use of nanoparticles (NPs) for drug delivery is one approach that has been enticingly studied owing to the fact that nanomedicine has demonstrated to transport active drugs to the tumor site in both *in vivo* and *in vitro* studies. [5, 6]

NPs for drug delivery and as imaging molecules are organic and inorganic colloidal materials in the nanoscale range (1nm ( $10^{-9}$ m) -  $\frac{1}{4}$   $\mu$ m), which include particles made of polymers, [7, 8] lipids, [9-12] non-ionic surfactants, [13-17] metal oxides, [18-20] and gold and platinum [17, 21-27]. NPs offer several advantages over traditional methods of drug delivery such as stability of the drugs, containment of off-target delivery, higher activity onsite and inside the tumors, and controlled release kinetics.

In most instances, direct visualization of NPs' position, structure, and effects have also been improved with high resolution imaging techniques such as confocal microscopy, magnetic resonance imaging (MRI), Xenogen imaging, scanning and transmission electron microscopy (SEM and TEM), atomic force microscopy (AFM), scanning tunneling microscopy (STM), Fourier transform infrared spectroscopy (FTIR), and X-ray diffraction. [3, 28-30] While the BBB and blood-tumor barrier (BTB) are permeable to certain NPs such as spherical NPs between 7 nm to 10 nm in diameter (12-20 nm is upper limit of pore size in BBB/BTB), the CNS entry of polymeric NPs delivered systemically is dependent on their receptor-mediated transcytosis. [31, 32] Certain surface modifications of polymeric NPs can enhance this process. For example, it is believed that polysorbate 80, also known as Tween 80 (emulsifier and surfactant), applied in formulation of some NPs interacts with endothelial cells of BBB enhancing CNS penetration of systemically delivered polymeric NPs. [33, 34]

The surface characteristics of NPs can be readily modified by attaching specific receptor ligands allowing targeted delivery and recognition of and binding to receptors on the cell surface. [35] Receptors such as integrins and cadherins are known to have a strong interaction with biological ligands allowing for specific targeting of the cancer cell membrane. [10, 36] Other tumor targeting molecules include chlorotoxin (CTX), a peptide derived from scorpion venom. [35] Similarly, the membrane adhesive properties of some NPs made of biocompatible materials such as chitosan have been investigated for their specific affinity to tumor cell membrane proteins e.g., cell surface associated Mucin1, (MUC1). [6, 37]

The presence of the BBB is an obstacle for many therapeutic agents to reach glioma. Over time several methods for the systemic or localized delivery have been developed as illustrated in Figure 1. These include the systemic delivery following osmotic alterations of

the BBB previously used clinically [38, 39] and studied *in vivo* by our laboratory, [40–43] as well as localized delivery e.g., minipumps, catheter infusions or Gliadel<sup>®</sup> wafers used clinically [44] and evaluated *in vivo* and *in vitro*. [41] Often these methods were associated with serious side effects and did not result in a significant increase in patient survival time.

Liposomal preparations have been used mostly for CNS delivery. However, recently polymer NPs were shown to be effective for the localized intracerebral delivery to intracranial tumors in animal models using convection enhanced delivery (CED). [45]

In general, a polymer NP delivery system for optimal CNS delivery should include the following characteristics: biocompatible/biodegradable, non-immunogenic, non-toxic, in the range of 100 nm in diameter, ready for surface modifications, and cost-effective.

In addition, for systemic delivery, the NP system should be stable in blood for prolonged times and for localized delivery, the drug delivery system should be neutral or negatively charged and infused in a slightly viscous/hypertonic solution. [45]

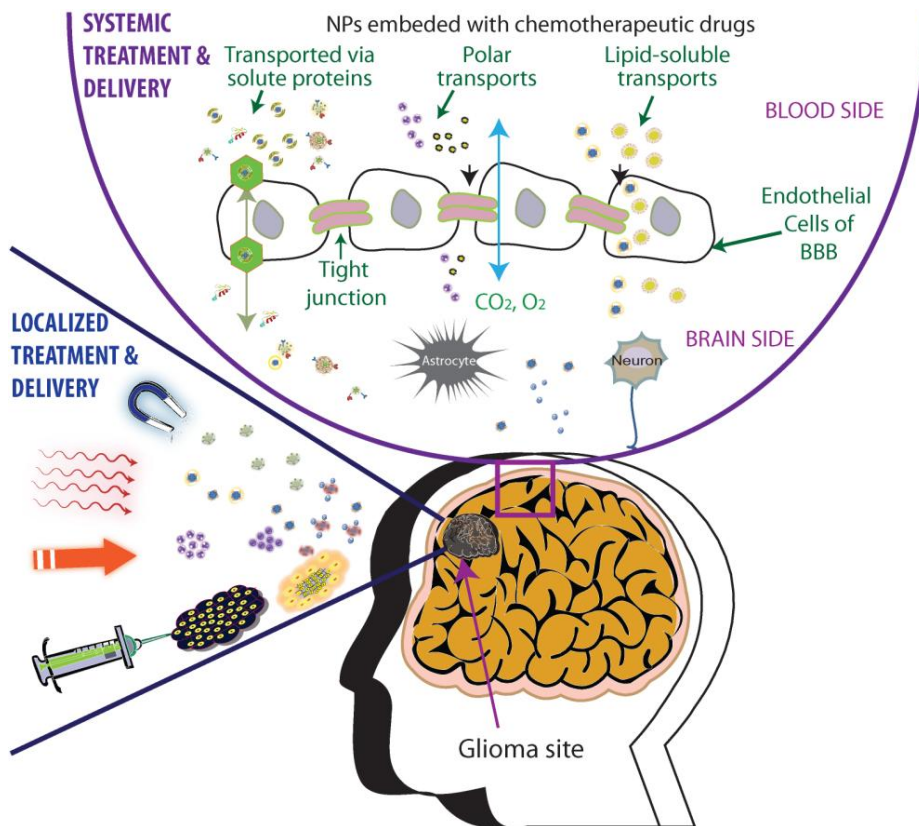


Figure 1. Schematic representation of the protocols for delivering chemotherapeutic drugs and inducing treatment for glioma. Both systemic and localized treatments using nanoparticles (NPs) delivery systems are depicted. NPs are introduced systemically across the blood-brain-barrier (BBB) via osmotic alterations, specific targeting or using convection enhanced delivery (CED). Similarly, NPs can also be used to treat glioma locally utilizing magnetic fields, external energetic radiation or injecting therapeutic viscous solutions (niosomes/chitosan “smart packaging” system) directly to glioma sites.

In this chapter, several protocols for NP delivery currently evaluated for the treatment of glioma are reviewed (Figure 1). These protocols include NP systems utilized either as carriers or directly targeting the tumors. They include magnetic NPs, photodynamic therapy (PDT), zinc oxide NPs, chlorotoxin coated NPs, gold NPs, and a “nano-smart packaging” drug delivery system using niosomes and a chitosan polymeric network. Depending on the available data, this review describes *in vitro* and *in vivo* studies and in some instances it refers to clinical trials.

## 2. MAGNETIC NANOPARTICLES (MNPs)

Magnetic nanoparticles (MNPs) consist of a magnetic core encapsulated within a polymeric or metal outer shell. The outer shells can further be modified for tumor targeting and/or therapeutic carrier multifunctionality. [46] MNPs may also consist of a porous polymer with precipitated MNPs within the pores. [46, 47] In most instances, MNPs are injected into the blood stream near the target site and a magnetic field is introduced over the target site resulting in an accumulation of NPs at the tumors. [47, 48]

The size of these particles plays an important role in systemic delivery since they can pass readily through the capillaries and can also penetrate the tumor cell membranes. [46-48] As a general guideline, the entire size of the NPs should be sufficiently small to elude splenic filtration and sufficiently large to evade renal clearance. It has been reported that the optimal sizes are more than 5.5 nm and less than 200 nm. [46]

Materials that make up the magnetic core typically contain iron oxides like magnetite  $\text{Fe}_3\text{O}_4$  and maghemite  $\text{Fe}_2\text{O}_3$ . [49] Iron metal oxides such as  $\text{CoFe}_2\text{O}_4$ ,  $\text{NiFe}_2\text{O}_4$  and  $\text{MnFe}_2\text{O}_4$  [46] and metal oxides as  $\text{MnO}^y$  are also used as materials for the NP core.

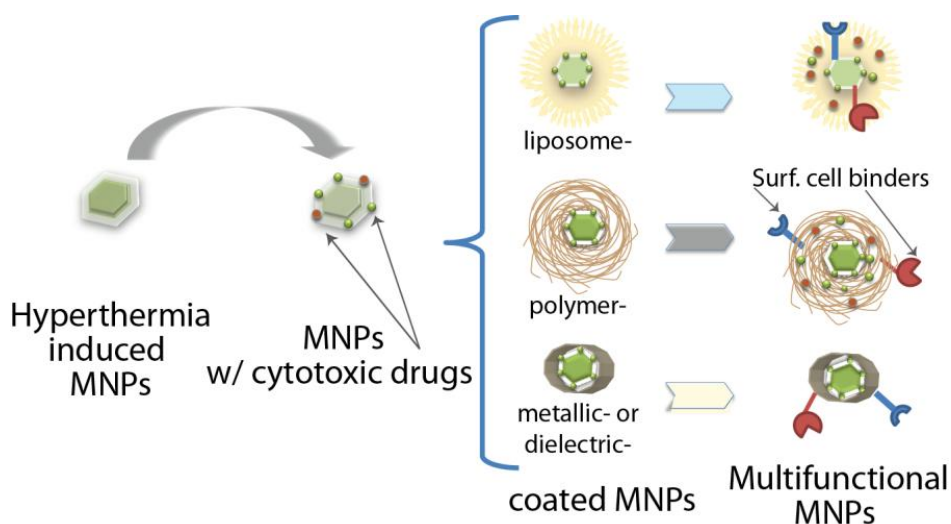


Figure 2. Schematic illustrations of MNPs. Layers of liposomes, polymers, metals or dielectrics enhance their uptake or solubility. These layers can be conjugated with tumor cell targeting molecules to directly attach to cell receptors, and/or engulf therapeutic agents to provide multi-functionality.



However, since cobalt (Co), nickel (Ni) and manganese (Mn) can cause toxicity if delivered over extended periods of time, they have to be coated with an impermeable material e.g., phospholipids, polyethylene glycol (PEG), polyvinyl alcohol (PVA), and poly-L-lactic acid (PLA), to prevent leaching and to make them feasible for biomedical applications. [46, 47, 49] Coating of the magnetic metal core is essential to prevent the toxicity arising from the compounds formed during oxidation of the metals. [18, 51] The choice of the coating material depends on the application and the functionality required for specific treatments.

Natural polymers such as dextran, starch, albumin and synthetic polymers such as PVA, PEG are common candidates for coating of the magnetic metal core. [18, 51-53] Materials like silica and gold are also used to provide impervious coating. [52] Figure 2 shows a schematic representation of the topography of MNPs.

## 2.1. *In Vitro* Studies with MNPs

For the *in vitro* studies, the type of magnetic particles commonly employed are magnetite or super-paramagnetic NPs of iron oxide (SPION) and the coating materials used are liposomes [54, 55] and PEG [53, 56]. Although the core structure of MNPs employed for *in vivo* studies and clinical trials are the same as those in *in vitro* studies, the coating materials are different. For the *in vivo* studies, SPION coated with carboxydextran [52], aminosilane [52], and carboxymethylcellulose [57] are commonly utilized whereas for clinical trials SPION coated with aminosilane [52] are used. MNPs can be used to destroy cancer cells in a number of ways. They can either be used as carriers to attach cytotoxic drugs or they can be used to induce hyperthermia at the tumor sites. [57, 58] Attachment of cytotoxic drugs to the MNPs enables their release at the tumor site, [47, 49, 58] and is akin to other NP delivery methods with the exception that the targeting is accomplished through the application of an alternating magnetic field (AMF).

Magnetic hyperthermia (MHT) is a technique wherein the tumors are subjected to high temperatures emitted by MNPs through the application of an AMF. This therapy utilizes the fact that tumor cells are more sensitive to increased temperature than normal cells. [51, 59] Hyperthermia induces changes in the cell function, differentiation and cell growth by altering the activity of enzymes and proteins. Since tumor cells are less resistant to sudden increases in the temperature than normal cells, an increase in the temperature to 41-42°C can induce their necrosis or apoptosis. [51] Other cell functions affected by MHT include reduction in transmembrane transport and inhibition of repair enzymes. [47, 49, 51, 58] Hyperthermia can be induced by heat sources such as electricity, electromagnetic waves, infrared and ultrasound waves. The properties of the MNPs crucial to induce necrosis/apoptosis are size, magnetization and the ability to reach the desired temperature. All play an important role in the selection of appropriate NPs for tumor targeting. [4, 47, 58, 59]

MNPs hold a promising future in the treatment regimen for glioma. However, the magnetic attractive force is greatest at the edge of the body area directly in contact with the magnet.[58] Hence the local position of the tumor and its proximity to the external magnet directly influences the accumulation of MNPs at the tumor site. [58] Deep intracranial locations might hinder the effectiveness of the external magnetic field in the targeting of MNPs to the tumor site.

The first *in vitro* glioma study which used MHT was reported by Shinkai et al. (1996). [60] In this study MNPs i.e., magnetite coated with cationic liposomes, (MCL) were evaluated in the treatment of rat T-9 glioma cells. The study showed that the NPs were readily taken up by the T-9 cells and 80% absorption was reached in 60 minutes. In addition, the application of an AMF increased the temperature of the MCL NPs to 42.6°C within 20 minutes. Viability of these T-9 cells decreased with time after AMF application and total cell death was achieved in 40 minutes. In another similar study also using T-9 cells, Ito et al. (2003) reported that MHT was responsible for not only the lysis of these tumor cells, but additionally resulted in the activation of heat shock proteins (HSP70) inducing an antitumor immune response.[54] A separate study evaluating MHT in human glioblastoma M059 cells was reported by Meenach et al. (2010) using PEG coated magnetite NPs. [8] The authors reported cell death in the region of the plate containing the NPs where a temperature of 63°C was reached, whereas no cell death was reported in the surrounding regions.

In another *in vitro* study by Corem- Salkmon (2011), SPION maghemite NPs conjugated with methotrexate (MTX) coated either with human serum albumin (HSA) or PEG for convection enhanced delivery (CED). [61] The HSA coated NPs were found superior to PEG coated NPs when tested *in vitro* and *in vivo* showing no toxicity, good distribution and longer clearance time. *In vitro* the cell-kill efficacy of HSA coated MTX-loaded particles was the same as that of with MTX alone while PEG coated MTX-loaded particles had low efficacy. Trapani et al. (2011) used MTX-conjugated NPs coated with chitosan or glycol chitosan (GCS) and additionally with a layer of Tween 80 and evaluated their efficiency *in vitro* in C6 rat glioma.[62] They found that the distribution efficiency of the NPs was increased by the addition of Tween 80 surfactant even when used at low concentrations. In a recent study by Baskin et al. (2012), “nanosyringes” composed of hydrophilic carbon cluster (HCC), which are nanovectors that could be magnetized for directing therapeutics delivery and are conjugated with antibodies for enhancing drug delivery, were used in combination with chemotherapeutics for direct application to glioma cells. [63] The 20 nm “nanosyringe” was capable of selectively killing malignant glioma cells by delivering combined chemotherapy drugs directly to these cells *in vitro*.

## 2.2. *In Vivo* Studies with MNPs

One of the first *in vivo* studies studies by Yanase et al. (1997) evaluated the antitumor activity of MCL NPs, where the NPs were preincubated with T-9 glioma cells before subcutaneous (sc) implantation in rat model.[64] MHT treatment was used followed by AMF showing that a temperature of 45°C could be reached in the tumor tissue within 20 minutes after the magnetic field was applied. Interestingly, the area surrounding the tumors remained at 35-36.5°C. Complete inhibition of tumor growth *in vivo* occurred after three AMF applications of 60 minutes each at 12 hour intervals. In addition, in the follow up study it was also shown that even higher tumor inhibition could be obtained when MCL was injected intratumorally. [55] Although the number of AMF applications remained the same as in the initial study, the duration of each AMF application was reduced to 30 minutes at 24 hour intervals resulting in complete tumor regression. Histochemistry revealed that MCL was evenly distributed in the tumor tissue and their location coincided with necrotic regions. The same authors showed in a subsequent *in vivo* study in T-9 rat glioma model, that 3 sessions as

short as 30 minutes of AMF application resulted in the presence of immunocytes (CD4, CD8 and NK cells) in the MHT treated tissue indicating immune response. [12] A complete tumor regression with no recurrence for a period of 3 months was also observed [12]. Further, increase in heat shock protein 70 (HSP70) expression was reported for the *in vivo* system as previously shown *in vitro* by Ito et al. (see section 2.1). [54]

An improved type of MCL was reported by Le et al. (2001) who conjugated MCL with G22 monoclonal antibody and fragment of antibody (Fab), which specifically binds to MUC1 antigen on glioma cells therefore increasing specificity of the NPs. [65] This resulted in a 7 fold higher uptake of MCL injected intratumorally in a sc grown U251 human glioma in nude mouse model. Additionally, a rapid increase in temperature to 40°C was noticed within 5 minutes of the application of AMF and tumor growth was inhibited for 2 weeks. In 2002, Ohno et al. used rat intracerebral T-9 glioma model to investigate the effect of intracerebrally (stereotactically) delivered SPION coated with stick-type carboxymethylcellulose (CMC). [57] Application of an AMF increased the temperature of the NPs to 44°C within the tumor without affecting the surrounding normal cells. Additionally, three sessions of AMF application resulted in ~3 fold increased survival of tumor bearing animals compared to controls. Similar studies by Jordan et al. (2006) where SPION coated with aminosilane were used in RG -2 intracerebral rat glioma model also resulted in increased animal survival ~4.5 fold over controls. [66]

A different approach was employed by Rabias et al. (2010) who used NPs made of maghemite and coated with dextran to bestow a high surface charge. [67] The high surface charge aided in colloidal NP stability and further assisted in their complete dispersion. These NPs were injected intratumorally into the exocranial rat model of C6 glioma. In this study, histochemistry showed again damage of the tumor tissues with no discernible changes to the surrounding normal cells.

### 2.3. Clinical Studies with MNPs

Maier-Hauff et al. (2007) have been the pioneers in Phase I clinical studies using MNPs. They used SPION coated MNPs with aminosilane in 14 patients with glioblastoma multiforme. [68] The aim of this study was to evaluate the feasibility and tolerability of this new thermotherapy. Each patient received neuronavigationally controlled intratumoral instillation of approximately 0.2 mL of magnetic fluid per cm<sup>3</sup> of the tumor volume and was subjected to six semi-weekly sessions with each thermotherapy session of 1 hour. MHT was combined with fractionated stereotactic radiotherapy in these patients as combined hyperthermia and radiotherapy have shown benefits in previous clinical studies. [69] In general, this new thermotherapy was tolerated well with only minimal side effects in some patients. The patient survival time was not an endpoint of this study, but it was noted that one patient (out of 14) was still in remission with a follow-up of 28 months after treatment. The other 13 patients died either due to the tumor progression (9 out of 14) or due to the side effects related to the long term glucocorticoid prescription (4 out of 14) which is commonly used in glioma patients. Since the NPs did not produce significant side effects, it was concluded that coated MNPs are safe for patients with glioblastoma. [68]

In the follow up, single-arm Phase II clinical study, a total of 65 patients with the recurrent glioblastoma were enrolled. [70] The patient overall survival (OS) rate after

diagnosis of first recurrence of the tumor (OS2) was evaluated. The second objective for this study was to investigate patient survival after diagnosis of primary tumor (OS1). A median overall survival of 13.4 months and 23.2 months was reported respectively for OS2 and OS1. These OS were significantly increased when compared to the historical reference for median survival time in glioblastoma patients treated with chemotherapy, specifically temozolomide. [71] The extension for OS2 of 13.4 months in this study compared with 6.2 months in the reference group (gain of 7.2 months) and extension for OS1 of 23.2 months compared with 14.6 months (gain of 8.6 months) were reported. [71] Due to the increase in overall median survival combined with moderate side effects and no complications, magnetic thermotherapy was concluded to be a recommendable treatment option for patients with glioblastoma multiforme. The postmortem pathological examination of three patients who had undergone MHT in Phase I clinical studies by Maier-Hauff et al. (2007) [68] revealed that necrosis was confined to the local region of instillation of NPs. [72]

In summary, magnetic particles showed greater absorbance by glioma cells than normal cells resulting in glioma cell death. In addition, coated MNPs have already been successfully used in *in vivo* and are currently being developed for clinical applications.

### 3. PHOTODYNAMIC THERAPY (PDT) USING NPs

Photodynamic therapy (PDT) involves selective uptake of a photosensitizer molecule to the tumor tissue immediately followed by irradiation with light of appropriate wavelength for a given photosensitizer. [73-75]

The irradiation leads to activation of the photosensitizer followed by its binding to molecular oxygen resulting in production of a singlet oxygen species ( $^1\text{O}_2$ ), [73] The singlet oxygen, which is cytotoxic has a short lifetime of only around 40 nsec [76, 77] and the effects of  $^1\text{O}_2$  are confined within the cell at the place in which it was formed. [77, 78]

A number of mechanisms have been proposed for the photodynamic therapy-induced destruction of the tumors. Cell death due to cellular toxicity is the main contributor, [73, 77, 79] provided the photosensitizer produces lethal concentrations of singlet oxygen after irradiation. Figure 3 illustrates the simple mechanisms of action that NPs undergo via PDT. Commonly used photosensitizers in clinical studies of glioma are derivative of porphyrin such as Haematoporphyrin (HpD), [73, 78] Photofrin, [75] Photosan, [77, 78] and Photocan. [77, 78] Photosensitizers administered alone often have the adverse effect of causing prolonged cutaneous photosensitization, [7, 9, 73, 80, 81] which leads to skin damage when exposed to direct sunlight. Photosensitizers are also known to cause vascular damage and inflammation. [73, 78, 79]

To minimize the potential for causing above mentioned side effects and to achieve the targeted delivery of high amounts of the photosensitizers to the tumor, the photosensitizers are encapsulated within liposomes. [9, 80] Most commonly reported in glioma studies are polymeric NPs made of polyacrylamide. [7, 73, 81]

Another approach in PDT is the application of two-photon photodynamic therapy (TP-PDT), a method that uses a dye molecule which can absorb two less energetic photons leading to the state of excitation. [82] The advantage of this method is the ability to reach deeper tissue regions. The excitation of the photons occurs in the near infra-red region, preventing

their absorbance by tissues. This approach was evaluated by Gao et al. in 2006, who demonstrated that rat C6 glioma cells exposed to TP-PDT showed cell damage within 10 minutes. [82] Although PDT has been used in clinical trials since the 1980's, and several drugs have been approved by the FDA like Photofrin (porfimer sodium) Levulan (5-aminolevulinic acid or ALA), and Metvix (methyl aminolevulinate [MAOP]) the PDT approach applying NPs is still in the preclinical phase for glioma. In the first reported study using PDT NPs, Kopelman et al. used polyacrylamide NPs coated with PEG and with encapsulated Photofrin, which were evaluated in 9L rat glioma cells. [87]

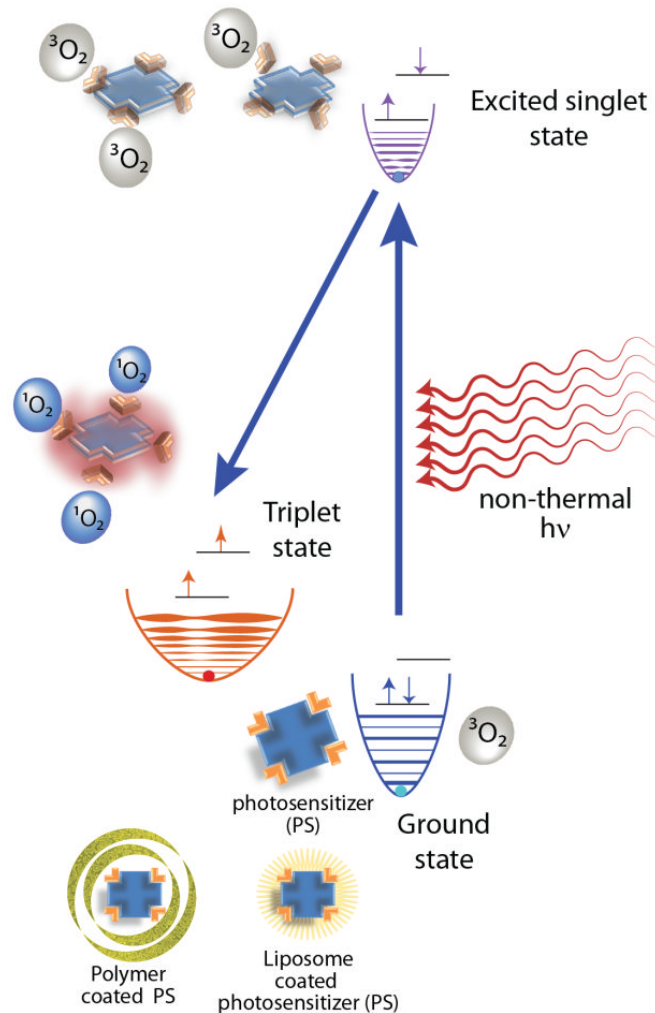


Figure 3. PDT is a technique where photosensitive (PS) compounds are encapsulated in either liposomal or polymeric coatings to enhance their uptake and solubility across the BBB and in the CNS. Non-thermal light at a specific wavelength is used to activate the PS, which is already closed by the tumor site. The PS is excited to a highly unstable singlet state that rapidly moves to a PS triplet state. In this energy level, any of the ground-state triplet oxygen molecules ( $^3\text{O}_2$ ) nearby the PS receive the energy transferred from the PS triplet state passage, and transform into a cytotoxic, reactive singlet oxygen species ( $^1\text{O}_2$ ). These singlet oxygen species are capable of killing tumor cells by the induction of apoptosis or necrosis. [83-86]

This *in vitro* study showed that the photosensitizers produced sufficient levels of  $^1\text{O}_2$  to induce glioma cell death. Additionally, in the rat model of intracerebral 9L glioma, intravenous injection of the NPs followed by photoirradiation resulted in massive regional necrosis of the tissue when compared to the photoirradiated controls.

In 2006 Reddy et al. used the same *in vivo* glioma model to analyze the effect of intravenously delivered iron oxide/Photofrin-encapsulated F3-targeted NPs. [73] Tumor vasculature targeting F3 peptide used in this study enhanced the delivery of the NPs to glioma resulting in increased animal survival. The study showed an average survival of 33 days in PDT - Photofrin NPs treated mice as compared to an average of 7 days in the control group.

In summary, PDT-NPs have shown selective uptake by glioma in both *in vitro* and *in vivo* studies and have demonstrated their ability to significantly decrease glioma growth. The usefulness of this new approach should be further investigated in clinical trials of suitable formulations of photosensitizer containing NPs.

## 4. ZINC OXIDE (ZNO) NPs

Zinc oxide (ZnO) NPs have found use in biomedical applications because of their photocatalytic and photo-oxidizing capacity. [88] They work by bombarding the outer surface of malignant cells through the release of reactive oxygen species (ROS) generated by a redox reaction when NPs are exposed to UV light (Figure 4). [89, 90] The oxygen species generated by the NPs lead to the destruction of cellular proteins, DNA and lipids hence resulting in apoptosis. [89, 91] Additionally, the size of the ZnO NPs was reported to correlate with their toxicity, [92] e.g., smaller sizes showing higher toxicity. [88] However, ZnO NPs have a tendency to form clusters in aqueous solution. This occurs due to the presence of high superficial energy on the surface of the particles which leads to increased aggregation, and therefore, decreased interactions with the surrounding environment. [19, 20, 88] Some studies have reported that narrowed size distributions of dispersed NPs in an aqueous solution, are needed to prevent cluster formation. However, such dispersions contained residual toxic materials which limited their biomedical applications. [20, 88] This limitation was overcome by Taccola et al. (2011) who studied the aggregation of ZnO NPs in various reagents and found that ZnO NPs dispersed in Arabic gum resulted in a highly stable and concentrated dispersion. [20] In general, ZnO NPs have been shown to be non-toxic for normal dividing cells whereas their toxicity is prominent in rapidly dividing cancerous cells. For example, Hanley et al. (2008) reported that rapidly dividing tumor T-cells showed preferential binding and selectivity towards the ZnO NPs as compared to normal cells. [93] Ostrovsky et al. reported in 2009 high binding of the ZnO NPs resulting in toxicity for human glioma cell lines (A172, U87, LN2308, LN18, and LN229), [88] while there was no cytotoxicity observed in normal human astrocytes (RWPE-1) even at 10 mM of ZnO NPs. Similar findings by Wahab et al. in 2011 using human U87 glioma cell line showed that treatment with ZnO NPs increased their propensity for cytogenic damage in these cells. [94] In addition, the studies by Ostrovsky et al. showed that the amount of reactive oxygen species produced by the ZnO NPs within glioma cells far exceeded that detected within normal cells again confirming ZnO's mode of action through generating ROS. [88]

In summary, ZnO NPs have demonstrated selective toxicity for glioma *in vitro* suggesting further evaluations *in vivo* to determine their potential as treatment modalities. Their mechanism of action produces a reactive amount of oxidizing species capable of destroying the surface of a tumor cell in close proximity. This action is due to the effect of UV radiation capable of triggering the photocatalytic activity of this particular non-toxic, biocompatible metal oxide NPs.

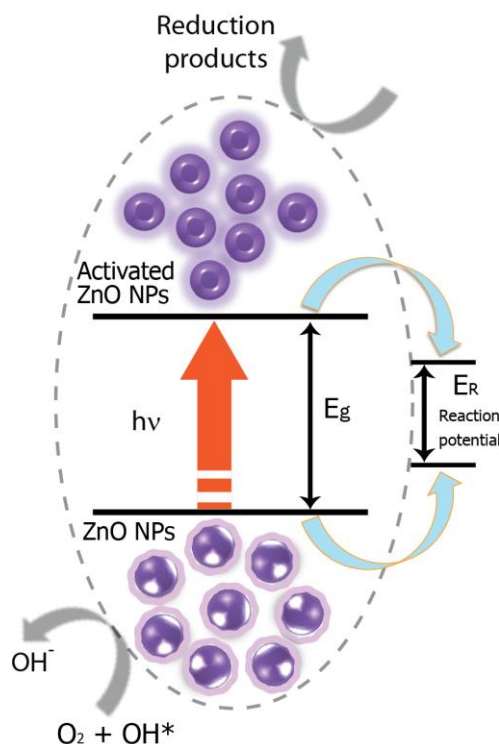


Figure 4. Schematic representation of the overall reduction mechanism for attacking glioma cells via photocatalysis of ZnO nanosized particles. The ZnO NPs oxidize neighboring  $O_2$  atoms into reacting  $OH^-$  anions. This action produces enough energy ( $E_R$ ) to reduce any carbon species (i.e., organic compounds) in their vicinity in which the components of the tumor cell surfaces are the most vulnerable.

## 5. CHLOROTOXIN (CTX) CONJUGATED NPs FOR TARGETED DELIVERY

Chlorotoxin (CTX), found in the venom of the scorpion *Leiurus quinquestratus*, is a 36 amino acid peptide. [35, 95, 96] It was first shown that CTX is an inhibitor of small conductance chloride channels in the cell membrane, [95, 97] and therefore, inhibits glioma cell invasion. [98]

CTX also inhibits the enzymatic activity of metalloproteinase-2 (MMP-2) endopeptidase, which is involved in cell migration. [35, 96, 99] It is believed that CTX preferentially binds to the membrane bound MMP-2 upregulated on the cell surface of glioma and other tumor cells. [27, 96] Recently, annexin A2 was identified as another possible binding partner in many

human tumor cell lines. [100] CTX was shown to bind selectively to human glioma and other tumors of neuroectodermal origin as well as to many other transformed cells. However, it did not bind to over 15 normal human tissues and normal non-transformed cells such as neurons, astrocytes and fibroblasts. [101] In addition, it was recently reported that CTX' cellular entry and localization in human glioma cells differs from that found in normal cells. [102] Targeted delivery of radiolabeled synthetic chlorotoxin (TM-601) was already evaluated in Phase I in patients with recurrent glioblastoma and other solid tumors and Phase II clinical studies in patients with recurrent glioblastoma. [103]

Combining chlorotoxin and NPs has emerged as a novel approach which combines the advantages of both entities resulting in enhancing the efficiency of drug delivery. [30, 35] Antibodies have traditionally been used for cellular targeting, but their application in glioma has been limited because of their large size, limited tissue penetration and cellular uptake. [104] In contrast, a smaller oligopeptide such a CTX can be readily taken up by the cells. [97, 102] In recent years, CTX has been conjugated to a wide variety of NPs while retaining its targeting properties and used for drug delivery to glioma based on its selectivity for tumor cells. [30, 97] Figure 5 lists some of the applications related to nanoscaled systems that have benefited by conjugating CTX. In a study by Sun et al. in 2008, iron oxide NPs conjugated to the chemotherapeutic agent methotrexate (MTX)-CTX was reported to have greater accumulation in 9L rat glioma *in vitro* than controls and eliciting a more efficient toxic effect on rapidly dividing glioma cells. [30] A similar observation was made *in vivo* in the flank xenograft model of 9L rat glioma when the intravenously delivered CTX-bound chemotherapeutics targeted tumor cells, and remained inside the tumor for almost two weeks. [30]

Another example of a chlorotoxin conjugated NPs is a magnetic and fluorescent NP system (MFNP-CTX) developed by Wan et al. in 2010 by incorporating magnetite NPs and a fluorescent dye conjugated to CTX. This NP system combines magnetic properties with cellular targeting (through CTX) leading to destruction of glioma cells. [106] The study reported high binding specificity of the NPs to human U251-MG glioma cells by a CTX-receptor-mediated endocytosis mechanism.

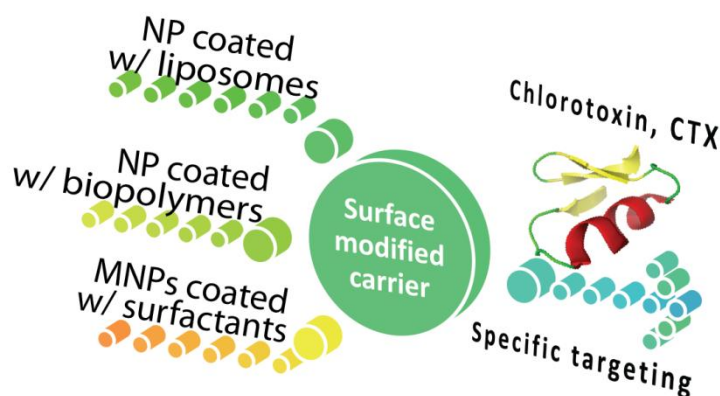


Figure 5. Chlorotoxin (CTX) has been drafted onto NPs surface via peptide affined binding with synthetic biopolymers (e.g., PVA, PEG or chitosan), liposome surface modification (using phospholipid polar groups), and on surfactant modified NPs (using peptide-receptor spacers). [105]



In addition, Huang et al. in 2011 reported the selective uptake by the human glioblastoma C6 cell line of CTX NPs (made from polyimidoamine, PAMAM and PEG and evaluated for gene therapy). [107] CTX itself was taken up by C6 glioma cells, but not normal cell controls [human embryonic kidney (HEK) 293 cells]. In addition, the conjugation of NPs with CTX was reported to significantly increase the cellular uptake of PAMAM-PEG-CTX in C6 glioma cells when compared to uptake of NPs without CTX. Mice with intracerebral C6 glioma were treated systemically via tail vein with CTX NPs containing DNA. Histochemistry analysis of intracerebral tumor tissue showed a greater distribution of CTX containing PAMAM-PEG-CTX/DNA NPs compared to NP controls. Furthermore, median survival time of mice administered with PAMAM-PEG-CTX/DNA was reported to be 59.5 days compared to 49 days for the control group that received commercial temozolomide only.

In another study, Kievet et al. in 2010 investigated intravenously (iv) administered CTX labeled NPs for targeted gene therapy in human C6 xenograft tumors in nu/nu mice. [108] The NPs consisted of an iron oxide core, which was coated with a copolymer of PEG and polyethylenimine (PEI) with bound green fluorescent protein encoding DNA and attached CTX. This study found that the uptake of NPs-CTX/DNA by C6 glioma *in vivo* was significantly higher compared to control NPs. Interestingly, the accumulation of the CTX NPs occurred specifically in the cells inside the tumor but not at its periphery occupied mostly by the normal cells. In another study, Veisheh et al. (2009) showed the ability of a nanoprobe conjugated with CTX to cross the BBB and specifically target a tumor in a transgenic ND2:SmoA1 mouse model that closely resembles human medulloblastoma. [109] This nanoprobe consisting of iron oxide coated with a copolymer of PEG and chitosan showed an innocuous toxicity profile and a sustained retention in tumors. Additionally, the NPs were reported to accumulate in neoplastic regions and not in healthy tissues. Overall, current findings show that CTX is suitable for targeting NPs to brain tumors.

## 6. GOLD NANOPARTICLES (GNPs)

GNPs have been repeatedly evaluated in recent years for labeling, delivery and sensing. Their biocompatibility, surface properties and the ease of surface modification makes them suitable drug delivery vehicles. [110-112] However, GNP size is a key parameter that needs close consideration because of toxicity concerns. GNP penetration across BBB is size dependent and has an upper limit of 12-20 nm. [31, 32] Particles with diameters of 1-2 nm were reported to be highly toxic due to their irreversible binding to key cellular components. [32] Particles with diameters from 3-100 nm do not have significant toxicity if they do not exceed  $10^{12}$  particles/mL. [32] Additionally, organ distribution in rats depends on GNPs size with 10 nm showing wide organ distribution and larger particles showing uptake only in blood, liver and spleen. [113]

Due to their biocompatibility and photo-optical characteristics, GNPs find use in a variety of applications such as in bioimaging, [114, 115] magnetic resonance imaging, [28] X-ray computed tomography, [18, 29] optical imaging, [116, 117] biosensing, [118, 119] photothermal therapy, [120, 121] protein delivery, [122, 123], peptide delivery. [124, 125] small molecule delivery, [126-128] and drug delivery to glioma. [110, 111] For example, Dhar et al. (2010) reported the preferential cellular uptake of GNPs conjugated with a

microbial polysaccharide, gellan gum, by human glioma cells LN-229, whereas no uptake by normal mouse embryonic fibroblast cells NIH3T3 was observed. [112] Furthermore, this new delivery system was shown to be well tolerated in normal animals.

Figure 6 includes two examples where GNPs have been surface modified to be utilized in the treatment of glioma.

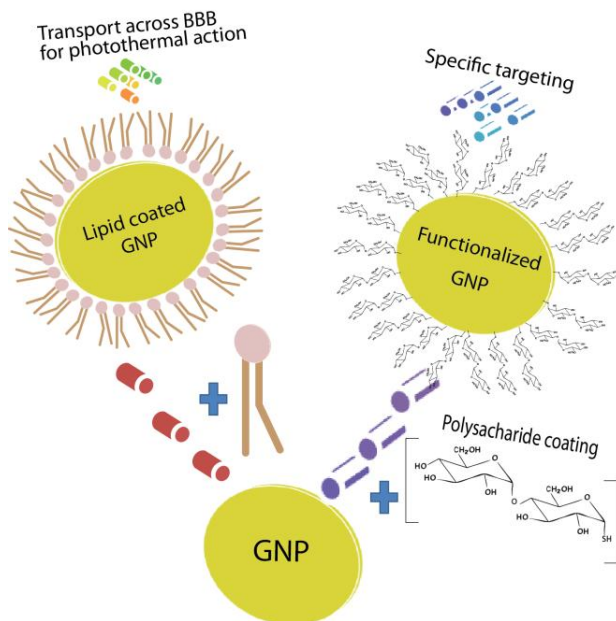


Figure 6. GNP can be readily modified by using thiol type specific binding interactions. Polysaccharides extracted from gellan gums can form a robust coating for specific surface targeting strategies. Likewise, sophorolipid layers can be deposited onto the surface of GNPs to enhance their uptake across BBB and therefore, these metallic NPs can be used for glioma treatment via photothermal activity.

In another study, Dhar et al. in 2011 described a conjugated form of GNP with Sophorolipids (SL), called sophorolipid-gellan, gum-gold NPs (SL-GG-AuNPs). [111] Gellan gum provided reducing and stabilizing effect to the GNP and conjugation of GG-AuNP with SL further increased its efficacy in killing human glioma LN-229 cells. [110] It was also reported that conjugation of doxorubicin hydrochloride (DOX) to SL-GG-AuNPs further increased the cytotoxicity of these NPs and a prominent increase in their efficacy was reported in LN-229 cells and the human glioma stem cell line HNGC-2. [111]

## 7. NIOSOME-CHITOSAN NPs AS A “SMART PACKAGING” DRUG DELIVERY SYSTEM FOR GLIOMA

Our laboratory, Williams et al. in 2010 reported the effects of “smart packaging” system which has a particular response to the physiological temperature, and drug delivery from biodegradable NPs tuned to specific release times and dosage in localized treatments. [129] These NPs are made of non-ionic surfactants (niosomes) embedded in a thermo-responsive

cross-linked chitosan network (Figure 7). [130] Chitosan, which is biodegradable, biocompatible naturally occurring polysaccharide, was used efficiently in niosome preparation and can be surface modified for CNS delivery. [45]

The NP niosome (sizes range between 80 and 300 nm), which is a non-ionic surfactant bilayer vesicle was reported to encapsulate multiple therapeutics with varied polarity in the same NPs. [131] Encapsulation of multiple chemotherapeutic drugs is advantageous especially for cancers requiring combination chemotherapy. [132]

Altering the cross-link mesh size in the chitosan network allowed for fine tuning of the release kinetics of NPs. When the niosomal NP size and the cross-link mesh size were optimized, it resulted in a prolonged, continuous and slow therapeutic release system over a period of 55 days. Transformation from solution to gel (sol-to-gel) with increase in temperature from 25-37°C proved advantageous for use as an injectable system. [129, 133]

Additionally, release kinetics *in vivo* showed a continuous steady release for over 14 days [134] Our most recent *in vitro* study has shown that smart packaging niosome-chitosan system encapsulating paclitaxel conjugated to Bodipy 564/570 has a significantly higher binding affinity to human epithelial ovarian carcinoma (OV2008) than to normal human ovarian epithelial cell lines (Ilow and MCC3). [37]

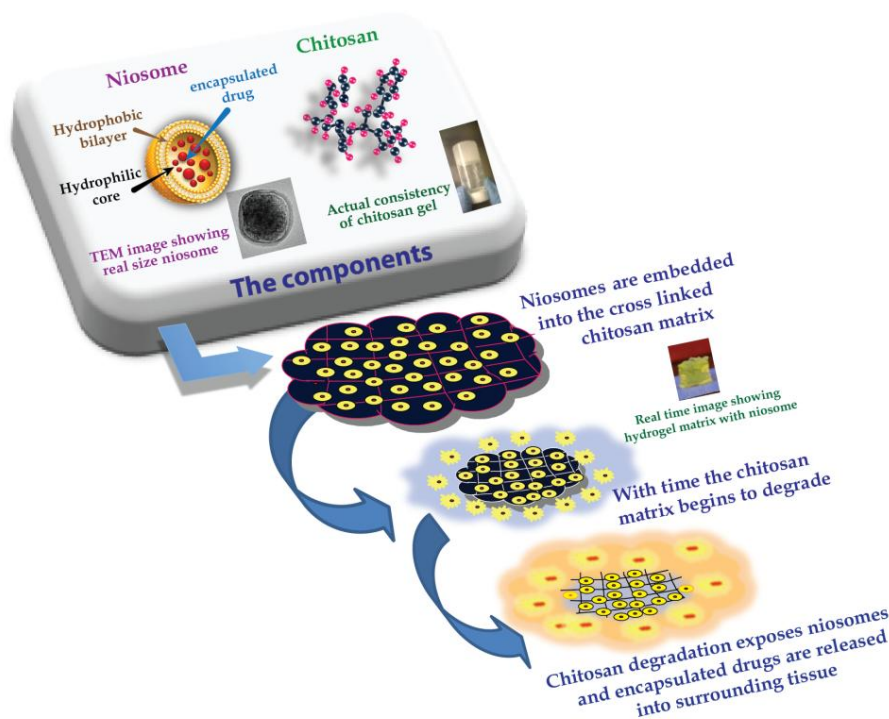


Figure 7. Schematic representation of the “smart packaging” drug delivery system. Chemotherapeutic drugs are encapsulated in either the hydrophobic bilayer or hydrophilic core of the niosomes. These niosomes are subsequently embedded into a thermo-sensitive cross-linked chitosan network, which transforms from a freely flowing liquid at 25°C to a non-flowing solid gel at 37°C. The “smart packing” system is used for localized injection to tumor tissues. With time chitosan matrix begins to degrade which exposes the niosomes leading to the release of encapsulated therapeutics to the surrounding tissues.

This increased affinity is due to interactions with the MUC1 protein, overexpressed on the surface of epithelial carcinoma cell lines. *In vitro* studies showed a higher uptake by the epithelial ovarian carcinoma of the therapeutic paclitaxel-bodipy 564/570 conjugate using this system. [17, 37, 134] We are currently evaluating this technology in human U373 and mouse G-26 glioma cell lines.

Preliminary results using attenuated total reflection Fourier transform infrared (ATR-FTIR) spectroscopy to monitor the interactions of G-26 glioma cells with our chitosan network show the following: (1) a distinct shift in wavelengths assigned to isolated OH groups (between 3700 and 3200  $\text{cm}^{-1}$ ), and (2) a change in the bending conformation for carboxylic groups (centered at 1450  $\text{cm}^{-1}$ ).

These observations are indicative of a specific chemical binding between G26 cells and chitosan compared to the spectra for chitosan alone (data not shown).

## CONCLUSION

New NP based delivery systems as described in this chapter have the potential to lead to the development of improved brain cancer therapies, especially for glioma. Their nanometer size allows for easy passage through the BBB. A wide variety of materials have been examined for the preparation of NPs. The choice of material critically determines their properties like tumor targeting and tumor suppression. Magnetic NPs have shown the greatest promise for a successful treatment technique. They are currently in the clinical studies and so far have shown clear therapeutic effects. Some of the other NPs such as ZnO NPs, PDT NPs, chlorotoxin conjugated NPs, and gold NPs are still in the preclinical phase. An expansion of our understanding and knowledge of NP pharmacology and therapeutic utility is highly desirable and critical for the development of novel nanoscaled systems for the treatment of brain tumor patients. New methodologies, where the whole NP is embedded in a polymeric network, are capable of localized, specific and controlled delivery of one or multiple chemotherapeutic drugs from biodegradable NPs (niosomes) packaged in a chitosan network.

Some *in vitro* and *in vivo* observations describe the toxicity caused by various NPs. [135] For example, CNS damage may be caused by elevated levels of ROS resulting from PDT therapy. Therefore, a controlled release targeted nanodelivery system containing suitable therapeutics for glioma is essential to minimize CNS toxicity, and to cause maximum damage to the tumor tissue.

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